



# AURAK

الجامعة الأمريكية في رأس الخيمة  
AMERICAN UNIVERSITY OF RAS AL KHAIMAH

## LIFE-CHANGING EDUCATION WITH A GLOBAL REACH



AMERICAN UNIVERSITY OF RAS AL KHAIMAH  
CATALOG ACADEMIC YEAR 2023-2024

## **Message from the President**

Dear AURAK Students,

Welcome to The American University of Ras Al Khaimah (AURAK).

I am so pleased that you are joining us here on campus, and it is a great honor to welcome you into the AURAK Community as you embark upon this incredible educational journey.

With the support of His Highness Sheikh Saud bin Saqr Al Qasimi, Ruler of Ras Al Khaimah, AURAK has a rich history of academic excellence. Over the past few years, the university has grown substantially with the development of new academic programs, the hiring of new faculty and staff, and a significant increase of its student population. The trajectory of this growth is still continuing upwards.

The university's mission is to provide its students a transformational, student-centered learning experience. This approach prepares students to become future leaders and entrepreneurs for meaningful, fulfilling and successful lives. AURAK has set a new standard in the Gulf Region for excellence that empowers its graduates to change the world through inspired teaching, research, creative work, and community service.

The AURAK Community is proud of the highly qualified and committed global faculty and staff who are dedicated to serving our students. Those faculty and staff join me in welcoming you as student members of that community.

At AURAK, we strive to create an environment where students can awaken curiosity, expand knowledge, and strengthen skills. In the years ahead, you will have many opportunities to interact with our dedicated faculty and staff and with your classmates. I encourage you to become engaged on campus in as many ways as possible and take advantage of all that AURAK has to offer.

I am delighted that you chose to make AURAK your educational home, and I look forward to seeing you on campus.

Sincerely,

Dr. David A. Schmidt

**President, AURAK**

# **AMERICAN UNIVERSITY OF RAS AL KHAIMAH**

## **ACADEMIC CATALOG**

**2023-2024**

**Effective September 1, 2023**

### **Board of Trustees**

The Board of Trustees is the governing board for American University of Ras Al Khamiah. Members of the Board are:

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Ms. Rania Hafiz Hussein Ali – University Registrar



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## ACADEMIC CALENDAR 2023 – 2024

### Fall 2023

| Month     | Day       | Week    | Activity/Event                                                                |
|-----------|-----------|---------|-------------------------------------------------------------------------------|
| August    | 1-27      |         | Placement Tests / Continuing Student Registration                             |
| August    | 1-27      |         | New Student Course Registration                                               |
| August    | 7         |         | Joining Date for New Faculty                                                  |
| August    | 21        |         | Returning Date for Continuing Faculty                                         |
| August    | 21-27     |         | Faculty Professional Development Workshops, Including New Faculty Orientation |
| August    | 21-27     |         | Continuing Advising & Registration for Returning Students                     |
| August    | 24        |         | New Student Orientation                                                       |
| August    | 25        |         | International Student Orientation                                             |
| August    | 28        |         | First day of Class                                                            |
| August    | 28-Sept 3 | Week 1  |                                                                               |
| September | 4-10      | Week 2  |                                                                               |
| September | 9         |         | Graduation Date for Summer 2023                                               |
| September | 11        |         | AURAK Admissions Application Deadline                                         |
| September | 11        |         | Last day to Add/Drop                                                          |
| September | 11-17     | Week 3  |                                                                               |
| September | 18-24     | Week 4  |                                                                               |
| September | 24        |         | Census Date for Fall                                                          |
| September | 25-Oct 1  | Week 5  |                                                                               |
| September | 29        |         | Prophet's Birthday*                                                           |
| October   | 2-8       | Week 6  |                                                                               |
| October   | 9-15      | Week 7  |                                                                               |
| October   | 16-22     | Week 8  |                                                                               |
| October   | 22        |         | Mid Term Grades Report Due                                                    |
| October   | 23-29     | Week 9  |                                                                               |
| October   | 29        |         | Last date to withdraw from a course with a grade of "W"                       |
| October   | 30-Nov 5  | Week 10 |                                                                               |
| October   | 30        |         | Advisement Begins For Spring 2024                                             |
| November  | 6-12      | Week 11 |                                                                               |
| November  | 6         |         | Early-Registration For Spring Semester begins                                 |
| November  | 13-19     | Week 12 |                                                                               |
| November  | 20-26     | Week 13 | Student Course Evaluations                                                    |
| November  | 27-Dec 3  | Week 14 |                                                                               |
| December  | 1         |         | UAE Commemoration Day                                                         |
| December  | 2-3       |         | UAE National Day                                                              |

|          |          |         |                                                                  |
|----------|----------|---------|------------------------------------------------------------------|
| December | 4-10     | Week 15 |                                                                  |
| December | 6        |         | Commencement Ceremony (2022 – 2023)                              |
| December | 10       |         | Last Day of Classes                                              |
| December | 11-20    | Week 16 | Final Examination Period                                         |
| December | 24       |         | Grades Available and Student Course Evaluation Results Available |
| December | 21-Jan 7 |         | Winter Break                                                     |
| January  | 25       |         | Graduation Date For Fall 2023                                    |

## Spring 2024

| Month    | Day         | Week   | Activity/Event                                                    |
|----------|-------------|--------|-------------------------------------------------------------------|
| January  | 2-7         |        | Continuing Advising & Registration for Returning Students         |
| January  | 2           |        | Placement Tests begin                                             |
| January  | 4           |        | New Student Orientation and New Student Registration begins       |
| January  | 4           |        | New Faculty Joining Date<br>New Faculty Orientation               |
| January  | 5           |        | International Student Orientation                                 |
| January  | 6           |        | Continuing Faculty Return Date                                    |
| January  | 8           |        | First day of Class                                                |
| January  | 8-14        | Week 1 |                                                                   |
| January  | 12          |        | Faculty Professional Development Session for Spring Semester 2024 |
| January  | 15-21       | Week 2 |                                                                   |
| January  | 21          |        | AURAK Admissions Application Deadline                             |
| January  | 21          |        | Last day to Add/Drop                                              |
| January  | 22-28       | Week 3 |                                                                   |
| January  | 29 -Feb 4   | Week 4 |                                                                   |
| February | 4           |        | Census Date for Spring                                            |
| February | 5-11        | Week 5 |                                                                   |
| February | 12-18       | Week 6 |                                                                   |
| February | 16          |        | Faculty Professional Development Session for Spring Semester 2024 |
| February | 19-25       | Week 7 |                                                                   |
| February | 26 -March 3 | Week 8 |                                                                   |
| March    | 1           |        | Faculty Professional Development Session for Spring Semester 2024 |
| March    | 3           |        | Mid Term Grades Report Due                                        |
| March    | 4-10        | Week 9 |                                                                   |

|       |           |         |                                                                          |
|-------|-----------|---------|--------------------------------------------------------------------------|
| March | 11        |         | Last date to withdraw from a course with a grade of “W”                  |
| March | 11-17     | Week 10 |                                                                          |
| March | 18-24     | Week 11 |                                                                          |
| March | 25-31     | Week 12 |                                                                          |
| April | 1-14      |         | Spring Break                                                             |
| April | 8 - 11    |         | Eid Al-Fitr*                                                             |
| April | 15        |         | Classes Resume                                                           |
| April | 15-21     | Week 13 |                                                                          |
| April | 15        |         | Fall Advisement Begins                                                   |
| April | 22-28     | Week 14 |                                                                          |
| April | 22        |         | Summer Registration begins – Early Registration For Fall Semester begins |
| April | 22- May 5 |         | Student Course Evaluations                                               |
| April | 26        |         | Faculty Professional Development Session for Spring Semester 2024        |
| April | 29- May 5 | Week 15 |                                                                          |
| May   | 5         |         | Last Day of Classes                                                      |
| May   | 6-16      |         | Final Examination Period                                                 |
| May   | 19        |         | Grades and Student Course Evaluation Results Available                   |
| June  | 25        |         | Graduation Date For Spring 2024                                          |
| June  | 24        |         | Faculty Summer Break Begins                                              |

## Summer Session 2024

| Summer Session Regular Courses (6 weeks) |           |                                                    |
|------------------------------------------|-----------|----------------------------------------------------|
| May                                      | 20- 26    | Course Registration                                |
| May                                      | 27        | Classes Begin                                      |
| May                                      | 27-June 2 | Week 1                                             |
| June                                     | 2         | Last day to Add / Drop                             |
| June                                     | 3-9       | Week 2                                             |
| June                                     | 5         | Census Date                                        |
| June                                     | 10-16     | Week 3                                             |
| June                                     | 16-19     | Eid Al Adha*                                       |
| June                                     | 20        | Classes Resume                                     |
| June                                     | 20-21     | Makeup classes for graduate programs               |
| June                                     | 24 -30    | Week 4                                             |
| June                                     | 30        | Last date to withdraw from a course with grade “W” |



|                                                    |           |                                                                  |
|----------------------------------------------------|-----------|------------------------------------------------------------------|
| July                                               | 1-7       | Week5                                                            |
| July                                               | 1-10      | Student Course Evaluations                                       |
| July                                               | 7         | Islamic New Year*                                                |
| July                                               | 8-14      | Week 6                                                           |
| July                                               | 14        | Last Day of Summer Session for undergraduate programs            |
| July                                               | 20-21     | Makeup classes for graduate programs                             |
| July                                               | 21        | Last Day of Summer Session for graduate programs                 |
| July                                               | 24        | Grades Available and Student Course Evaluation Results Available |
| September                                          | 9         | Graduation Date for Summer 2024                                  |
| <b>Summer Session Internship Courses (8 weeks)</b> |           |                                                                  |
| May                                                | 20- 26    | Internship Course Registration                                   |
| May                                                | 27        | Internship Begins                                                |
| May                                                | 27-June 2 | Week 1                                                           |
| June                                               | 2         | Last day to Add / Drop                                           |
| June                                               | 3-9       | Week 2                                                           |
| June                                               | 5         | Census Date                                                      |
| June                                               | 10-16     | Week 3                                                           |
| June                                               | 16-19     | Eid Al Adha*                                                     |
| June                                               | 20-21     | Possible make-up days for Eid Holiday                            |
| June                                               | 24 -30    | Week 4                                                           |
| July                                               | 1-7       | Week5                                                            |
| July                                               | 7         | Last date to withdraw from a course with grade “W”               |
| July                                               | 7         | Islamic New Year*                                                |
| July                                               | 8-14      | Week 6                                                           |
| July                                               | 8-21      | Student Course Evaluations                                       |
| July                                               | 15-21     | Week 7                                                           |
| July                                               | 22-28     | Week 8                                                           |
| July                                               | 28        | Internship Ends                                                  |
| July                                               | 31        | Grades Available and Student Course Evaluation Results Available |
| September                                          | 9         | Graduation Date for Internship Session 2024                      |

**Note: All Islamic Lunar Holidays will be fixed per announcement by the relevant Ministry.**

## **ABOUT AMERICAN UNIVERSITY OF RAS AL KHAIMAH**

### **Licensure and Accreditation:**

The American University of Ras Al Khaimah (AURAK), located in the Emirate of Ras Al Khaimah, is officially licensed from 1 August 2009 to 15 September 2026 by the Ministry of Education of the United Arab Emirates to award degrees/qualifications in higher education.

AURAK is accredited by the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC) to award baccalaureate and masters, degrees. Questions about the accreditation of AURAK may be directed in writing to the Southern Association of Colleges and Schools Commission on Colleges at 1866 Southern Lane, Decatur, GA 30033-4097, by calling (404) 679-4500, or by using information available on SACSCOC's website ([www.sacscoc.org](http://www.sacscoc.org)).

AURAK has earned accreditation through the Commission for Academic Accreditation of the Ministry of Education for all programs offered. In addition, the School of Engineering has the following program accreditations:

BS in Civil and Infrastructure Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Computer Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Computer Sciences is accredited by the Computing Accreditation Commission of ABET, <http://www.abet.org>

BS in Electronics and Communications Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Industrial Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Mechanical Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Chemical Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Electrical Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Petroleum Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

AURAK's School of Business is accredited by AACSB International—The Association to Advance Collegiate Schools of Business, <https://www.aacsb.edu/accredited>

### **Location and Contact Information:**

AURAK maintains an extensive campus located on American University of Ras Al Khaimah Road in the northern Emirate of Ras Al Khaimah. The university may be contact by mail at PO Box 10021; Ras Al Khaimah, UAE or by phone at +971 7 2210 900.

## **History:**

With the support of His Highness Sheikh Saud bin Saqr Al Qasimi, Crown Prince of Ras Al Khaimah, the university was initially established as George Mason University, Ras Al Khaimah campus (GMU-RAK) in 2005. After George Mason University decided to discontinue its operations in Ras Al Khaimah in 2009, the Crown Prince decreed the establishment of the American University of Ras al Khaimah (AURAK) in the same location. Students who had enrolled in GMU-RAK were given the opportunity to continue at AURAK, transfer to GMU's main campus in Virginia, USA, or transfer to another institution.

Following the short-term presidencies of Professor Sharon Siverts (2009-10) and Dr. Shaukat Mirza (2010-11), AURAK's Board of Trustees appointed Professor Hassan Hamdan Al Alkim as Vice Chancellor in August 2011. Subsequent changes in the organizational structure saw Professor Al Alkim named as President in 2014. In August, 2023, Professor Al Alkim retired from AURAK, and Dr. David A. Schmidt succeeded him as President.

Since its establishment, the university has continued to develop new academic programs, attract new faculty and staff, and significantly increase its student enrollment. This rapid change has required an adaptable organizational structure to accommodate the effective and efficient delivery of programs and services. Consequently, in 2013, Vice President positions were added resulting in the organization of units in three divisions under the President, Senior Vice President Academic Affairs and Student Success and Vice President of Operational Excellence and Financial Management.

Today, AURAK's strategy focuses on cultivating excellence within the existing curricular offerings while improving policies and procedures across the institution. Student enrollment is projected to grow over the next several years. It is also expected that, with an increasingly strong global economy, AURAK will capture the attention of educators, economists, business entrepreneurs, and other professionals who want to take advantage of the programs and services the institution offers, meeting national and regional workforce needs and earning the university regional and global recognition.

## **Vision:**

Through inspired teaching, research, creative work, and community engagement, AURAK sets a new standard in the Gulf Region for student-centered excellence that empowers graduates to change the world.

## **Mission:**

To be a leading institution in the Gulf Region, the American University of Ras Al Khaimah:

- Provides a transformational, student-centered learning experience based on the North American model of higher education enriched by attention to the culture of the region.
- Engages students, faculty, and staff in innovative undergraduate and graduate programs and high-impact research.
- Prepares future leaders and entrepreneurs through community outreach and creative initiatives involving local, regional, and global partners.

The American University of Ras Al Khaimah pursues this mission as an independent, public, state-owned, non-profit, coeducational institution.

## **Strategic Goals:**

The following strategic goals guide all aspects of AURAK's operation:

- Implement exemplary governance and organizational processes that ensure student excellence;
- Attract talented students from around the world through a student-centered culture of success;
- Offer leading-edge academic programs and community engagement initiatives that meet local, regional, and international needs and generate high-impact research;
- Enhance faculty and staff excellence through improved recruitment, retention, and professional development; AND
- Provide stellar support services that contribute to the institution's excellence and maximize benefits to university stakeholders.

## **Values:**

The following values guide all aspects of AURAK's operation:

### **Integrity:**

To model adherence to a code of honesty, transparency, and accountability.

### **Excellence and Continuous Improvement:**

To set high expectations for every unit and every constituent of the university.

### **Collaborative Engagement:**

To ensure continual involvement of students, faculty, and staff in activities that contribute to the success of the institution and to the solution of local, regional, and global problems.

### **Diversity, Inclusiveness, and Cultural Authenticity:**

To facilitate, through respectful dialogue, social and multicultural understanding that contributes to the success of all.

### **Scholarship and Research:**

To advance knowledge through discovery, dissemination, application, and community engagement.

### **Empowerment:**

To celebrate the power of inspired teaching to ignite transformational learning that shapes students into agents of positive change.

### **Academic Freedom:**

To cultivate an environment of critical thought and enquiry, tolerance for divergent views and beliefs, and an open and respectful exchange of ideas.

### **Innovation and Entrepreneurship:**

To encourage creative, enterprising ideas and the translation of those ideas into sustainable initiatives.

### **Global Interdependence:**

To promote engagement with the international community in ways that encourage constituents of the university to see themselves as citizens of the world.

### **Sustainability:**

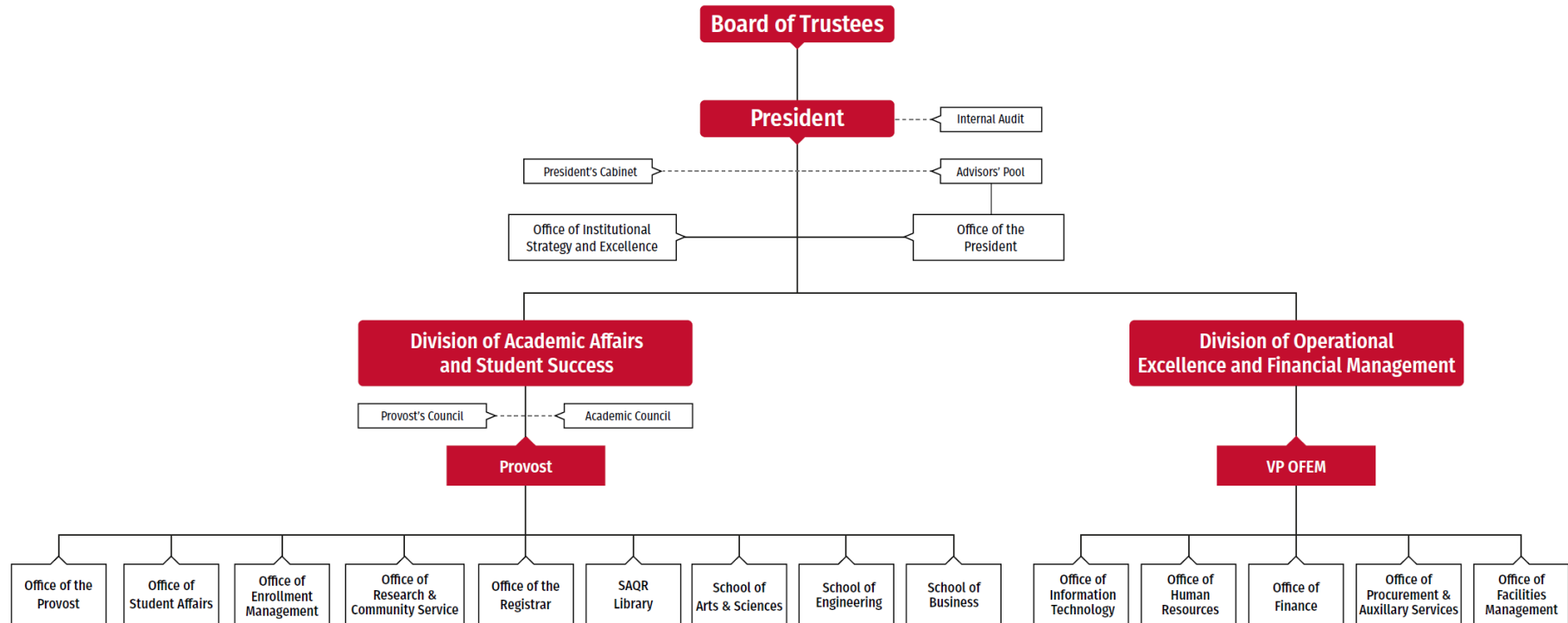
To nurture an appreciation of the shared responsibility of humankind to preserve planet Earth for future generations.



# AURAK Organizational Chart



## AURAK ORGANIZATIONAL CHART



## AURAK CAMPUS MAP



# AURAK

الجامعة الأمريكية في رأس الخيمة  
AMERICAN UNIVERSITY OF RAS AL KHAIMAH

### Campus Map



- |                |                              |                     |
|----------------|------------------------------|---------------------|
| 1 Building A   | 8 Building G                 | 15 Store            |
| 2 Building B   | 9 Building H                 | 16 Workshop         |
| 3 Building C   | 10 Building J (Saqr Library) | 17 Basketball Court |
| 4 Building D   | 11 Building K                | 18 Tennis Court     |
| 5 Building E   | 12 Dormitories               | 19 Football Pitch   |
| 6 Building F   | 13 Mosque                    | 20 Parking          |
| 7 Building F 2 | 14 Gymnasium                 |                     |

## **ACADEMIC PARTNER INSTITUTIONS**

Al Farabi-Kazakh National University, Republic of Kazakhstan

Appalachian State University, USA

Aspen Institute (Stevens Initiatives), USA

Coastal Carolina University, USA

Coventry University, UK

Hong Kong University of Science and Technology, Hongkong

Incheon National University – INU, South Korea

Knowledge Exchange Institute, USA

Management and Science University, Malaysia

Narxoz University, Republic of Kazakhstan

RAK Medical & Health Sciences University, RAK, UAE

Sudan University of Science & Technology (SUST), Sudan

Sungkyunkwan University Business School, South Korea

Temple University, USA

University of Texas Arlington, USA

University at Albany, State University of New York, USA

University of Barcelona, Spain

University of Nebraska at Kearney, USA

Universiti Teknologi Petronas, Malaysia

University of Wyoming, USA

Wayne State University (WSU), USA

Windesheim University of Applied Sciences, Netherlands



## STUDENT RESOURCES AND SUPPORT SERVICES

## STUDENT RESOURCES AND SUPPORT SERVICES

AURAK provides a range of student resources and support services. The following narratives outlines six major areas of delivery: 1) university student resources and services, 2) the Saqr Library, 3) the Office of Student Affairs, 4) the Office of the Registrar, 5) the Office of Research and Community Service, and 6) the university bookstore.

### UNIVERSITY STUDENT RESOURCES AND SERVICES

#### Computer Facilities and Resources

AURAK provides computing resources to support the needs of the students and faculty with their academic work. This includes Wi-Fi coverage throughout the campus, audio/video system in classrooms/laboratories equipped with overhead projectors or interactive touch panels, and a dedicated email account for each student, faculty, and staff member.

#### Laboratories

- ***Engineering Laboratories***

AURAK has well equipped labs aligned with international standards for all of their programs. These include: Architecture Digital Studios Labs, Surveying Laboratory, Construction Materials Laboratory, Geotechnical Engineering Laboratory, Fluid and Hydraulic Laboratory, Highway Engineering Laboratory, Computer Architecture and Microprocessors Laboratory, Computer Networking Laboratory, Computer Programming Laboratory, Embedded Systems Laboratory, Space Lab, AI Lab, Electrical/Electronics/Communications Labs, Thermal Sciences Laboratory, Manufacturing Process and Material Laboratory, Fluid & Hydraulics Laboratory, Chemical Engineering Lab that includes Reaction Lab, Unit Operation Lab, and Characterization Lab, Petroleum Engineering Lab that includes Samples Preparation Lab, Drilling Engineering Lab, Flow in Porous Media Lab, Reservoir Rock and Fluid Properties Lab, and a Computer Applications Lab. A new Engineering Laboratory Building to house all Engineering laboratories is under construction and is slated to open during the 2023-2024 academic year.

- ***Science Laboratories***

To support the laboratory-based modules of the Biotechnology Program, there are dedicated biotechnology, chemistry, and physics laboratories. In these labs, student learning is facilitated through practical exercises. The biotechnology, chemistry, and physics laboratories are fully equipped with standard and specialized instrumentation, including: microscopes, PCR and Real-Time PCR (qRT-PCR), Denovix spectrophotometer, Gel Electrophoresis and Western Blot apparatus, ELISA plate reader, High Performance Liquid Chromatography, cell culture incubator, laminar hood, freezers, liquid nitrogen tank, automatic cell counter, UV-Vis Spectrophotometers, pH meter, melting point apparatus, FTIR, Gas Chromatograph, structure modelling software for performing various experiments, PASCO software, free fall apparatus, force tables, momentum tracks, and rotational motion devices.



- ***Mass Communication Laboratory***

The Mass Communication Program has a Computer Lab with 15 Apple computers installed with software that meet the producing and editing needs for photos and videos. The software includes Full Adobe Suite and Final Cut Pro. The Mass Communication Program has also a Radio and TV studio facility including HD studio camcorders, Cinema HD Display 23-inch Widescreen, studio grade multi-purpose tripods and studio sound monitors, and electronic news-gathering equipment including 4K Field video camera and sound recording gear.

### **Mosque (Masjid) and Multi-faith Room**

The mosque, located on campus between the Cordoba and Granada Residence Halls (also known as Buildings 2 & 3) is open for prayers at all times. There is a separate female prayer room within the mosque. The campus also has a multi-faith room in Building H. This is an open, accepting space where all AURAK community members are invited to freely practice their various personal religions.

### **Parking**

Parking services are offered to AURAK members on campus. Students will find information on “AURAK Parking Rules and Regulations” displayed on signs throughout the campus. Students must park only in the student designated parking, and they must display the appropriate parking permit on their vehicle. Permits can be obtained from the Office of Human Resources and Administrative Services.

Students are expected to drive carefully and follow parking rules on campus. This includes allowing pedestrians the right of way; not parking on fire lanes, curbs, special needs parking areas, or designated parking areas for university vehicles; parking within the white lines; and maintaining speeds between fifteen to twenty km/hour. Students who drive recklessly through the university parking lot and/or campus are subject to disciplinary action.

### **Photocopy Facilities**

A coin-operated photocopier is available in the library. When using a photocopier, students are reminded that AURAK campus adheres to UAE copyright laws.

### **Security**

The university provides twenty-four hours security services. Security guards monitor the entire campus premises including the student residence halls and all university buildings to ensure the safety and security of the campus community. CCTV cameras are installed in all campus buildings.

### **Student Center**

The Student Center, located in Building C is the heart of the student community, serving students and the campus community by offering several ways to relax and socialize. The Student Center, open every day of the week from 8:00 am to 8:00 pm, provides wireless internet access, pool tables, table tennis, a foosball table, video games, and chess. If you would like more information, please contact the Department of Student Life located in Building D.

## **Food Services**

Various restaurants and food serving facilities are available either on campus or immediately off campus. These services are available to ensure the health and wellbeing of AURAK's student population.

## **THE SAQR LIBRARY**

The spacious 3-floor new library building with 35,000 sq. ft. facility comprises the library stacks, Reading Areas, Individual and Group Study Rooms, Staff Work Areas, a Computer Laboratory for library applications, Meeting Rooms, Training Room, Conference Rooms, Auditorium, Faculty Reserved Room, and Special Collection Room. AURAK's SAQR Library plays a significant role in supporting student academic success. The library continuously upgrades information resources and services that support the instructional and research activity of the university. The library collection consists of many different types of library materials and information sources, either in print or electronic formats.

The SAQR Library is a key participant in the academic development of the university and actively responds to the needs of the learning community. The spacious 5,350 sq. ft. facility provides room for the library stacks, places for reading, faculty, staff, and student work areas, and a computer laboratory for library applications. Additionally, the SAQR library has sixteen individual private study carrels, a group study room, and a conference room.

The SAQR Library facilitates access to information through the sharing of resources at national, regional, and international levels, providing circulation, reference, Internet, printing, scanning services, and online access catalog services. The hours of operation, posted on the AURAK website and at the library entrance, provide students with optimal time for study and research.

SAQR Library hours of operation provides generous opportunities for students to utilize services and resources. Hours of operation vary depending on the university academic schedule and may be found on the AURAK library webpage and at the library entrance.

In addition to textbooks and reserve collections, SAQR Library houses a wide range of information sources including reference collections, periodicals, multimedia presentations, and electronic databases.

## **Library Circulation Services**

### **Borrowing Service**

Borrowing privileges are provided to all registered students and currently employed faculty and staff. A university ID card is required to check-out circulating items. Students, faculty, and staff have the privilege of borrowing books based on the following policy:

### **Loan Period**

- Students – ten items for three weeks
- Faculty – twenty items for one semester
- Staff – three items for three weeks

## **Renewals**

Circulating items may be renewed online or at the Library Circulation Desk. The borrowers may renew items twice in succession unless the item has been requested by another user.

## **Fines**

All borrowers pay overdue fines for late return of the borrowed items. Days on which the library is closed are not counted. The current overdue fine is:

AED 10.00 per day per item (circulating and reserve materials)

## **Lost and/or Damaged Items**

Replacement fines for lost or damaged library materials are the actual price of the material.

## **Reference Services**

Upon request, the library provides reference services for students, faculty, staff, and library visitors to use and identify proper resources for their work. Patrons are encouraged to contact the librarian on duty for help in using reference services and the print and online resources of books and journals.

## **Internet Service**

Workstations within the library lab have internet access, as well as access to common productivity software applications. The library is serviced by wireless internet to facilitate the use of laptops. The wireless network will allow users to access both the library catalog, as well as, the electronic databases inside and outside the university campus.

## **Online Public Access Catalog (OPAC)**

The library catalog contains all library holdings and serves as a search guide for easy retrieval of library materials. The Online Public Access Catalog allows users to browse all existing library resources of the university online. The catalog can be accessed through the library webpage within the university homepage.

## **Printing and Scanning Services**

Students can use the networked printer located in the library computer lab through their accounts (IDs and Password). The library also provides a scanner and a coin-operated self-service photocopier.

## **Library General Rules and Regulations**

The Saqr Library exists to serve students. Accordingly, some rules are in place to ensure that all students are provided a positive learning environment. The library usage rules are as follows:

- Silence should be maintained in the library.
- An identification card must be presented for borrowing of library materials.
- Eating, drinking, and other activities that distract other readers are strictly prohibited.
- Online chatting and playing games are not allowed in the library computer laboratory.
- The use of mobile phones inside the library is strictly prohibited.

- Patrons should not leave valuables in the library. The library is not responsible for the loss of personal items or possessions.
- Patrons should not reshelved used library materials.
- Patrons should handle all library materials with care.
- Patrons should keep the library clean.

## **THE OFFICE OF THE REGISTRAR**

The Office of the Registrar is committed to enhancing a culture of high academic performance and enhancing the student experience. The Registrar's team holds specialized expertise in supporting admissions and registration, maintaining student records, certifying degrees, and overseeing the university catalog. The Office of the Registrar is located in Building D.

## **THE OFFICE OF STUDENT AFFAIRS**

The Office of Student Affairs (OSA) provides services and facilities necessary to promote the academic, professional, and social well-being of AURAK students and encourages students to become actively involved in campus life. Out-of-classroom learning experiences play a significant role in students' professional and personal development by enhancing, their social, leadership, and organizational skills. OAS offers comprehensive programs, orientation, activities, workshops, and initiatives that revolve around student needs, requests, and suggestions. OSA, in cooperation with the Student Government Association (SGA), organizes a wide range of activities for students throughout the year. These activities are carefully planned with the purpose of exposing students to a diverse variety of cultural, social, and educational experiences. In addition to their entertainment value, these activities enhance students' creativity, collegiate, humanitarian, and social skills.

There are a number of traditional activities that OSA conducts on an annual or semester basis. These include the UAE National Day Celebration, Global Day, annual IFTAR meal, start and end of semester celebrations, and academic and student recognition award ceremonies. These activities are supplemented by cultural events – museum and theatre trips, art exhibitions, hiking trips, dhow cruises, movie nights, and beach barbecues. AURAK's very active sports program completes the comprehensive repertoire of student activities. Tournaments and leagues are organized in most sports for both men and women.

### **Academic Recovery, Disability Support, and Counseling Services**

Academic Recovery, Disability Support, and Counseling Services assist students in both academic and personal success. Students may request the free services from the point of entry into the university through graduation. Staff are available to provide one-on-one attention to assist in meeting students' unique needs and challenges. Services include individual counseling, the loan of self-help DVD's, workshops, educational abilities testing, long-term and short-term disability

services, and assisting students who are experiencing challenges both personally and academically.

AURAK provides personal and academic counseling services for individual students requiring assistance with a variety of issues. All counseling services are free and are provided to students enrolled in three or more credit hours. These services include counseling, loan of self-help DVDs, workshops, and, when necessary, referral services to qualified professionals and organizations. The service provides students the opportunity to discuss matters causing them concern or stress, with the objective of providing assistance to and support for the student so that the student can reach his or her highest potential, personally and academically. All services will be provided in a confidential, private setting.

Students with verified disabilities may request support through Disability Support Services in the Office of Student Affairs. The university will support students with special needs where resources are available. Requests are confidential and will be handled with complete discretion. The Office of Student Affairs will verify disabilities and determine the needed accommodations to provide equal access to educational facilities, programs, services, and activities. With the student's written consent, the departmental staff will communicate with appropriate faculty, staff, and other individuals regarding needed accommodations.

#### **Office of Enrollment Management/Admissions Department**

The Office of Enrollment Management/Admissions Department team leads a comprehensive, institution-wide, research-driven, systemic process committed to providing educational access for the local, regional, and international community. Admissions supports AURAK's Strategic Plan and aims to place AURAK as the first institution of choice for the pursuit of quality post-secondary education. The purpose of the Office of Enrollment Management is to achieve the university's desired enrollment objectives, engage top-quality student applicants, and guide the delivery of a superior student experience to ensure student success and satisfaction with the University. Accordingly, the admissions team works with marketing, recruitment, and retention of UAE nationals, UAE expatriates, and international students. Admissions staff also provide important information resources for students, providing them with an introduction, clarification, and guidance to the university environment, policies, and procedures. The Office of Enrollment Management is located in Building D.

#### **Scholarships and Financial Aid Department**

The Scholarships and Financial Aid Department exists to dismantle those financial obstacles standing between students and their dreams. Every student is important to us. The University actively recruits the quality of student capable of raising the institution to a higher level. To facilitate the student's entry into the university, AURAK offers a host of scholarships. Students are encouraged to explore the many options available to assist them in achieving their educational goals. For questions related to financial aid and scholarships visit the Admissions team in Building D to meet with an Admissions staff member.



## **Athletics**

We believe that engaging in physical activity plays a significant role in promoting a healthy physical condition, self-confidence, a high quality of life, and serving as a focal point for more opportunities to sustain community interaction and building social networks.

AURAK offers athletic scholarships to qualified, outstanding student-athletes., subject to terms and conditions of the AURAK Athletic Scholarship.

## **Recreation and Sports Facilities**

The University sports facilities include a multipurpose hall, a well-equipped gymnasium with workout machines, and several outdoor sports fields. The sports activities aim to develop and maintain physical fitness among all members of the University community by offering a variety of sports programs. Students participating in these sports activities have the opportunity to develop leadership and teamwork skills. Games, tournaments and friendly matches with other colleges/universities are regularly organized. There are also two gyms available for student use—one for men and one for women in the Sports Complex.

To know more and to participate in the sports activities contact the Student Life Department: [student.life@aurak.ac.ae](mailto:student.life@aurak.ac.ae)

## **Wellness Education and First Aid Station**

AURAK contracts with a medical provider to deliver an on-campus Wellness Education and First Aid station located in Building D on the first floor. It is open weekly from 8:00 am to 4:00 pm, Monday through Friday. Students requiring emergency assistance at other times should contact Campus Security.

## **Health Insurance**

All students (undergraduate and graduate) enrolled at AURAK are required to have health insurance. International students, under sponsorship by AURAK, will automatically have the health insurance policy ordered for them and the annual expense of the insurance premium will be billed to their student account. UAE-based students who are covered by their parent's health insurance policy (or other comparable health insurance policy) are required to provide proof of valid insurance coverage by submitting details (declaration of coverage page and insurance card) in the EUMS student portal. The declaration page submitted must specify the extent of coverage and include the AED upper limit of coverage provided by the policy. This confirmation of health insurance coverage must be received by the last day of the class drop/add period each fall and spring semester. Refunds are not granted for proof of insurance that is submitted after the last scheduled day to drop/add classes each semester. The required two documents must be uploaded on the AURAK EUMS portal by accessing the following link: <https://eums.aurak.ac.ae/SisStudentPortal/InsuranceForm>

By entering this information before the deadline date, each semester, UAE-based students will be exempted from the purchase of the University offered insurance policy. If a student does not enter proof of insurance coverage the University will automatically purchase health insurance on behalf of the student and the policy fee will be billed to the student's account. UAE nationals and citizens of other GCC countries who are covered by the UAE Government Health Insurance policy are also

exempted from the purchase of the University offered insurance policy.

For questions regarding Health Insurance, contact the Office of Student Affairs: [studentaffairs@aurak.ac.ae](mailto:studentaffairs@aurak.ac.ae)

### **Student Clubs & Student Life Activities**

The Department of Student Life facilitates student growth through a variety of clubs. It supports activities to complement courses such as academic, civic, social, or recreational clubs for fun and leisure. Membership in these clubs not only helps students make new friends and meet people with common interests, but it also helps them to acquire important skills such as leadership, organization, teamwork, and event planning. All clubs work very closely with the SGA. All student clubs must be authorized and registered annually with the Department of Student Life. The university encourages all students to establish or join clubs to explore or further their interests.

### **Student Government Association (SGA)**

The SGA represents the student body, provides an opportunity for students to demonstrate leadership, and serves as a liaison between the students and the university administration. The SGA also plays an important role in planning and sponsoring student activities and events. The representative of the SGA serves on the different committees in the university.

The SGA at AURAK follows a constitution, which defines SGA's goals and objectives and establishes the roles and responsibilities of each SGA member. The constitution also defines the policies and procedures to follow for planning and organizing university events. The SGA constitution also establishes the minimum qualifications for students who want to run for office. Further information about the SGA can be obtained by contacting the Department of Student Life.

### **International Learning Opportunities: International Student Exchange and Study Trips**

The Office of Student Affairs also supports visiting study abroad and exchange students to our campus as well as assisting current AURAK students with plans to study abroad/exchange. The study abroad and exchange programs provide students rich experiences to encourage tolerance, acceptance, and discovery. Students participating in the study abroad or exchange programs have the opportunity to build character, adaptability, increase knowledge of international issues, and foster multicultural understanding needed to successfully navigate an ever-changing global society. The Office of Student Affairs also assists in broadening AURAK's outreach and world standing by assisting in the development of university linkages and partnerships that enhance the academic and study abroad/exchange experience for the campus community.

### **Center for Learning Support Services, Advising, and eLearning**

AURAK's Center for Learning Support Services, Advising, and eLearning operates to ensure that every AURAK student is provided the opportunity to maximize his or her academic success. Specifically, the Center seeks to help students achieve their academic goals by: 1) overseeing the Math and Science Center, the Writing Center and Peer Tutoring Center, 2) supporting the first-year experience, and 3) guiding the improvement of AURAK's advising and educational experiences.

The Center for Learning Support Services, Advising, and eLearning facilitates the application of the latest pedagogical theories and research to enhance knowledge instruction and acquisition at

AURAK for all students, faculty, and staff. The center strives to identify and respond to the dynamic nature of student needs and learning environments. Students who request academic assistance are provided free tutoring by qualified peer tutors in various subject areas.

The Center for Learning Support Services, Advising, and eLearning, located in Building D, provides qualified peer tutors to aid students in a wide array of subjects including English, mathematics, biology, chemistry, physics, and a variety of computer courses. Tutoring services provided are at no cost to the students. All students are strongly encouraged to take advantage of this key academic support service. Students interested in tutoring are asked to visit the Center or contact the Director of this Center.

### **Internship, Career Development and Alumni Department**

AURAK is committed to providing students with a wide range of career services. Students visit with a career counselor who can assist in self-awareness exploration (values, skills, abilities, and personal style), career exploration (job descriptions, educational requirements, salary range, and advancement possibilities), and employment and employability skills (writing cover letters, curriculum vitas/resume, interviewing skills, and employee decorum). Students access job listings, internship opportunities, and career fairs.

AURAK celebrates the success of its distinguished alumni and encourages alumni to stay in contact with the University. AURAK's alumni are also encouraged to utilize the University's student support services and the Saqr Library.

## **OFFICE OF RESEARCH AND COMMUNITY SERVICE**

The Office of Research and Community Service promotes research and community engagement activities among AURAK's faculty and students. The Office also supervises AURAK's research-oriented programs and makes available small seed grants to faculty on a competitive basis. These seed grants often support students engaged in research under a research assistant contract. The Office promotes a culture of research and scholarship. Students are encouraged to attend sponsored events in order to broaden their knowledge and gain awareness of the different research activities of the AURAK faculty.

The Office of Research and Community Service manages the Ras Al Khaimah Research Innovation Center (RAKRIC), the Center of Executive and Professional Education (CEPE), the Community and Business Development Department, and the RAK Center for Outdoor Comfort. RAKRIC is a state-of-the-art research and development center specializing in sustainable system solution development related to solar technologies. RAKRIC accommodates several senior research projects of AURAK's students under the supervision of the faculty and with the assistance of the RAKRIC staff. CEPE provides professional training courses and English language training/testing including the EmSAT, TOEFL, and IELTS tests. The Community and Business Development Center provides leadership in implementing the university's Community Engagement Strategy, and the RAK Center for Outdoor Comfort focuses on improving outdoor comfort in hot and humid climates.

## **COST OF ATTENDANCE – TUITION & FEES**

### **TUITION FEES for Fall 2023 - Spring 2024**

**PLEASE NOTE:** AURAK reserves the right to increase tuition fees and other charges up to a maximum of 10% per annum.

| <b>Undergraduate Program</b>                                                | <b>Less than 12 Credit Hours</b> | <b>12-16 Credit Hours</b> | <b>More than 16 Credit Hours</b> | <b>Payment Terms</b> |
|-----------------------------------------------------------------------------|----------------------------------|---------------------------|----------------------------------|----------------------|
| All undergraduate programs except otherwise mentioned for specific programs | 2,200 Per Credit Hour            | 26,565                    | 1,730 per Credit Hour            | Per Semester         |
| Architecture Engineering                                                    | 2,200 per Credit Hour            | 27,720                    | 1,730 per Credit Hour            | Per Semester         |
| Petroleum Engineering                                                       | 2,500 Per Credit Hour            | 30,975                    | 1,935 per Credit Hour            | Per Semester         |
| Chemical Engineering                                                        | 2,500 Per Credit Hour            | 30,975                    | 1,935 per Credit Hour            | Per Semester         |
| Artificial Intelligence                                                     | 2,310 per Credit Hour            | 27,885                    | 1,815 per Credit Hour            | Per Semester         |
| Biotechnology                                                               | 2,310 per Credit Hour            | 27,885                    | 1,815 per Credit Hour            | Per Semester         |
| Computer Engineering                                                        | 2,310 per Credit Hour            | 27,885                    | 1,815 per Credit Hour            | Per Semester         |
| Computer Science                                                            | 2,310 per Credit Hour            | 27,885                    | 1,815 per Credit Hour            | Per Semester         |
| Electrical and Electronics Engineering                                      | 2,310 per Credit Hour            | 27,885                    | 1,815 per Credit Hour            | Per Semester         |

**Tuition fees for Summer shall apply on per credit hour basis as per below table**

| <b>Undergraduate Programs – Tuition Fee (Summer)</b>                              | <b>Fee in AED</b>     |
|-----------------------------------------------------------------------------------|-----------------------|
| All undergraduate programs except where mentioned otherwise for specific programs | 1,730 per Credit Hour |
| Architecture Engineering                                                          | 1,730 per Credit Hour |
| Petroleum Engineering                                                             | 1,935 per Credit Hour |
| Chemical Engineering                                                              | 1,935 per Credit Hour |
| Artificial Intelligence                                                           | 1,815 per Credit Hour |
| Biotechnology                                                                     | 1,815 per Credit Hour |
| Computer Engineering                                                              | 1,815 per Credit Hour |
| Computer Science                                                                  | 1,815 per Credit Hour |
| Electrical and Electronics Engineering                                            | 1,815 per Credit Hour |

| <b>Undergraduate Program –<br/>Non Tuition Fees in AED</b> | <b>Fall/Spring</b> | <b>Summer</b> | <b>Additional Information</b>                |
|------------------------------------------------------------|--------------------|---------------|----------------------------------------------|
| Application                                                | 400                | 400           | Application Submission, Non-refundable       |
| Admission                                                  | 750                | 750           | Issuance of Admission Letter, Non-refundable |
| Technology Fees                                            | 1,000              | 400           | Per semester                                 |
| Service Fees                                               | 500                | 500           | Per Semester                                 |
| Health Insurance                                           | Actual Price       | Actual Price  | Subject to Market Price                      |
| Transportation                                             | Free               | Free          | N/A                                          |
| Medical Services                                           | 100                | 100           | Per Semester                                 |

| <b>Other Fees – Undergraduate &amp;<br/>Graduate – in AED</b>  | <b>Amount</b> | <b>Additional Information</b>                                    |
|----------------------------------------------------------------|---------------|------------------------------------------------------------------|
| Visa Fees                                                      | 2,000         | Non-refundable, due at the time of admission                     |
| Visa Change Status                                             | 900           | Non-refundable, due at the time of admission                     |
| Visa Security Deposit                                          | 2,000         | Due at the time of admission, Refundable after Visa cancellation |
| Security Deposit                                               | 1,500         | One time – Refundable – if no damages                            |
| Graduation Fee                                                 | 1,500         | Non-refundable                                                   |
| Late Payment (if tuition and fees are not settled by due date) | 500           | Applicable to first payment, if not paid by due date             |
| Reinstatement fee                                              | 1,500         | To reinstate access after block due to non-payment               |
| Cheque Returned Fee                                            | 500           | Per Returned Cheque                                              |
| Transcript – Official                                          | 100           | Per Transcript                                                   |
| Transcript – True                                              | 50            | Per Transcript                                                   |
| ID Card Replacement Fee                                        | 100           | Per Instance                                                     |
| Official Letter Fee                                            | 50            | Per letter                                                       |
| Gymnasium                                                      | 500           | Per Month – Non-refundable                                       |
| Library Books overdue Fine                                     | 10            | Per day, Per item                                                |
| Library Books Lost/Damage Fine                                 | Actual Cost   | Per item                                                         |

## Tuition and Fees for graduate studies apply as per the below tables

(PLEASE NOTE: AURAK reserves the right to increase tuition fees and other charges up to a maximum of 10% per annum.)

| Graduate Programs – Tuition Fees                                          |              |                 |
|---------------------------------------------------------------------------|--------------|-----------------|
| Program                                                                   | Amount (AED) | Payment Terms   |
| Master of Business Administration (MBA – 36 total credit hours)           | 2,520        | Per Credit Hour |
| Executive Master of Business Administration (EMBA -36 total credit hours) | 3,150        | Per Credit Hour |
| Master of Education in Educational Leadership (33 total credit hours)     | 2,100        | Per Credit Hour |
| Master of Engineering Project Management (MEPM 36 total credit hours)     | 2,730        | Per Credit Hour |
| Master in Sustainable and Renewable Energy (36 total credit hours)        | 3,000        | Per Credit Hour |

| Graduate Program – Non Tuition Fees | Fall/Spring Fees in AED | Summer Fees in AED | Due by:                                      |
|-------------------------------------|-------------------------|--------------------|----------------------------------------------|
| Application                         | 500                     | 500                | Application Submission, Non-refundable       |
| Admission                           | 2,000                   | 2,000              | Issuance of Admission Letter, Non-refundable |
| Technology                          | 750                     | 400                | Per Semester Non-Refundable                  |
| Service Fees                        | 500                     | 500                | Per Semester                                 |
| Health Insurance                    | Actual Cost             | Actual Cost        | Subject to Market Price                      |
| Transportation                      | Free                    | Free               | N/A                                          |
| Graduation Fee                      | 1,500                   | 1,500              | One Time, Non-Refundable                     |
| Medical Service                     | 100                     | 100                | Per Semester                                 |

The Housing Accommodations for both undergraduate and graduate students are outlined in the following table.

| Housing and Accommodation                                                                      |                         |             |               |
|------------------------------------------------------------------------------------------------|-------------------------|-------------|---------------|
| Accommodation Type (Amount in AED)                                                             | Fall 2023 & Spring 2024 | Summer 2024 | Payment Terms |
| Housing (Large Room) Single Occupancy                                                          | 11,000                  | 2,400       | Per Semester  |
| Housing (Small Room) Single Occupancy                                                          | 10,000                  | 2,200       | Per Semester  |
| Housing (Large Room) Shared Occupancy                                                          | 6,500                   | 1,600       | Per Semester  |
| Housing (Small Room) Shared Occupancy                                                          | 5,500                   | 1,400       | Per Semester  |
| Security Deposit                                                                               | 2,000                   | 2,000       | Refundable    |
| VAT shall apply wherever applicable as per UAE Law and Federal Tax Authority (FTA) guidelines. |                         |             |               |



## **Payment Policies and Procedures**

Tuition and all other fees must be paid by the first day of classes unless necessary arrangements are made with the Office of Finance. Fees can be paid by cash, check, wire transfer, or credit card.

Students who do not pay their fees by the specified deadline will be put on financial hold. Students on financial hold are not allowed to register for classes or take final exams and will not be issued official transcripts until all fees are paid.

Students are responsible for full tuition payment for all courses in which they are registered unless their registration is canceled for nonpayment; canceled administratively due to academic suspension, dismissal, or termination; or the course is dropped before the tuition liability begins.

## **Refund Policies**

Tuition fees are refundable subject to the following rate for the Fall and Spring semesters:

| <b>Withdraw by</b>                                       | <b>% of Refund</b> |
|----------------------------------------------------------|--------------------|
| The end of the <b>First Week</b> of the semester         | 100%               |
| The end of the <b>Second Week</b> of the semester        | 75%                |
| The end of the <b>Third Week</b> of the semester         | 50%                |
| The end of the <b>Fourth Week</b> of the semester        | 25%                |
| Beyond the end of the <b>Fourth Week</b> of the semester | <b>No Refund</b>   |

*Please check the university Academic Calendar for dates of Drop/Add.*

Tuition fees are refundable subject to the following rate for the Summer terms:

| <b>Withdraw by</b>                                  | <b>% of Refund</b> |
|-----------------------------------------------------|--------------------|
| The end of the <b>First Week</b> of the term        | 50%                |
| Beyond the end of the <b>First Week</b> of the term | <b>No Refund</b>   |

## **Payment by Installment**

If full fees cannot be paid immediately, installments can be arranged with the Office of Finance. Deferment of tuition and fees is approved only if the following conditions are met:

- The first installment shall be paid through cash, current dated cheque, credit card or bank/wire transfer and the remaining installments through post-dated cheques (PDCs).
- The remaining installments must be paid within the same semester.
- If any installment is not paid on a due date, the student shall be placed on financial hold.
- For Fall and spring, registration shall be approved only if the first installment of 25% or 50% is paid and PDCs are received for remaining installments.
- For Summer, installments shall not be allowed, and students must settle full payment before the start of classes.

Failure to attend a course (no show) does not result in a refund of charges. Students not planning to attend must drop their registration during drop/add period. Students reported as “no show” in class will be withdrawn from the class without a refund. These students shall be placed on a drop/delete hold and will not be able to register in future semesters unless the fees are paid in full in advance.

For detailed guide on payment deadlines please refer the University website payment guide section.

### **Payment and Plan Violations**

Where the student does not meet a scheduled payment, the University will place the student on a financial hold and block the student’s access to the portal. **This may also result in the student not being able to attend the exams.**

For any clarification, please visit the Office of Finance or send an email at:

[student.finance@aurak.ac.ae](mailto:student.finance@aurak.ac.ae).

### **Financial Responsibilities**

The following outline the financial responsibilities relating to attendance at AURAK.

- Students are responsible for maintaining current addresses with AURAK and for activating and checking their personal AURAK e-mail accounts.
- Registration shall not be considered complete unless all outstanding balances from the prior term are paid in full.
- By registering for classes, students accept responsibility for charges for the entire semester.
- Failure to receive a reminder bill confirming charges does not waive the requirement for payment when due.
- Students are responsible for dropping unwanted courses by the drop dates and using the drop and withdrawal procedures published in each term’s Schedule of Classes. Full or partial tuition liability may apply.
- Non-returning students are responsible for submitting a written withdrawal to the Associate Provost of Academic Affairs. Penalties may apply.
- Students may not attend classes without completing the registration process.

### **Residence Halls**

AURAK residence halls are located on campus. These residence halls are managed by the university’s Student Residential Life team under the supervision of the Office of Student Affairs. EDRAK. For information about AURAK’s residence halls, please contact the Office of Student Affairs, [studentaffairs@aurak.ac.ae](mailto:studentaffairs@aurak.ac.ae)

### **Issues Relating to AURAK Representatives**

AURAK utilizes representatives to assist and recruit applicants from countries external to the UAE. These representatives are held to the same standards as all AURAK employees. Applicants should never pay money to any representative. All funds are to be paid to the Office of Finance for which official university receipts will be issued. **Should an issue ever arise in dealing with any AURAK**

representative, concern should be addressed with the Office of Student Affairs for investigation and resolution as appropriate.

## **SCHOLARSHIPS AND FINANCIAL AID**

AURAK is committed to providing incentives in the form of scholarships and financial grants to attract and retain high academic achievers. In addition to meeting the expectations set for each scholarship/grant, students must meet other conditions set including, at a minimum, meeting all requirements for admission to the program of interest and being enrolled as a full-time student. To this end, AURAK will establish a range of scholarships and financial grants for new and continuing students. The eligibility criteria for such awards will be benchmarked against other institutions to ensure that the scholarships and financial grants will be useful in attracting and retaining students who might otherwise choose to enroll at other institutions. These criteria will also be crafted to promote the recruitment and retention of students who will contribute positively to the academic reputation of the university and who will bring talents, skills, and diversity to the campus that contribute to achievement of the university's strategic objectives.

### **Categories of Scholarships and Financial Aid**

The following institutional scholarships will be in effect for students joining AURAK beginning in AY 2023-2024 and onward. The awards will be determined by the following standards. All of the scholarships and grants described below will be awarded to eligible admitted students upon their signing a formal Agreement of Undertaking that will be subject to renewal for up to four years (five years if admitted to the Bachelor of Architecture), contingent on the students meeting the performance expectations for continuation of the scholarship or discount.

### ***New Undergraduate Student Scholarships***

New Undergraduate Student Scholarships will be awarded to full-time students beginning their undergraduate studies at AURAK in the following categories:

- **Scholastic achievement scholarships:**

| Scholarship              | High School Percentage | Scholarship Amount | Basis for Award | CGPA Requirement for Continuation of Scholarship in Subsequent Years |
|--------------------------|------------------------|--------------------|-----------------|----------------------------------------------------------------------|
| Presidential Scholarship | 99% or Higher          | 70%                | Competitive     | 3.75                                                                 |
| Provost's Scholarship    | 97% to 98.9%           | 40%                | Competitive     | 3.60                                                                 |
| Dean's Scholarship       | 94% to 96.9%           | 30%                | Competitive     | 3.25                                                                 |
| Faculty Scholarship      | 90% or Higher          | 20%                | Competitive     | 3.00                                                                 |

For all scholarships based on high school scholastic achievement, the high school average to qualify will be adapted to be equivalent for different curricula with different grading system.

Subject to the CGPA requirements shown in the table above, the Scholastic Achievement Scholarships will be renewed annually up to the number of years specified for the completion of a program of study as long as recipients abide by the Academic Integrity Policy and the Student Code of Conduct.

- **Talent-based and skills-based scholarships:**

- Creative Expression Scholarships – for students with talent and achievement in Art or Music or other forms of creative expression documented through submission of a portfolio of work.
  - Promising Creative Expression Scholarship – 30% scholarship
  - Notable Creative Expression Scholarship – 40% scholarship
  - Exemplary Creative Expression Scholarship – 50% scholarship
- Athletic Scholarships – for students with a history of achievement in athletics and sports documented through submission of a portfolio of achievement awards – 30% scholarship.
- Entrepreneurship Scholarships – for students with a history of entrepreneurial achievements documented through submission of a portfolio of entrepreneurial accomplishments – 30% scholarship.
- Community Service Scholarships – for students with a history of community service documented through submission of a portfolio of community service activities – 30% scholarship.

The Talent-based and Skills-based Scholarships will be renewed annually up to the number of years specified for the completion of a program of study as long as recipients remain in good academic standing and abide by the Academic Integrity Policy and the Student Code of Conduct.

- **First-year recruitment grant:**

- AURAK Recruitment Grant – For first-year students not eligible for a scholarship, a 15% reduction in tuition and associated instruction-related fees may be available for the first semester of study. To retain the grant for the second semester, students must have a CGPA of 2.75 or higher at the end of the first semester.

- **Other scholarships:**

- Sibling Scholarships – for students admitted who have a sibling currently enrolled at AURAK – 20% scholarship.
- Partnership Scholarships – for students admitted who are employees of entities that have been designated as “AURAK Scholarship Partners” – 20% scholarship.
- Scholarships for employees and their relatives in accordance with the relevant remuneration and benefits policy.

For continuation of these “other” scholarships in subsequent years, scholarship recipients must be in good academic standing and must abide by the Academic Integrity Policy and the Student Code of Conduct.

- **Need-Based Financial Grants:**

- For students who can demonstrate that their family's financial resources are not sufficient to cover tuition and fees at AURAK. Recipients of Need-Based Financial Grants who remain unable to pay the full cost of tuition and fees at the time of registration for subsequent academic years and who are not eligible for one of the other scholarships available must re-apply for a Need-based Financial Grant. The availability of Need-based Financial Grants cannot be guaranteed beyond the initial award of a Need-based Financial Grant. A Need-Based Financial Grant can be combined with a scholastic achievement scholarship or a talent/skill-based scholarship if the scholarship recipient demonstrates that the family is unable to pay the tuition and fees not covered by the scholarship.

### **Undergraduate Continuing Student Scholarships**

Students completing their first or second year of full-time study at AURAK who were not previously awarded one of the following scholarships may apply for the following scholarships or a Needs-based Financial Grant subject to meeting the eligibility criteria specified:

- Presidential Scholarship – the student has a CGPA of 3.75 or higher and who has appeared at least once on the President's List for Academic Excellence.
- Provosts' Scholarship – the student has a CGPA of 3.60 to 3.7449 and who has appeared at least once on the Dean's List for Academic Excellence.
- Dean's Scholarship – the student has a CGPA of 3.40 to 3.599 and appeared at least once on the Dean's List for Academic Excellence.
- Faculty Scholarship – the student has a CGPA of 3.25 to 3.399.
- Need-based Financial Grants – the student is in good academic standing and can demonstrate that the family's financial resources are not sufficient to cover tuition and fees at AURAK. A Need-Based Financial Grant can be combined with a scholastic achievement scholarship if the scholarship recipient demonstrates that the family is unable to pay the tuition and fees not covered by the scholarship.

In order to retain the scholarship the following criteria will be applied:

- Presidential Scholarship recipients must have a CGPA of 3.75 or higher at the end of each year of study following award of the scholarship.
- Provost's Scholarship recipients must have a CGPA of 3.60 to 3.7449 at the end of each year of study following award of the scholarship.
- Dean's Scholarship recipients must have a CGPA of 3.40 to 3.599 at the end of each year of study following award of the scholarship.
- Faculty Scholarship recipients must have CGPA of 3.25 to 3.399 at the end of each year of study following award of the scholarship.
- Recipients of Need-Based Financial Grants who remain unable to pay the full cost of tuition and fees at the time of registration for subsequent academic years must re-apply for a Need-

based Financial Grant. The availability of Need-based Financial Grants cannot be guaranteed beyond the initial award of a Need-based Financial Grant.

### **New Graduate Student Scholarships**

Scholarships will be awarded to students beginning their graduate studies at AURAK in the following categories:

- Scholastic Achievement Scholarship
  - Graduate Study Scholarship (20%) – for students with an undergraduate CGPA of 3.25 or above.
- Other scholarships:
  - Alumni Scholarship (20%) – for AURAK alumni admitted to pursue an additional graduate degree.

In order to retain a Graduate Study Scholarship or Alumni Scholarship recipients must have a CGPA of 3.50 or higher at the end of the year of study following the award of the scholarship.

### **External Scholarships**

External scholarships are available for UAE Nationals from various entities such as the Sheikh Saqr Program for Government Excellence (SSPGE) and the Telecommunications Regulatory Authority (TRA). Other external scholarships may be available for students of all nationalities. All external scholarships are subject to the selection criteria of the indicated entities.

### **Obtaining Additional Information Concerning Scholarships and Financial Aid**

Information relative to the process for applying for Scholarships and Financial Aid can be obtained by contacting the Office of Enrollment Management.

## **STUDENT RIGHTS AND RESPONSIBILITIES**

AURAK adheres to the following values that underlie the philosophy and goals of the university:

- The importance of personal integrity.
- The right of every individual to be treated with respect and dignity.
- The freedom of intellectual inquiry in the pursuit of truth.
- The freedom of speech and the open exchange of ideas.
- The acceptance and appreciation of diversity with regards to race, gender, religion, age, disability, and ethnicity.
- The recognition that community service is an important component of the intellectual development of students.

### **STUDENT RIGHTS**

AURAK students have the right to pursue their education in a free and fair manner so long as they maintain their eligibility to remain as a member of the student community by meeting its academic



and behavioral standards. It is the responsibility of students to observe the regulations imposed by the university for the governance of the academic community.

- Students have a right to expect AURAK to maintain a climate conducive to thinking and learning. University teaching should reflect consideration for the dignity of students and their rights as persons. Students have a right to be treated with courtesy and respect.
- Students have a right to reasonable notice of the general content of the course, what will be required of them, and the criteria upon which their performance will be evaluated. Students have a right to have their performance evaluated promptly, conscientiously, without prejudice or favoritism, and consistently by the criteria stated at the beginning of the course.
- Students have a right to perform student evaluations considered in the retention and promotion reviews of faculty members.
- Students have a right to be free from discrimination and sexual harassment. Campus policy prohibits discrimination and harassment of a student because of his/her race, color, religion, national origin, sex, age, or status as an individual with a disability.
- Students have a right to privacy and confidentiality subject to reasonable campus rules and regulations, right to protect against unauthorized disclosures of confidential information contained in their educational records. Students have a right to examine the information contained in their educational records.
- Students have a right to form student organizations for any lawful purpose consistent with the goals and mission of AURAK.
- Students have the right to excellent faculty, academic technology, classrooms, libraries, presentations and other resources necessary for the learning process.

### **Student Responsibilities**

AURAK students are required to abide by the expectations within the Code of Student Conduct, maintain current contact information with the University, including permanent and local addresses, telephone numbers, student ID number, and legal name (as written in their passport). Each student must also maintain the University e-mail account assigned at the time of admission. Students are responsible for official communications directed to AURAK e-mail accounts. Students are responsible for providing a copy of their University ID when requested by AURAK administration, faculty staff members and/or security guards and to comply with directives of a university official.

### **Student Academic Grievance Policy and Procedures**

Academic grievance include grievance students may have against a faculty member, procedures implemented in class, or related academic issues. All students have the right to express their dissatisfaction. If the student has a disagreement with the faculty, grades, or the classroom the grievance is a Student Academic Grievance. To appeal a grade, the Grade Appeal Form, obtained from the Office of the Registrar and filled out by the student is the most expeditious and efficient

way to address a grade appeal with a professor. The following is the process for a student academic grievance.

If any student wishes to file a formal grievance, these steps apply:

1. The grievant must file a formal written complaint with the Associate Provost for Student Affairs.
2. The Associate Provost for Student Affairs will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
3. If such action is deemed appropriate, the Associate Provost for Student Affairs will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
4. The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.
5. The Committee will recommend a cause of action to the Associate Provost for Student Affairs within seven working days.
6. The Associate Provost for Student Affairs will forward to the student the recommendation of the Committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
7. If the student does not agree with the decision, the student has the right to file an appeal with the Provost. The Provost will review the decision and render a final decision in the matter within three working days. The Provost is the final authority to whom appeal may be made.
8. If a grievance is with the Provost, then the student may submit the grievance in writing to the President to initiate the process.

### **Student Non-Academic Grievance Policy and Procedures**

All students have the right to express their dissatisfaction and misconduct issues are handled under the non-academic grievance policy and procedures. Incidents that occur outside the classroom or disagreements with the staff fall under the non-academic grievance policy. If a student wishes to file a grievance, these steps apply.

1. The grievant must file a formal written complaint with the Associate Provost for Student Affairs.
2. The Associate Provost for Student Affairs will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
3. If such action is deemed appropriate, the Associate Provost will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
4. The committee will review the complaint, interview the complainant and any witnesses

cited in the complaint.

5. The committee will recommend a cause of action to the Associate Provost for Student Affairs within seven working days.
6. The Associate Provost for Student Provost will forward to the student the recommendation of the committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
7. If the student does not agree with the decision, the student has the right to file an appeal with the Provost. The Provost will review the decision and render a final decision in the matter within three working days. The Provost is the final authority to whom appeal may be made.
8. If a student's grievance is with the Associate Provost for Student Affairs, it may be submitted to the Provost to begin the process.
9. If a grievance is with the VPAASS or other Vice-President, the student may submit the grievance in writing to the President to begin the process.

### **Obtaining Additional Information**

The Office of Student Affairs is available to answer any questions that may arise concerning AURAK's grievance policies and procedures. Please feel free to visit should you require additional information.

## **STUDENT RESPONSIBILITIES**

AURAK students are expected to conduct themselves according to AURAK policy. The following policies are of particular interest in maintaining these expectations.

### **AURAK Communication Policy**

AURAK students are required to maintain current contact information with the university, including permanent and local addresses, telephone numbers, student ID number, and legal name (as written in their passport). Each student must also maintain the university e-mail account assigned at the time of admission. Students are responsible for official communications directed to AURAK e-mail accounts.

### **Student Publication Policy**

Any student publication at AURAK will be required to conform to the norms of responsible journalism and avoid libelous, indecent, or harassing material. The same publication policies will apply when deciding what material can be included on the website, or any other social media, developed and managed by the students. All the publications should particularly keep in mind the cultural norms of Ras al Khaimah and the UAE.

Students should get all articles, stories, or any other related materials reviewed by the editor to ensure the correctness of the English language. Students should take prior approval from the Associate Provost of Student Affairs before distributing, posting, or publishing any printed/online material in the name of AURAK.

### **Guidelines for Using AURAK Name, Logo, and Mark**

Students may use the AURAK name in association with their university-sanctioned activities. Recognized student groups producing merchandise for sale that incorporates AURAK's name, marks, or logos must comply and obtain permission from the Associate Provost of Student Affairs. Students may use the name of a school, department, or other AURAK programs outside the university (other than on a resume) only with approval of the appropriate authorized personnel.

### **STUDENT CONDUCT**

The American University of Ras Al Khaimah adopts the following Student Code of Conduct to ensure that students conduct themselves appropriately at all times as members of the University community. Students will:

- Attend classes regularly and punctually and complete assignments on time.
- Participate actively in class discussions and other course-related classroom activities.
- Be knowledgeable of and plan appropriately to meet the requirements for course and program completion.
- Adhere to the published test or examination rules and regulations.
- Demonstrate sensitivity to the cultural and religious norms and customs of the UAE and ensure that their conduct is in line with those cultural expectations.
- Display principles of integrity at all times, including those associated with academic integrity, ethics, and honesty.
- Treat others as they would themselves wish to be treated.
- Show proper respect to fellow students as well as to AURAK employees and representatives. This includes security personnel. Respectful behavior includes refraining from disruptive behavior such as:
  - Talking during speeches or classes when the speaker or instructor is talking
  - Arriving late for classes or leaving early
  - Receiving or initiating telephone calls on mobile phones during classes
  - Sending text messages during class
  - Making negative or rude comments during class about other students, faculty members, or the opinions of other students
- Have knowledge of and observe the regulations of AURAK.
- Use University property/facilities responsibly.
- Provide security guards with personal identification (ID card) and appropriate documentation when requested.
- Seek permission for using someone else's property, and refrain from using another person's name and/or ID number for any reason.

- Plan and organize all extra-curricular activities on the AURAK campus in collaboration with the Department of Student Life, obtaining all necessary permissions.
- Obtain permission from the Department of Student Life to organize events in the name of AURAK.
- Obtain permission from the Department of Student Life to distribute leaflets or journals, post notices, or collect signatures on the premises of AURAK.
- Use the internet responsibly, observing all applicable IT policies.
- Refrain from the possession, use, or sale of illegal substances or materials.
- Refrain from exchanging any gifts of significant monetary value with members of the faculty, staff, or administration.
- Refrain from carrying mobile phones or any other electronic devices inside classrooms in which examinations are being administered.

*Students should note that during the Covid pandemic (or similar occurrences) the university reserves the right to impose protective measures to secure a safe environment for students, staff, faculty, and administrators. Failure to comply with any of these protective measures constitutes a violation of the student code of conduct.*

### **Student Behavioral Misconduct**

The Associate Provost for Student Affairs is responsible for the administration of disciplinary procedures related to student behavioral misconduct. Reports of alleged student violations of university regulations may be submitted by any individual to the Associate Provost of Student Affairs. Allegations of violations of the AURAK Code of Conduct are accepted for investigation only when the apparent infractions are observed on AURAK property, at AURAK student events and activities off campus or at other locations at which AURAK provides services. Infractions of UAE or Ras Al Khaimah laws occurring off campus shall be the sole concern of the civil authorities, except when such actions: 1) directly affect the health, safety, or security of AURAK community, 2) affect AURAK's pursuit of its educational goals, or 3) occur as a direct result of an AURAK-related activity.

All student behavioral misconduct will, if possible, be resolved by the Associate Provost for Student Affairs through mediation or through verbal and written warning. For examples of behavioral misconduct, please see the *AURAK Student Handbook*.

The resolution of cases of behavioral misconduct by the Associate Provost may involve issuance of a verbal or written warning. However, the Associate Provost may judge that the conduct violation requires the student's probation, suspension or dismissal. In this situation, the case will be referred to the Disciplinary Committee, following any immediate action warranted by the Associate Provost. The following summarizes possible violations and disciplinary action to be taken by the Disciplinary Committee.

| Student Behavioral Misconduct         | Possible Disciplinary Action to be taken by the Disciplinary Committee                                                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inappropriate Student Conduct Level 1 | Disciplinary Written Warning<br>Disciplinary Probation                                                                                                                                                           |
| Inappropriate Student Conduct Level 2 | Disciplinary Probation<br>Dismissal                                                                                                                                                                              |
| Damage to University Property         | Orders by committee to pay actual repair or replacement costs – Failure to pay may lead to additional disciplinary actions and/or<br>Registration Hold<br>Transcript Hold<br>Disciplinary Probation<br>Dismissal |

For all levels of behavioral misconduct the Disciplinary Committee may remove the privilege of participating in or representing the university in extracurricular activities, including athletics, as well as the privilege of running for or holding office in any student organization that is allowed to use university facilities or receives university funds, or other restrictions as determined by the Disciplinary Committee.

### **Academic Integrity Violations**

Behavior that violates the Student Academic Integrity Policy is addressed through specific procedures associated with that policy.

### **Release of Student Course Evaluation Results**

Each semester AURAK conducts course evaluations. Students are assured of anonymity under Board Policy, which does not allow release of the results until all grades are posted. Should any student feel that anonymity of the course evaluation has been violated, a grievance may be filed with the Associate Provost of Student Affairs.

## **POLICIES RELATING TO STUDENT CONDUCT AND SAFEGUARDS**

### **Drug and Alcohol Policy**

The AURAK campus adheres to the laws of the UAE concerning alcohol and drug use. Any student found using drugs will face automatic dismissal. Any student found drinking alcohol or having it in their possession on campus will be automatically dismissed.

### **Individuals with Disabilities Policy**

The campus is committed to providing equal access to employment and educational opportunities for people of determination. AURAK recognizes that people of determination may need reasonable accommodations to have equally effective opportunities to participate in or benefit from campus educational programs, services, and activities and to have equal employment opportunities. The campus will adhere to all applicable UAE federal and state laws, regulations, and guidelines with respect to providing reasonable accommodations as necessary to afford equal employment opportunity and equal access to programs for qualified people of determination.



Students who seek reasonable accommodations for a disability should contact the Office of Student Affairs.

### **Non-discrimination Policy**

AURAK's non-discrimination policy prohibits discrimination based on race, color, religion, national origin, disability, sexual orientation, gender, or age. AURAK is committed to providing faculty, staff, and students with an environment in which they may pursue their careers or studies free from discrimination.

### **Statement Relating to Sexual Assault**

The following policy applies equally to all members of the AURAK community: students, faculty, administrators, staff, contract employees, and visitors.

AURAK and the AURAK campus is committed to providing an institutional environment where all people may pursue their studies, careers, duties, and activities in an atmosphere free of threat of unwelcome and unwanted sexual actions. It strongly condemns sexual offenses, will not tolerate sexual offenders, and supports those who have been victimized.

The campus will respond promptly, fairly, and decisively to all reports of sexual assault. Members of the campus community accused of sexual assault will be subject to campus disciplinary procedures when the alleged incident has occurred, on or off-campus, and affects the learning environment or operations of the campus. For those seeking assistance, please contact the Office of Student Affairs.

### **Statement Relating to Sexual Harassment and Associated Grievances**

The university is committed to providing an institutional environment in which all persons may pursue their studies, careers, duties, and activities in an atmosphere free of threat of unwelcome and unwanted sexual advances. The university strongly condemns sexual offenses, will not tolerate offenders, and supports those who have been harassed. Students who believe they have been the victim of sexual harassment should report the incident to the Office of Student Affairs. This policy applies equally to all members of university community: students, faculty, administrators, staff, contract employees, and visitors.

Unwelcome sexual advances, requests for sexual favors, and other verbal or physical conduct of a sexual nature constitute harassment when:

- The submission to or rejection of such conduct is made either explicitly or implicitly a term or condition of an individual's academic performance or employment.
- The submission to or rejection of such conduct by an individual is used as the basis for decisions about academic evaluation, employment, promotion, transfer, selection for training, performance evaluation, or selection for academic awards or benefits.
- The conduct has the purpose or effect of creating an intimidating, hostile, or offensive educational or work environment or substantially interferes with a student's academic or an employee's work performance.

Students who believe they have been the victim of sexual harassment should immediately report the incident to the Associate Provost for Student Affairs. Every effort will be taken to provide protection should any student feel threatened in any way.

### **Statement Relating to Stalking**

Stalking is prohibited and will not be tolerated. Stalking is defined as behaviors or activities, which, when taken as a whole, give the victim reason to believe that his or her safety is at risk or affects the learning experience or participation at AURAK. This policy applies to students, faculty, staff, contractors, and visitors, and applies to behaviors and activities occurring both on and off campus. Victims of stalking are encouraged to report incidents to the Office of Student Affairs.

### **Pets on Campus**

No pets, except those assisting people of determination, are permitted on campus at any time. Pets on campus grounds must be on a leash and under supervision at all times.

### **Facilities Management & Maintenance Policy - Smoke-free Environment**

Smoking is not permitted in any building on the AURAK campus including the student residences. There are designated areas for smoking outside the buildings.

## **STUDENT ATTENDANCE POLICY**

AURAK'S Student Code of Conduct requires that students attend classes regularly and punctually, complete assignments on time, and participate actively in class discussions and other course-related classroom activities. This section of the catalog outlines the University's expectations of regular student attendance in class meeting in all courses offered by the institution.

AURAK students are expected to:

- Attend all learning and teaching sessions associated with the courses in which they are enrolled.
- Notify their course instructors in advance that they will be absent from scheduled class sessions.
- Obtain prior permission from their instructors for planned absences of two or more consecutive class sessions during the semester.
- Provide a medical certificate or other corroborating evidence to explain an absence based on a health or medical condition.

Unsatisfactory student attendance includes failure to regularly attend class meetings without providing a satisfactory reason to instructors for absence and failure to observe protocols for online course attendance (e.g., having camera turned on). Unsatisfactory student attendance also includes persistent late arrival at, or early departure from, class meetings. At their discretion, instructors may count students absent if they arrive late for or depart early from scheduled class meetings as long as the information is included in the syllabus.

Where students fail to attend 10% of the scheduled classes for the semester without the University's permission, students will receive a "non-attendance warning," (NAW) and will be

required to provide satisfactory explanation for their non-attendance. Where students fail to attend 15% of the scheduled classes for the semester without the University's permission, a second NAW will be issued. With each NAW issued, a formal report on the student's non-attendance is made to the student's academic advisor and to the course instructor.

Where students fail to attend 20% of the scheduled classes for the semester, students will receive a notice that they have been withdrawn from the course with a grade of W if the 20% non-attendance is recorded prior to the deadline for withdrawing with a grade of W. If the 20% non-attendance is recorded after the deadline for withdrawing with a grade of W, students will receive a notice that they have failed the course and will receive an "F (Fail)" or "U (Unsatisfactory)" grade, as appropriate.

All students must adhere to AURAK expectations regarding class attendance.

## **ACADEMIC INTEGRITY**

### **AURAK HONOR CODE**

To promote a stronger sense of mutual responsibility, respect, trust, and fairness among all members of the AURAK community and with the desire for greater academic and personal achievement, we, the student members of the university community, have set forth this honor code:

***"As a member of the AURAK family, I pledge to be reliable, responsible, honorable, and embrace only the success and recognition I earn and deserve. I will act with integrity and honesty and will strive daily to become the best version of myself – a decent human being, a conscientious student, a good citizen, and a successful professional."***

The role of the Honor Code and associated Academic Integrity Policy is to protect the academic integrity of the university, encourage consistent ethical behavior among students, and foster a climate of honorable academic achievement. The Honor Code is an integral part of university life and students are responsible, therefore, for understanding and abiding by the code's provisions. While a student's commitment to honesty and personal integrity is assumed and expected, this Code and associated policy and procedures provides clarity of expectations. To maintain an academic community according to these standards, students and faculty follow the reporting policy and procedures herein.

At the beginning of each semester, faculty members have the responsibility to explain the Honor Code and their expectations for students in each class with special focus on plagiarism as well as specifics related to anticipated coursework and assignments. They must also explain the extent to which aid from others, if any, is permitted on academic work. Students should request an explanation of any aspect of an instructor's expectations regarding the Honor Code that they do not fully understand. Students have an obligation not only to follow the code themselves, but to encourage respect for and compliance with provisions of the code by other students. This includes the student obligation to report potential violations by fellow students and to understand that not doing so may also violate the Honor Code.

Faculty members are responsible for maintaining the integrity of the learning and evaluation process. Faculty members are expected to actively proctor exams, and should enforce the prohibition on possession of mobile phones during exams. Faculty may also take any additional action they believe is warranted to assure academic integrity.

While the internet and easy access to information and knowledge is commonplace, the University is committed to students understanding and demonstrating the ethical foundational values of scholarship. For all academic assignments, project work, and presentations, students need to ensure that due acknowledgement is given to the source of any information which they incorporate in their work.

### **Expectations**

Cheating, plagiarism, and all other forms of academic fraud are unacceptable; they are serious violations of university policy. AURAK expects all students to be familiar with university policies on academic integrity. The university will not accept a claim of ignorance – either of the policy itself or of what constitutes academic fraud – as a valid defense against such a charge.

### **Violations of Academic Integrity**

Violations of academic integrity constitute academic fraud. Academic fraud consists of any actions that serves to undermine the integrity of the academic process or that gives the student an unfair advantage, including:

- Inspecting, duplicating or distributing test materials without authorization.
- Cheating, attempting to cheat, or assisting others to cheat – relevant here is the prohibition on being in possession of a mobile telephone or similar electronic device during a test or examination. In case such devices are found with a student, the student will be deemed to have attempted to cheat and will be subject to disciplinary action under the Student Academic Integrity Policy.
- Altering work after it has been submitted for a grade.
- Plagiarizing.
- Using or attempting to use anything that constitutes unauthorized assistance. PLEASE NOTE: Faculty members may prohibit the use of generative AI, including though not limited to, generative AI such as Open AI ChatGPT and Canva, in completing assignments. When such prohibitions have been communicated by the faculty member, incorporating information from such sources into your assignment submission will be treated as a serious violation of academic integrity expectations.
- Fabricating, falsifying, distorting, or inventing any information, documentation, or citation.

### **Statement on Plagiarism**

One of the most common violations of academic integrity is plagiarism. Plagiarism can be intentional or unintentional. However, since each student is responsible for knowing what constitutes plagiarism, unintentional plagiarism is as unacceptable as intentional plagiarism and thus will bring the same penalties.

Plagiarism – submitting the work of others as one’s own - is a serious offense. This includes submitting work obtained from AI writers such as Open AI Chat GPT, as well as other forms of generative AI. In the academic world, plagiarism is theft. Information from sources – whether quoted, paraphrased, or summarized – must be given credit through specific citations. When a student paraphrases a work, it is still necessary to cite the original source, even when the information has been provided by generative AI writers and/or sources. Merely rearranging a sentence or changing a few words is not sufficient. The citation style should be appropriate for the discipline and should clearly indicate the beginning and ending of the referenced material. All sources used in the preparation of an academic paper must also be listed with full bibliographic details at the end of the paper, as appropriate in the discipline. **PLEASE NOTE:** Faculty members may prohibit the use of generative AI in completing assignments. When such prohibitions have been communicated by the faculty member, incorporating information from such sources into your assignment submission will be treated as a serious violation of academic integrity expectations.

While plagiarism detection software can assist identifying plagiarism, there is no “percentage of matching content” threshold for determining that content in a written assignment has been plagiarized. Indeed, the presentation of a single striking phrase originally written by another without attribution to the original source can constitute plagiarism, even though the percentage of matching content found by plagiarism-checking software might be very small.

### **Faculty and Student Responsibilities**

- Every student, faculty member, and administrator is responsible for upholding the highest standards of academic integrity. Every member of the AURAK community shall honor the spirit of this policy by refusing to tolerate academic fraud.
- When expectations for a course are not addressed in this policy, it is the responsibility of the instructor to provide students with additional guidelines for what constitutes “authorized” and “unauthorized” assistance.
- It is the responsibility of every student to seek clarification if in doubt about what constitutes “authorized” and “unauthorized” assistance. In cases involving collaborative work, all students within the collaborative group may be held responsible for violating the code if any member of the group receives, accepts, or utilizes “unauthorized” assistance.
- Students are required to obtain permission prior to submitting work, any part of which was previously or will be submitted in another course. The instructor has the option of accepting, rejecting, or requiring modification of the content of previously or simultaneously submitted work. A student who suspects that a violation of academic integrity has occurred should report that violation to the dean or to the Office of the Provost. In this report, the student should describe any action taken, such as talking with the person involved or with a faculty or staff member. Every effort will be made to preserve the anonymity of the student reporting the incident; however, confidentiality cannot be guaranteed.

## Resolution at the Faculty/Student Level

Where a faculty member suspects a student of a possible violation of academic integrity, the faculty member writes formally to the student, using the Student Academic Integrity Violation Submission Form, with a copy of the communication sent to the Academic Integrity Officer. The communication provides the student with five working days to respond, prior to the faculty member making a final determination regarding an academic integrity violation.

Taking into account any response provided by the student, the faculty member decides, following the five-day period for the student's response, if the student has breached academic integrity and whether a penalty is warranted. The faculty member then prepares a report detailing the alleged academic fraud, with a copy sent to the:

- Dean of the School in which the affected course is located.
- Dean of the School in which the student is enrolled.
- Student
- Academic Integrity Office (as Office of Record).

The suggested penalty for academic fraud in any course is failure in the course. However, faculty members may impose lesser penalties. Penalties available to faculty members include:

- Formal warning – appropriate only for early assignments in first-year undergraduate courses.
- Reduction in grade for the assignment or assessment (test) up to and including a grade of zero for the assignment or assessment – a reduction in grade less severe than a grade of zero is appropriate only for early assignments/assessments in first-year undergraduate courses.
- Reduction in the grade for the course up to and including a grade of F for the course. A reduction in grade for an assignment or assessment is likely to result in a reduction in the grade for the course. However, in instances in which the assignment or assessment is weighted such that a grade of zero does not result in a grade of F for the course, the faculty member may impose the penalty of a grade of F in the course. A faculty member can also choose to impose this penalty for an egregious act of academic dishonesty, such as attempting to steal an examination or submitting the work of another person as one's own.

## Review by the Academic Review Board

All reports of violations of academic integrity are reviewed by the Academic Integrity Officer to determine whether there is a record of previous academic integrity violations and if the violation warrants a penalty different from that imposed by the faculty member. If the student has multiple academic integrity violations or the Academic Integrity Officer determines the violation under review warrants a penalty different from that imposed by the faculty member, the case may be referred to the Academic Review Board.

Penalties for academic integrity violations available to the Academic Review Board include:

- **An F grade in the course.** An F grade assigned for a course as a result of academic fraud will replace any other grade previously posted for the course.

- **Academic Dishonesty Probation.** When a student is found to have a second academic integrity violation but the offenses are not sufficiently egregious to warrant dismissal, a student will be placed on Academic Dishonesty Probation for a designated period of time, e.g., until graduation.
- **Academic Dishonesty Suspension.** Any academic integrity violation by a student while on Academic Dishonesty Probation will result in suspension from the university for at least one semester and possibly up to one academic year. Students suspended for academic dishonesty cannot transfer into AURAK any credits earned at another institution during the suspension. To qualify for readmission to AURAK following suspension, a suspended student will be asked to complete online training related to academic integrity. Readmission is not assured, but will be based on approval by the Academic Integrity Officer in consultation with the Dean of the School in which the student is enrolled and the Provost.
- **Academic Dishonesty Dismissal.** Any additional instance of academic dishonesty following Academic Dishonesty Suspension will result in Academic Dishonesty Dismissal. A student dismissed for academic dishonesty will not be readmitted to the university.

### Appeal to the Academic Review Board

Students may appeal a decision on academic integrity made by either a faculty member or the Provost to the Academic Review Board within ten (10) working days of notice of the decision. For decisions made by the Academic Review Board, the student may request reconsideration of the decision by the Academic Review Board within ten (10) working days of notice of the decision. Grounds for appeal are limited to:

- failure to follow university procedure for cases of alleged academic integrity violations;
- presentation of evidence not considered when the original decision was reached.

At an appeal before the Academic Review Board, the student has the opportunity to be heard and the right to produce witnesses or introduce evidence subject to the reasonable discretion of the Academic Review Board. The student may also be accompanied by a representative of his or her choosing who may not participate in the appeal.

The results of all appeals to the Academic Review Board, permitted in the Academic Integrity Policy, shall be final. The Academic Review Board will communicate its final decision to the student, the dean of the school in which the student is pursuing a program of study, the Provost, and the Department of Registrar.

Any matter submitted to a faculty member or the university for decision or review under this policy shall be decided in a timely manner. The parties shall use their best efforts to conclude the proceedings within the semester in which the alleged offense has occurred or the appeal is received. However, any delay in the proceedings shall in no way operate as a waiver of the university's right to assess any or all of the sanctions permitted under the Academic Integrity Policy.



# **STUDENT RECORDS**

## **CUSTODY OF RECORDS**

All transcripts and other documents submitted from other institutions at time of admission, or later, are the property of AURAK, and are part of the student record that is under the custody of the AURAK Registrar. The university is not required to provide copies of these documents. Transcripts submitted to AURAK for admission or credit transfer cannot be returned to the student or forwarded to other institutions.

## **Students Privacy Rights**

Students have the right to inspect and review information contained in their educational records. The university is not required to provide copies of these documents. The university allows, nevertheless, copies to be made of specific documents included in a student's record. Copies of such documents will only be provided upon submission of a signed request from the student concerned.

## **Academic Transcripts**

A permanent record reflecting the academic achievements of each student who registers at the university (i.e., transcript) is maintained by the Office of the Registrar. Official transcripts are available on request.

## **Records on Academic Integrity Code Violations**

The retention of records on academic integrity code violations is governed by the following:

All records pertaining to the infringement of the code are maintained by the student's school. If the student does not graduate from AURAK, the records are retained for five years after the student's last registration. If the student graduates from AURAK, these records are destroyed by the school upon the student's graduation.

## **Official Verifications and Certifications**

All certificates related to a student's status at AURAK must be requested from the Office of the Registrar; a nominal fee applies.

## **RELEASE OF INFORMATION**

Access to a student's education records is available to authorized officials of AURAK for academic purposes, such as recording grades, attendance, advising, or review of academic progress.

The confidentiality of student information is maintained, and periodic reviews are made in order to check the efficacy of the system. Student records are not disclosed to any third party unless a student gives permission for this in writing. The only exceptions regarding release of student information include the release of information to official accreditation bodies conducting a review or to law enforcement officials conducting pertinent individual case reviews.



## UNDERGRADUATE PROGRAMS

## UNDERGRADUATE ADMISSIONS

AURAK seeks to admit qualified students who wish to become part of an innovative global academic institution based on the American model. It is the goal of AURAK to admit those students most likely to benefit from its academic programs and diverse student body. Applications are welcome from all students without regard to race, color, religion, national origin, disability, sex, or age. To be considered for admission, each applicant must present sufficient evidence of appropriate qualifications as outlines by AURAK Board Policy and AURAK Presidential Procedure. The factors considered include, but are not limited to, scores on school leaving certificates and on standardized exams, English proficiency, and level of difficulty and appropriateness of coursework, essays, and recommendations. An offer of admission is valid only for the semester for which the student applies but may be extended at the discretion of the university. ***As AURAK must comply with the expectations of licensure and accreditation, the university reserves the right to modify admissions criteria at any time by action of the Board of Trustees or the President.***

### English Language Proficiency

The language of instruction at AURAK is English, therefore prospective undergraduate students must demonstrate a reasonable level of proficiency when admitted to a degree program. AURAK accepts the following assessments at the indicated levels as evidence of English proficiency.

| Type of English Proficiency Exam | Minimum Score |
|----------------------------------|---------------|
| EmSAT                            | 1100          |
| TOEFL Paper-based                | 500           |
| TOEFL iBT                        | 61            |
| TOEFL CBT                        | 173           |
| IELTS Academic                   | 5.0           |

Institutional TOEFL scores from other institutions are not valid except if the TOEFL was completed in AURAK or AMIDEAST offices in the UAE. International TOEFL scores should be sent directly to AURAK's Department of Admissions or ordered online through the official ETS website at [www.ets.org](http://www.ets.org). AURAK's TOEFL school code is 4574.

IELTS (Academic) scores are valid for admission to the university, but IELTS (General) scores are not. The exam can be conducted in any of the test centers listed on [www.ielts.org](http://www.ielts.org). Once the IELTS result is received, a copy of the results must be provided to the Department of Admissions by the student.

When reviewing applications, the Department of Admissions considers the academic rigor of the stream of study.

| AURAK Academic Program   | Admission Criteria Based on High School Performance and EmSAT Test Scores |                       |     |  |
|--------------------------|---------------------------------------------------------------------------|-----------------------|-----|--|
| BA in Mass Communication | UAE Curriculum                                                            | Advanced (Scientific) | 70% |  |
|                          |                                                                           | General (Literary)    | 70% |  |

| AURAK Academic Program   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Admission Criteria Based on High School Performance and EmSAT Test Scores                               |     |                                                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------|
|                          | Non UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The University recognizes all the other certificates and converts their grades to the equivalent grade. |     |                                                                     |
| BA in Interior Design    | UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Advanced (Scientific)                                                                                   | 70% |                                                                     |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | General (Literary)                                                                                      | 70% |                                                                     |
|                          | Non UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The University recognizes all the other certificates and converts their grades to the equivalent grade. |     |                                                                     |
|                          | ➤ Once accepted into the Interior Design program, students will be placed into an appropriate initial course in Mathematics based either on their score on the EmSAT in Mathematics or their score on the Accuplacer Quantitative Reasoning, Algebra and Statistics Test. The Accuplacer Quantitative Reasoning, Algebra and Statistics Test will be administered on the AURAK campus through the Center for Executive and Professional Education.                          |                                                                                                         |     |                                                                     |
| Bachelor of Architecture | UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Advanced (Scientific)                                                                                   | 70% |                                                                     |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | General (Literary)                                                                                      | 70% |                                                                     |
|                          | Non UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The University recognizes all the other certificates and converts their grades to the equivalent grade. |     |                                                                     |
|                          | ➤ Achieving an average of 70% or higher in the secondary school certificate.                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                         |     |                                                                     |
|                          | ➤ Once accepted into the program Architecture, students will be placed into an appropriate initial course in Mathematics based either on their score on the EmSAT in Mathematics or their score on the Accuplacer Advanced Algebra and Functions Placement Test. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education.                                         |                                                                                                         |     |                                                                     |
|                          | ➤ Once accepted into the program in Architecture, students will be placed into an appropriate initial course in Physics based either on their score on the EmSAT in Physics or their score on the AURAK Physics Placement Test. The AURAK Physics Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education.                                                                                                      |                                                                                                         |     |                                                                     |
| BS in Biotechnology      | UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Advanced (Scientific)                                                                                   | 70% |                                                                     |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | General (Literary)                                                                                      | 70% | Successful completion of BIOL 095, MATH 095, and CHEM 095 at AURAK. |
|                          | Non UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The University recognizes all the other certificates and converts their grades to the equivalent grade. |     |                                                                     |
|                          | ➤ Passing EmSAT in Mathematics with a score of 700, or achieving on the Accuplacer Advanced Algebra and Functions Placement Test a score corresponding to placement in AURAK’s Mathematics course MATH 111 – Calculus for the Life Sciences. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed. |                                                                                                         |     |                                                                     |

| AURAK Academic Program                                  | Admission Criteria Based on High School Performance and EmSAT Test Scores                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                         |     |  |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----|--|
|                                                         | <p>➤ Passing EmSAT in Chemistry or EmSAT in Biology with a score of 700 or equivalent, or achieving a score on the AURAK Chemistry Placement Test corresponding to placement in AURAK’s Chemistry Course CHEM 211 – General Chemistry 1. The AURAK Chemistry Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                         |     |  |
| BS in Artificial Intelligence<br>BS in Computer Science | UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Advanced (Scientific)                                                                                   | 70% |  |
|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | General (Literary)                                                                                      | 70% |  |
|                                                         | Non UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The University recognizes all the other certificates and converts their grades to the equivalent grade. |     |  |
|                                                         | <p>➤ Passing EmSAT in Mathematics with a score of 700, or achieving on the Accuplacer Advanced Algebra and Functions Placement Test a score corresponding to placement in AURAK’s Mathematics course MATH 113 – Calculus 1. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.</p> <p>➤ Passing EmSAT in Physics with a score of 700 or equivalent, or achieving a score on the AURAK Physics Placement Test corresponding to placement in AURAK’s Physics Course PHYS 110 – University Physics 1. The AURAK Physics Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.</p> |                                                                                                         |     |  |
| All Business Programs                                   | UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Advanced and Elite (Scientific)                                                                         | 70% |  |
|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | General (Literary)                                                                                      | 75% |  |
|                                                         | Non UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The University recognizes all the other certificates and converts their grades to the equivalent grade. |     |  |
|                                                         | <p>➤ Passing EmSAT in Mathematics with a score of 600, or achieving on the Accuplacer Quantitative Reasoning, Algebra and Statistics Test a score corresponding to placement in AURAK’s Mathematics course STAT 100 – Introductory Probability &amp; Statistics. The Accuplacer Quantitative Reasoning, Algebra and Statistics Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.</p> <p>➤ Personal interview is also required.</p>                                                                                                                                                                                                                                                                                                                               |                                                                                                         |     |  |
| All Engineering Programs                                | UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Elite Track                                                                                             | 75% |  |
|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Advanced Track                                                                                          | 80% |  |
|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Or Equivalent in Standardized International Systems                                                     |     |  |
|                                                         | Non UAE Curriculum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The University Recognizes all the other certificates and converts their grades to the equivalent grade. |     |  |

| AURAK Academic Program | Admission Criteria Based on High School Performance and EmSAT Test Scores                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <ul style="list-style-type: none"> <li>➤ Passing EmSAT in Mathematics with a score of 900, or achieving on the Accuplacer Advanced Algebra and Functions Placement Test a score corresponding to placement in AURAK's Mathematics course MATH 113 – Calculus 1. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.</li> <li>➤ Passing EmSAT in Physics with a score of 800 or equivalent, or achieving a score on the AURAK Physics Placement Test corresponding to placement in AURAK's Physics Course PHYS 110 – University Physics 1. The AURAK Physics Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.</li> <li>➤ Passing EmSAT in Chemistry with a score of 700 or equivalent, or achieving a score on the AURAK Chemistry Placement Test corresponding to placement in AURAK's Chemistry Course CHEM 211 – General Chemistry 1. The AURAK Chemistry Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.</li> </ul> |

### **Minimum EmSAT Scores in Subjects Other than English**

For students from school systems that use standardized and/or centralized examinations (such as the British, the International Baccalaureate, or the Emirati School system), achieving 80% or higher (or the equivalent based on the Scholaro GPA Calculator) in required subjects can substitute for the required EmSAT score for all subjects except English.

### **Secondary School Completion**

For admission, applicants must provide a Secondary School Certificate confirming completion of 12 years of secondary school at a Ministry of Education School, or an equivalence certificate issued by the Ministry of Education for private schools.

### **Transfer Admission Requirement**

AURAK will award credit transfer where a student is transferring from:

- a UAE institution recorded in the National Register of Licensed Higher Education Institutions (HEIs) (or approved registry), or
- other organizations in the UAE approved by the Commission for Academic Accreditation, or
- recognized institutions of higher learning located outside the UAE, which are eligible for transfer admission.

Students will only be eligible for the above credit transfer arrangements, where they:

- Provide valid evidence that they meet the English language competency scores for full admission.
- Are in good academic standing and have a CGPA of 2.0 on a 4.0 scale, or equivalent.



Transfer students not in good academic standing at their previous institution, but who meet the English language competency scores for full admission, may receive credit transfer to a program that is not related to the program that they previously studied.

In all cases, students seeking credit transfer may be admitted as per the Ministry of Education requirements, and credit transfer will be restricted to:

- Related courses where students have earned a grade of C (i.e., 2.0 on a 4.0 scale (or equivalent)).
- Courses that deliver similar outcomes.
- A maximum of fifty-percent (50%) of the credit for the receiving program.
- Courses other than graduation projects or thesis.

Where credit transfer is awarded for courses, it will only be awarded once for a given course and there will be no duplication of credit for a course. In addition, credit awarded by credit transfer will not contribute to the calculation of the CGPA.

AURAK will not award credit for experiential or other non-classroom based learning unless the student has undertaken a challenge examination and the process for awarding such credit has been pre-approved by the Commission for Academic Accreditation.

### **Other Transfer Requirements**

The Department Chair/Program Coordinator, with the assistance of faculty specializing in the field covered by courses, will review the application and supporting documentation to identify whether credit transfer can be applied for the courses. The following will be applied:

- The courses for which credit transfer is being sought must have the same number of credits and equivalent learning outcomes as the AURAK courses.
- The courses for which credit transfer is being sought must be at the same level (upper or lower).
- Credit cannot be given for two courses which are substantially the same course.
- For Undergraduate credit transfer will only be permitted for up to 50% of the program credits and only courses graded as “C” or above are considered for transfer credit.
- Graduation projects and thesis are not eligible for transfer credit.

The Department Chair/Program Coordinator will complete an Evaluation Form setting out which courses and credits are recommended for transfer credit, and will forward this to the Dean of School who will validate the appropriateness of the transfer credit recommended by the Department Chair/Program Coordinator.

Where there is disagreement the Dean of School will discuss with the Department Chair/Program Coordinator but the Dean of School decision is final. The Dean of School will sign the evaluation form with the final decision and notify the Department of Admissions and the Registrar.

The Registrar will finalize the number of credits to be transferred and will ensure that the:



- Transferred courses are listed on the student's transcript courses.
- Transferred credits and grades are not used in calculating the cumulative grade point average (CGPA).

The Admissions Department will notify the transfer applicant of the final decision.

The Registrar will place the documentation related to the transfer credit on the student's file.

### **Application Procedures (For Both Freshman and Transfer Applicants)**

Prospective students can apply online or in person. The online application is available on <http://apply.aurak.ac.ae>

The following documents are required with every application:

- Passport copy
- Copy of health card
- Copy of valid Emirates ID (UAE residents only)
- Health History Form (Completed and signed by a physician)
- Four (4) recent passport-size photographs
- Exemption letter from the National and Reserve Service Authority (UAE male applicants between the ages of 18 and 30 only)
- Birth Certificate
- Family Book (UAE nationals only)
- Army Exemption/Completion Letter (UAE nationals only)

### **Attestation of Documents**

Admitted students are required to submit officially attested education certificates by the deadline mentioned in their admission letter. Students with certificates obtained outside the UAE **may** be given an extension until the end of the first semester of their studies subject to the approval of the Manager of Admissions. Subsequent extensions are subject to approval of the Associate Provost of Student Affairs. Failure to provide appropriate attested education certificates will lead to expulsion from the institution. AURAK transcripts are not provided until all admission requirements are satisfied. For more information regarding attestations, contact the Enrollment Management Department: [admissions@aurak.ac.ae](mailto:admissions@aurak.ac.ae).

### **Admission Interview**

The Enrollment Management Department reserves the right to require an applicant to attend an admissions interview as part of the application process when required by the prospective program

or there is a need for additional information. Interviews may take place on campus, in person, by digital means, or by phone where appropriate.

### **Admission Categories**

#### **Full Admission**

Applicants must meet all outlined requirements to obtain full admission status. To qualify for full admission a new undergraduate student must hold a UAE valid secondary stage completion certificate (SSCC) or attested equivalent along with a detailed statement of the grades for each subject. Applicants may be granted full admission on the basis of unofficial copies of such documents but must provide the official copies of required documents within two semesters following initial enrollment or face withdrawal from the program of study.

#### **Conditional Admission**

Applicants considered marginal may be conditionally admitted subject to a recommendation from the program coordinator requesting exceptional consideration.

Applicants lacking the following may also be conditionally admitted:

- English language proficiency.
- Required EmSat score in Mathematics, Physics, or Chemistry for the desired program of study.

During the initial semester of study as conditionally admitted students, students may take up to 12 semester credits of General Education in addition to possible foundational coursework corresponding to the missing EmSAT score(s). The foundational coursework will not count toward the undergraduate degree or the students' CGPA.

### **Admission of Visiting Students**

A visiting student is one who is not formally admitted to AURAK and who is not participating in a formal exchange agreement between AURAK and a partner university. A visiting student is typically enrolled in a degree program at another university. This student may take courses at AURAK for transfer to the student's home institution. Visiting students are advised to check with their home institutions to confirm the transferability of AURAK credits.

Applicants seeking visiting student status must submit to AURAK's Department of Admissions the following documents:

- Copy of high school certificate
- An official letter from the university allowing the student to enroll for the course(s) at AURAK
- English proficiency scores (EmSAT, TOEFL, IELTS or Equivalent).
- Passport copy, Emirates ID, and residence visa (If Applicable)
- Four (4) passport-size photographs
- Proof of acceptable health insurance or purchase AURAK health insurance

### **Admission Requirements Subject to Program Admission Specifications as Outlined by the Ministry of Education**

From time to time, the Ministry of Education establishes expectations for admission to specific programs. These expectations are communicated by MOE Letters of Reform. In all cases, AURAK conforms to the program admission standards outlined in MOE Letters of Reform.

### **Application Deadlines**

The application deadline for fall admission is notified through newspapers or through university notice boards for freshmen and transfer applicants. Admission decisions for first year students are usually made after receipt of the first-semester grades of the senior year of secondary school and all appropriate test results. Admission is contingent upon satisfactory completion of in-progress course work and graduation from the secondary school. Transfer decisions are made as files become complete.

AURAK Admissions Application Deadline for the Fall 2023 Semester is **11 September 2023**.

AURAK Admissions Final Application Deadline for the Spring 2024 Semester is **21 January 2024**.

Admission deadlines are subject to extension by the Department of Admissions where circumstances warrant.

Note: International students must apply two weeks in advance of the stated deadline to meet visa timeline requirements.

### **Acceptance of Admission Offer**

Students will be required to confirm their acceptance of an admission offer by completing the enrollment confirmation form in the admission packet and submitting it to AURAK with the enrollment deposit, admission fee, visa and health insurance fee (if applicable), and residence hall fee (if applicable).

### **Application for a Second Bachelor's Degree**

Existing and former AURAK students may apply to undertake a second degree at AURAK provided such degree will enable the completion of at least thirty (30) credit hours beyond the first degree undertaken at the institution. For existing students, this will only be permitted where they have met all the requirements of their first degree.

### **New Student Placement Exams**

An English language placement exam is administered to entering students. The local English language placement exam is utilized for placement in the appropriate level English course. AURAK uses a math placement exam to determine which mathematics course new students may take. Students who were awarded transfer credit for a math course taken at another institution are not required to take the math placement exam.

Scores from the EmSAT can be utilized for placement into mathematics, physics, and English courses.

### Student Residence Visa

The Human Resources Department arranges student visas under the sponsorship of AURAK. Visas must be renewed every year. Students on an AURAK visa who discontinue their studies or transfer to another university must submit their passport for visa cancellation prior to their departure.

The visa process takes approximately two weeks during which students cannot leave the country as they must hand over their passports temporarily. The visa application requires:

- Eight (8) passport photos
- Passport copy
- Copy of the secondary school certificate
- Visa Request Form
- Payment of visa fee and security deposit for visa

All student applicants for a residence visa must undergo medical tests conducted by the Preventive Health Department of the RAK Government and fingerprinting by the Police Department. The medical tests include a blood test and an X-ray. Failure to pass either one of the medical tests leads to deportation to country of origin.

### Right to Withdraw Offer

AURAK may on the approval of the Registrar withdraw offers of admission if an applicant fails to satisfy all requirements, or if it is determined that admission was obtained through the use of falsified, altered, or embellished information (or similar). In the case of withdrawal of admission from a matriculated student, credit earned may be withheld.

### Readmission

Students who have missed two or more consecutive semesters (excluding Summer Semester) of enrollment at AURAK will be required to apply for re-admission on the following basis:

- A student returning after any absence during which he or she studied at another institution without prior written permission of the Vice President of Academic Affairs and Student Success (VPAASS) or designee will be treated as a **transfer student**.
- A student who failed to demonstrate good academic standing during their previous studies at AURAK, will be required to reapply for **full admission** in order to be considered for admission.
- A student who was in good academic standing during the previous studies at AURAK, may be readmitted on completion of a Re-Enrollment Form available through the Office of the Registrar.

## **UNDERGRADUATE ACADEMIC POLICIES**

Each student is responsible for knowing AURAK's rules, regulations, requirements, and academic policies. The Student Handbook, the University Catalog, and the institutional website are repositories of policy statements. Corrections, changes, or interpretations may be communicated by other means, including electronically. Any student in doubt about an academic matter should consult the Office of the Registrar or their assigned faculty advisor. Students are subject to the university's stated policies regarding patents and copyrights.

### **Academic Advising**

AURAK is committed to helping students achieve success in their course of study. Ensuring that students receive appropriate academic advising is part of that commitment.

New students are first advised during a required orientation program in which they learn about university policies and procedures and receive a current university catalog and a student handbook. At the initial meeting with their faculty advisor during orientation, each student is informed of program procedures, course of studies, and the importance of understanding the information in the university catalog. Advisors also answer specific questions concerning the academic program.

Each student is responsible for his or her progress toward degree and for meeting all requirements for graduation. The advisement process is available to assist the student in completing his or her studies, but responsibility for doing so lies ultimately with each student.

Faculty advisors are required to post office hours for advising that coincide with students' availability. The faculty advisor is to maintain advising files for all advisees that contain a log of meetings with a brief description of major issues discussed or advice given. Any information contained in these files shall remain confidential.

Students on probation are required to meet with their advisor on a regular basis to assess their progress and to report their use of AURAK's various resources for academic support. Students violating AURAK's attendance policy or doing poorly at mid-term are also required to meet with their advisors.

If students have concerns, comments, or recommendations about their educational experiences at AURAK, they should contact the dean of the school in which their degree program is located.

### **Courses and Course Schedules**

Each course offered by the university has a designated course prefix (or code) and number. The course prefix represents the discipline or field of study; the number indicates the level of the course content. MATH 108 thus indicates that the course is Mathematics and is appropriate for students in their first year; CHEN 472 indicates that the course is in Chemical Engineering and is intended for fourth year students.

Courses are offered at the discretion of the Senior Vice President for Academic Affairs and Student Affairs and the respective deans and department chairs. Not every course is offered every semester

but are offered at a frequency that enables students to make satisfactory progress toward their degrees.

### **Course Credit**

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the sixteen-week semester. Normally, one credit represents fifty minutes of class instruction per week per semester or 150 minutes of laboratory experience per week per semester.

### **Class Periods**

Classes normally meet two times a week in one hour fifteen-minute periods or two. In some cases, usually advanced courses, classes may meet once per week for 150 minutes. The university traditionally holds classes four days a week from Monday through Thursday. Graduate courses are usually held on Friday, Saturday or Sunday. Though university administrative offices are usually closed on Saturday and Sunday, appointments may be arranged to meet students during the weekends.

All classes are expected to meet on the days and times published in the course schedule. Changes may only be made with the approval of the Senior Vice President for Academic Affairs and Student Affairs.

Independent study or research, internships, other programs for experimental learning, and other study opportunities may follow a different, approved time frame and schedule.

### **Course Prerequisites and Corequisites**

Course prerequisites or corequisites state requirements for student entry into courses and reflect necessary preparation for enrolling in a course. It is the student's responsibility to be aware of these requirements as stated in the university catalog and to register for courses only if stated prerequisites have been completed. Similarly, if a course has a corequisite, a student can only register for the course if the corequisite is also selected for registration in the same semester. The Student Information System has been designed to prevent students from registering for courses for which prerequisites have not been completed or for which corequisites have not been selected for simultaneous registration. However, if a student succeeds in enrolling in a course without having completed re-requisites or corequisites for the course, the student will be dropped from the courses as soon as the registration error is detected. No waivers of prerequisites or corequisites will be granted.

### **Course Selection and Registration**

Typically, by mid-semester, the schedule of classes for the following semester is available through the Office of the Registrar and on the website. Students may select courses in consultation with their faculty/academic advisor, and they can then register online or submit a completed "Course Request Form" to the Office of the Registrar by the published deadlines.

### **Registration Criteria**

Only registered students may participate in AURAK classes. Full-time status requires that an undergraduate student enroll in 12 or more semester credit hours in Fall and Spring semesters.

Undergraduate students enrolling in fewer than 12 semester credit hours are considered to be enrolled on a part-time basis, but no undergraduate student may enroll for fewer than 6 semester credit hours, except in cases in which students need fewer than 6 credit hours to complete the degree. Undergraduate students whose semester credit hour loads are limited due to probation requirements are considered to hold full-time status. All undergraduate students are accepted to the university on the assumption that they will be studying full time, and the timeframe for completing the program of study described in this catalog is based on full-time study. Registering as a part-time student results in a significantly higher per credit tuition fee, makes students ineligible for scholarships, and lengthens the program of study.

### **Independent Study**

Occasionally, students may have the opportunity to benefit from an independent study experience that will significantly add to their preparation for a career or ensure timely completion of their program of study. Only one course (maximum of 4 credits) may be completed as an independent study in a student's program of study. The independent study must be approved by the instructor(s) involved. Students must have completed a minimum of sixty credits, must have a 3.0 (B) or better average, and seek approval through completion of the appropriate "Permission for Independent Study" Form. The instructor must hold a regular rank faculty appointment at AURAK within the department or program requesting the independent study and is responsible for submitting the final grade, and ensuring adherence to academic standards, policies, and procedures. In addition to the individual effort of the student, a student enrolled for independent study must meet weekly with the instructor for guidance in research, analysis, and composition of the project report or presentation. Independent study courses may not duplicate available course offerings in the semester in which the independent study is being taken.

The independent study must provide a rigorous academic experience equivalent to that of any other undergraduate course at the same level at AURAK. Grading of an independent study adheres to the established grading system.

### **Concurrent Enrollment at Other Institutions**

Students who apply for admission to AURAK usually do not seek simultaneous enrollment with another institution. Should students seek concurrent enrollment, they must obtain written approval in advance from the VPAASS/Provost. Such approval enables a student to enroll elsewhere in a course unavailable at AURAK.

Catalog numbers and descriptions of courses to be taken elsewhere must be submitted with the request for approval. As for any transferred course, students must submit an official transcript to the Office of the Registrar to earn credit, and grades are not included in the GPA. Students who enroll elsewhere without advance written permission while enrolled at AURAK may not receive transfer credit for the course work taken.



### **Summer Term**

During the summer term, students may register for one or two courses. All attendance, grading, and other academic policies apply to the summer term as well as to the two academic semesters.

### **Student Coursework Assessment Policy**

Academic assessment of students must be robust, reliable, fair, accurate, aligned with learning outcomes and program goals, and undertaken at an appropriate level. As such, undergraduate students will be required to undergo a minimum of three (3) assessments distributed evenly throughout the semester. Acceptable assessment tools include:

- (a) Exam
- (b) Project
- (c) Paper
- (d) Presentation
- (e) Homework assignment
- (f) In class quiz
- (g) Case study/Analysis

'In class participation' will also be acceptable as an assessment tool, but will only be used to calculate the final grade where it does not exceed ten percent (10%) of the final grade.

Irrespective of the assessment tool used, it must be appropriate for the learning outcome(s) under assessment. Assessments may be adjusted to accommodate people of determination, however, such special adjustments must not invalidate the assessment.

### **Weighting of Assessment**

- No single assessment tool can count for more than forty percent (40%) of the total grade.
- "In class participation" may be used in calculating final grade provided it does not exceed ten percent (10%) of the total grade.
- Attendance has no weight in the final grade as attendance requirements are explained under "attendance policy."

### **Midterm Exams**

Midterm progress is reported for all semester-long courses so that students will have a clear idea of their standing in each course midway through the semester. The reporting period extends from the fifth through the ninth week of the semester, allowing flexibility as to when individual faculty provide reports for their classes. Midterm grades are reported as S (satisfactory - C+ and above) or U (unsatisfactory - C and below). Students should check with their instructors as to when reports will be complete and available for viewing. These progress reports, which appear as 'Midterm Grades,' are not part of the student's official record. They are not calculated in the GPA, nor do they appear on any official or unofficial transcript. The midterm progress report is

distributed to the student, his or her faculty advisor, and the Office of Student Affairs.

### **Final Exams**

Final exams are usually given at the end of all undergraduate courses. Except for project submissions, laboratory courses, and studio courses, no exam or assessment tools of any kind may be given during the last week of classes. Exams may not exceed the scheduled length of two (2) hours for most courses. The days following the last day of classes each semester that comprise the final exam period are identified in the Academic Calendar found on the university website. Students must plan to be available for final exams during the entire final exam period. The schedule showing the specific days and times for final exams for each course is posted each semester on the university website six weeks prior to the last day of classes.

An instructor who is assigning a take-home exam or significant end-of-semester paper or project should so inform the students on the syllabus at the beginning of the semester. Take home exams should be distributed by the beginning of the last week of classes so that students can coordinate them with preparation for other exams. Students must not be required to submit exams before the date of the regularly scheduled exam for a course. Accommodations will be made for students with more than one final exam scheduled in the same time period. Retaking (or re-sitting) a final exam is not permitted. Students are not expected to take more than two final exams in one day.

### **Absence from Final Exams**

Absences from final exams are not to be excused except for sickness on the day of the exam or for other causes approved by the dean of the respective school. The effect of an unexcused absence from an undergraduate final exam shall be determined by the weighted value of the exam as stated in the course syllabus provided by the instructor. If an absence from a final exam is unexcused, a grade of zero will be entered for the final exam.

### **Final Semester Grades**

Final grades are available through the student portal on the university website and/or collected from the Office of the Registrar. Official transcripts and official semester grade reports for tuition reimbursement are obtained through the Office of the Registrar.

Students may access and print an unofficial record of their semester grades and other academic information through the student portal on the university website by logging onto their Student Information System (SIS) account. Official transcripts are available upon request.

### **Change of Grade/Grade Appeal**

Once a final grade has been recorded by the faculty member, it can be changed only through the Grade Appeal process. Where students believe their final course grades to be in error, students must request a meeting with the member of faculty to ask that the faculty member reconsider the grade and ensure that the appeal is submitted within the required timeline. Completed Grade Appeal Forms must be submitted within the following timeframe:

- For Fall Semester courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Spring Semester.

- For Spring Semester courses, completed Grade Appeal Forms must be submitted within three weeks of the “Grades Available” date listed in the Academic Calendar.
- For Summer Session courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Fall Semester.

***The grade appeal will only be considered where:***

- There was an error or unfairness in the assessment of the exam grade.
- There was an error or unfairness in the evaluation process for determining the course grade.

The faculty member will meet formally with the student and a record of the meeting will be recorded on the Grade Appeal Review Form. In the meeting with the student, the faculty member must explain how the grade in question was arrived at. If the faculty member decides, based on the discussion with the student, that a change of grade is warranted, a Change of Grade Form will be completed by the faculty member, explaining the basis for the grade change and this will be forwarded to the Department Chair.

If the faculty member decides a change of grade is not warranted, the faculty member will complete the Grade Appeal Review Form, explaining why the original grade should not be changed and this will be forwarded to the Department Chair. The Change of Grade Form/Grade Appeal Review Form (as applicable) will then be reviewed in succession by the Department Chair, the Dean, and the Associate Provost for Academic Affairs, and a decision will be made.

The Associate Provost for Academic Affairs will render a decision only after the faculty member has responded to any requests for further consideration made by the Department Chair and the Dean. The decision by the Associate Provost for Academic Affairs will be final and not subject to further appeal. The Registrar will not change a student's final grade in a course for any other reason or through any other procedure.

Where an appeal is successful, the Associate Provost for Academic Affairs will forward the Change of Grade Form to the Registrar's Department so that the student's grade is changed on the Student Information System (SIS). A copy of the Grade Change Form will be filed in the student's record.

**Pending Grade Appeal**

A student may request the VPAASS/Provost to delay imposing an academic suspension because of a pending grade appeal that could change the student's status. An approved delay allows the student to register. If the grade appeal is successful, the official transcript is corrected, and the student continues in classes as a student in good academic standing. If the grade appeal is not successful, the student is required to stop attending all classes and settle the case with VPAASS/Provost appropriately.

**Grading System**

University course work is measured in terms of quantity and quality. A credit normally represents one hour per week of lecture or recitation or not fewer than two hours per week of independent or laboratory work throughout a semester. The number of credits is a measure of quantity. The grade is a measure of quality. The university system for undergraduate grading is as follows:

| Grade | Quality Points | Percentage Scores |
|-------|----------------|-------------------|
| A     | 4.0            | 95-100            |
| A-    | 3.7            | 90-94             |
| B+    | 3.3            | 86-89             |
| B     | 3.0            | 83-85             |
| B-    | 2.7            | 80-82             |
| C+    | 2.3            | 76-79             |
| C     | 2.0            | 73-75             |
| C-    | 1.7            | 70-72             |
| D+    | 1.3            | 66-69             |
| D     | 1.0            | 60-65             |
| F     | 0              | 0-59              |

No credit toward graduation is received from a failing grade or additional credit gained from a grade that is replaced by a repeated course.

### Grade Point Average (GPA)

Grade or quality point values are assigned to letter grades as indicated in the grading system table. A quality or grade point score is computed by multiplying the value of a letter grade by the number of credits for the course. For example, a student receiving an A in a three-credit course earns twelve quality points. The GPA is computed by dividing the quality points earned by the number of credits graded "A" through "F" (GPA hours).

For undergraduates, the GPA computed for the current semester gives the current GPA, which is the measure of academic performance in one semester and affects eligibility for the honor's list. The GPA computed for all institutional credit gives the CGPA, which is the basis for the university's retention policies, including good standing, warning, probation, suspension, and dismissal. The CGPA also determines students' eligibility to graduate and to have university honors posted to their record upon graduation.

### Additional Grade Notations

#### *Satisfactory/Unsatisfactory (S/U):*

An "S" grade reflects satisfactory, or passing, work in a course; a "U" grade reflects unsatisfactory, or failing, work in a course. S and U will have no effect on GPA.

#### *Incomplete (I):*

This grade is given to students who are passing a course but are unable to complete all the course work or the final exam for a verified reason beyond their control or the instructor agrees to allow the student additional time to complete substantial aspects of the course's required work. The student must complete the final exam or the required work for the course by the end of the subsequent registered semester, not including summer terms, and the instructor must turn in the final grade by the end of that semester's grading deadline. Unless an explicit written extension is filed, the grade of "I" is changed to an "F" if the course requirements have not been fulfilled.

Students who have filed their intention to graduate have six weeks from the date of degree conferral to resolve any incomplete grades. An “I” is not calculated in a student’s GPA.

#### *In Progress (IP):*

The grade of “IP” indicates that the student is making progress in a course, writing a thesis or dissertation, or participating in an internship that extends beyond one semester or summer term. The grade of “IP” has no effect on the student’s GPA.

#### *Withdrawal (W):*

The grade of “W” on a course indicates that the student has dropped the course. While it has no effect on the GPA, dropped courses are part of attempted course credits that serve as the basis for the student’s credit level. A “W” for all courses in a given semester and the transcript notation “withdrawn” indicates that the student withdrew from AURAK.

### **Repeating a Course**

Undergraduate students may repeat courses for which they seek a higher grade. A grade received in a repeated course replaces the previous grade in the same course in the calculation of the GPA, even if the more recent grade is lower. Every registration for a course and its grade remains as part of a student’s transcript. Duplicate credit is not given. No adjustment to the CGPA will be made if the grade in the repeated course is a W. No more than three courses can be repeated at AURAK over the course of a given degree program.

A grade in an AURAK course will not be excluded from the CGPA based on the subsequent taking of an equivalent course at a transfer institution. Note that individual programs may disallow students from retaking certain high-demand courses simply for the purpose of improving their grade.

### **Transfer Credits**

Courses taken at other institutions of higher education may be counted toward a degree but are not computed in the AURAK GPA.

### **Academic Progress**

#### ***Academic Status***

Only registered students may participate in AURAK classes. Fulltime status requires that an undergraduate student enroll in 12 or more semester credit hours in Fall and Spring semesters. Undergraduate students enrolling in less than 12 semester credit hours are considered to be enrolled on a part-time basis. Undergraduate students whose semester credit hour loads are limited due to probation requirements are considered to hold fulltime status. All undergraduate students are expected to enroll on a fulltime basis unless granted permission by the dean of the respective school to enroll on a part-time basis.

All students are required to demonstrate sufficient academic performance to maintain enrollment with the university. Accordingly, AURAK establishes the following to recognize high academic performance and establish parameters for improving academic performance falling below minimal expectations.

### ***Academic Excellence***

An undergraduate student will be named to the Dean's List for a fall or spring semester if during that semester the student has all of the following:

- Completed twelve or more semester credit hours in courses counted in the CGPA
- Received no grade below B
- Received no grade of I (Incomplete) in any course
- Attained a semester's GPA of 3.60 or above

The dean of the school will notify students who have been named to the Dean's List.

An undergraduate student who qualifies for the Dean's List in a given semester with a GPA of 3.90 or above will also be named to the President's List.

Please note that these lists are only compiled for fall and spring semesters.

### ***Satisfactory Academic Progress***

In order to be considered in good standings, an undergraduate student must maintain a CGPA of 2.0 or higher. To be considered in good standing, a graduate student must maintain a CGPA of 3.0 or higher. A student, undergraduate or graduate, must be in good standing to be eligible for graduation.

Every student, undergraduate or graduate, must evidence sufficient academic progress towards the completion of his or her program requirements. Academic progress is evaluated based on the student's maintenance of a satisfactory CGPA and satisfaction of a minimum course completion pace. All undergraduate students enrolled in mathematics/science intensive programs (e.g. Engineering, Biotechnology, Computer Science) are required to meet academic progression expectations relating to the required pace for completing mathematics and science courses.

### ***Academic Probation, Suspension, and Dismissal***

If the CGPA of an undergraduate student falls below 2.0 in any semester, the student will be given an Academic Warning and will be required to enroll in UPAS 100, a study skills course, in the subsequent semester. If the student's CGPA remains below 2.0 at the end of the subsequent semester, the student will be placed on Academic Probation 1. All students on academic probation will be required to sign the AURAK Probation Contract before being allowed to register for any coursework.

A student on Probation 1 will be allowed to take a maximum of 13 semester credit hours. Students earning a CGPA of 2.0 or higher in the subsequent semester will be removed from Probation 1.

Students on Probation 1 earning a CGPA of 1.5 and higher but less than 2.0 will be placed on Probation 2. A student on Probation 2 is allowed a maximum of 10 semester credit hours. Students earning a CGPA of 2.0 or higher in the subsequent semester will be removed from Probation 2. Students completing a semester of Probation 2 with a CGPA less than 2.0 will be academically suspended.

Suspended students will not be allowed to enroll in AURAK courses for the subsequent semester unless they change their program of study to a distinctly different academic discipline, utilizing the appropriate procedure. Suspended students who elect not to change their program of study to a different academic discipline can only request re-admission once they have been out for one full academic fall or spring semester and have completed during that time not-for-credit remedial coursework. Students readmitted following suspension will be dismissed from the university if they are suspended for a second time unless they change their program of study to a different academic discipline.

Academic dismissal from the university is permanent.

### ***New Beginnings for Suspended Students***

1. A student who has been academically suspended from the university may apply for re-admission through the Department of the Registrar as part of the New Beginnings Initiative immediately following suspension from the original program of study. The New Beginnings Initiative will only enable the student to undertake an alternative program that is distinctly different from the program from which the student was suspended. Students suspended from a math/science intensive program such as an Engineering program or Biotechnology cannot undertake an alternative program in another math/science intensive program.
2. The Department of the Registrar will:
  - Verify that the student has not previously been re-admitted under the New Beginnings Initiative.
  - Seek the approval of the Department Chair for the program in which the student will be enrolled.
  - Seek the approval of the Dean of School for the program in which the student will be enrolled.
  - Seek approval from the Provost or his or her designate.
3. Once a student is re-admitted to a different program, the Registrar will remove from the calculation of the CGPA:
  - Grades below C in the major/program for which the student was not suited.
  - Grades for courses that do not meet requirements in the new degree program and having a grade below C.
4. The following will not be removed from the calculation of the CGPA:
  - Courses in which a student was guilty of academic dishonesty i.e., cheating, plagiarism, etc.
  - General education courses that continue to meet requirements in the new degree program.
  - Courses with grades C and above, even if they are not a part of the student's new program.
5. The Registrar will recalculate both the semester GPA and CGPA based on the above and the adjusted courses will remain on the transcript and will receive an "E" notation, meaning the grades and the credit hours will be "Excluded from GPA calculations."

### **Intellectual Property (Student Work)**

Intellectual property (IP) is property (an idea, invention, or process) that derives from the work of the mind or intellect. IP is also an application, right or registration of an idea, invention, or prose. Copyrightable works, including publications and patentable works developed in connection with course work by students who are not AURAK employees, are deemed to be intellectual property



that belongs to the student. However, AURAK may claim copyright ownership of a work or ownership of a patentable invention when extraordinary use of university facilities, personnel, or resources is made in the development of the materials or invention, especially when unrelated to course work. Students are subject to the university's stated policies regarding patents and copyrights.

### **University Withdrawal**

Official withdrawal removes students from any academic program and cancels student status at AURAK. In order to return to the university, students need to apply for readmission through the Department of Admissions. Students who wish to withdraw have to complete a withdrawal form available at the Office of the Registrar and get clearance from all university departments. If the withdrawal process is completed satisfactorily and all financial obligations to the university are cleared, the effective date of withdrawal is noted on the student's permanent academic record. The effective date is the date used for calculating billing or refunds. No grades other than a W are recorded for the current semester.

### **Leave of Absence**

Occasionally, students are forced to leave the university for a semester or two because of circumstances beyond their control. Others find they simply need a break from studying. In such circumstances, taking a leave of absence might be wise. Students who have an approved leave of absence for a semester or a year may register for the semester in which they plan to return without applying for readmission. Unless there are extenuating circumstances such as illness, a leave of absence is not normally given to a student who leaves the university during a term.

### **Fields of Study - Degree Offerings**

American University of Ras Al Khaimah has three schools that offer both undergraduate and graduate degree programs. Undergraduate programs are listed below:

- Bachelor of Architecture
- Bachelor of Arts in Interior Design
- Bachelor of Arts in Mass Communication
- Bachelor of Science in Artificial Intelligence
- Bachelor of Science in Biotechnology
- Bachelor of Science in Business Administration
- Bachelor of Science in Chemical Engineering
- Bachelor of Science in Civil and Infrastructure Engineering
- Bachelor of Science in Computer Engineering
- Bachelor of Science in Computer Science
- Bachelor of Science in Electrical and Electronics Engineering

- Bachelor of Science in Mechanical Engineering
- Bachelor of Science in Petroleum Engineering

### **Concentrations**

Undergraduate and graduate degree programs may offer students one or more concentrations within the major from which to choose. A concentration is a set of related courses that focus on a specific sub-field within the general discipline. A concentration must consist of no fewer than fifteen (15) credit hours of study.

The concentration is specified on the student's diploma and transcript. To be recognized as completing an official concentration, a student must complete and submit the Request for Concentration Form before having completed 105 credit hours towards the degree.

### **Double Major**

An undergraduate student may elect to complete a double major, which is a single degree with two majors.

A double major may be undertaken under the following conditions:

- A student must have a CGPA of 2.5 or above at the time of electing the second major.
- A student must have completed no more than 105 credit hours when electing to pursue a second major.
- A student must have completed two full years of study toward the first major (sixty credit hours) before applying for a double major.
- The second major may not be in the same academic department as the first.

To complete a double major, the following conditions must be met:

- A student must meet all the degree requirements for both majors.
- A student must complete at least twelve credits in addition to those required by the first major.
- No course substitutions will be available to a student in order to complete the requirements for the second major.

A student who completes a double major will receive one degree that indicates both majors on the transcript.

### **Change of Major**

A student may change his or her major at any point during matriculation at AURAK. At the time of the major change, the student must indicate which major and the General Education requirements that will be met: those in the catalog stipulated at the time the student entered AURAK as a matriculating student or those stipulated at the time of the change of major. The student must choose one set of General Education requirements or the other; they may not split the General Education requirements listed in either catalog.

Unless the student invokes the GPA Adjustment Policy, all previous courses taken remain on the student's transcript and grades received are calculated in the student's CGPA.

To take effect, the student must complete and submit the Change of Major form with all signatures.

### **Course Substitution**

All degree programs, concentrations, and minors have published requirements for graduation, including GPA requirements applicable to all university students. All requirements are met by successful completion of specific courses.

The need occasionally arises for a course substitution in a student's program, which AURAK permits under certain circumstances:

- If a required course has not been offered in at least two different semesters during the student's period of full-time matriculation

OR

- Under exceptional circumstances (e.g., certain disabilities make it impossible for a student to enroll in a required course)

In such cases, a course substitution is possible only if:

- The substitute course covers material similar in content and skills to the initial requirement.
- The substitute course is at the same level as, or more advanced than, the course for which it is being substituted.
- The replacement is approved by the student's academic advisor, a content specialist, the chair of the department of the program for which the substitution is being made, and the dean of the appropriate school, or, in the case of General Education, by the chair of the General Education Committee.

A course being transferred from another university may be used to substitute for an AURAK requirement if all other conditions are met.

Permission for a course substitution is granted only rarely, and a student may receive no more than two course substitutions in any program, including General Education. For a double major, no course substitutions are permissible in the second major.

Among others, the following will not be considered appropriate justifications for applying for a course substitution:

- A desire to complete the program of study quickly
- Failure to enroll and complete a required course that is offered regularly
- Dislike of a professor teaching a required course
- Substitution for a required course which the student has failed previously
- An independent choice to diversify courses taken for personal preference

### Undergraduate Enrollment in Graduate Courses

AURAK recognizes that there are occasions where it is academically valuable and appropriate for an undergraduate student to enroll in and receive credit for a graduate level course. To be eligible to undertake such courses the undergraduate student must gain the approval of their Academic Advisor and Department Chair and must have:

- senior status
- a CGPA of no less than 3.0

Students granted permission to undertake graduate level courses must complete the same assignments as the graduate students enrolled on the course(s) and will only be permitted to enroll in **two** graduate level courses.

The credit achieved may count toward the student's Bachelor degree as determined by the Program's Specification and contribute to the relevant graduate program should the student later matriculate into the graduate program.

This policy is subject to approval by the Commission for Academic Accreditation (CAA).

### Internship Policy

AURAK considers Internships to be an important education-related experience in the workplace and an opportunity for learning that integrates theory learned in the classroom with practical application. The Internship is also seen as an opportunity for skills development on the job that contribute to the development of personal and professional maturity as well as ethical decision-making.

Undergraduate students will only be permitted to register for the Internship if they:

- (a) have completed a minimum of two years of coursework (or higher if determined by the applicable School). *In exceptional circumstances a student who has not met this condition, may be permitted to undertake the Internship on the recommendation of the Dean of School and with approval of the Provost.*
- (b) are in good standing (at least 2.0 CGPA). *In exceptional circumstances a student who has not met this condition, may be permitted to undertake the Internship on the recommendation of the Dean of School and with the approval of the Provost.*
- (c) have the written approval of their Academic Advisor and the School's Internship Coordinator.

### Internship Arrangements

In compliance with accrediting body requirements, AURAK will ensure that the Office/Department responsible for Internships:

- (a) Has an effective method for matching students to potential placement employers.
- (b) Prepares an Internship Manual which covers the following:
  - i. Policies and procedures related to Internships and other experiential learning opportunities.

- ii. A delineation of the responsibilities of the faculty or academic supervisor, the provider and the student.
  - iii. A delineation of the placement of the internship in the curriculum or program of the student.
- (c) Prepares official agreements that clearly specify the responsibilities of AURAK, its student and the employer during the Internship, obtains the signature of all parties to the agreement in advance of the placement of students within an organization and retains a copy on file.
  - (d) Organizes an Internship for each undergraduate student of a duration that is compliant with the relevant accrediting body(ies) for the program in which the student is enrolled.
  - (e) Provides an orientation for students as to the skills, behaviors and knowledge which they need to succeed in the work environment.
  - (f) Has an effective process in place to monitor the employment site and gather feedback from students and the provider.

## **Student Petitions and Appeals**

### **Students' Responsibility**

All official university communications are distributed through the AURAK-issued email address. These are considered official notifications. Students are responsible for checking their AURAK email accounts and for acting upon messages accordingly.

Students should keep their own records of all transactions with the university (e.g., registration schedules and forms, grade reports, payment records, etc.). It is also advisable to keep copies of all tests, digital files, papers, and so forth submitted in fulfillment of course work. Students should keep copies of all course syllabi.

### **Appeal of Academic-Related Issues**

Academic grievances include grievances students may have against a faculty member, procedures implemented in class, or related academic issues. Note that grade appeals follow a different procedure and are described elsewhere in the Student Handbook and the AURAK catalog. For information regarding grievances please contact the Associate Provost of Student Affairs.

- The grievant must file a formal, written complaint with the VPAASS/Provost.
- The VPAASS/Provost will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
- If such action is deemed appropriate, the VPAASS/Provost will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
- The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.

- The committee will recommend a cause of action to the VPAASS/Provost within seven working days.
- The VPAASS/Provost will forward to the student the recommendation of the Committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
- If the student does not agree with the decision, the student has the right to file an appeal with the President. The President will review the decision and render a final decision in the matter within three working days. The President is the final authority to whom appeal may be made.
- If a grievance is with the VPAASS/Provost, then the student may submit the grievance in writing to the President to initiate the process.

## **Graduation**

### **Undergraduate Graduation Requirements**

The graduation requirements for any individual student are normally determined by the catalog that was effective when the student began matriculated studies in the degree program. Every individual student is personally responsible for meeting all graduation requirements as detailed in his or her catalog year.

If a required course within an undergraduate program changes its number of credits, then the number of credits required by the program for graduation may, at the discretion of the school, change by the same amount provided the minimum total number of credits for graduation is 120 and the CGPA is at least 2.00. In case of substantial changes, in course offerings, equivalent graduation requirements are determined by the dean of the student's school.

*Important: Course information, content, and prerequisites may be subject to change as a result of the university's commitment to a process of continual improvement in academic programs. Students must comply with the most up-to-date course requirements.*

### **Academic Standing Requirement**

A student must be in good academic standing to be eligible for graduation.

### **Application for Graduation**

Candidates for undergraduate degrees must file an Application for Graduation form with the Office of the Registrar during the registration period of the last expected term of study. Only after an Application for Graduation form has been filed can the Office of the Registrar begin processing the necessary information for final certification for graduation.

Students who fail to complete all degree requirements by the end of the term for which they apply to graduate need not reapply for graduation. Their previous application will be automatically moved to the following semester.

### **Conferral of Degrees**

Only students who have successfully completed degree requirements by the end of the term for which they have applied to graduate qualify for conferral of a degree. Degrees are conferred at the end of the semester in which requirements have been met. Conferral of the degree is noted on the academic transcript of the graduate with the date of graduation.

### **Names on Degrees**

The names of AURAK students will be spelled in English exactly as they appear on the student's passport or identity card when printed on degrees. If a name on a passport or an identity card does not appear in English, then the spelling of the name will be printed according to the personal preference of the student.

### **Attestation of Degrees and Transcripts**

The Office of the Registrar provides information relevant to the attestation of degrees and transcripts to the UAE Ministry of Education.

## **ACADEMIC PROGRAMS**

A degree program—the major or field—is a program of study that normally requires at least thirty credits of coursework in a specific field. The degree and major, or field, appear on the diploma. The Bachelor of Science degree programs at AURAK are professionally oriented, and all require the completion of a minimum of 120 credits, approximately forty courses, including General Education courses. Some programs require additional coursework, and all require an internship.

Degrees normally take four years to complete with a minimum of three years, assuming summer enrollment. If a degree is not completed within six years, all coursework in the major is re-evaluated for its current relevance. The Bachelor of Architecture degree is a five-year program.

AURAK currently offers the following thirteen undergraduate majors:

- Bachelor of Architecture
- Bachelor of Arts in Interior Design
- Bachelor of Arts in Mass Communication
- Bachelor of Science in Artificial Intelligence
- Bachelor of Science in Biotechnology
- Bachelor of Science in Business Administration
  - Major in Business Administration
  - Major in Accounting
  - Major in Business Analytics
  - Major Finance



- Major in Hospitality and Tourism Management
- Major in Human Resource Management
- Major in Marketing
- Bachelor of Science in Chemical Engineering
- Bachelor of Science in Civil and Infrastructure Engineering
- Bachelor of Science in Computer Engineering
- Bachelor of Science in Computer Science
- Bachelor of Science in Electrical and Electronics Engineering
- Bachelor of Science in Mechanical Engineering
- Bachelor of Science in Petroleum Engineering

### **AURAK Allocates Academic Credit According to the Following Criteria**

AURAK follows the standard US calibration of academic credit, such that 1 semester credit hour implies 15 hours of class time per semester. Two (2) semester credit hours of credit implies 30 hours of class time. Three (3) semester credit hours implies 45 hours of class time. Four (4) semester credit hours implies 60 hours of class time. The class meetings are typically distributed evenly across a 15-week semester in fall and spring, excluding vacation days. Summer sessions and accelerated (short) sessions/courses vary in length.

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the 15-week semester. Normally one credit represents 50 minutes of class instruction per week per semester.

- Lectures carry one academic credit for 50 minutes of class instruction per week for 15 weeks.
- Laboratories are usually 2.5 contact hours per week and carry one credit.
- Courses in drawing and design, currently taught in the Bachelor of Architecture, carry one academic credit for three contact hours in class.
- Independent Study courses carry 1-4 credits, depending on the amount of work required for the specific project as determined by the supervisor of the project.
- Internships carry 3 credits for a full-time experience.

### **Degree Completion Requirements**

- Successful completion of all credit hours as specified by the program curriculum with a minimum CGPA of 2.0
- Completion of the minimum enrollment period and not exceeding the maximum enrollment period

- Transfer or concurrently enrolled students are required to complete a minimum of fifty percent of the required credit hours at AURAK including the majority of the final thirty credit hours (please refer to this catalog for additional information on transfer credit policies)

### **Enrollment Periods**

Degrees normally take four years to complete with a minimum of three years, assuming summer enrollment. At AURAK the degree must be completed within a specified period in order to remain current and the maxima for programs is as follows:

- Programs requiring up to 140 credits must be completed within seven (7) years from initial registration (inclusive of periods of non-enrollment).
- Programs requiring more than 140 credits must be completed within eight (8) years from initial registration (inclusive of periods of non-enrollment).

AURAK will permit an appeal for extenuating circumstances to extend the period by up to two additional semesters.

## **GENERAL EDUCATION COMPONENT**

### **Introduction**

The General Education courses required of each baccalaureate degree at AURAK are based on a coherent rationale. Courses offered to meet the General Education Component requirements are designed to provide students with the foundational knowledge necessary to engage in higher order thinking, analysis of information, and problem-solving. These courses are intended to cultivate an informed, capable, and progressive citizenry. All courses included in the General Education Component are university-level courses and emphasize experiential and activity-based learning. Specifically, the General Education Component examines the application of knowledge in concrete situations to foster critical thinking skills, communication, teamwork, empirical and quantitative skills, and personal and social responsibility. While the General Education Component is not a degree program, all undergraduate degree students must complete the University's General Education Component requirements as a part of their academic program of study.

### **Mission**

The General Education Component is the fundamental constituent of all degree-granting programs. It provides students with fundamental knowledge that broadens their acquaintance with disciplines. As the component providing transferable skills of cortical thinking, numeracy, and synthesis, the General Education Component supports the University mission by preparing students to face the challenges of a globalized, multicultural, and rapidly changing world. The General Education Component complements the students' major program of study by examining achievements in areas of human inquiry and creative endeavor to cultivate those abilities so essential for the acquisition of knowledge.

## **Purpose**

The purpose of the General Education Component is to explore broad disciplinary areas in order to expand students' perspectives and improve their skills in analytical and critical thinking. The General Education Component is intended to facilitate a well-rounded and informed educational experience established on a holistic awareness of a broad spectrum of knowledge areas to ensure that students are exposed to a wide range of disciplines and topics. The General Education Component is designed to provide a relevant and challenging educational experience.

## **Completion of General Education Requirements**

AURAK's General Education Component embodies the vision, mission, and purpose of the University. The courses are required of all graduates as the intellectual foundation upon which AURAK's educational framework is built. They are an essential element in the University's educational process and lay the foundation for professional success, personal growth, and responsible citizenship. Students have the opportunity to acquire knowledge, learn communication skills, and examine their goals, values, and potentialities. Students are encouraged to complete their General Education Component requirements by the end of their second year of study.

## **Opportunities for Students to Develop and Demonstrate Competencies in Four Areas**

The General Education Component provides opportunities for students to develop and demonstrate competencies in the following four areas:

**Critical Thinking Skills:** including reasoning, inquiry, reflection, creativity, and innovation, and the ability to analyze and synthesize information.

**Communication and Teamwork:** including the development, interpretation, and expression of ideas, the ability to consider and integrate multiple points of views and to work with others on shared objectives.

**Empirical and Quantitative Skills:** including the ability to derive knowledge from observable facts, the manipulation and analysis of data, resulting in informed, verifiable solutions and conclusions. Effective understanding of technological concepts and the appropriate use of technology as a problem-solving tool.

**Personal and Social Responsibility:** including the ability to consider ethical and moral aspects of actions and consequences, intercultural awareness, and knowledge of regional, national, and global responsibility.

## **General Education Component Learning Outcomes**

Upon completion of the General Education Component requirements, completers will be able to:

**Learning Outcome One:** Solve abstract, familiar, and non-routine problems without assistance using appropriate information drawn from relevant fields of work or disciplines.

**Learning Outcome Two:** Present collaborative group analyses of topics that include complex ideas and integrate multiple point of view.

**Learning Outcome Three:** Analyze empirical data using appropriate information retrieval and analysis methods, tools and techniques to arrive at informed, verifiable solutions to abstract, familiar, and non-routine problems.

**Learning Outcome Four:** Articulate the ethical and moral aspects of possible courses of action in particular situations and explain the consequences that might follow with reference to intercultural issues at regional, national, and global levels.

### **Required Pathway to Completion of General Education Component Learning Outcomes:**

AURAK ensures that graduates acquire a holistic awareness of a broad range of knowledge areas. The General Education Component addresses these needs in a diverse, connect, globalized environment through courses that are not only broad and deep in content but use up-to-date pedagogical methods. The diverse range of topics in the General Education Component, the interdisciplinary and globalized focus of the content, the pedagogical shift to experiential and activity-based learning, the focus on research, innovation and enhanced aesthetic sensitivity all combine to aid in the development of responsible, self-directed individuals who will engage in life-long learning.

The General Education Component at AURAK fulfills the requirements of a liberal arts and sciences education by providing a comprehensive foundation in the competencies that enable students to grow academically as well as professionally. To achieve the General Education Component learning outcomes, students must complete 31-33 semester credit hours consisting of:

- Orientation Courses (O) – 13-14 semester credit hours
- Humanities/Fine Arts (H) – minimum of 6 semester credit hours
- Social and Behavioral Sciences (S) – minimum of 6 semester credit hours
- Natural Sciences/Mathematics (N) – minimum of 6 semester credit hours

AURAK distinguishes between categories of **Orientation Courses** (O) that provide a coherent foundation and academic orientation, and **Knowledge Domains** that offer students opportunities to develop and pursue a variety of academic interests. The Knowledge Domains are delineated into **Humanities/Fine Arts** (H), **Social and Behavioral Sciences** (S), and **Natural Sciences/Mathematics** (N). In addition, the General Education Component requires that of the courses taken to fulfil General Education Component requirements, five (5) courses must qualify as **Writing Intensive** courses. The codes “O”, “H”, “S”, and “N” identify the category to which a course belongs.

All AURAK undergraduate students must complete all General Education Component requirements to graduate. Completion of these requirements will amount to a total of 31-33 credit hours. All students must complete the Orientation Course requirements (amounting to 13-14 credits), and a minimum of 18 credits from the three remaining Knowledge Domains (amounting to a 18-19 credits). The courses for each of these are as follows.

### **ORIENTATION COURSES** (13-14 semester credit hours required)

Students must complete from 13 to 14 semester credit hours from the following list of courses.

| Orientation Courses                 |                                                                                                                            |                       |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Required Courses                    | Title                                                                                                                      | Semester Credit Hours |
| ENGL 101 (O)                        | Composition ( <b>Writing Intensive</b> )                                                                                   | 3                     |
| UNIV 100 (O)                        | University First-Year Transition                                                                                           | 1                     |
| UNIV 200 (O)                        | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                              | 3                     |
| ITEC 103 (O) or<br>CSCI 112/113 (O) | Fundamentals of Information Technology<br>Introduction to Computer Programming                                             | 3-4                   |
| ARAB 101 (O) or<br>ARAB 110 (O)     | Arabic Language and Culture for Non-Native<br>Learners I or<br>Arabic Language and Culture for Native Arabic<br>Speakers I | 3                     |
| Subtotal =                          |                                                                                                                            | 13 to 14 Credit Hours |

**KNOWLEDGE DOMAIN** (18-19 semester credit hours required)

Students must complete a minimum of 18 semester credit hours in this area. The requirements are divided into the following three categories: 1) Humanities/Fine Arts (H), 2) Social and Behavioral Sciences (S), and 3) Natural Sciences/Mathematics (N). Expectations are outlined below.

| Humanities/Fine Arts                                                                            |                                                                                                                                                                                                   |                       |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Required Courses                                                                                | Title                                                                                                                                                                                             | Semester Credit Hours |
| PHIL 100 (H) OR<br>ENGL 200 (H)                                                                 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> )<br>Advanced Composition ( <b>Writing Intensive</b> )                                                                                 | 3                     |
| MEST 100 (H)                                                                                    | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                                                                                                               | 3                     |
| Subtotal =                                                                                      |                                                                                                                                                                                                   | 6 Credit Hours        |
| Social and Behavioral Sciences                                                                  |                                                                                                                                                                                                   |                       |
| Required Courses                                                                                | Title                                                                                                                                                                                             | Semester Credit Hours |
| UAES 200 (S)                                                                                    | Survey of United Arab Emirates Studies ( <b>Writing Intensive</b> )                                                                                                                               | 3                     |
| PSYC 100 (S)<br>ECON 103 (S)<br>POLI 100 (S)<br>POLI 101 (S)<br>GEOG 100 (S) or<br>COMM 101 (S) | Introduction to Psychology<br>Principles of Microeconomics<br>Contemporary Global Issues<br>Politics of Scarcity<br>World Regional Geography<br>Interpersonal Communication and Group Interaction | 3                     |
| Subtotal =                                                                                      |                                                                                                                                                                                                   | 6 Credit Hours        |

| Natural Sciences/Mathematics                                                 |                                                                                                                                                |                     |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Required Courses                                                             | Title                                                                                                                                          | Credit Hours        |
| MATH 101 (N)<br>STAT 100 (N)<br>MATH 108 (N)<br>MATH 111 (N)<br>MATH 113 (N) | Numbers and Data Interpretation<br>Statistics<br>Calculus with Business Applications<br>Calculus with Life Sciences Applications<br>Calculus I | 3                   |
| BIOL 100 (N)<br>CHEM 100/101 (N)<br>CHEM 211/212 (N)<br>ENVS 102 (N)         | Humankind in a Biological World<br>Chemistry in Everyday Life<br>General Chemistry I<br>Sustainability and Human-Environment Relations         | 3-4                 |
| Subtotal =                                                                   |                                                                                                                                                | 6 to 7 Credit Hours |
| <b>TOTAL = 31 to 33 Credit Hours</b>                                         |                                                                                                                                                |                     |

Academic programs are encouraged to include additional courses focused on promoting creative and aesthetic understanding from *COMM 102 Reading Image and Film* and *ENGL 102 Public Speaking*. Academic programs are also encouraged to include additional courses focused on promoting cross-cultural and ethical understanding from *HIST 100 Contemporary Middle Eastern History*, *HIST 101 Ancient History of the Arabian Peninsula*, *PHIL 101 Ethics in Today's World*, and *PHIL 102 World Philosophies*.

### **Writing Intensive Courses**

To promote students' achievement of the Written Communication competency, the University's General Education Component also requires that students complete five writing-intensive courses. One of these writing-intensive courses is ENGL 101. Of the remaining four required writing-intensive courses, three must be other writing-intensive General Education courses, and the fifth writing-intensive course must be a course in the major program of study. Programs are encouraged to require students to take more than the university minimum of five writing-intensive courses.

The criteria for writing-intensive courses are provided below:

- The course includes a sustained focus on individual student writing, as demonstrated in the syllabus through multiple writing assignments.
- Students receive actionable feedback on their writing from the course instructor.
- Students substantially revise and resubmit written drafts, incorporating feedback from the course instructor and using their own critical thought to improve their writing.
- The final versions of written assignments total to at least 14 double-spaced, printed pages.

To be designated as "Writing-Intensive," courses must be approved by the General Education Committee as satisfying the criteria listed above. Courses that have been approved as Writing-

Intensive are clearly designated as such in the catalog, on transcripts, and in study plans for all programs.

### **The General Education Component Requirements and Course Substitutions**

The degree requirements for the individual undergraduate programs outline the course options from which a student may select courses to meet the General Education requirements. Other General Education courses that fulfill the requirements listed in the General Education Component may be substituted to complete respective General Education requirements in accordance with the established Course Substitution Policy and Procedure. Although students are encouraged to identify and select courses that meet their needs, faculty advisors must approve these course selections to ensure that they satisfy General Education Component requirements.





## SCHOOL OF ARTS & SCIENCES

## **SCHOOL OF ARTS AND SCIENCES**

### **Bachelor of Arts in Mass Communication**

The Bachelor of Arts degree in Mass Communication provides students with a high-quality education in line with emerging market trends in the media industry within the UAE, the Middle East, and the world. Rapid advancements in the field of communication and new media have created demand for qualified professionals and leaders who possess the knowledge to address global issues and, by so doing, shape the future of the media industry.

***The Bachelor of Arts in Mass Communication is overseen by the School of Arts and Sciences and is designed to satisfy QF Emirates Level 7 requirements.***

#### **Program Mission**

The Bachelor of Arts in Mass Communication with concentrations in Digital Media or Public Relations will produce competent graduates with the ability to apply both knowledge and skills of their area of concentration to engage in research to benefit the community and the nation, to ensure quality in the learners' professional lives.

#### **Program Educational Objectives**

The goals and objectives of the Bachelor of Arts in Mass Communication are simultaneously aligned with the mission statement of the university and seek to achieve the objectives embedded in the university mission.

#### **The Goals of the Mass Communication Program are to:**

1. Provide the student with a broad and deep knowledge about the study of communication, the similarities and differences between communication delivery modes, including their structure, processes, an ethical practice thereof, and the significance in social and professional uses;
2. Foster an understanding of the interdisciplinary nature and the cultural, ethical, and historical contexts of human communication;
3. Enhance the student's proficiency in research skills, the use of technology and media, critical thinking, creativity, and analytical abilities to pursue knowledge independently in the pursuit of advanced higher education;
4. Produce graduates who possess advanced practical and interpersonal skills that are grounded in truth and ethically practiced who demonstrate the capacity to establish and evaluate communication relationships in order to lead and serve the community; and
5. Produce responsible and competent professionals with practical skills, social commitment, and ethical standards that satisfy the demands of employment in the UAE, the region, and the world.

#### **Program Learning Outcomes**

AURAK Mass Communication graduates should demonstrate the ability to:

1. Explain communication theory and demonstrate applications of relevant theoretical concepts for different mass communication fields;

2. Recognize key skills and employ contemporary tools for communicating effectively in realistic media environments;
3. Utilize appropriate messages and techniques for communicating effectively;
4. Apply research methodologies and models and implement these in communication work;
5. Identify factors characterizing the global nature of modern media systems;
6. Distinguish cultural issues in planning various forms of communication; and
7. Analyze principles of ethics, fairness, and regulation as they relate to media practices.

### **Goals for the Digital Media Concentration**

The goals of the Digital Media Concentration are to:

1. Provide a strong foundation in the theoretical principles, approaches, techniques, and practices of communication through digital media;
2. Produce graduates that have proficiency in communication and communication delivery modes that is applicable in diverse contexts;
3. Enhance the student's proficiency in research skills by developing their critical thinking skills, creativity, analytical abilities, and their ability to work with communication technology and communication media; and
4. Produce responsible professionals competent in communication in digital media who satisfy the demands of employment and the needs of the community in the UAE and the wider region.

### **Program Learning Outcomes for the Digital Media Concentration**

Upon completing the Bachelor of Arts in Mass Communication with a concentration in Digital Media, graduates will demonstrate the ability to:

1. Employ convergent technologies for production and expression of communication pieces for news, documentaries, entertainment, and persuasive communication; and
2. Apply latest trends in multimedia practices and uses for professional digital communication to a local or international context.

### **Goals for the Public Relations Concentration**

The goals of the Public Relations Concentration are to:

1. Foster an understanding of the interdisciplinary nature and the historical and cultural contexts of human communication in English and in another modern language in all public relations roles;
2. Enhance the learner's proficiency in research skills, the use of technology and the media, critical thinking, creativity, and analytical abilities to pursue public relations communication knowledge independently in advanced higher education;
3. Demonstrate how contemporary public relation approaches and strategies may be used to improve communication with the audience in both public and private spheres;

4. Listen to and to speak, and to read and write public relations texts thoughtfully and to demonstrate an ability to support any communication with truthful and honest responses that increase the trust of the audience; and
5. Write effective public relations texts and to create public relations in media.

### **Program Learning Outcomes for the Public Relations Concentration**

Upon completing the program in Mass Communication with a concentration in Public Relations, graduates will demonstrate the ability to:

1. Discuss the characteristics of unique publics and design strategies to best establish symmetrical relationships with them; and
2. Summarize modern principles of public relations practices and analyze how these can be used in the UAE and globally.

### **Degree Requirements**

To earn a BA degree in Mass Communication, students must satisfactorily complete at least 120 approved credits, fulfill all the requirements for the BA in Mass Communication degree, and achieve a GPA of 2.00 or higher.

Thirty-one credit hours from approved courses are required to fulfill the general education requirements. Thirty-five credit hours from the program's core courses are similarly required. The BA in Mass Communication offers concentrations in Digital Media and Public Relations; these courses are offered in the second, third, and fourth years of study. Students complete eighteen credits in their concentration, an additional fifteen credits through the completion of concentration electives, and then fifteen credits in free electives. After completing 90 credits, students are required to complete an internship (3 credit hours) which is required for their concentration (Digital Media or Public Relations).

Students also take an Arabic for Media course. This three credit-hour course is offered in two versions: one for native-Arabic Speakers and a one for non-native Arabic speakers. No native Arabic speaker is permitted to take Arabic courses that are designed for non-native learners.

The BA degree in Mass Communication is designed to be completed in four years, assuming students do not interrupt their study. Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in the AURAK Catalog. Students must meet specific standards to progress, as well as the maximum time allowed to complete the program, which are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the major will be re-evaluated for its current relevance.

## BA MASS COMMUNICATION DEGREE COMPLETION REQUIREMENTS

The Bachelor of Arts in Mass Communications degree requires the completion of the following requirements:

| Requirement                                                               | Credits    |
|---------------------------------------------------------------------------|------------|
| I. University General Education Requirements                              | 31         |
| II. Program Core Course Requirements                                      | 35         |
| III. Concentration Course Requirements: Digital Media or Public Relations | 18         |
| IV. Mass Communication Electives                                          | 15         |
| V. Free Electives                                                         | 15         |
| VI. Arabic for Media (native speaking or non-native speaking)             | 3          |
| VII. Internship                                                           | 3          |
| <b>Total</b>                                                              | <b>120</b> |

### University General Education Requirements

**31 Credits**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. ***The fifth writing intensive course for the BA in Mass Communication is COMM 450 Selected Topics in Communication.***

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 OR<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition ( <b>Writing Intensive</b> )                                                                             | 3       |
| ITEC 103                | Fundamentals of Information Technology                                                                               | 3       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive Course</b> )                                 | 3       |
| PHIL 100 OR<br>ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR<br>Advanced Composition ( <b>Writing Intensive</b> ) | 3       |
| MEST 100                | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                                  | 3       |
| UAES 200                | Survey of United Arab Emirates Studies                                                                               | 3       |
| Gen Ed Course           | Social and Behavioral Science Course                                                                                 | 3       |
| Gen Ed Course           | Mathematics Course                                                                                                   | 3       |
| Gen Ed Course           | Natural Science Course                                                                                               | 3       |

**PROGRAM CORE COURSE REQUIREMENTS****35 Credits**

| Course No. | Title                                                | No. of Credits |
|------------|------------------------------------------------------|----------------|
| COMM 111   | Introduction to Mass Communication                   | 3              |
| COMM 112   | Introduction to Public Relations                     | 3              |
| COMM 113   | Introduction to Digital Media                        | 3              |
| COMM 212   | Media Writing                                        | 3              |
| COMM 215   | Feature Writing                                      | 3              |
| COMM 222   | Intercultural Mass Communication                     | 3              |
| COMM 223   | Globalization and Media Culture                      | 3              |
| COMM 311   | Media Law and Ethics                                 | 3              |
| COMM 321   | Theories of Media Communication                      | 3              |
| COMM 391   | Communication Research Methodology                   | 2              |
| COMM 492   | Mass Media and Public Opinion                        | 3              |
| COMM 450   | Selected Topics in Communication (Writing Intensive) | 3              |

**CONCENTRATION COURSE REQUIREMENTS****Concentration in Digital Media****18 Credits**

| Course No. | Title                             | No. of Credits |
|------------|-----------------------------------|----------------|
| COMM 214   | New Media and Digital Culture     | 3              |
| COMM 224   | Visual Storytelling               | 3              |
| COMM 323   | News Reporting                    | 3              |
| COMM 334   | Broadcast Journalism              | 3              |
| COMM 423   | Interactive Multimedia            | 3              |
| COMM 424   | Capstone: Multimedia Storytelling | 3              |

**Concentration in Public Relations****18 Credits**

| Course No. | Title                                | No. of Credits |
|------------|--------------------------------------|----------------|
| COMM 213   | Public Relations Writing             | 3              |
| COMM 334   | Broadcast Journalism                 | 3              |
| COMM 337   | Public Relations Cases               | 3              |
| COMM 344   | Public Relations & Event Management  | 3              |
| COMM 421   | Social Media: Audiences and Messages | 3              |
| COMM 427   | Capstone: Public Relations Campaigns | 3              |

**MASS COMMUNICATIONS ELECTIVES****15 Credits**

| Course No. | Title                                             | No. of Credits |
|------------|---------------------------------------------------|----------------|
| COMM 221   | Communication Analysis and Criticism              | 3              |
| COMM 225   | Visual Communication and Digital Photography      | 3              |
| COMM 226   | Graphic Communication Design and Print Production | 3              |
| COMM 227   | Film and Video Editing                            | 3              |
| COMM 322   | Digital Resources and Content                     | 3              |

|          |                                                  |   |
|----------|--------------------------------------------------|---|
| COMM 422 | Streaming Media, Podcasting and Radio Production | 3 |
| COMM 425 | Media Management and Entrepreneurship            | 3 |

\* Note: Students may choose one course from the other concentration as an elective course.

### **MODERN LANGUAGE COMPETENCY REQUIREMENT**

**3 Credits**

#### **Arabic for Non-Native Arabic Learners**

| Course No. | Title                                 | Prerequisites and Corequisites | No. of Credits |
|------------|---------------------------------------|--------------------------------|----------------|
| ARAB 211   | Arabic for Media: Non-Native Speakers | ARAB 101                       | 3              |

#### **Arabic for Native Arabic Learners**

| Course No. | Title                             | Prerequisites and Corequisites | No. of Credits |
|------------|-----------------------------------|--------------------------------|----------------|
| ARAB 210   | Arabic for Media: Native Speakers | ARAB 110                       | 3              |

### **FREE ELECTIVES**

**15 Credits**

#### **Internship**

**3 Credits**

| Course No. | Title      | No. of Credits |
|------------|------------|----------------|
| COMM 390   | Internship | 3              |

**TOTAL PROGRAM COURSE REQUIREMENTS = 120 CREDIT HOURS**



## B.A. Degree in Mass Communication with Concentration in Digital Media Four-Year Schedule

| Year 1 Semester 1           |                                                                                                                      |    | Year 1 Semester 2          |                                                                               |    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------|----|----------------------------|-------------------------------------------------------------------------------|----|
| Course No.                  | Description                                                                                                          | CR | Course No.                 | Description                                                                   | CR |
| COMM 111                    | Introduction to Mass Communication                                                                                   | 3  | COMM 112                   | Introduction to Public Relations                                              | 3  |
| ENGL 101                    | Composition                                                                                                          | 3  | COMM 113                   | Introduction to Digital Media                                                 | 3  |
| ITEC 103                    | Fundamentals of Information Technology                                                                               | 3  | MEST 100                   | Introduction to Islam in World Culture                                        | 3  |
| UNIV 100                    | University First-Year Transition                                                                                     | 1  | UNIV 200                   | Innovation, Entrepreneurship, and Sustainability                              | 3  |
| ARAB 101<br>or<br>ARAB 110  | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3  | PHIL 100<br>or<br>ENGL 200 | Critical Thinking and Reasoning<br>or<br>Advanced Composition                 | 3  |
| Gen Ed                      | Social and Behavioral Science                                                                                        | 3  |                            |                                                                               |    |
| Subtotal =                  |                                                                                                                      | 16 | Subtotal =                 |                                                                               | 15 |
| Year 2 Semester 1           |                                                                                                                      |    | Year 2 Semester 2          |                                                                               |    |
| Course No.                  | Description                                                                                                          | CR | Course No.                 | Description                                                                   | CR |
| COM 212                     | Media Writing                                                                                                        | 3  | COMM 215                   | Feature Writing                                                               | 3  |
| COMM 222                    | Intercultural Mass Communication                                                                                     | 3  | COMM 223                   | Globalization and Media Culture                                               | 3  |
| COMM 214                    | New Media and Digital Culture                                                                                        | 3  | COMM 224                   | Visual Storytelling                                                           | 3  |
| UAES 200                    | Survey of United Arab Emirates Studies                                                                               | 3  |                            | Mass Communication Elective                                                   | 3  |
| Gen Ed                      | Mathematics Course                                                                                                   | 3  | ARAB 210 OR<br>ARAB 211    | Arabic for Media: Native Speakers OR<br>Arabic for Media: Non-Native Speakers | 3  |
| Subtotal =                  |                                                                                                                      | 15 | Subtotal =                 |                                                                               | 15 |
| Year 3 Semester 1           |                                                                                                                      |    | Year 3 Semester 2          |                                                                               |    |
| Course No.                  | Description                                                                                                          | CR | Course No.                 | Description                                                                   | CR |
| COMM 311                    | Media Law and Ethics                                                                                                 | 3  | COMM 391                   | Communication Research Methodology                                            | 2  |
| COMM 321                    | Theories of Media Communication                                                                                      | 3  | COMM 334                   | Broadcast Journalism                                                          | 3  |
| COMM 323                    | News Reporting                                                                                                       | 3  |                            | Mass Communication Elective                                                   | 3  |
| Gen Ed                      | Natural Science Course                                                                                               | 3  |                            | Free Elective                                                                 | 3  |
|                             | Mass Communication Elective                                                                                          | 3  |                            | Free Elective                                                                 | 3  |
| Subtotal =                  |                                                                                                                      | 15 | Subtotal =                 |                                                                               | 14 |
| Year Three, Summer Semester |                                                                                                                      |    |                            |                                                                               |    |
| Course No.                  | Description                                                                                                          | CR |                            |                                                                               |    |
| COMM 390                    | Internship for Mass Communication                                                                                    | 3  |                            |                                                                               |    |
| Year 4 Semester 1           |                                                                                                                      |    | Year 4 Semester 2          |                                                                               |    |
| Course No.                  | Description                                                                                                          | CR | Course No.                 | Description                                                                   | CR |
| COMM 492                    | Mass Media and Public Opinion                                                                                        | 3  | COMM 424                   | Capstone: Multimedia Storytelling                                             | 3  |
| COMM 423                    | Interactive Multimedia                                                                                               | 3  |                            | Free Elective                                                                 | 3  |
| COMM 450                    | Selected Topics in Communication (Writing Intensive)                                                                 | 3  |                            | Free Elective                                                                 | 3  |
|                             | Mass Communication Elective                                                                                          | 3  |                            | Mass Communication Elective                                                   | 3  |
|                             | Free Elective                                                                                                        | 3  |                            |                                                                               |    |
| Subtotal =                  |                                                                                                                      | 15 | Subtotal =                 |                                                                               | 12 |
| TOTAL CREDIT HOURS 120      |                                                                                                                      |    |                            |                                                                               |    |

## B.A. Degree in Mass Communication with Concentration in Public Relations (Four-Year Schedule)

| Year 1 Semester 1           |                                                                                                                   |    | Year 1 Semester 2    |                                                                            |    |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|----|----------------------|----------------------------------------------------------------------------|----|
| Course No.                  | Description                                                                                                       | CR | Course No.           | Description                                                                | CR |
| COMM 111                    | Introduction to Mass Communication                                                                                | 3  | COMM 112             | Introduction to Public Relations                                           | 3  |
| ENGL 101                    | Composition                                                                                                       | 3  | COMM 113             | Introduction to Digital Media                                              | 3  |
| ITEC 103                    | Fundamentals of Information Technology                                                                            | 3  | MEST 100             | Introduction to Islam in World Culture                                     | 3  |
| UNIV 100                    | University First-Year Transition                                                                                  | 1  | UNIV 200             | Innovation, Entrepreneurship, and Sustainability                           | 3  |
| ARAB 101 OR ARAB 110        | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3  | PHIL 100 OR ENGL 200 | Critical Thinking and Reasoning OR Advanced Composition                    | 3  |
| Gen Ed                      | Social and Behavioral Science                                                                                     | 3  |                      |                                                                            |    |
| Subtotal =                  |                                                                                                                   | 16 | Subtotal =           |                                                                            | 15 |
| Year 2 Semester 1           |                                                                                                                   |    | Year 2 Semester 2    |                                                                            |    |
| Course No.                  | Description                                                                                                       | CR | Course No.           | Description                                                                | CR |
| COMM 212                    | Media Writing                                                                                                     | 3  | COMM 215             | Feature Writing                                                            | 3  |
| COMM 222                    | Intercultural Mass Communication                                                                                  | 3  | COMM 223             | Globalization and Media Culture                                            | 3  |
|                             | Mass Communication Elective                                                                                       | 3  | COMM 213             | Public Relations Writing                                                   | 3  |
| UAES 200                    | Survey of United Arab Emirates Studies                                                                            | 3  |                      | Mass Communication Elective                                                | 3  |
| Gen Ed                      | Mathematics Course                                                                                                | 3  | ARAB 210 OR ARAB 211 | Arabic for Media: Native Speakers OR Arabic for Media: Non-Native speakers | 3  |
| Subtotal =                  |                                                                                                                   | 15 | Subtotal =           |                                                                            | 15 |
| Year 3 Semester 1           |                                                                                                                   |    | Year 3 Semester 2    |                                                                            |    |
| Course No.                  | Description                                                                                                       | CR | Course No.           | Description                                                                | CR |
| COMM 311                    | Media Law and Ethics                                                                                              | 3  | COMM 391             | Communication Research Methodology                                         | 2  |
| COMM 321                    | Theories of Media Communication                                                                                   | 3  | COMM 334             | Broadcast Journalism                                                       | 3  |
| COMM 337                    | Public Relations Cases                                                                                            | 3  | COMM 344             | Public Relations & Event Management                                        | 3  |
| Gen Ed                      | Natural Science Course                                                                                            | 3  |                      | Free Elective                                                              | 3  |
|                             | Mass Communication Elective                                                                                       | 3  |                      | Free Elective                                                              | 3  |
| Subtotal =                  |                                                                                                                   | 15 | Subtotal =           |                                                                            | 14 |
| Year Three, Summer Semester |                                                                                                                   |    |                      |                                                                            |    |
| Course No.                  | Description                                                                                                       | CR |                      |                                                                            |    |
| COMM 390                    | Internship for Mass Communication                                                                                 | 3  |                      |                                                                            |    |
| Year 4 Semester 1           |                                                                                                                   |    | Year 4 Semester 2    |                                                                            |    |
| Course No.                  | Description                                                                                                       | CR | Course No.           | Description                                                                | CR |
| COMM 492                    | Mass Media and Public Opinion                                                                                     | 3  | COMM 427             | Capstone: Public Relations Campaigns                                       | 3  |
| COMM 421                    | Social Media: Audiences and Message                                                                               | 3  |                      | Free Elective                                                              | 3  |
| COMM 450                    | Selected Topics in Communication (Writing Intensive)                                                              | 3  |                      | Free Elective                                                              | 3  |
|                             | Mass Communication Elective                                                                                       | 3  |                      | Mass Communication Elective                                                | 3  |
|                             | Free Elective                                                                                                     | 3  |                      |                                                                            |    |
| Subtotal =                  |                                                                                                                   | 15 | Subtotal =           |                                                                            | 12 |
| TOTAL CREDIT HOURS 120      |                                                                                                                   |    |                      |                                                                            |    |

## **Bachelor of Science in Biotechnology**

Biotechnology is the modern face of the biological sciences. It applies technological advances, particularly those involving the sciences of molecular genetics, recombinant DNA techniques, and molecular diagnostics to solve biological problems through the production of materials, processes, and services to benefit our environment.

In medicine, Biotechnology aids in the manufacture of therapeutics, enzymes, antibiotics, vitamins, and vaccines. In agriculture, Biotechnology helps improve food quality, quantity, and processing. Additionally, Biotechnology has applications in green manufacturing technologies. It minimizes contamination and pollution and locates alternative energy sources. In the past decade, industry has increasingly turned to biotechnology to improve product quality and to make the production process easier, cleaner, and more cost-effective.

Biotechnology graduates work in fields such as molecular biology, genetics, forensics, plant and animal sciences, food production, therapeutics, agriculture, and environmental sciences.

***The Bachelor of Science in Biotechnology is overseen by the School of Arts and Sciences and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The Bachelor of Science in Biotechnology Program, with separate concentrations in Cell and Molecular Biotechnology, Medical Biotechnology, and Forensic Biology, aims to produce skilled and knowledgeable biotechnologists who are capable of applying their acquired knowledge to scientific research and development in medicine, science, agriculture, forensics and the environment for the benefit of their societies, while maintaining the highest standards of professional ethics in their careers.

### **Program Educational Objectives**

The goals of the Biotechnology Program are to:

1. Supply society with biologically literate citizens who are capable of advising and making decisions;
2. Produce ambitious, creative graduates who are interested in continuing their education in the biosciences;
3. Practice scientific inquiry and appreciate its role in the development of research, science, technology, and society;
4. Produce responsible biotechnology professionals to fulfill the employment and research needs in the biotechnology industry in the UAE and the region;
5. Contribute to the advancement of agriculture, medicine, and environmental sciences through the application of biotechnology theory and recombinant DNA technology; and
6. Enhance the students' ability to integrate their acquired math, computer, and bioscience knowledge and skills to investigate and solve biological problems.

### **Program Learning Outcomes**

On completion of the program graduates will be able to:

1. **PLO1:** Identify, formulate, and solve broadly defined technical or scientific problems by applying knowledge of mathematics and science and/or technical topics to areas relevant to the discipline.
2. **PLO2:** Formulate or design a system, process, procedure or program to meet desired needs.
3. **PLO3:** Develop and conduct experiments or test hypotheses, analyze and interpret data and use scientific judgment to draw conclusions.
4. **PLO4:** Communicate effectively with a range of audiences.
5. **PLO5:** Discuss ethical and professional responsibilities and the impact of technical and/or scientific solutions in global, economic, environmental, and societal contexts.
6. **PLO6:** Function effectively on teams that establish goals, plan tasks, meet deadlines, and analyze risk and uncertainty.

### **Program Concentration Learning Outcomes – Cell and Molecular Biotechnology**

7. **PLO7:** Lead and implement appropriate research strategies to produce verifiable data in Cell and Molecular Biotechnology.
8. **PLO8:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Cell and Molecular Biotechnology graduate programs and employment sectors.

### **Program Concentration Learning Outcomes – Medical Biotechnology**

7. **PLO9:** Lead and implement appropriate research strategies to produce verifiable data in Medical Biotechnology.
8. **PLO10:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Medical Biotechnology graduate programs and employment sectors.

### **Program Concentration Learning Outcomes – Forensic Biology**

7. **PLO11:** Lead and implement appropriate research strategies to produce verifiable data in Forensic Biology.
8. **PLO12:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Forensic Biology graduate programs and employment sectors.

### **Degree Requirements**

The Bachelor of Science in Biotechnology degree is a four-year degree program with concentration in Cell and Molecular Biotechnology, Medical Biotechnology, and Forensic Biology. For the first three years of the Bachelor of Science in Biotechnology program, students follow a common sequence of courses, with courses specific to the respective concentration taken in the fourth year.

of the program. To earn a bachelor's degree, students must satisfactorily complete at least 128 approved credits, fulfill all the requirements for the Bachelor of Science in Biotechnology degree, and achieve a GPA of 2.00 or higher. Students must apply for at least forty-eight credits of upper-level courses (designated 300- level or higher) toward graduation requirements. Thirty-one to Thirty-four credit hours from approved courses are required to fulfill the General Education Program requirements. The degree is designed to be completed in four years, assuming students do not interrupt their study. Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in AURAK catalog. Students are required to meet specific standards to progress, as well as the maximum time allowed to complete the program, which are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the major will be re-evaluated for its current relevance.

The Bachelor of Science in Biotechnology degree necessitates the completion of the following requirements:

| Requirements                                                                                                     | Credits    |
|------------------------------------------------------------------------------------------------------------------|------------|
| University General Education Requirements                                                                        | 31         |
| Biotechnology Program Core Courses                                                                               | 63         |
| Biotechnology Program Elective courses                                                                           | 7          |
| Concentration (Cell and Molecular Biotechnology, Medical Biotechnology, or Forensic Biology) Course Requirements | 21         |
| Free Electives                                                                                                   | 6          |
| <b>TOTAL</b>                                                                                                     | <b>128</b> |

#### University General Education Requirements

**31 Credit Hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The 5<sup>th</sup> writing intensive course for the BS in Biotechnology is BIOL 450 Bioethics.**

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ENGL 101                | Composition <b>(Writing Intensive)</b>                                                                               | 3       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| ITEC 103                | Fundamentals of Information Technology                                                                               | 3       |
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability <b>(Writing Intensive)</b>                                          | 3       |
| MEST 100                | Introduction to Islam in World Culture <b>(Writing Intensive)</b>                                                    | 3       |
| PHIL 100 or<br>ENGL 200 | Critical Thinking and Reasoning <b>(Writing Intensive)</b> OR<br>Advanced Composition <b>(Writing Intensive)</b>     | 3       |
| UAES 200                | Survey of United Arab Emirates Studies                                                                               | 3       |

|                         |                                                                                      |   |
|-------------------------|--------------------------------------------------------------------------------------|---|
| MATH 111                | Calculus for Life Sciences                                                           | 3 |
| Gen Ed                  | General Education Social Sciences Course                                             | 3 |
| BIOL 100 or<br>ENVS 102 | Humankind in a Biological World or<br>Sustainability and Human-Environment Relations | 3 |

### Biotechnology Program Core Courses

63 Credit Hours

| Course Code | Title                                             | No. of Credits |
|-------------|---------------------------------------------------|----------------|
| BIOL 112    | University Biology I                              | 3              |
| BIOL 113    | University Biology I Lab                          | 1              |
| BIOL 114    | University Biology II                             | 3              |
| BIOL 115    | University Biology II Lab                         | 1              |
| BIOL 230    | General Microbiology                              | 3              |
| BIOL 231    | General Microbiology Lab                          | 1              |
| BIOL 270    | General Genetics                                  | 3              |
| BIOL 271    | General Genetics Lab                              | 1              |
| BIOL 330    | Applied and Industrial Microbiology               | 3              |
| BIOL 331    | Techniques in Applied and Industrial Microbiology | 1              |
| BIOL 250    | Biochemistry I                                    | 3              |
| BIOL 251    | Biochemistry I Lab                                | 1              |
| BIOL 350    | Biochemistry II                                   | 3              |
| BIOL 356    | Virology                                          | 3              |
| BIOL 380    | Biotechnology and Genetic Engineering             | 3              |
| BIOL 381    | Biotechnology Lab Methods and Techniques          | 1              |
| BIOL 450    | Bioethics ( <b><i>Writing Intensive</i></b> )     | 3              |
| BIOL 491    | Senior Seminar                                    | 2              |
| BIOL 493    | Research Methodology in the Biological Sciences   | 2              |
| CHEM 111    | Principles of General Chemistry                   | 3              |
| CHEM 112    | Principles of General Chemistry Lab               | 1              |
| CHEM 120    | Analytical Chemistry                              | 3              |
| CHEM 121    | Analytical Chemistry Laboratory                   | 1              |
| CHEM 215    | Organic Chemistry I                               | 3              |

|          |                                         |   |
|----------|-----------------------------------------|---|
| CHEM 216 | General Organic Chemistry Lab I         | 1 |
| PHYS 110 | University Physics I                    | 3 |
| PHYS 111 | University Physics I Lab                | 1 |
| STAT 100 | Introductory Probability and Statistics | 3 |
| BIOL 390 | Internship                              | 3 |

#### Biotechnology Program Elective Courses

**7 Credit Hours**

| Course Code | Title                                                                             | No. of Credits |
|-------------|-----------------------------------------------------------------------------------|----------------|
| BIOL 322    | Microbial Genetics                                                                | 3              |
| BIOL 360    | Developmental Biology                                                             | 3              |
| BIOL 480    | Food Biotechnology                                                                | 3              |
| BIOL 481    | Bioprocessing Technology in the Pharmaceutical Industry                           | 3              |
| BIOL 492    | Special Topics in Biotechnology                                                   | 2              |
|             | Students may choose one course from the other concentration as an elective course | 3-4            |

#### Concentration Course Requirements

**21 Credit Hours**

#### Concentration in Cell and Molecular Biotechnology

**21 Credit Hours**

| Course Code | Course Title                                       | No. of Credits |
|-------------|----------------------------------------------------|----------------|
| BIOL 420    | Molecular Biology                                  | 3              |
| BIOL 421    | Molecular Biology and Recombinant DNA Lab          | 1              |
| BIOL 425    | Advanced Genetics                                  | 3              |
| BIOL 430    | Cell Culture Theory and Technology                 | 3              |
| BIOL 431    | Cell Culture Techniques Lab                        | 1              |
| BIOL 434    | Cell Communication and Signal Transduction         | 3              |
| BIOL 473    | Bioinformatics and Computational Biology           | 3              |
| BIOL 494    | Senior Project in Cell and Molecular Biotechnology | 4              |

#### Concentration in Medical Biotechnology

**21 Credit Hours**

| Course Code | Course Title                           | No. of Credits |
|-------------|----------------------------------------|----------------|
| BIOL 411    | Drug Discovery, Design and Development | 3              |



|          |                                             |   |
|----------|---------------------------------------------|---|
| BIOL 435 | Stem Cell Biology and Regenerative Medicine | 3 |
| BIOL 436 | Immunology                                  | 3 |
| BIOL 440 | Medical Microbiology                        | 3 |
| BIOL 441 | Medical Microbiology Lab                    | 1 |
| BIOL 442 | Molecular Diagnostics                       | 3 |
| BIOL 443 | Molecular Diagnostics Lab                   | 1 |
| BIOL 495 | Senior Project in Medical Biotechnology     | 4 |

### Concentration in Forensic Biology

**21 Credit Hours**

| Course Code | Course Title                                     | No. of Credits |
|-------------|--------------------------------------------------|----------------|
| CHEM 420    | Forensic Chemistry                               | 3              |
| CHEM 421    | Forensic Chemistry Lab                           | 1              |
| BIOL 423    | Advanced Molecular Biology and Forensic Sciences | 3              |
| BIOL 424    | DNA Fingerprinting and Serology Lab              | 1              |
| BIOL 462    | Crime Scene Investigation                        | 3              |
| BIOL 464    | Criminal Justice                                 | 3              |
| BIOL 473    | Bioinformatics and Computational Biology         | 3              |
| BIOL 496    | Senior Project in Forensic Biology               | 4              |

### Program Free Electives

**6 Credits Hours**

| Course Code | Course Title   | No. of Credits |
|-------------|----------------|----------------|
|             | Free Electives | 6              |

## B.S. Degree in Biotechnology with Concentrations Proposed 4 Year Schedule

| Year 1 Semester 1 |                                                |    | Year 1 Semester 2    |                                                         |    |
|-------------------|------------------------------------------------|----|----------------------|---------------------------------------------------------|----|
| Course No.        | Description                                    | CR | Course No.           | Description                                             | CR |
| BIOL 112          | University Biology I                           | 3  | BIOL 114             | University Biology II                                   | 3  |
| BIOL 113          | University Biology I Lab                       | 1  | BIOL 115             | University Biology II Lab                               | 1  |
| MATH 111          | Calculus for the Life Sciences                 | 3  | MEST 100             | Introduction to Islam in World Culture                  | 3  |
| ENGL 101          | Composition                                    | 3  | STAT 100             | Introductory Probability and Statistics                 | 3  |
| CHEM 111          | Principles of General Chemistry                | 3  | CHEM 120             | Analytical Chemistry                                    | 3  |
| CHEM 112          | Principles of General Chemistry Lab            | 1  | CHEM 121             | Analytical Chemistry Lab                                | 1  |
| UNIV 100          | University First-Year Transition               | 1  | Gen Ed               | Social Sciences Course                                  | 3  |
| Subtotal =        |                                                | 15 | Subtotal =           |                                                         | 17 |
| Year 2 Semester 1 |                                                |    | Year 2 Semester 2    |                                                         |    |
| Course No.        | Description                                    | CR | Course No.           | Description                                             | CR |
| BIOL 230          | General Microbiology                           | 3  | BIOL 270             | General Genetics                                        | 3  |
| BIOL 231          | General Microbiology Lab                       | 1  | BIOL 271             | General Genetics Lab                                    | 1  |
| UNIV 200          | Innovation, Entrepreneurship, & Sustainability | 3  | BIOL 250             | Biochemistry I                                          | 3  |
| CHEM 215          | Organic Chemistry I                            | 3  | BIOL 251             | Biochemistry I Lab                                      | 1  |
| CHEM 216          | General Organic Chemistry I Lab                | 1  | PHYS 110             | University Physics I                                    | 3  |
| ITEC 103          | Fundamentals of Information Technology         | 3  | PHYS 111             | University Physics I Lab                                | 1  |
|                   |                                                |    | PHIL 100 or ENGL 200 | Critical Thinking and Reasoning or Advanced Composition | 3  |
| Subtotal =        |                                                | 14 | Subtotal =           |                                                         | 15 |
| Year 3 Semester 1 |                                                |    | Year 3 Semester 2    |                                                         |    |
| Course No.        | Description                                    | CR | Course No.           | Description                                             | CR |
| BIOL 350          | Biochemistry II                                | 3  | UAES 200             | Survey of United Arab Emirates Studies                  | 3  |
| BIOL 380          | Biotechnology and Genetic Engineering          | 3  | BIOL 330             | Applied and Industrial Microbiology                     | 3  |
| BIOL 381          | Biotechnology Lab Methods and Techniques       | 1  | BIOL 331             | Techniques in Applied and Industrial Microbiology       | 1  |
| BIOL 356          | Virology                                       | 3  | Program Electives    | Biotechnology Elective                                  | 4  |
| Elective          | Free Elective                                  | 3  | BIOL 100 Or          | Humankind in a Biological World OR                      | 3  |

|                                                    |                                                                                                                         |    |                    |                                                |    |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----|--------------------|------------------------------------------------|----|
|                                                    |                                                                                                                         |    | ENVS 102           | Sustainability and Human Environment Relations |    |
| ARAB 101<br>or<br>ARAB 110                         | Arabic Language and Culture for Non-Native Learners I<br>OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3  | Elective           | Free Elective                                  | 3  |
| Subtotal =                                         |                                                                                                                         | 16 | Subtotal =         |                                                | 17 |
| Year 3, Summer Semester                            |                                                                                                                         |    |                    |                                                |    |
| Course No.                                         | Description                                                                                                             | CR |                    |                                                |    |
| BIOL 390                                           | Internship                                                                                                              | 3  |                    |                                                |    |
| Year 4 with Concentration in Medical Biotechnology |                                                                                                                         |    |                    |                                                |    |
| Year 4 Semester 1                                  |                                                                                                                         |    | Year 4, Semester 2 |                                                |    |
| Course No.                                         | Description                                                                                                             | CR | Course No.         | Description                                    | CR |
| BIOL 435                                           | Stem Cell Biology and Regenerative Medicine                                                                             | 3  | BIOL 440           | Medical Microbiology                           | 3  |
| BIOL 442                                           | Molecular Diagnostics                                                                                                   | 3  | BIOL 441           | Medical Microbiology Lab                       | 1  |
| BIOL 443                                           | Molecular Diagnostics Lab                                                                                               | 1  | BIOL 436           | Immunology                                     | 3  |
| BIOL 493                                           | Research Methodology in the Biological Sciences                                                                         | 2  | BIOL 411           | Drug Discovery, Design and Development         | 3  |
| BIOL 450                                           | Bioethics                                                                                                               | 3  | BIOL 495           | Senior Project Medical Biotechnology           | 4  |
| BIOL 491                                           | Senior Seminar                                                                                                          | 2  |                    |                                                |    |
| Program Elective                                   | Biotechnology Elective                                                                                                  | 3  |                    |                                                |    |
| Subtotal =                                         |                                                                                                                         | 17 | Subtotal =         |                                                | 14 |

| Year 4 with Concentration in Cell and Molecular Biotechnology |                                                 |    |                    |                                                 |    |
|---------------------------------------------------------------|-------------------------------------------------|----|--------------------|-------------------------------------------------|----|
| Year 4 Semester 1                                             |                                                 |    | Year 4, Semester 2 |                                                 |    |
| BIOL 425                                                      | Advanced Genetics                               | 3  | BIOL 420           | Molecular Biology                               | 3  |
| BIOL 430                                                      | Cell Culture Theory and Technology              | 3  | BIOL 421           | Molecular Biology Recombinant DNA Lab           | 1  |
| BIOL 431                                                      | Cell Culture Theory and Technology Lab          | 1  | BIOL 434           | Cell Communication and Signal Transduction      | 3  |
| BIOL 493                                                      | Research Methodology in the Biological Sciences | 2  | BIOL 473           | Bioinformatics and Computational Biology        | 3  |
| BIOL 450                                                      | Bioethics                                       | 3  | BIOL 494           | Senior Project Cell and Molecular Biotechnology | 4  |
| BIOL 491                                                      | Senior Seminar                                  | 2  |                    |                                                 |    |
| Program Elective                                              | Biotechnology Elective                          | 3  |                    |                                                 |    |
| Subtotal =                                                    |                                                 | 17 | Subtotal =         |                                                 | 14 |

| Year 4 with Concentration in Forensic Biology |                                                  |    |                    |                                          |    |
|-----------------------------------------------|--------------------------------------------------|----|--------------------|------------------------------------------|----|
| Year 4 Semester 1                             |                                                  |    | Year 4, Semester 2 |                                          |    |
| BIOL 423                                      | Advanced Molecular Biology and Forensic Sciences | 3  | CHEM 420           | Forensic Chemistry                       | 3  |
| BIOL 424                                      | DNA Fingerprinting and Serology Lab              | 1  | CHEM 421           | Forensic Chemistry Lab                   | 1  |
| BIOL 462                                      | Crime Scene Investigation                        | 3  | BIOL 464           | Criminal Justice                         | 3  |
| BIOL 493                                      | Research Methodology in the Biological Sciences  | 2  | BIOL 473           | Bioinformatics and Computational Biology | 3  |
| BIOL 450                                      | Bioethics                                        | 3  | BIOL 496           | Senior Project Forensic Biology          | 4  |
| BIOL 491                                      | Senior Seminar                                   | 2  |                    |                                          |    |
| Program Elective                              | Biotechnology Elective                           | 3  |                    |                                          |    |
| Subtotal =                                    |                                                  | 17 | Subtotal =         |                                          | 14 |
| TOTAL CREDIT HOURS 128                        |                                                  |    |                    |                                          |    |



# SCHOOL OF BUSINESS

# **SCHOOL OF BUSINESS**

## **Bachelor of Science in Business Administration**

### **Program Description**

The Bachelor Science in Business Administration (BSBA) at AURAK aims to provide a broad background of study for students who wish to pursue a career in one of many business sectors - retail, hospitality, information technology, manufacturing, or non-profit organizations.

The BSBA program provides students with tools and knowledge valuable to any business and with practice in developing skills such as teamwork, analysis of numerical data, and communicating in writing and orally.

BSBA majors may choose to specialize in accounting, finance, human resource management, marketing, business analytics or hospitality and tourism management by taking seven major required courses and three major electives in one of these areas or choose to be a generalist, by taking a combination of courses from all the six specialist areas.

***The Bachelor of Science in Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The mission of the Business Administration Program is to prepare graduates who are knowledgeable, skilled, globally oriented, and ethical for entrance into business, government, other professions, and/or graduate programs.

### **Program Educational Objectives**

The overall aims of the program are to:

1. Develop a breadth of knowledge across business functions
2. Create an understanding of ethical and social responsibility
3. Demonstrate clarity of written and oral communication using appropriate technology
4. Develop teamwork and leadership skills
5. Enable Professional reflection and self-development
6. Develop critical thinking, analytical and problem-solving skills
7. Develop proficiency in a chosen business discipline

### **Program Learning Outcomes**

**On completion of the program graduates will be able to demonstrate**

#### **PL01: (CORE) Breadth of Business Knowledge**

Critically expand on the basic principles of economics, accounting, finance, management, information systems, marketing and operations in the context of a global economy.

**PLO2: (CORE) Ethical Reasoning**

Apply concepts and theories of ethics and social responsibility to practical business dilemmas, recognizing the implications of management decisions for the interests of key internal and external stakeholders.

**PLO3: (CORE) Communication Skills**

Demonstrate competency in presentation and writing skills using the latest business communication tools.

**PLO4: (CORE) Teamwork**

Effectively work in teams and take the lead in team initiatives.

**PLO5: (CORE) Professional Reflection**

Engage in critical self-analysis and professional reflection about potential roles in a business context.

**PLO6: (CORE) Critical Thinking and Problem Solving**

Evaluate business situations and critique managerial decisions, using financial statements, statistical tools, and other appropriate methods to organize, analyze and present data.

**Major Learning Outcomes (MLOs): Disciplinary Proficiency**

Achievement of the learning outcomes specified for one of the BSBA majors.

**Degree Requirements**

The Bachelor of Science in Business Administration (BSBA) program includes forty-eight (48) credits of core courses in business administration covering a broad knowledge of business areas and functions, emphasizing global business environment and practices.

BSBA majors may choose to specialize in accounting, finance, human resource management, marketing, business analytics or hospitality and tourism management by taking 21 credits of major required courses and nine credits of major electives in one of these areas.

Additionally, students must fulfill the General Education requirements with thirty-one credits and complete 9 credits of general business electives and a three-credit general electives to complete the 121-credit program.

The Bachelor of Science in Business Administration degree requires the completion of the following degree requirements:

|                                           | <b>BSBA Major in Accounting/<br/>Business Analytics/ Finance/<br/>HRM/ Marketing/ Hospitality &amp;<br/>Tourism Management</b> | <b>BSBA –<br/>Generalist</b> |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Program Requirements</b>               | <b>No. of Credits</b>                                                                                                          | <b>No. of<br/>Credits</b>    |
| University General Education Requirements | 31                                                                                                                             | 31                           |
| School Requirements                       | 48                                                                                                                             | 48                           |



|                             |            |            |
|-----------------------------|------------|------------|
| Major Requirements          | 21         | -          |
| Major Elective Requirements | 9          | -          |
| Business Electives          | 9          | 39         |
| Free Electives              | 3          | 3          |
| <b>Total</b>                | <b>121</b> | <b>121</b> |

### University General Education Component Requirements

**31 Credits**

The General Education Component requires undergraduate students to take 31-34 credits in designated categories. To review the General Education Component requirements, please refer to the section of the catalog entitled, "General Education Component." ***The fifth writing intensive course for the BS in Business Administration is BUSN 301 Business Communication.***

Note that those seeking the BS of Business Administration are required to *take ECON 103 Principles of Economics* to satisfy the Social and Behavioral Sciences requirements *and STAT 100 Statistics* to satisfy the mathematics requirement for the Natural Sciences. For further guidance in selecting courses to satisfy the General Education Component requirements, please consult your academic advisor.

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition ( <b>Writing Intensive</b> )                                                                             | 3       |
| ITEC 103                | Fundamentals of Information Technology                                                                               | 3       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                        | 3       |
| PHIL 100 or<br>ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR<br>Advanced Composition ( <b>Writing Intensive</b> ) | 3       |
| MEST 100                | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                                  | 3       |
| UAES 200                | Survey of United Arab Emirates Studies                                                                               | 3       |
| ECON 103                | Principles of Microeconomics                                                                                         | 3       |
| STAT 100                | Introduction to Probability & Statistics                                                                             | 3       |
| Gen Ed Course           | Natural Science Course                                                                                               | 3       |

### School of Business Core Course Requirements

**48 Credits**

| Course Code | Title                                                   | Prerequisites   | Credits |
|-------------|---------------------------------------------------------|-----------------|---------|
| ACCT 204    | Principles of Financial Accounting                      | ECON 103        | 3       |
| ACCT 205    | Principles of Managerial Accounting                     | ECON 103        | 3       |
| BUAN 301    | Analytics for Decision Making                           | OPMT 311        | 3       |
| BUSN 301    | Business Communication (A) ( <b>Writing Intensive</b> ) | ENGL 101        | 3       |
| BUSN 304    | Business Law                                            | Junior Standing | 3       |

|          |                                              |                    |   |
|----------|----------------------------------------------|--------------------|---|
| BUSN 306 | International Business (A)                   | ECON 104           | 3 |
| BUSN 307 | Research Methodology (A)                     | Junior Standing    | 3 |
| BUSN 312 | Public Policy and Business Ethics (A)        | ECON 103           | 3 |
| BUSN 498 | Strategic Management (A)                     | Senior Standing    | 3 |
| ECON 104 | Macroeconomic Policy                         | ECON 103           | 3 |
| FNAN 300 | Principles of Business Finance               | ACCT 204           | 3 |
| MGMT 301 | Principles of Management                     | Sophomore Standing | 3 |
| MGMT 401 | Organizational Behavior                      | Senior Standing    | 3 |
| MIST 301 | Introduction to Business Information Systems | Sophomore Standing | 3 |
| MKTG 301 | Principles of Marketing                      | ECON 103           | 3 |
| OPMT 311 | Methods and Models of Management Sciences    | STAT 100           | 3 |

(A) – Applied Course

### Major Requirements

**21 credits**

### BSBA Major Specialization

To complete a major, graduates must complete thirty (30) credits in the chosen discipline, twenty-one (21) credits from the major required courses, and nine (9) credits from major elective courses

### BSBA (Major in Accounting)

The BSBA (Major in Accounting) is a specialized major within the BSBA program that complements general education and core courses taken by all students within the BSBA by providing the student tools for business accounting used by financial managers. The degree is useful for students who wish to pursue careers in business, especially for company accountants, or auditors.

### Accounting Major Program Learning Outcomes (AMLO)

On completion of a BSBA (Major in Accounting) students will be able to:

**AMLO 1:** Evaluate, aggregate, and synthesize knowledge relating to the accounting function from diverse sources for planning, controlling and performance evaluation;

**AMLO 2:** Assess how auditing and fiscal control measures can be implemented in manual and electronic accounting information systems: and

**AMLO 3:** Prepare, analyze, and interpret financial statements using International Financial Reporting Standards on accounting practice for business organizations and other corporate entities.

### BSBA (Major in Accounting) Required Courses

**21 Credits**

| Course   | Course Title                            | Credits | Prerequisites   |
|----------|-----------------------------------------|---------|-----------------|
| ACCT 311 | Managerial and Cost Accounting (A)      | 3       | ACCT 205        |
| ACCT 331 | Intermediate Accounting I (A)           | 3       | ACCT 204        |
| ACCT 332 | Intermediate Accounting II (A)          | 3       | ACCT 331        |
| ACCT 361 | Accounting Information Systems (A)      | 3       | ACCT 331        |
| ACCT 390 | Internship in Accounting (A)            | 3       | Senior Standing |
| ACCT 412 | Auditing (A)                            | 3       | ACCT 331        |
| ACCT 444 | Accounting Project Based Internship (A) | 3       | ACCT 331        |

(A) – Applied Course

**BSBA (Major in Accounting) Elective courses (Any 3 courses to be taken)**

**9 Credits**

| Course Code | Course Title                                       | Credits | Prerequisites |
|-------------|----------------------------------------------------|---------|---------------|
| ACCT 352    | Taxation and Managerial Decisions (A)              | 3       | ACCT 331      |
| ACCT 472    | International Accounting (A)                       | 3       | ACCT 331      |
| ACCT 491    | Special Topics in Accounting (A)                   | 3       | ACCT 331      |
| FNAN 303    | Financial Analysis, Forecasting, and Valuation (A) | 3       | ACCT 204      |

(A) – Applied Course

**BSBA (Major in Finance)**

The BSBA (Major in Finance) is a specialized major within the BSBA program that complements general education and core courses taken by all students within the BSBA by providing a sound base of knowledge of the field of Finance. The BSBA (Major in Finance) provides tools for analysis of business, and investment. The degree is useful for students who wish to pursue careers in government, services, and manufacturing in corporate finance.

**Finance Major Program Learning Outcomes (FMLO)**

On completion of a BSBA (Major in Finance) students will be able to:

**FMLO 1:** Assess the salience to specific decision scenarios of the structures and regulatory frameworks encompassing financial institutions and markets

**FMLO 2:** Evaluate, aggregate and synthesize knowledge relating to financial operations from diverse sources for the benefit of the organization; and

**FMLO 3:** Formulate and craft effective solutions to finance-related business problems, based on sound application of underlying financial principles.

**BSBA (Major in Finance) Required Courses**

**21 Credits**

| Course Code | Title                                          | Credits | Prerequisite       |
|-------------|------------------------------------------------|---------|--------------------|
| FNAN 303    | Financial Analysis, Forecasting, and Valuation | 3       | ACCT 204           |
| FNAN 304    | Risk Management and Insurance (A)              | 3       | FNAN 303           |
| FNAN 321    | Financial Institutions (A)                     | 3       | FNAN 300           |
| FNAN 390    | Internship in Finance (A)                      | 3       | Senior Standing    |
| FNAN 401    | Advanced Financial Management (A)              | 3       | FNAN 303           |
| FNAN 411    | Investment Analysis and Portfolio Management   | 3       | STAT 100, FNAN 303 |
| FNAN 444    | Finance Project Based Internship (A)           | 3       | FNAN 303           |

(A) – Applied Course

**BSBA (Major in Finance) Elective Courses (Any 3 courses to be taken)**

**9 Credits**

| Course   | Title                           | Credits | Prerequisites                |
|----------|---------------------------------|---------|------------------------------|
| ACCT 331 | Intermediate Accounting I (A)   | 3       | ACCT 204                     |
| FNAN 412 | Futures and Options Markets (A) | 3       | FNAN 303, FNAN 411, OPMT 311 |

|          |                                    |   |          |
|----------|------------------------------------|---|----------|
| FNAN 421 | Money and Capital Markets (A)      | 3 | FNAN 303 |
| FNAN 431 | International Financial Management | 3 | FNAN 303 |

(A) – Applied Course

### **BSBA (Major in Human Resource Management)**

The BSBA (Major in Human Resource Management) is a specialized major within the BSBA program that is intended to provide a broad background of study for students who wish to pursue a human resource career in one of the many business sectors: retail, hospitality, information technology, manufacturing, or non-profit organizations. The BSBA (Major in Human Resource Management) provides students with tools and knowledge valuable to any business and with practice in developing skills such as teamwork, analysis of numerical data, and communicating in writing and orally.

### **Human Resource Management Major Program Learning Outcomes (HMLO)**

On completion of a BSBA (Major in Human Resource Management) students will be able to

**HMLO 1:** Assess sustainable paths to develop the human resource within an organization;

**HMLO 2:** Evaluate, aggregate, and synthesize knowledge relating to people management functions from diverse sources for the benefit of the organization; and

**HMLO 3:** Formulate and craft viable solutions to people management problems at both higher strategic and operational levels.

### **BSBA (Major in Human Resource Management) Required Courses**

**21 Credits**

| Course Code | Course Title                                                   | Credits | Prerequisites   |
|-------------|----------------------------------------------------------------|---------|-----------------|
| MGHR 301    | Human Resource Management                                      | 3       | MGMT 301        |
| MGHR 302    | Compensation Management                                        | 3       | MGHR 301        |
| MGHR 304    | Human Resource Recruitment & Selection (A)                     | 3       | MGHR 301        |
| MGHR 390    | Internship in Human Resource Management (A)                    | 3       | Senior Standing |
| MGHR 403    | Cross Cultural and International Human Resource Management (A) | 3       | MGHR 301        |
| MGHR 431    | Employee Relations                                             | 3       | MGHR 301        |
| MGHR 444    | HRM Project Based Internship (A)                               | 3       | MGHR 301        |

(A) – Applied Course

### **BSBA (Major in Human Resource Management) Elective Courses**

**9 Credits**

| Course Code | Course Title                            | Credits | Prerequisites |
|-------------|-----------------------------------------|---------|---------------|
| MGHR 461    | Diversity in organization               | 3       | MGMT 301      |
| MGHR 462    | Strategic Human Resource Management (A) | 3       | MGHR 301      |
| MGHR 463    | Negotiations in Organizations (A)       | 3       | MGHR 301      |
| MGHR 464    | Training and Development (A)            | 3       | MGMT 301      |

(A) – Applied Course

### **BSBA (Major in Marketing)**

The BSBA (Major in Marketing) exposes students to the latest research and theory in global and multicultural marketing and examines research driven strategic advertising and promotion based on established analytics. Students are equipped with marketing knowledge and skills to develop specialized marketing professionals.

#### **Marketing Major Program Learning Outcomes (MMLO)**

On completion of a BSBA (Major in Marketing) students will be able to

**MMLO 1:** Apply salient marketing techniques to product/service positioning within market segments;

**MMLO 2:** Evaluate, aggregate, and synthesize knowledge relating to the marketing function from diverse sources for the benefit of the organization; and

**MMLO 3:** Formulate and craft integrated market plans and marketing strategies for various business domains.

#### **BSBA (Major in Marketing) Required Courses**

**21 Credits**

| Course Code | Course Title                                     | Credits | Prerequisites         |
|-------------|--------------------------------------------------|---------|-----------------------|
| MKTG 312    | Consumer behavior                                | 3       | MKTG 301              |
| MKTG 313    | Integrated Marketing Communications (A)          | 3       | MKTG 301              |
| MKTG 351    | Marketing Research Techniques & Applications (A) | 3       | MKTG 301,<br>OPMT 311 |
| MKTG 390    | Internship in Marketing (A)                      | 3       | Senior Standing       |
| MKTG 412    | Marketing of Services (A)                        | 3       | MKTG 301              |
| MKTG 471    | Marketing Strategy                               | 3       | Senior Standing       |
| MKTG 444    | Marketing Project Based Internship (A)           | 3       | MKTG 301              |

(A) – Applied Course

#### **BSBA (Major in Marketing) Elective Courses**

**9 Credits**

| Course Code | Course Title                 | Credits | Prerequisites |
|-------------|------------------------------|---------|---------------|
| MKTG 311    | Sales Management             | 3       | MKTG 301      |
| MKTG 332    | Retailing and E-Commerce (A) | 3       | MKTG 301      |
| OPMT 405    | Supply Chain Management      | 3       | OPMT 311      |
| MKTG 407    | International Marketing (A)  | 3       | MKTG 301      |

(A) – Applied Course

### **BSBA (Major in Business Analytics)**

The BSBA (Major in Business Analytics) will equip the students to use the power of data science, big data, statistics, and machine learning to optimize business performance and to make the right decisions, organizations need people with expert skills and an analytical approach. The students will combine statistics, information technology, computer science, and business administration to come up with innovative solutions to problems in the business world.

### **Business Analytics Major Program Learning Outcomes (BMLO)**

On completion of a BSBA (Major in Business Analytics) students will be able to

**BMLO 1:** Integrate information technologies with quantitative analysis and data science methods to extract business value from data sets, and to enable data-driven business decisions.

**BMLO 2:** Identify opportunities, needs and constraints for data analytics within specific organizational contexts.

**BMLO 3:** Demonstrate proficiency and creativity with the integrated use of analytics tools and techniques to formulate useful data analytics solutions.

### **BSBA (Major in Business Analytics) Required Courses**

**21 Credits**

| Course   | Course Title                                       | Credits | Prerequisites      |
|----------|----------------------------------------------------|---------|--------------------|
| BUAN 311 | Business Data Mining (A)                           | 3       | BUAN 301           |
| BUAN 312 | Visual Analytics (A)                               | 3       | BUAN 301           |
| BUAN 390 | Internship in Business Analytics (A)               | 3       | Senior Standing    |
| BUAN 401 | Business Data Engineering (A)                      | 3       | MIST 301, BUAN 311 |
| BUAN 411 | Business Intelligence & Big Data Analytics         | 3       | MIST 301, BUAN 311 |
| MKTG 351 | Marketing Research Techniques and Applications (A) | 3       | MKTG 301, OPMT 311 |
| BUAN 444 | Business Analytics Project Based Internship (A)    | 3       | BUAN 301           |

(A) – Applied Course

### **BSBA (Major in Business Analytics) Elective Courses**

**9 Credits**

| Course Code | Course Title                                       | Credits | Prerequisites      |
|-------------|----------------------------------------------------|---------|--------------------|
| ACCT 361    | Accounting Information Systems (A)                 | 3       | ACCT 331           |
| FNAN 303    | Financial Analysis, Forecasting, and Valuation (A) | 3       | ACCT 204           |
| OPMT 405    | Supply Chain Management                            | 3       | OPMT 311           |
| OPMT 301    | Operations Management                              | 3       | Sophomore Standing |

(A) – Applied Course

### **BSBA (Major in Hospitality and Tourism Management)**

The BSBA (Major in Hospitality and Tourism Management) aims to provide a broad background of study for students who wish to pursue a career in hospitality and tourism management. The BSBA (Major in Hospitality and Tourism Management) program provides students with valuable tools and knowledge required in business. It also aims to provide students with skills such as teamwork, analysis of numerical data, and communicating in writing and orally. The BSBA (Major in Hospitality and Tourism Management) is offered in collaboration with Appalachian State University. Students will undertake one semester in a partner organization in the United States of America (USA).

### **Hospitality and Tourism Management Major Program Learning Outcomes (HTMLO)**

On completion of a BSBA (Major in Hospitality and Tourism Management) students will be able to

**HTMLO 1:** Apply core hospitality and tourism management principles to an evaluation of hospitality and tourism systems, structures and processes;

**HTMLO 2:** Apply practical considerations to an evaluation of hospitality and tourism systems, structures and processes; and

**HTMLO 3:** Integrate an analysis of the external environment into an evaluation of management decisions in the hospitality and tourism sector.

**BSBA (Major in Hospitality and Tourism Management) Required Courses** **21 Credits**

| Course Code | Course Title                                                    | Credits | Prerequisites            |
|-------------|-----------------------------------------------------------------|---------|--------------------------|
| HTMT 311    | Survey of Hospitality and Tourism                               | 3       | None                     |
| HTMT 321    | Hospitality Operations I – F&B                                  | 3       | HTMT 311                 |
| HTMT 322    | Hotel Operations (A)                                            | 3       | HTMT 311                 |
| HTMT 390    | Internship in Hospitality & Tourism Management (A)              | 3       | Senior Standing          |
| HTMT 411    | Destination Management (A)                                      | 3       | HTMT 311                 |
| HTMT 412    | Meetings and Conventions (A) (C)                                | 3       | Completion of 84 Credits |
| HTMT 444    | Hospitality and Tourism Management Project Based Internship (A) | 3       | HTMT 311                 |

(A) – Applied Course, (C) Compulsory Course, (O) Optional Course

**BSBA (Major in Hospitality and Tourism Management) Elective Courses** **9 Credits**

| Course Code | Course Title                  | Credits | Prerequisites            |
|-------------|-------------------------------|---------|--------------------------|
| HTMT 312    | Sustainable Tourism           | 3       | HTMT 311                 |
| BUSN 391    | International Internship (A)  | 3       | Completion of 90 Credits |
| MKTG 313    | Integrated Marketing          | 3       | MKTG 301                 |
| MKTG 412    | Marketing of Services (A)     | 3       | MKTG 301                 |
| MGHR 301    | Human Resource Management (O) | 3       | MKTG 301                 |

(A) – Applied Course, (C) Compulsory Course, (O) Optional Course

**Business Elective Courses** **9 Credits**

Any three courses from the major requirements and major electives listed above other than the declared major required and major elective courses taken to complete a major.

**Free Elective Courses** **3 Credits**

Any Course from other Programs and not already taken.

**Other Paths to complete the 121 credit hours requirement for a BSBA degree**

**The Generalist Option**

Instead of completing thirty (30) credits (21 + 9) of major elective courses in a specific major, students can complete thirteen (13) courses, 39 credit hours of business electives (including 2 internship courses in any of the majors) provided these comprise major required and major elective courses in any of the 6 majors.

**TOTAL DEGREE REQUIREMENT** **121 Credit Hours**



**Bachelor of Business Administration Four-Year Sample Schedule of Delivery**  
**(Major in Accounting/Business Analytics/Finance/HRM/Marketing)**

| Year One, First Semester   |                                          |    | Year One, Second Semester   |                                                                                                                   |    |
|----------------------------|------------------------------------------|----|-----------------------------|-------------------------------------------------------------------------------------------------------------------|----|
| Course Code                | Course Title                             | CR | Course Code                 | Course Title                                                                                                      | CR |
| ENGL 101                   | Composition                              | 3  | ECON 104                    | Principles of Macroeconomics                                                                                      | 3  |
| ECON 103                   | Principles of Microeconomics             | 3  | UNIV 200                    | Innovation, Entrepreneurship, and Sustainability                                                                  | 3  |
| ITEC 103                   | Introduction to Computing                | 3  | ARAB 101<br>or<br>ARAB110   | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3  |
| STAT 100                   | Introductory Probability & Statistics    | 3  | PHIL 100                    | Critical Thinking & Reasoning                                                                                     | 3  |
| UNIV 100                   | University First-Year Transition         | 1  | UAES 200                    | Survey of United Arab Emirates Studies                                                                            | 3  |
| MEST 100                   | Introduction to Islam in World Culture   | 3  |                             |                                                                                                                   |    |
| Subtotal =                 |                                          | 16 | Subtotal =                  |                                                                                                                   | 15 |
| Year Two, First Semester   |                                          |    | Year Two, Second Semester   |                                                                                                                   |    |
| Course Code                | Course Title                             | CR | Course Code                 | Course Title                                                                                                      | CR |
| GEN ED                     | Any Natural Science Course               | 3  | ACCT 205                    | Principles of Managerial Accounting                                                                               | 3  |
| ACCT 204                   | Principles of Financial Accounting       | 3  | MGMT 301                    | Principles of Management                                                                                          | 3  |
| MKTG 301                   | Principles of Marketing                  | 3  | BUAN 301                    | Analytics for Decision Making                                                                                     | 3  |
| BUSN 301                   | Business Communication                   | 3  | FNAN 300                    | Principles of Business Finance                                                                                    | 3  |
| OPMT 311                   | Methods and Models of Management Science | 3  | MIST 301                    | Introduction to Business Information Systems                                                                      | 3  |
| Subtotal =                 |                                          | 15 | Subtotal =                  |                                                                                                                   | 15 |
| Year Three, First Semester |                                          |    | Year Three, Second Semester |                                                                                                                   |    |
| Course Code                | Course Title                             | CR | Course Code                 | Course Title                                                                                                      | CR |
| BUSN 304                   | Business Law                             | 3  | BUSN 307                    | Research Methodology                                                                                              | 3  |
| BUSN 312                   | Public Policy and Ethics                 | 3  | MAJOR                       | PROGRAM MAJOR 3                                                                                                   | 3  |
| BUSN 306                   | International Business                   | 3  | MAJOR                       | PROGRAM MAJOR 4                                                                                                   | 3  |
| MAJOR                      | PROGRAM MAJOR 1                          | 3  | Major Elective              | MAJOR ELECTIVE 1                                                                                                  | 3  |
| MAJOR                      | PROGRAM MAJOR 2                          | 3  | Elective                    | BUSINESS ELECTIVE 1                                                                                               | 3  |
| Subtotal =                 |                                          | 15 | Subtotal =                  |                                                                                                                   | 15 |
| Year Three, Summer Session |                                          |    |                             |                                                                                                                   |    |
| Major Internship 390       | Major Specific Internship                | 3  |                             |                                                                                                                   |    |

| Year Four, First Semester     |                                         |    | Year Four, Second Semester |                     |    |
|-------------------------------|-----------------------------------------|----|----------------------------|---------------------|----|
| Course Code                   | Course Title                            | CR | Course Code                | Course Title        | CR |
| BUSN 498                      | Strategic Management                    | 3  | Major Elective             | MAJOR ELECTIVE 3    | 3  |
| MGMT 401                      | Organizational Behavior                 | 3  | Elective                   | BUSINESS ELECTIVE 2 | 3  |
| MAJOR                         | PROGRAM MAJOR 5                         | 3  | Elective                   | BUSINESS ELECTIVE 3 | 3  |
| Major Elective                | MAJOR ELECTIVE 2                        | 3  | Elective                   | FREE ELECTIVE       | 3  |
| Major Specific Internship 444 | Major Specific Project Based Internship | 3  |                            |                     | 3  |
| Subtotal =                    |                                         | 15 | Subtotal =                 |                     | 12 |
| 121 Semester Credit Hours     |                                         |    |                            |                     |    |

**Bachelor of Business Administration Four-Year Sample Schedule of Delivery**  
**(Major in Hospitality and Tourism Management)**

| Year One, First Semester   |                                                |    | Year One, Second Semester   |                                                                                                                   |    |
|----------------------------|------------------------------------------------|----|-----------------------------|-------------------------------------------------------------------------------------------------------------------|----|
| Course Code                | Course Title                                   | CR | Course Code                 | Course Title                                                                                                      | CR |
| ENGL 101                   | Composition                                    | 3  | ECON 104                    | Principles of Macroeconomics                                                                                      | 3  |
| ECON 103                   | Principles of Microeconomics                   | 3  | UNIV 200                    | Innovation, Entrepreneurship, and Sustainability                                                                  | 3  |
| ITEC 103                   | Introduction to Computing                      | 3  | ARAB 101<br>or<br>ARAB110   | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3  |
| STAT 100                   | Introductory Probability & Statistics          | 3  | PHIL 100                    | Critical Thinking & Reasoning                                                                                     | 3  |
| UNIV 100                   | University First-Year Transition               | 1  | UAES 200                    | Survey of United Arab Emirates Studies                                                                            | 3  |
| MEST 100                   | Introduction to Islam in World Culture         | 3  |                             |                                                                                                                   |    |
| Subtotal =                 |                                                | 16 | Subtotal =                  |                                                                                                                   | 15 |
| Year Two, First Semester   |                                                |    | Year Two, Second Semester   |                                                                                                                   |    |
| Course Code                | Course Title                                   | CR | Course Code                 | Course Title                                                                                                      | CR |
| GEN ED                     | Any Natural Science Course                     | 3  | ACCT 205                    | Principles of Managerial Accounting                                                                               | 3  |
| ACCT 204                   | Principles of Financial Accounting             | 3  | MGMT 301                    | Principles of Management                                                                                          | 3  |
| MKTG 301                   | Principles of Marketing                        | 3  | BUAN 301                    | Analytics for Decision Making                                                                                     | 3  |
| BUSN 301                   | Business Communication                         | 3  | FNAN 300                    | Principles of Business Finance                                                                                    | 3  |
| OPMT 311                   | Methods and Models of Management Science       | 3  | MIST 301                    | Introduction to Business Information Systems                                                                      | 3  |
| Subtotal =                 |                                                | 15 | Subtotal =                  |                                                                                                                   | 15 |
| Year Three, First Semester |                                                |    | Year Three, Second Semester |                                                                                                                   |    |
| Course Code                | Course Title                                   | CR | Course Code                 | Course Title                                                                                                      | CR |
| BUSN 304                   | Business Law                                   | 3  | BUSN 307                    | Research Methodology                                                                                              | 3  |
| BUSN 312                   | Public Policy and Ethics                       | 3  | BUSN 498                    | Strategic Management                                                                                              | 3  |
| BUSN 306                   | International Business                         | 3  | MGMT 401                    | Organizational Behavior                                                                                           | 3  |
| HTMT 311                   | Survey of the Hospitality & Tourism Industry   | 3  | Elective                    | BUSINESS ELECTIVE 2                                                                                               | 3  |
| Elective                   | BUSINESS ELECTIVE 1                            | 3  | Elective                    | BUSINESS ELECTIVE 3                                                                                               | 3  |
| Subtotal =                 |                                                | 15 | Subtotal =                  |                                                                                                                   | 15 |
| Year Three, Summer Session |                                                |    |                             |                                                                                                                   |    |
| HTMT 390                   | Internship in Hospitality & Tourism Management | 3  |                             |                                                                                                                   |    |

| Year Four, First Semester |  |  | Year Four, Second Semester |  |  |
|---------------------------|--|--|----------------------------|--|--|
|---------------------------|--|--|----------------------------|--|--|

|                           |                                                           |    |                |                          |    |
|---------------------------|-----------------------------------------------------------|----|----------------|--------------------------|----|
| Major Elective            | MAJOR ELECTIVE 1                                          | 3  | HTMT 412       | Meetings and Conventions | 3  |
| Major Elective            | MAJOR ELECTIVE 2                                          | 3  | HTMT 411       | Destination Management   | 3  |
| HTMT 322                  | Hotel Operations                                          | 3  | Major Elective | MAJOR ELECTIVE 3         | 3  |
| HTMT 321                  | Hospitality Operations I – F&B Management                 | 3  |                |                          |    |
| HTMT 444                  | Hospitality & Tourism Management Project Based Internship | 3  | Elective       | FREE ELECTIVE            | 3  |
| Subtotal =                |                                                           | 15 | Subtotal =     |                          | 12 |
| 121 Semester Credit Hours |                                                           |    |                |                          |    |

## Bachelor of Business Administration Four-Year Sample Schedule of Delivery (Generalist)

| Year One, First Semester          |                                          |    | Year One, Second Semester   |                                                                                                                   |    |
|-----------------------------------|------------------------------------------|----|-----------------------------|-------------------------------------------------------------------------------------------------------------------|----|
| Course Code                       | Course Title                             | CR | Course Code                 | Course Title                                                                                                      | CR |
| ENGL 101                          | Composition                              | 3  | ECON 104                    | Principles of Macroeconomics                                                                                      | 3  |
| ECON 103                          | Principles of Microeconomics             | 3  | UNIV 200                    | Innovation, Entrepreneurship, and Sustainability                                                                  | 3  |
| ITEC 103                          | Introduction to Computing                | 3  | ARAB 101<br>or<br>ARAB110   | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3  |
| STAT 100                          | Introductory Probability & Statistics    | 3  | PHIL 100                    | Critical Thinking & Reasoning                                                                                     | 3  |
| UNIV 100                          | University First-Year Transition         | 1  | UAES 200                    | Survey of United Arab Emirates Studies                                                                            | 3  |
| MEST 100                          | Introduction to Islam in World Culture   | 3  |                             |                                                                                                                   |    |
| Subtotal =                        |                                          | 16 | Subtotal =                  |                                                                                                                   | 15 |
| Year Two, First Semester          |                                          |    | Year Two, Second Semester   |                                                                                                                   |    |
| Course Code                       | Course Title                             | CR | Course Code                 | Course Title                                                                                                      | CR |
| GEN ED                            | Any Natural Science Course               | 3  | ACCT 205                    | Principles of Managerial Accounting                                                                               | 3  |
| ACCT 204                          | Principles of Financial Accounting       | 3  | MGMT 301                    | Principles of Management                                                                                          | 3  |
| MKTG 301                          | Principles of Marketing                  | 3  | BUAN 301                    | Analytics for Decision Making                                                                                     | 3  |
| BUSN 301                          | Business Communication                   | 3  | FNAN 300                    | Principles of Business Finance                                                                                    | 3  |
| OPMT 311                          | Methods and Models of Management Science | 3  | MIST 301                    | Introduction to Business Information Systems                                                                      | 3  |
| Subtotal =                        |                                          | 15 | Subtotal =                  |                                                                                                                   | 15 |
| Year Three, First Semester        |                                          |    | Year Three, Second Semester |                                                                                                                   |    |
| Course Code                       | Course Title                             | CR | Course Code                 | Course Title                                                                                                      | CR |
| BUSN 304                          | Business Law                             | 3  | BUSN 307                    | Research Methodology                                                                                              | 3  |
| BUSN 312                          | Public Policy and Ethics                 | 3  | Elective 3                  | Business Elective 3                                                                                               | 3  |
| BUSN 306                          | International Business                   | 3  | Elective 4                  | Business Elective 4                                                                                               | 3  |
| Elective 1                        | Business Elective 1                      | 3  | Elective 5                  | Business Elective 5                                                                                               | 3  |
| Elective 2                        | Business Elective 2                      | 3  | Elective 6                  | Business Elective 6                                                                                               | 3  |
| Subtotal =                        |                                          | 15 | Subtotal =                  |                                                                                                                   | 15 |
| Year Three, Summer Session        |                                          |    |                             |                                                                                                                   |    |
| Major Internship 390 (Elective 7) | Major Specific Internship                | 3  |                             |                                                                                                                   |    |

| Year Four, First Semester        |                                         |    | Year Four, Second Semester |                      |    |
|----------------------------------|-----------------------------------------|----|----------------------------|----------------------|----|
| Course Code                      | Course Title                            | CR | Course Code                | Course Title         | CR |
| BUSN 498                         | Strategic Management                    | 3  | Elective 11                | Business Elective 11 | 3  |
| MGMT 401                         | Organizational Behavior                 | 3  | Elective 12                | Business Elective 12 | 3  |
| Elective 8                       | Business Elective 8                     | 3  | Elective 13                | Business Elective 13 | 3  |
| Elective 9                       | Business Elective 9                     | 3  | Free Elective              | Free Elective        | 3  |
| Major Specific 444 (Elective 10) | Major Specific Project Based Internship | 3  |                            |                      | 3  |
| Subtotal =                       |                                         | 15 | Subtotal =                 |                      | 12 |
| 121 Semester Credit Hours        |                                         |    |                            |                      |    |



# SCHOOL OF ENGINEERING



# **SCHOOL OF ENGINEERING**

## **Bachelor of Architecture**

### **Program Description**

Bachelor of Architecture program is designed to prepare well educated, critical-thinkers and problem-solvers who possess and can effectively apply highly refined aesthetic, technical knowledge, are capable of facing professional and global challenges. Focused on sustainability challenges, which are as broadly cultural as they are technical, the program graduates are prepared to become credible stewards of the built and surrounding environment. The program leads students towards the application of principles of architecture as a science to analyze physical environments, components, and processes to identify problems and propose solutions through design and modeling. It navigates graduates to integrate into multidisciplinary teams as architects, to respect professional ethics, and through highly professional attitude contribute to the local, regional, and global sustainability.

***The Bachelor of Architecture is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

Bachelor of Architecture program is focused on sustainability challenges and prepares graduates that are critical-thinkers, problem-solvers and capable of facing professional and global challenges. Equipped with highly refined aesthetic and technical knowledge, capable of respecting professional ethics and work in multidisciplinary teams, graduates can easily integrate into architectural practices for built environment problem solving.

### **Program Educational Objectives**

The Bachelor of Architecture Program at AURAK produces graduates who are armed with knowledge and skills as articulated in its student outcomes to become:

1. Industrious architects who create healthy and safe spaces of different scale (buildings, cities) that are responsive to the environment and users.
2. Practitioners who value professional ethics and architecture business as influencing the built environment and human wellness.
3. Professionals capable of continuous learning to meet the requirements of altering built environment and comply with the regional/ national laws and regulations.
4. Designers experienced in evaluating solutions as it relates to building systems design, materials, technologies against project objectives, cost and building performance.
5. Team leaders capable of making decisions within architectural projects for universal design with minimal impact on environment.
6. Active contributors to the local community and the world through an integrative design process with a multidisciplinary team.

## **Program Learning Outcomes**

Graduates of the Bachelor of Architecture program will have the ability to:

1. Employ critical thinking that takes into consideration alternative perspectives by analyzing, evaluating, and synthesizing ideas and information gathered through research grounded in acquired information.
2. Implement complex two and three-dimensional graphic representation techniques using a wide variety of manual and digital media as well as oral and written forms to reflect on and convey the architectural design process to a wide range of stakeholders.
3. Demonstrate a solid base of knowledge in history and theory grounded in the intellectual understanding of social, cultural, and economic aspects and through applied research by raising clear and precise questions to clarify and express information to reach logical conclusions and evaluate options against relevant design criteria, building standards, and program requirements.
4. Utilize and integrate technical skills and professional knowledge in comprehensive application of life safety, accessibility, and sustainability issues in integrating the principles of environmental stewardship and making appropriate design decisions of varying scales and levels of complexity through a technical design process based on appropriate documentation in a manner that is client-centered, sustainable, aesthetic, cost effective, and socially responsible.
5. Develop the capability to synthesize variable components into an integrated design solution through the employment of appropriate building materials, building systems, and construction practices which are environmentally friendly and based on research and design decisions for a variety of systems and levels of complexity.
6. Show the ability to lead teams of stakeholders collaboratively within the process of understanding, developing, and applying solutions to problems in the built as well as the natural environments, utilizing knowledge and parameters of professional practice associated with ethical, legal, financial and social responsibilities.
7. Apply reasoning skills including math, physics, and logic to investigate problems related to gravity and other forces in structural systems, thermal heat gain and loss in buildings, material quantity estimates, budget management, and life-cycle cost analysis.

## **Degree Requirements**

Students complete the Bachelor of Architecture by completing twelve (12) courses (32 credit hours) of University General Education, five (5) course (13 credits) of Faculty Requirements, three (3) course (9 credits) of Department of Architecture requirements, thirty-one (31) courses (101 credits) of Architecture Major requirements and two (2) courses (6 credits) of Architecture Major Electives. Total number of credits – 161.

The required courses are:

**The University General Education Component Requirements****32 Credit Hours**

| Course Code                                                                | Title                                                                                                                | Credit |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|
| MATH 113                                                                   | Calculus I                                                                                                           | 3      |
| ENGL 101                                                                   | Composition ( <b>Writing Intensive</b> )                                                                             | 3      |
| UNIV 100                                                                   | University First-Year Transition                                                                                     | 1      |
| CSCI 112                                                                   | Introduction to Computer Programming                                                                                 | 3      |
| CSCI 113                                                                   | Introduction to Computer Programming Lab                                                                             | 1      |
| ARAB 101 or<br>ARAB 110                                                    | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3      |
| UNIV 200                                                                   | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                        | 3      |
| <b>Required Electives – The field of Art and Humanities (6 credits)</b>    |                                                                                                                      |        |
| MEST 100                                                                   | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                                  | 3      |
| PHIL 100 or<br>ENGL 200                                                    | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR<br>Advanced Composition ( <b>Writing Intensive</b> ) | 3      |
| <b>Electives – The field of Social and Behavioral Sciences (6 credits)</b> |                                                                                                                      |        |
| UAES 200                                                                   | Survey of United Arab Emirates Studies (Required)                                                                    | 3      |
| PSYC 100                                                                   | Introduction to Psychology (Or another course from the same list)                                                    | 3      |
| <b>Required Elective – The field of natural Sciences (3 credits)</b>       |                                                                                                                      |        |
| ENVS 102                                                                   | Sustainability and Human Environment Relations                                                                       | 3      |

The fifth writing intensive course for the Bachelor of Architecture is ARCH 591 Graduation Project Thesis Research.

**School of Engineering Requirements****13 Credit Hours**

| Course Code | Title                     | Credit |
|-------------|---------------------------|--------|
| PHYS 110    | University Physics I      | 3      |
| PHYS 111    | University Physics I Lab. | 1      |
| MENG 323    | Engineering Economy       | 3      |
| ARCH 390    | Internship I              | 3      |
| ARCH 391    | Internship II             | 3      |

**Department Requirements****9 Credit Hours**

| Course   | Title                                                       | Credit |
|----------|-------------------------------------------------------------|--------|
| INDS 112 | Design Communication 1: Sketching and Drawing for Interiors | 3      |
| INDS 121 | Interior Design Process, Human Factors and Ergonomic        | 3      |
| INDS 122 | Design Communication 2: Digital Media for Interiors         | 3      |

**Major Compulsory Requirements****101 Credit Hours**

| Course   | Title                          | Credit |
|----------|--------------------------------|--------|
| ARCH 121 | Basic Design 1                 | 4      |
| ARCH 122 | Basic Design 2                 | 4      |
| ARCH 223 | Computer Architectural Drawing | 3      |
| ARCH 221 | Architectural Design Studio 1  | 4      |

|          |                                                                                  |   |
|----------|----------------------------------------------------------------------------------|---|
| ARCH 222 | Architectural Design Studio 2                                                    | 4 |
| ARCH 231 | Building Construction 1                                                          | 3 |
| ARCH 232 | Building Construction 2                                                          | 3 |
| ARCH 241 | History and Theory of Architecture 1                                             | 3 |
| ARCH 242 | History and Theory of Architecture 2                                             | 3 |
| ARCH 321 | Architectural Design Studio 3                                                    | 4 |
| ARCH 322 | Architectural Design Studio 4                                                    | 4 |
| ARCH 331 | Building Construction 3                                                          | 3 |
| ARCH 336 | Construction Drawing                                                             | 3 |
| ARCH 341 | History and Theory of Architecture 3                                             | 3 |
| ARCH 342 | History and Theory of Contemporary Architecture                                  | 3 |
| ARCH 372 | Sustainable Architecture                                                         | 3 |
| ARCH 421 | Architectural Design Studio 5                                                    | 4 |
| ARCH 422 | Architectural Design Studio 6                                                    | 4 |
| ARCH 454 | Urban Planning                                                                   | 3 |
| ARCH 457 | Urban Design                                                                     | 3 |
| ARCH 463 | Landscape Architecture                                                           | 3 |
| ARCH 466 | Building Utilities II: Illumination, Acoustics, and Electrical Building Services | 3 |
| ARCH 485 | Professional Practice                                                            | 3 |
| ARCH 487 | Project Management                                                               | 3 |
| ARCH 521 | Architectural Design Studio 7                                                    | 4 |
| ARCH 591 | Graduation Project Thesis Research ( <b>Writing Intensive</b> )                  | 2 |
| ARCH 592 | Graduation Project Design                                                        | 5 |
| CIEN 216 | Structures for Architecture 1                                                    | 3 |
| CIEN 261 | Surveying                                                                        | 1 |
| CIEN 316 | Structures for Architecture 2                                                    | 3 |
| MENG 468 | Building Utilities: HVAC and Mechanical Building Services                        | 3 |

### Major Electives

**6 Credit Hours**

| Course   | Title                                     | Credits |
|----------|-------------------------------------------|---------|
| ARCH 449 | Technology of the Built Environment       | 3       |
| ARCH 456 | Sustainable Housing                       | 3       |
| ARCH 459 | Conservation of Historic Environment      | 3       |
| ARCH 473 | Environmental Control                     | 3       |
| INDS 351 | Parametric Design                         | 3       |
| INDS 353 | Landscape and Garden Design               | 3       |
| INDS 451 | Portfolio Design                          | 3       |
| INDS 453 | Advanced BIM: Scheduling and coordination | 3       |

## Bachelor of Architecture Five-Year Sample Schedule

| Year 1 Semester 1      |                                                             |         | Year 1 Semester 2    |                                                                                                                   |    |
|------------------------|-------------------------------------------------------------|---------|----------------------|-------------------------------------------------------------------------------------------------------------------|----|
| Course No.             | Description                                                 | CR      | Course No.           | Description                                                                                                       | CR |
| ARCH 121               | Basic Design 1                                              | 4       | ARCH 122             | Basic Design 2                                                                                                    | 4  |
| MATH 113               | Calculus I                                                  | 3       | INDS 121             | Interior Design Process, Human Factors and Ergonomics                                                             | 3  |
| INDS 112               | Design Communication 1: Sketching and Drawing for Interiors | 3       | INDS 122             | Design Communication 2: Digital Media for Interiors                                                               | 3  |
| ENGL 101               | Composition                                                 | 3       | PHYS 110             | University Physics I                                                                                              | 3  |
| UNIV 100               | University First-Year transition                            | 1       | PHYS 111             | University Physics I Lab.                                                                                         | 1  |
|                        |                                                             |         | CSCI 112             | Introduction to Computer Programming                                                                              | 3  |
|                        |                                                             |         | CSCI 113             | Introduction to Computer Programming Lab.                                                                         | 1  |
| Subtotal =             |                                                             | 14      | Subtotal =           |                                                                                                                   | 18 |
| Year 2 Semester 1      |                                                             |         | Year 2 Semester 2    |                                                                                                                   |    |
| Course No.             | Description                                                 | CR      | Course No.           | Description                                                                                                       | CR |
| ARCH 221               | Architectural Design Studio 1                               | 4       | ARCH 222             | Architectural Design Studio 2                                                                                     | 4  |
| ARCH 231               | Building Construction 1                                     | 3       | ARCH 232             | Building Construction 2                                                                                           | 3  |
| ARCH 241               | History & Theory of Architecture 1                          | 3       | ARCH 242             | History & Theory of Architecture 2                                                                                | 3  |
| CIEN 216               | Structures for Architecture 1                               | 3       | CIEN 261             | Surveying                                                                                                         | 1  |
| ARCH 223               | Computer Architectural Drawing                              | 3       | ARAB 101 or ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3  |
|                        |                                                             |         | PSYC 100             | Introduction to Psychology (or other from the same list)                                                          | 3  |
| Subtotal =             |                                                             | 16      | Subtotal =           |                                                                                                                   | 17 |
| Year 3 Semester 1      |                                                             |         | Year 3 Semester 2    |                                                                                                                   |    |
| Course No.             | Description                                                 | CR      | Course No.           | Description                                                                                                       | CR |
| ARCH 321               | Architectural Design Studio 3                               | 4       | ARCH 322             | Architectural Design Studio 4                                                                                     | 4  |
| ARCH 331               | Building Construction 3                                     | 3       | ARCH 336             | Construction Drawings                                                                                             | 3  |
| ARCH 341               | History and Theory of Architecture 3                        | 3       | ARCH 342             | History and Theory of Contemporary Architecture                                                                   | 3  |
| CIEN 316               | Structures for Architecture 2                               | 3       | ARCH 372             | Sustainable Architecture                                                                                          | 3  |
| ENVS 102               | Sustainability and Human-Environment Relations              | 3       | ARCH 000             | Technical Elective 1                                                                                              | 3  |
| Subtotal =             |                                                             | 16      | Subtotal =           |                                                                                                                   | 16 |
| Summer Semester        |                                                             |         |                      |                                                                                                                   |    |
| Code                   | Course Title                                                | Credits |                      |                                                                                                                   |    |
| ARCH 390               | Internship I                                                | 3       |                      |                                                                                                                   |    |
| Year 4 Semester 1      |                                                             |         | Year 4 Semester 2    |                                                                                                                   |    |
| Course No.             | Description                                                 | CR      | Course No.           | Description                                                                                                       | CR |
| ARCH 421               | Architectural Design Studio 5                               | 4       | ARCH 422             | Architectural Design Studio 6                                                                                     | 4  |
| ARCH 463               | Landscape Architecture                                      | 3       | ARCH 466             | Building Utilities II: Illumination, Acoustics, and Electrical Building Services                                  | 3  |
| MENG 468               | Building Utilities I: HVAC and Mechanical Building Services | 3       | ARCH 000             | Technical Elective 2                                                                                              | 3  |
| MEST 100               | Introduction to Islam in World Culture                      | 3       | UAES 200             | Survey of United Arab Studies                                                                                     | 3  |
| ARCH 457               | Urban Design                                                | 3       | ENGL 200 or PHIL 100 | Advanced Composition or Critical Thinking and Reasoning                                                           | 3  |
| Subtotal =             |                                                             | 16      | Subtotal =           |                                                                                                                   | 16 |
| Summer Semester        |                                                             |         |                      |                                                                                                                   |    |
| Course No.             | Description                                                 | CR      |                      |                                                                                                                   |    |
| ARCH 391               | Internship II                                               | 3       |                      |                                                                                                                   |    |
| Year 5 Semester 1      |                                                             |         | Year 4 Semester 2    |                                                                                                                   |    |
| Course No.             | Description                                                 | CR      | Course No.           | Description                                                                                                       | CR |
| ARCH 521               | Architectural Design Studio 7                               | 4       | ARCH 592             | Graduation Project Design                                                                                         | 5  |
| ARCH 591               | Graduation Project Thesis Research                          | 2       | MENG 323             | Engineering Economy                                                                                               | 3  |
| ARCH 454               | Urban Planning                                              | 3       | ARCH 485             | Professional Practice                                                                                             | 3  |
| ARCH 487               | Project Management                                          | 3       | UNIV 200             | Innovation, Entrepreneurship, & Sustainability                                                                    | 3  |
| Subtotal =             |                                                             | 12      | Subtotal =           |                                                                                                                   | 14 |
| TOTAL CREDIT HOURS 161 |                                                             |         |                      |                                                                                                                   |    |

## **Bachelor of Arts in Interior Design**

### **Program Description**

The Bachelor of Art in Interior Design Degree is a comprehensive experience that provides those who are creative excellent learning opportunities in Interior Design theory and practice. The study for the degree last nine semesters (four academic years) and equips graduates with substantial knowledge and skills to work in multidisciplinary teams for creating healthy indoor environments within diverse historical, social, cultural, geographical and economic contexts. Capable of critical thinking, continuous self-development and lifelong learning, graduates are well prepared to meet the challenges of the ever-changing world and building sector, evaluate design theories and solve the problems related to indoor spaces, human and environmental wellbeing.

*The Bachelor of Arts in Interior Design is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.*

### **Program Mission**

Bachelor of Art in Interior Design at AURAK prepares students for continuous self-development and lifelong learning and equips them with professional, competitive skills and capability of producing contextually appropriate design solutions that meet the requirements of the rapidly developing building sector and in accordance with the cultural, social, and political sensibility, human and environmental wellbeing, safety and aesthetic values.

### **Program Educational Objectives**

The Bachelor of Arts in Interior Design at AURAK produces graduates who are armed with knowledge and skills as articulated in its student outcomes to become:

1. Assiduous designers who engage in lifelong learning, continuous self-development and value business ethics to meaningfully develop their careers and contribute to the human and environmental wellbeing.
2. Problems-solvers with advanced knowledge and specialized skills in interior design who are capable of working in multidisciplinary team and in diverse historical, social, cultural, geographical and economic contexts.
3. Designers capable of evaluating design theories and philosophies conserving indoor spaces, built and natural environment and make data-driven design decisions.
4. Wellness practitioners that are well equipped with knowledge in codes and regulations related to the built environment and can face the challenges of the interior design industry and positively contribute to it.
5. Professionals armed with skills in latest design technologies and communication media capable of producing interior design projects that satisfy the requirements of the clients and users.

## **Program Learning Outcomes**

Graduates of the program will have the ability to:

1. Value business ethics and interior design practice as influenced by the regional and global market and integral part of multidisciplinary teamwork for building environment problem solving, space quality enhancement and human wellbeing.
2. Appraise global context and its impact on interior design practice and construction decisions, using knowledge and theories related to the history of interior design, decorative arts, material culture and architecture in relationship to the social, political and physical factors that affect the changes of the built environment.
3. Evaluate theories related to design elements and principles, space organization and produce design problems solutions throughout the phases of design process for their two- and three-dimensional representation via various media and communication techniques.
4. Apply philosophies concerning interrelations between building, nature, virtual and technological environments as influencing human experience, behavior, wellbeing, and performance, considering human factors, ergonomics and recommend design solutions for building systems to contribute human wellbeing and environmental safety.
5. Develop appropriate color and natural/artificial lighting schemes for supporting design ideas with minimal impact on the environment through use of color and light doctrines that are dependent on context and influencing space perception and human wellbeing.
6. Compare properties, safety, and life cycle cost of furnishings, objects, materials and finishes, fabrication, installation, and maintenance methods for supporting design project criteria while minimizing the impact on human and environmental wellbeing.
7. Assess the impact of construction on the environment and recommend best design solutions; develop interior construction drawings and documents for the given project scope; appraise legislations concerning built and natural environments human wellbeing and universal design.

## **Degree Requirements**

Student complete Bachelor of Art in Interior design by completing eleven (11) courses (31 credit hours) of University General Education, two (2) course (6 credits) of Faculty Requirements, four (4) courses (14 credits) of Department of Architecture requirements, twenty-two (22) courses (72 credits) of Interior Design Major requirements and two (2) courses (6 credits) of Interior Design Major Electives. The total number of credits is 129.

### **University General Education Requirement**

**31 Credits**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BA in Interior Design is INDS 412 Graduation Final Thesis Research.**



| Course Code                                                                | Title                                                                                                                | Credits |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| <b>Core – (16 credits)</b>                                                 |                                                                                                                      |         |
| STAT 100                                                                   | Introductory Probability and Statistics                                                                              | 3       |
| ENGL 101                                                                   | Composition <b>(Writing Intensive)</b>                                                                               | 3       |
| UNIV 100                                                                   | University First-Year Transition                                                                                     | 1       |
| ITEC 103                                                                   | Fundamentals of Information Technology                                                                               | 3       |
| ARAB 101 or<br>ARAB 110                                                    | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| UNIV 200                                                                   | Innovation, Entrepreneurship, and Sustainability <b>(Writing Intensive)</b>                                          | 3       |
| <b>Electives – The field of Arts and Humanities (6 credits)</b>            |                                                                                                                      |         |
| MEST 100                                                                   | Introduction to Islam in World Culture <b>(Writing Intensive)</b>                                                    | 3       |
| PHIL 100 or<br>ENGL 200                                                    | Critical Thinking and Reasoning <b>(Writing Intensive)</b> OR<br>Advanced Composition <b>(Writing Intensive)</b>     | 3       |
| <b>Electives – The field of Social and Behavioral Sciences (6 credits)</b> |                                                                                                                      |         |
| UAES 200                                                                   | Survey of United Arab Emirates Studies                                                                               | 3       |
| PSYC 100                                                                   | Introduction to Psychology                                                                                           | 3       |
| <b>Electives – The field of Natural Sciences – (3 credits)</b>             |                                                                                                                      |         |
| ENVS 102                                                                   | Sustainability and Human Environment Relations                                                                       | 3       |

### School of Engineering Requirements

**6 Credits**

| Course Code | Title         | Credits |
|-------------|---------------|---------|
| INDS 390    | Internship I  | 3       |
| INDS 391    | Internship II | 3       |

### Department (Architecture) Requirements

**14 Credits**

| Course Code | Title                                | Credits |
|-------------|--------------------------------------|---------|
| ARCH 121    | Basic Design 1                       | 4       |
| ARCH 122    | Basic Design 2                       | 4       |
| ARCH 223    | Computer Architectural Drawing       | 3       |
| ARCH 341    | History and Theory of Architecture 3 | 3       |

### Major (Interior Design) Requirements

**72 Credits**

| Course Code | Title                                                       | Credits |
|-------------|-------------------------------------------------------------|---------|
| INDS 111    | Color Theory and Rendering                                  | 3       |
| INDS 112    | Design Communication 1: Sketching and Drawing for Interiors | 3       |
| INDS 121    | Interior Design Process: Human Factors and Ergonomics       | 3       |
| INDS 122    | Design Communication 2: Digital Media for Interiors         | 3       |
| INDS 211    | Interior Design Studio 1                                    | 4       |
| INDS 212    | History of Art and Interior Design 1                        | 3       |
| INDS 213    | Interior Construction Methods, Materials, and Finishes 1    | 3       |
| INDS 221    | Interior Design Studio 2                                    | 4       |
| INDS 222    | History of Art and Interior Design 2                        | 3       |

|          |                                                                      |   |
|----------|----------------------------------------------------------------------|---|
| INDS 223 | Interior Construction Methods, Materials, and Finishes 2             | 3 |
| INDS 224 | Sustainable Approaches to Interior Design                            | 3 |
| INDS 311 | Interior Design Studio 3                                             | 4 |
| INDS 312 | Construction Drawings and Interior Detailing                         | 3 |
| INDS 313 | Environmental Systems 1: Indoor Comfort and Human Wellbeing          | 3 |
| INDS 321 | Interior Design Studio 4                                             | 4 |
| INDS 322 | Furniture Design                                                     | 3 |
| INDS 323 | Environmental Systems 2: Design with Light and Sound                 | 3 |
| INDS 411 | Interior Design Studio 5                                             | 4 |
| INDS 412 | Graduation Final Thesis Research ( <b><i>Writing Intensive</i></b> ) | 2 |
| INDS 414 | Professional Practice 1: Codes and Regulations                       | 3 |
| INDS 421 | Graduation Project Design                                            | 5 |
| INDS 422 | Professional Practice II: Ethics and Project Management              | 3 |

### Major (Interior Design) Electives

**6 Credits**

| Course Code | Title                                     | Credits |
|-------------|-------------------------------------------|---------|
| INDS 351    | Parametric Design                         | 3       |
| INDS 352    | Sketching and Painting                    | 3       |
| INDS 353    | Landscape and Garden Design               | 3       |
| INDS 451    | Portfolio Design                          | 3       |
| INDS 452    | Photography                               | 3       |
| INDS 453    | Advanced BIM: Scheduling and Coordination | 3       |
| INDS 454    | Research Methods for Interior Design      | 3       |

## Bachelor of Arts in Interior Design Four-Year Study Plan

| Year 1 Semester 1      |                                                             |    | Year 1 Semester 2    |                                                                                                                   |    |
|------------------------|-------------------------------------------------------------|----|----------------------|-------------------------------------------------------------------------------------------------------------------|----|
| Course No.             | Description                                                 | CR | Course No.           | Description                                                                                                       | CR |
| ARCH 121               | Basic Design 1                                              | 4  | ARCH 122             | Basic Design 2                                                                                                    | 4  |
| INDS 111               | Color Theory and Rendering                                  | 3  | INDS 121             | Interior Design Process, Human Factors and Ergonomics                                                             | 3  |
| INDS 112               | Design Communication 1: Sketching and Drawing for Interiors | 3  | INDS 122             | Design Communication 2: Digital Media for Interiors                                                               | 3  |
| STAT 100               | Introductory Probability and Statistics                     | 3  | ITEC 103             | Fundamental of Information Technology                                                                             | 3  |
| ENGL 101               | Composition                                                 | 3  | ARAB 101 or ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3  |
| UNIV 100               | University First-Year Transition                            | 1  |                      |                                                                                                                   |    |
| Subtotal =             |                                                             | 17 | Subtotal =           |                                                                                                                   | 16 |
| Year 2 Semester 1      |                                                             |    | Year 2 Semester 2    |                                                                                                                   |    |
| Course No.             | Description                                                 | CR | Course No.           | Description                                                                                                       | CR |
| INDS 211               | Interior Design Studio 1                                    | 4  | INDS 221             | Interior Design Studio 2                                                                                          | 4  |
| INDS 212               | History of Art and Interior Design 1                        | 3  | INDS 222             | History of Art and Interior Design 2                                                                              | 3  |
| INDS 213               | Interior Construction Methods, Materials and Finishes 1     | 3  | INDS 223             | Interior Construction Methods, Materials and Finishes 2                                                           | 3  |
| ARCH 223               | Computer Architectural Drawing                              | 3  | INDS 224             | Sustainable Approaches in Interior Design                                                                         | 3  |
| ENVS 102               | Sustainability and Human-Environment Relations              | 3  | MEST 100             | Introduction to Islam in World Culture                                                                            | 3  |
| Subtotal =             |                                                             | 16 | Subtotal =           |                                                                                                                   | 16 |
| Year 2 Summer Semester |                                                             |    |                      |                                                                                                                   |    |
| Code                   | Course Title                                                | CR |                      |                                                                                                                   |    |
| INDS 390               | Internship I                                                | 3  |                      |                                                                                                                   |    |
| Year 3 Semester 1      |                                                             |    | Year 3 Semester 2    |                                                                                                                   |    |
| Course No.             | Description                                                 | CR | Course No.           | Description                                                                                                       | CR |
| INDS 311               | Interior Design Studio 3                                    | 4  | INDS 321             | Interior Design Studio 4                                                                                          | 4  |
| INDS 312               | Construction Drawing and Interior Detailing                 | 3  | INDS 322             | Furniture Design                                                                                                  | 3  |
| INDS 313               | Environmental Systems 1: Indoor Comfort and Human Wellbeing | 3  | INDS 323             | Environmental Systems 2: Design with Light and Sound                                                              | 3  |
| ARCH 341               | History and Theory of Architecture                          | 3  | PHIL 100 Or ENGL 200 | Critical Thinking and Reasoning or Advanced Composition                                                           | 3  |
| UAES 200               | Survey of United Arab Emirates Studies                      | 3  | INDS                 | Elective 1                                                                                                        | 3  |
| Subtotal =             |                                                             | 16 | Subtotal =           |                                                                                                                   | 16 |
| Summer Semester        |                                                             |    |                      |                                                                                                                   |    |
| Course No.             | Description                                                 | CR |                      |                                                                                                                   |    |
| INDS 391               | Internship II                                               | 3  |                      |                                                                                                                   |    |
| Year 4 Semester 1      |                                                             |    | Year 4 Semester 2    |                                                                                                                   |    |
| Course No.             | Description                                                 | CR | Course No.           | Description                                                                                                       | CR |
| INDS 411               | Interior Design Studio 5                                    | 4  | INDS 421             | Graduation Project Design                                                                                         | 5  |
| INDS 412               | Graduation Final Thesis Research                            | 2  | INDS 422             | Professional Practice: Ethics and Project Management                                                              | 3  |
| INDS 414               | Professional Practice 1: Codes and Regulations              | 3  | UNIV 200             | Innovation, Entrepreneurship, and Sustainability                                                                  | 3  |
| PSYC 100               | Introduction of Psychology                                  | 3  | INDS                 | Elective 2                                                                                                        | 3  |
| Subtotal =             |                                                             | 12 | Subtotal =           |                                                                                                                   | 14 |
| TOTAL CREDIT HOURS 129 |                                                             |    |                      |                                                                                                                   |    |

## **Bachelor of Science in Artificial Intelligence**

### **Program Description**

The Bachelor of Science in Artificial Intelligence (BSAI) degree program at AURAK is designed to equip learners with an in-depth knowledge of computer science theory and a thorough understanding of artificial intelligence (AI) and data science technologies. Students are exposed to state-of-the-art AI-related methods relevant to knowledge representation and reasoning. Such techniques will enhance human capabilities through automated analysis of complex inputs and facilitate automated decision making. The program includes techniques and topics such as machine learning, computer vision, robotics, data analysis and visualization, natural language processing, and data mining.

The global demand for AI and data science professionals is on the rise due to the integration of the field into most industries. This need is further exacerbated by a worldwide shortage of AI talent.

***The Bachelor of Science in Artificial Intelligence is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The mission of the Bachelor of Science in Artificial Intelligence program is to graduate students with the knowledge and skills to enable them to design and develop computer systems and data models using the latest advances in the field and hence become effective leaders capable of utilizing artificial intelligence and data science locally and globally across various domains.

### **Program Educational Objectives**

A few years after earning their degree, our graduates will be:

1. Successful professionals and innovators in theoretical and practical areas of computer science, artificial intelligence, and data science;
2. Engaged in creating a positive technological impact with a sufficient awareness of the ethical, legal, and security issues related to computing, artificial intelligence, and data science; and
3. Equipped with the skills required for professional practice including functioning in teams and communicating effectively.

### **Program Learning Outcomes**

Graduates of the program will have the ability to:

1. Analyze a complex computing problem and to apply principles of computing, artificial intelligence, statistics, and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate computer science or artificial intelligence solutions to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.

6. Apply computer science and artificial intelligence theory and software development fundamentals to produce computing-based solutions.
7. Build, apply, and evaluate data science models to solve problems using theoretical fundamentals.

### Degree Requirements

The BS in Artificial Intelligence requires the completion of 128 credits in the following areas:

| Area                                         | Credit                                        |
|----------------------------------------------|-----------------------------------------------|
| University General Education Requirements    | 32                                            |
| School of Engineering Requirements           | 26                                            |
| Artificial Intelligence Program Requirements | 70<br>(64 compulsory & 6 technical electives) |

### General Education Requirements

**32 Credits**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, “General Education Component.” You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Artificial Intelligence is CSAI 451 Machine Learning Lab.**

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition ( <b>Writing Intensive</b> )                                                                             | 3       |
| CSCI 112                | Introduction to Computer Programming                                                                                 | 3       |
| CSCI 113                | Introduction to Computer Programming Lab                                                                             | 1       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                        | 3       |
| PHIL 100 or<br>ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR<br>Advanced Composition ( <b>Writing Intensive</b> ) | 3       |
| MEST 100                | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                                  | 3       |
| UAES 200                | Survey of United Arab Emirates Studies                                                                               | 3       |
| MATH 113                | Calculus I                                                                                                           | 3       |
| Gen Ed Course           | Social and Behavioral Sciences Course                                                                                | 3       |
| ENVS 102                | Sustainability and Human-Environment Relations                                                                       | 3       |

### School of Engineering Requirements

**26 Credits**

| Course Code | Title                    | Credits |
|-------------|--------------------------|---------|
| PHYS 110    | University Physics I     | 3       |
| PHYS 111    | University Physics I Lab | 1       |
| MATH 114    | Calculus II              | 3       |

|          |                           |   |
|----------|---------------------------|---|
| MATH 213 | Calculus III              | 3 |
| PHYS 220 | University Physics II     | 3 |
| PHYS 221 | University Physics II Lab | 1 |
| MATH 203 | Linear Algebra            | 3 |
| ENGR 200 | Engineering Statistics    | 3 |
| CSAI 390 | Internship I              | 3 |
| CSAI 391 | Internship II             | 3 |

### Artificial Intelligence Program Requirements

**70 Credits**

#### Core Course

**64 Credit Hours**

| Course Code | Title                                             | Credits |
|-------------|---------------------------------------------------|---------|
| MATH 225    | Discrete Mathematics                              | 3       |
| CSCI 104    | Introduction to Computing                         | 3       |
| CSCI 211    | Object Oriented Programming                       | 3       |
| CSCI 215    | Data Structures and Algorithms                    | 3       |
| CSCI 232    | Computer Organization                             | 3       |
| CSAI 350    | Introduction to Artificial Intelligence           | 3       |
| CSCI 315    | Design and Analysis of Algorithms                 | 3       |
| CSCI 326    | Database Systems                                  | 3       |
| CSCI 312    | Operating System Fundamentals                     | 3       |
| CSAI 351    | Data Science                                      | 3       |
| EEEN 331    | Digital Systems Design                            | 3       |
| EEEN 332    | Digital Systems Design Lab                        | 1       |
| CENG 411    | Software Engineering                              | 3       |
| CSCI 415    | Introduction to Parallel Programming              | 3       |
| CSAI 484    | Internet of Things System                         | 3       |
| CSAI 450    | Machine Learning                                  | 3       |
| CSAI 451    | Machine Learning Lab ( <b>Writing Intensive</b> ) | 1       |
| CSCI 462    | Data Communications and Computer Networks         | 3       |
| CSAI 452    | Natural Language Processing                       | 3       |
| CSAI 453    | Data Visualization                                | 3       |
| CSAI 490    | Professional Software Practice                    | 2       |
| CSCI 492    | Senior Design Project I                           | 2       |
| CSCI 493    | Senior Design Project II                          | 4       |

#### Technical Electives

**6 Credit Hours**

| Course Code | Title                            | Credits |
|-------------|----------------------------------|---------|
| CSCI 450    | Information Security and Privacy | 3       |
| CSAI 480    | Big Data                         | 3       |
| CSAI 481    | Computer Vision                  | 3       |
| CSAI 482    | Data Mining                      | 3       |

|          |                                           |   |
|----------|-------------------------------------------|---|
| CSAI 483 | Information Retrieval                     | 3 |
| CSAI 485 | Introduction to Deep Learning             | 3 |
| CSCI 416 | Human Computer Interaction                | 3 |
| CENG 431 | Embedded Systems                          | 3 |
| CENG 432 | Embedded Systems Lab                      | 1 |
| CSAI 486 | Special Topics in Artificial Intelligence | 3 |
| CSAI 487 | Introduction to Robotics                  | 3 |
| ENGR 399 | Undergraduate Research Project            | 3 |



## Bachelor of Science in Artificial Intelligence Four-Year Study Plan

| Year 1 Semester 1     |                                                  |    | Year 1 Semester 2          |                                                                                                                   |    |
|-----------------------|--------------------------------------------------|----|----------------------------|-------------------------------------------------------------------------------------------------------------------|----|
| Course No.            | Description                                      | CR | Course No.                 | Description                                                                                                       | CR |
| CSCI 104              | Introduction to Computing                        | 3  | ARAB 101<br>Or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3  |
| ENGL 101              | Composition                                      | 3  | CSCI 112                   | Introduction to Computer Programming                                                                              | 3  |
| MATH 113              | Calculus I                                       | 3  | CSCI 113                   | Introduction to Computer Programming Lab                                                                          | 1  |
| PHYS 110              | University Physics I                             | 3  | MATH 114                   | Calculus II                                                                                                       | 3  |
| PHYS 111              | University Physics I Lab                         | 1  | PHYS 220                   | University Physics II                                                                                             | 3  |
| UNIV 100              | University First-Year Transition                 | 1  | PHYS 221                   | University Physics II Lab                                                                                         | 1  |
|                       |                                                  |    | MEST 100                   | Introduction to Islam in World Culture                                                                            | 3  |
| Subtotal =            |                                                  | 14 | Subtotal =                 |                                                                                                                   | 17 |
| Year 2 Semester 1     |                                                  |    | Year 2 Semester 2          |                                                                                                                   |    |
| Course No.            | Description                                      | CR | Course No.                 | Description                                                                                                       | CR |
| CSCI 211              | Object Oriented Programming                      | 3  | CSCI 215                   | Data Structures and Algorithms                                                                                    | 3  |
| MATH 213              | Calculus III                                     | 3  | EEEN 331                   | Digital Systems Design                                                                                            | 3  |
| MATH 225              | Discrete Mathematics                             | 3  | EEEN 332                   | Digital Systems Design Lab                                                                                        | 1  |
| CSCI 232              | Computer Organization                            | 3  | ENGR 200                   | Engineering Statistics                                                                                            | 3  |
| GEN ED                | Social and Behavioral Sciences                   | 3  | PHIL 100<br>Or<br>ENGL 200 | Critical Thinking and Reasoning or Advanced Composition                                                           | 3  |
|                       |                                                  |    | MATH 203                   | Linear Algebra                                                                                                    | 3  |
| Subtotal=             |                                                  | 15 | Subtotal=                  |                                                                                                                   | 16 |
| Year 2 Summer Session |                                                  |    |                            |                                                                                                                   |    |
| Course No.            | Description                                      | CR |                            |                                                                                                                   |    |
| CSAI 390              | Internship I                                     | 3  |                            |                                                                                                                   |    |
| Year 3 Semester 1     |                                                  |    | Year 3 Semester 2          |                                                                                                                   |    |
| Course No.            | Description                                      | CR | Course No.                 | Description                                                                                                       | CR |
| CSAI 350              | Introduction to Artificial Intelligence          | 3  | CSCI 315                   | Design and Analysis of Algorithms                                                                                 | 3  |
| ENVS 102              | Sustainability and Human-Environment Relations   | 3  | CSAI 351                   | Data Science                                                                                                      | 3  |
| CSCI 312              | Operating System Fundamentals                    | 3  | CSCI 462                   | Data Comm. and Computer Networks                                                                                  | 3  |
| CSCI 326              | Database Systems                                 | 3  | CSCI 415                   | Introduction to Parallel Programming                                                                              | 3  |
| UNIV 200              | Innovation, Entrepreneurship, and Sustainability | 3  | UAES 200                   | Survey of United Arab Emirates Studies                                                                            | 3  |
| Subtotal =            |                                                  | 15 | Subtotal =                 |                                                                                                                   | 15 |
| Year 3 Summer Session |                                                  |    |                            |                                                                                                                   |    |
| Course No.            | Description                                      | CR |                            |                                                                                                                   |    |
| CSAI 391              | Internship II                                    | 3  |                            |                                                                                                                   |    |

| Year 4 Semester 1      |                                |    | Year 4 Semester 2 |                             |    |
|------------------------|--------------------------------|----|-------------------|-----------------------------|----|
| Course No.             | Description                    | CR | Course No.        | Description                 | CR |
| CSAI 450               | Machine Learning               | 3  | CSCI 493          | Senior Design Project II    | 4  |
| CSAI 451               | Machine Learning Lab           | 1  | CSAI 452          | Natural Language Processing | 3  |
| CENG 411               | Software Engineering           | 3  | CSAI 453          | Data Visualization          | 3  |
| CSCI 492               | Senior Design Project I        | 2  | Elective          | Technical Elective I        | 3  |
| CSAI 490               | Professional Software Practice | 2  | Elective          | Technical Elective II       | 3  |
| CSAI 484               | Internet of Things System      | 3  |                   |                             |    |
| Subtotal =             |                                | 14 | Subtotal =        |                             | 16 |
| TOTAL CREDIT HOURS 128 |                                |    |                   |                             |    |

## **Bachelor of Science in Chemical Engineering**

### **Program Description**

The Bachelor of Science in Chemical Engineering (CHEN) program was established in 2014 and started accepting students in Fall 2014.

The CHEN program is a unique four-year undergraduate program. The program covers areas of sustainable engineering, global issues, and multidisciplinary problem solving and system analysis components across its curriculum to solve society and industrial most challenging problems. In addition, the program closely matches local professional opportunities and the direction of professions in chemical engineering.

The CHEN program has 133 credit hours over the duration of eight semesters including six credit hours of internship program. The curriculum of the program covers reaction engineering, process design and control, fluid and material transportation, renewable energy, mass and heat transfer and water and environmental engineering. The program was prepared taking into consideration criteria set by the Accreditation Board for Engineering and Technology (ABET) and the Commission for Academic Accreditation (CAA) at the Ministry of Higher Education and Scientific Research, United Arab of Emirates.

The Chemical Engineering faculty members have graduated from well-known universities in the world, and their expertise covers a broad range of technical areas, including reaction engineering, process design, renewable energy and management, environmental engineering, and wastewater treatment.

***The Bachelor of Science in Chemical Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The Bachelor of Science in Chemical Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Chemical Engineering. The graduates of the program will be well versed in technology and in social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of Chemical Engineering and technologies and a broad introduction into chemistry, engineering materials, thermal sciences, reaction and separation sciences, energy and environment protection, and chemical systems. The program provides a strong background for graduate study in the diverse areas branching out of the Chemical Engineering field. The technical focus is complemented with topics in general education leading to a well-rounded member of the global society.

### **Program Educational Objectives**

The Program Educational Objectives (PEOs) describe the expected accomplishments of the graduates of the Bachelor of Science in Chemical Engineering. The PEOs have been derived from and support the mission statement of AURAK. The chemical engineering program has a strong focus on the integration

of engineering, science, and process design with complementary areas of study in environmental engineering, fuels and energy, pharmaceutical, petrochemicals, materials, and renewable energy.

The graduates of the Chemical Engineering Program are expected to:

1. Pursue a successful professional career in engineering, scientific, or complementary disciplines through technical competence, effective communication, teamwork, and leadership;
2. Engaged in creating a positive impact with a sufficient awareness of the ethical, cultural, legal, and environmental issues; and
3. Maintain a lifelong interest in learning for personal and professional development.

### **Program Learning Outcomes**

AURAK has adopted ABET's student outcomes for the Chemical Engineering Program. The outcomes for the program are for the student to acquire:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

### **Degree Requirements**

The Bachelor of Science in Chemical Engineering program requires the completion of 133 credits in the following areas:

| Area                                      | Credit                                               |
|-------------------------------------------|------------------------------------------------------|
| University General Education Requirements | 33                                                   |
| School of Engineering Requirements        | 34                                                   |
| Chemical Engineering Program Requirements | 66 (60 hrs compulsory and 6 hrs technical electives) |

#### **University General Education Requirements**

#### **33 Credit Hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General

Education Component.” You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Chemical Engineering is CHEN 471 Unit Operation Lab.**

| Course Code             | Course Title                                                                                                         | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition ( <b>Writing Intensive</b> )                                                                             | 3       |
| CSCI 112                | Introduction to Computer Programming                                                                                 | 3       |
| CSCI 113                | Introduction to Computer Programming Lab                                                                             | 1       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                        | 3       |
| PHIL 100 or<br>ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR<br>Advanced Composition ( <b>Writing Intensive</b> ) | 3       |
| MEST 100                | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                                  | 3       |
| UAES 200                | Survey of United Arab Emirates Studies                                                                               | 3       |
| MATH 113                | Calculus I                                                                                                           | 3       |
| ECON 103                | Principles of Microeconomics                                                                                         | 3       |
| CHEM 211                | General Chemistry I                                                                                                  | 3       |
| CHEM 212                | General Chemistry I Lab                                                                                              | 1       |

### School of Engineering Requirements

**34 Credit Hours**

| Course Code | Course Title                      | Credits |
|-------------|-----------------------------------|---------|
| ENGR 107    | Introduction to Engineering       | 2       |
| PHYS 110    | University Physics I              | 3       |
| PHYS 111    | University Physics I Lab.         | 1       |
| MATH 114    | Calculus II                       | 3       |
| MATH 203    | Linear Algebra                    | 3       |
| MATH 214    | Elementary Differential Equations | 3       |
| PHYS 220    | University Physics II             | 3       |
| PHYS 221    | University Physics II Lab.        | 1       |
| ENGR 200    | Engineering Statistics            | 3       |
| ENGR 210    | Sustainability in Engineering     | 2       |
| ENGR 390    | Internship I                      | 3       |
| ENGR 391    | Internship II                     | 3       |
| ENGR 450    | Engineering Seminar               | 1       |
| MENG 323    | Engineering Economy               | 3       |

## Chemical Engineering Program Requirements

66 Credit Hours

Program Requirements are 66 Credit Hours as follow:

### Compulsory:

60 Credit Hours

Courses include the following:

| Course Code | Course Title                                    | Credits |
|-------------|-------------------------------------------------|---------|
| CHEM 213    | General Chemistry II                            | 3       |
| CHEM 214    | General Chemistry II Lab                        | 1       |
| BIOL 112    | University Biology I                            | 3       |
| MENG 241    | Engineering Materials                           | 3       |
| MENG 242    | Engineering Materials Lab                       | 1       |
| CHEM 215    | Organic Chemistry I                             | 3       |
| CHEM 216    | General Organic Chemistry Lab I                 | 1       |
| CHEM 315    | Physical Chemistry                              | 3       |
| CHEM 316    | Physical Chemistry Lab                          | 1       |
| CHEN 201    | Principles of Chemical Engineering              | 3       |
| CIEN 251    | Fluid Mechanics                                 | 3       |
| CHEN 302    | Computer Applications in Chemical Engineering   | 1       |
| CHEN 312    | Chemical Engineering Thermodynamic              | 3       |
| CHEN 351    | Chemical Reaction Engineering                   | 3       |
| CHEN 352    | Chemical Reaction Lab                           | 1       |
| CHEN 371    | Heat and Mass Transport                         | 3       |
| CHEN 372    | Heat and Mass Transport Lab                     | 1       |
| CHEN 404    | Safety and Environmental Impact                 | 3       |
| CHEN 471    | Unit Operation Lab ( <b>Writing Intensive</b> ) | 2       |
| CHEN 472    | Separation Processes                            | 3       |
| CHEN 481    | Process & Plant Design                          | 3       |
| CHEN 482    | Chemical Process Dynamics & Control             | 3       |
| CHEN 483    | Advance Simulation Lab                          | 1       |
| CHEN 484    | Process Integration and Optimization            | 3       |
| CHEN 491    | Senior Design Project I                         | 3       |
| CHEN 492    | Senior Design Project II                        | 3       |

**Technical Electives:****6 Credit Hours**

To be selected from the following list of courses:

| Course Code | Course Title                            | Credits |
|-------------|-----------------------------------------|---------|
| CHEN 403    | Fundamentals of Biochemical Engineering | 3       |
| CHEN 462    | Petroleum Refining Engineering          | 3       |
| CHEN 463    | Natural Gas Engineering                 | 3       |
| CHEN 464    | Industrial Catalysis                    | 3       |
| CHEN 473    | Water Desalination                      | 3       |
| CHEN 474    | Industrial and Wastewater Treatment     | 3       |
| CHEN 476    | Fundamentals of Nanotechnology          | 3       |
| CHEN 486    | Data Science in Chemical Engineering    | 3       |
| CHEN 493    | Special Topics in Chemical Engineering  | 3       |
| ENGR 399    | Undergraduate Research Project          | 3       |



## Chemical Engineering Four-Year Sample Schedule

| First Year, First Semester  |                                                               |         | First Year, Second Semester  |                                                                                                                      |         |
|-----------------------------|---------------------------------------------------------------|---------|------------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| Course Code                 | Course Title                                                  | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| PHYS 110                    | University Physics I                                          | 3       | Arab 101<br>OR<br>Arab 110   | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| PHYS 111                    | University Physics I Lab.                                     | 1       | MATH 114                     | Calculus II                                                                                                          | 3       |
| ENGL 101                    | Composition                                                   | 3       | MEST 100                     | Introduction to Islam in World Culture                                                                               | 3       |
| ENGR 107                    | Introduction to Engineering                                   | 2       | PHYS 220                     | University Physics II                                                                                                | 3       |
| MATH 113                    | Calculus I                                                    | 3       | PHYS 221                     | University Physics II Lab.                                                                                           | 1       |
| UNIV 100                    | University First-Year Transition                              | 1       | CSCI 112                     | Introduction to Computer Programming                                                                                 | 3       |
| CHEM 211                    | General Chemistry I                                           | 3       | CSCI 113                     | Introduction to Computer Programming Lab.                                                                            | 1       |
| CHEM 212                    | General Chemistry I Lab.                                      | 1       |                              |                                                                                                                      |         |
| Subtotal =                  |                                                               | 17      | Subtotal =                   |                                                                                                                      | 17      |
| Second Year, First Semester |                                                               |         | Second Year, Second Semester |                                                                                                                      |         |
| Course Code                 | Course Title                                                  | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| CHEM 213                    | General Chemistry II                                          | 3       | MENG 241                     | Engineering Materials                                                                                                | 3       |
| CHEM 214                    | General Chemistry II Lab.                                     | 1       | MENG 242                     | Engineering Materials Lab.                                                                                           | 1       |
| ENGR 210                    | Sustainability in Engineering                                 | 2       | CHEM 215                     | Organic Chemistry I                                                                                                  | 3       |
| MATH 203                    | Linear Algebra                                                | 3       | CHEM 216                     | General Organic Chemistry Lab. I                                                                                     | 1       |
| UNIV 200                    | Innovation, Entrepreneurship, and Sustainability              | 3       | MATH 214                     | Elementary Differential Equations                                                                                    | 3       |
| BIOL 112                    | University Biology I                                          | 3       | CHEN 201                     | Principles of Chemical Engineering                                                                                   | 3       |
| ECON 103                    | Principles of Microeconomics                                  | 3       | CIEN 251                     | Fluid Mechanics                                                                                                      | 3       |
| Subtotal =                  |                                                               | 18      | Subtotal =                   |                                                                                                                      | 17      |
| Second Year, Summer Session |                                                               |         |                              |                                                                                                                      |         |
| Course Code                 | Course Title                                                  | Credits |                              |                                                                                                                      |         |
| ENGR 390                    | Internship I                                                  | 3       |                              |                                                                                                                      |         |
| Third Year, First Semester  |                                                               |         | Third Year, Second Semester  |                                                                                                                      |         |
| Course Code                 | Course Title                                                  | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| PHIL 100<br>OR<br>ENGL 200  | Critical Thinking and Reasoning<br>OR<br>Advanced Composition | 3       | CHEN 302                     | Computer Applications in Chemical Engineering                                                                        | 1       |
| ENGR 200                    | Engineering Statistics                                        | 3       | CHEN 312                     | Chemical Engineering Thermodynamics                                                                                  | 3       |
| CHEM 315                    | Physical Chemistry                                            | 3       | CHEN 351                     | Chemical Reaction Engineering                                                                                        | 3       |
| CHEM 316                    | Physical Chemistry Lab.                                       | 1       | CHEN 352                     | Chemical Reaction Lab                                                                                                | 1       |
| MENG 323                    | Engineering Economy                                           | 3       | CHEN 371                     | Heat and Mass Transport                                                                                              | 3       |
| UAES 200                    | Survey of United Arab Emirates Studies                        | 3       | CHEN 372                     | Heat and Mass Transport Lab                                                                                          | 1       |
| Subtotal =                  |                                                               | 16      | Subtotal =                   |                                                                                                                      | 12      |

| Third Year, Summer Semester     |                           |         |                              |                                       |         |    |
|---------------------------------|---------------------------|---------|------------------------------|---------------------------------------|---------|----|
| Course Code                     | Course Title              | Credits | Course Code                  | Course Title                          | Credits |    |
| ENGR 391                        | Internship II             | 3       |                              |                                       |         |    |
| Fourth Year, First Semester     |                           |         | Fourth Year, Second Semester |                                       |         |    |
| Course Code                     | Course Title              | Credits | Course Code                  | Course Title                          | Credits |    |
| CHEN 471                        | Unit Operation Lab        | 1       | CHEN 404                     | Safety and Environment Impact         | 3       |    |
| CHEN 472                        | Separation Processes      | 3       | CHEN 482                     | Chemical Process Dynamics and Control | 3       |    |
| CHEN 481                        | Process & Plant Design    | 3       | CHEN 484                     | Process Integration and Optimization  | 3       |    |
| CHEN 483                        | Advance Simulation Lab    | 1       | CHEN 492                     | Senior Design Project II              | 3       |    |
|                                 | CHEN Technical Elective 1 | 3       |                              | CHEN Technical Elective II            | 3       |    |
| CHEN 491                        | Senior Design Project I   | 3       | ENGR 450                     | Engineering Seminar                   | 1       |    |
| Subtotal =                      |                           | 14      | Subtotal =                   |                                       |         | 16 |
| Total 133 Semester Hour Credits |                           |         |                              |                                       |         |    |

## **Bachelor of Science in Civil and Infrastructure Engineering**

### **Program Description**

The Civil and Infrastructure Engineering (CIEN) program is a unique undergraduate program in the United Arab Emirates and in the region as it concentrates on infrastructure engineering in addition to traditional civil engineering areas. The CIEN program prepares students to excel in several sectors including building design and construction, highways and transportation, planning and management, and consulting.

The program has 134 credit hours over the duration of eight semesters including hours at internship programs. The curriculum of the program covers construction engineering and management, structural and materials engineering, highway and transportation engineering, geotechnical engineering, and water and environmental engineering. The program was prepared according to the International Standards of Accreditation Board for Engineering and Technology (ABET) and has passed all the conditions and criteria set by the Commission for Academic Accreditation (CAA) at the Ministry of Higher Education and Scientific Research, United Arab of Emirates.

The faculty members of the Civil and Infrastructure Engineering program have graduated from top universities in the world and their expertise covers a broad range of technical areas, including structural engineering, geotechnical engineering, construction engineering and management, environmental systems, hydraulics and water resources, and transportation systems engineering.

***The Bachelor of Science in Civil and Infrastructure Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The Bachelor of Science in Civil and Infrastructure Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Civil and Infrastructure Engineering. The graduates of the program will be well versed in technology and social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of civil and infrastructure technologies and a broad introduction into structures, engineering materials, transport systems, soil engineering, and environment protection. The program provides a strong background for graduate study in the diverse areas branching out of the civil and infrastructure field. The technical focus is complemented with topics in general education leading to a well-rounded member of a global society.

### **Program Educational Objectives**

The Bachelor of Science in Civil and infrastructure Engineering program at AURAK, based on student outcomes, produces graduates, who are armed with knowledge and skills to engage in lifelong learning and application of sustainable technologies to become:

1. Leaders in their profession who serve society in the construction industry, the transportation sector, project management, and in shaping public environmental and infrastructure policies;
2. Competent civil engineers who are engaged in the design of civil engineering and infrastructure systems; and
3. Contributors to the advancement of the civil engineering discipline at the national and global levels.

### **Program Learning Outcomes**

AURAK has adopted ABET's student outcomes for the Civil and Infrastructure Engineering program. The outcomes for the program are for the student to acquire:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

### **Degree Requirements**

The Bachelor of Science in Civil and Infrastructure Engineering requires the completion of 134 credits in the following areas:

| Area                                            | Credit |
|-------------------------------------------------|--------|
| University General Education Requirements       | 32     |
| School of Engineering Requirements              | 30     |
| Civil and Infrastructure Engineering Department | 72     |

#### **University General Education Requirements**

**32 Credit Hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that

the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Civil and Infrastructure Engineering is CIEN 334 Civil Engineering Testing and Materials.**

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition <b>(Writing Intensive)</b>                                                                               | 3       |
| CSCI 112                | Introduction to Computer Programming                                                                                 | 3       |
| CSCI 113                | Introduction to Computer Programming Lab                                                                             | 1       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability <b>(Writing Intensive)</b>                                          | 3       |
| PHIL 100 or<br>ENGL 200 | Critical Thinking and Reasoning <b>(Writing Intensive)</b> OR<br>Advanced Composition <b>(Writing Intensive)</b>     | 3       |
| MEST 100                | Introduction to Islam in World Culture <b>(Writing Intensive)</b>                                                    | 3       |
| UAES 200                | Survey of United Arab Emirates Studies                                                                               | 3       |
| MATH 113                | Calculus I                                                                                                           | 3       |
| ECON 103                | Principles of Microeconomics                                                                                         | 3       |
| CHEM 211                | General Chemistry I                                                                                                  | 3       |

### School of Engineering Requirements

**30 Credit Hours**

| Course Code | Course Title                      | Credits |
|-------------|-----------------------------------|---------|
| ENGR 107    | Introduction to Engineering       | 2       |
| PHYS 110    | University Physics I              | 3       |
| PHYS 111    | University Physics I Lab.         | 1       |
| MATH 114    | Calculus II                       | 3       |
| MATH 213    | Calculus III                      | 3       |
| MATH 214    | Elementary Differential Equations | 3       |
| PHYS 220    | University Physics II             | 3       |
| PHYS 221    | University Physics II Lab.        | 1       |
| ENGR 200    | Engineering Statistics            | 3       |
| ENGR 390    | Internship I                      | 3       |
| ENGR 391    | Internship II                     | 3       |
| ENGR 450    | Engineering Seminar               | 1       |
| CHEM 212    | General Chemistry I Lab           | 1       |

**Civil and Infrastructure Engineering Department Requirements****72 Credit Hours****Core Courses****63 Credit Hours**

| Course Code | Course Title                                                                | Credits |
|-------------|-----------------------------------------------------------------------------|---------|
| CIEN 201    | Computer Aided Drawings                                                     | 3       |
| CIEN 211    | Statics                                                                     | 3       |
| CIEN 212    | Mechanics of Materials                                                      | 3       |
| CIEN 241    | Infrastructure Management                                                   | 3       |
| ENGR 210    | Sustainability in Engineering                                               | 2       |
| CIEN 251    | Fluid Mechanics                                                             | 3       |
| CIEN 261    | Surveying                                                                   | 1       |
| CIEN 265    | Geographical Information System                                             | 2       |
| CIEN 301    | Numerical Analysis                                                          | 3       |
| CIEN 311    | Structural Analysis                                                         | 3       |
| CIEN 321    | Reinforced Concrete Design                                                  | 3       |
| CIEN 331    | Construction Materials                                                      | 3       |
| CIEN 332    | Construction Materials Lab                                                  | 1       |
| CIEN 333    | Geotechnical Engineering                                                    | 3       |
| CIEN 334    | Civil Engineering Testing and Materials ( <b>Writing Intensive Course</b> ) | 1       |
| CIEN 351    | Environmental Engineering                                                   | 2       |
| CIEN 361    | Highway Engineering and Design                                              | 3       |
| CIEN 362    | Transportation Engineering                                                  | 3       |
| CIEN 421    | Structural Steel Design                                                     | 3       |
| CIEN 440    | Infrastructure Financing                                                    | 3       |
| CIEN 441    | Construction Management                                                     | 3       |
| CIEN 451    | Infrastructure Systems                                                      | 3       |
| CIEN 491    | Senior Design Project (1)                                                   | 2       |
| CIEN 492    | Senior Design Project (2)                                                   | 4       |

**Basic Science Electives****3 Credit Hours**

| Course Code | Course Title       | Credits |
|-------------|--------------------|---------|
| PENG 101    | Physical Geology   | 3       |
| BIOL 112    | University Biology | 3       |

**Technical Electives****6 Credit Hours**

| Course Code | Course Title                              | Credits |
|-------------|-------------------------------------------|---------|
| CIEN 422    | Advanced Reinforced Concrete Design       | 3       |
| CIEN 424    | Bridge Design                             | 3       |
| CIEN 431    | Foundation Engineering                    | 3       |
| CIEN 442    | Construction Planning and Scheduling      | 3       |
| CIEN 444    | Construction Cost Analysis and Estimating | 3       |
| CIEN 454    | Water and Wastewater Supply               | 3       |
| CIEN 455    | Solid Waste Management                    | 3       |
| CIEN 456    | Sustainable Urban Building Sites          | 3       |
| CIEN 462    | Advanced Pavement Design                  | 3       |
| CIEN 464    | Airport Planning and Design               | 3       |
| CIEN 493    | Special Topics in Civil Engineering       | 3       |
| ENGR 399    | Undergraduate Research Project            | 3       |

## Civil and Infrastructure Engineering Four-Year Sample Schedule

| First Year, First Semester      |                                                  |         | First Year, Second Semester  |                                                                                                                      |         |
|---------------------------------|--------------------------------------------------|---------|------------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| Course Code                     | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| ENGL 101                        | Composition                                      | 3       | ARAB 101<br>or<br>ARAB 110   | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGR 107                        | Introduction to Engineering                      | 2       | MATH 114                     | Calculus II                                                                                                          | 3       |
| MATH 113                        | Calculus I                                       | 3       | MEST 100                     | Introduction to Islam in World Culture                                                                               | 3       |
| UNIV 100                        | University First-Year Transition                 | 1       | CSCI 112                     | Introduction to Computer Programming                                                                                 | 3       |
| PHYS 110                        | University Physics I                             | 3       | CSCI 113                     | Introduction to Computer Programming Lab.                                                                            | 1       |
| PHYS 111                        | University Physics I Lab.                        | 1       | PHYS 220                     | University Physics II                                                                                                | 3       |
| CHEM 211                        | General Chemistry I                              | 3       | PHYS 221                     | University Physics II Lab.                                                                                           | 1       |
| CHEM 212                        | General Chemistry I Lab.                         | 1       |                              |                                                                                                                      |         |
| Subtotal =                      |                                                  | 17      | Subtotal =                   |                                                                                                                      | 17      |
| Second Year, First Semester     |                                                  |         | Second Year, Second Semester |                                                                                                                      |         |
| Course Code                     | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| CIEN 201                        | Computer Aided Drawings                          | 3       | CIEN 212                     | Mechanics of Materials                                                                                               | 3       |
| CIEN 211                        | Statics                                          | 3       | CIEN 251                     | Fluid Mechanics                                                                                                      | 3       |
| CIEN 241                        | Infrastructure Management                        | 3       | CIEN 261                     | Surveying                                                                                                            | 1       |
| ENGR 210                        | Sustainability in Engineering                    | 2       | CIEN 265                     | Geographical Information System                                                                                      | 2       |
| MATH 213                        | Calculus III                                     | 3       | MATH 214                     | Elementary Differential Equations                                                                                    | 3       |
| ECON 103                        | Principles of Microeconomics                     | 3       | PENG 101 or<br>BIOL 112      | Physical Geology or<br>University Biology I                                                                          | 3       |
| Subtotal =                      |                                                  | 17      | Subtotal =                   |                                                                                                                      | 15      |
| Second Year, Summer Session     |                                                  |         |                              |                                                                                                                      |         |
| Course Code                     | Course Title                                     | Credits |                              |                                                                                                                      |         |
| ENGR 390                        | Internship I                                     | 3       |                              |                                                                                                                      |         |
| Third Year, First Semester      |                                                  |         | Third Year, Second Semester  |                                                                                                                      |         |
| Course Code                     | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| CIEN 301                        | Numerical Analysis                               | 3       | CIEN 321                     | Reinforced Concrete Design                                                                                           | 3       |
| CIEN 311                        | Structural Analysis                              | 3       | CIEN 334                     | Civil Engineering Testing and Materials                                                                              | 1       |
| CIEN 331                        | Construction Materials                           | 3       | CIEN 351                     | Environmental Engineering                                                                                            | 2       |
| CIEN 332                        | Construction Materials Lab.                      | 1       | CIEN 361                     | Highway Engineering and Design                                                                                       | 3       |
| CIEN 333                        | Geotechnical Engineering                         | 3       | CIEN 362                     | Transportation Engineering                                                                                           | 3       |
| UNIV 200                        | Innovation, Entrepreneurship, and Sustainability | 3       | ENGR 200                     | Engineering Statistics                                                                                               | 3       |
| Subtotal =                      |                                                  | 16      | Subtotal =                   |                                                                                                                      | 15      |
| Third Year, Summer Semester     |                                                  |         |                              |                                                                                                                      |         |
| Course Code                     | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| ENGR 391                        | Internship II                                    | 3       |                              |                                                                                                                      |         |
| Fourth Year, First Semester     |                                                  |         | Fourth Year, Second Semester |                                                                                                                      |         |
| Course Code                     | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| CIEN 421                        | Structural Steel Design                          | 3       | CIEN ---                     | Technical Elective II                                                                                                | 3       |
| CIEN 440                        | Infrastructure Financing                         | 3       | CIEN 492                     | Senior Design Project (2)                                                                                            | 4       |
| CIEN 441                        | Construction Management                          | 3       | PHIL 100 or<br>ENGL 200      | Critical Thinking and Reasoning or<br>Advanced Composition                                                           | 3       |
| ENGR 450                        | Engineering Seminar                              | 1       | UAES 200                     | Survey of United Arab Emirates Studies                                                                               | 3       |
| CIEN 491                        | Senior Design Project (1)                        | 2       |                              |                                                                                                                      |         |
| CIEN 451                        | Infrastructure Systems                           | 3       |                              |                                                                                                                      |         |
| CIEN ---                        | Technical Elective I                             | 3       |                              |                                                                                                                      |         |
| Subtotal =                      |                                                  | 18      | Subtotal =                   |                                                                                                                      | 13      |
| Total 134 Semester Credit Hours |                                                  |         |                              |                                                                                                                      |         |



## **Bachelor of Science in Computer Engineering**

### **Program Description**

The Computer Engineering program integrates both electronics engineering and computer science in the development of computer systems. Computer Engineers are involved in many hardware and software aspects of computing including design of microcontrollers, computers, and circuit design. These engineers focus not just on how computers work but on how to integrate them into larger systems.

Students learn the fundamentals of circuits, systems, and software associated with the design of programmable systems as used for general purpose computing, communications, control, or signal processing. Because of their broad skills in both hardware and software, students in computer engineering are in high demand for employment after graduation. Computer engineering students are also positioned to pursue graduate programs in either computer science or electronics engineering.

***The Bachelor of Science in Computer Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The mission of the Computer Engineering program is to produce graduates who have the necessary theoretical background and the technical know-how and skills to create successful innovations and to solve problems in the field and to understand the social and environmental issues that may be involved. Graduates will have a thorough foundation in the basic principles of electrical engineering and information technology and a broad introduction to electronics, the design of digital computer systems, software development, data communications, and networking. Students will be prepared for employment in diverse areas of computer engineering or for further study in a graduate program.

### **Program Educational Objectives**

A few years after earning their degree, our graduates will be:

1. Successful professionals in the theoretical and practical areas of Computer Engineering;
2. Ethical and responsible engineers with sufficient awareness of the impact of engineering solutions in various contexts; and
3. Life-long learners equipped with the skills required for professional practice including functioning in teams and communicating effectively.

### **Program Learning Outcomes**

Graduates of the program will have the ability to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;

2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. Communicate effectively with a range of audiences;
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

### **Degree Requirements**

The BS in Computer Engineering (CE) requires the completion of 132 credits in the following areas:

| Area                                      | Credit |
|-------------------------------------------|--------|
| University General Education Requirements | 32     |
| School of Engineering Requirements        | 32     |
| CE Program Requirements                   | 68     |

### **University General Education Requirements**

**32 Credit Hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Computer Engineering is CSCI 463 Data Communication and Computer Networks Laboratory.**

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition ( <b>Writing Intensive</b> )                                                                             | 3       |
| CSCI 112                | Introduction to Computer Programming                                                                                 | 3       |
| CSCI 113                | Introduction to Computer Programming Lab                                                                             | 1       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                        | 3       |

|                      |                                                                                                                   |   |
|----------------------|-------------------------------------------------------------------------------------------------------------------|---|
| PHIL 100 or ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR Advanced Composition ( <b>Writing Intensive</b> ) | 3 |
| MEST 100             | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                               | 3 |
| UAES 200             | Survey of United Arab Emirates Studies                                                                            | 3 |
| MATH 113             | Calculus I                                                                                                        | 3 |
| Gen Ed               | Social and Behavioral Sciences Course                                                                             | 3 |
| ENVS 102             | Sustainability and Human-Environment Relations                                                                    | 3 |

### School of Engineering Requirements

**32 Credit Hours**

| Course Code | Course Title                      | Credits |
|-------------|-----------------------------------|---------|
| ENGR 107    | Introduction to Engineering       | 2       |
| PHYS 110    | University Physics I              | 3       |
| PHYS 111    | University Physics I Lab          | 1       |
| MATH 114    | Calculus II                       | 3       |
| MATH 213    | Calculus III                      | 3       |
| MATH 214    | Elementary Differential Equations | 3       |
| MATH 203    | Linear Algebra                    | 3       |
| PHYS 220    | University Physics II             | 3       |
| PHYS 221    | University Physics II Lab         | 1       |
| ENGR 200    | Engineering Statistics            | 3       |
| ENGR 450    | Engineering Seminar               | 1       |
| ENGR 390    | Internship I                      | 3       |
| ENGR 391    | Internship II                     | 3       |

### CE Program Requirements

**68 Credit Hours**

### Core Courses

**62 Credit Hours**

| Course Code | Course Title                            | Credits |
|-------------|-----------------------------------------|---------|
| CSCI 211    | Object Oriented Programming             | 3       |
| MATH 225    | Discrete Mathematics                    | 3       |
| CSCI 215    | Data Structures and Algorithms          | 3       |
| CSCI 312    | Operating System Fundamentals           | 3       |
| CENG 315    | Microprocessors                         | 3       |
| CENG 316    | Microprocessors Lab                     | 1       |
| CENG 335    | Computer Architecture                   | 3       |
| CENG 336    | Computer Architecture Lab               | 1       |
| CENG 411    | Software Engineering                    | 3       |
| CENG 431    | Embedded Systems Design                 | 3       |
| CENG 432    | Embedded Systems Design Lab             | 1       |
| CSAI 350    | Introduction to Artificial Intelligence | 3       |

| Course Code | Course Title                                                               | Credits |
|-------------|----------------------------------------------------------------------------|---------|
| CSAI 484    | Internet of Things System                                                  | 3       |
| EEEN 220    | Signal & Systems I                                                         | 3       |
| EEEN 221    | Signal & Systems I Lab                                                     | 1       |
| EEEN 280    | Electric Circuit Analysis I                                                | 3       |
| EEEN 281    | Electric Circuit Analysis I Lab                                            | 1       |
| EEEN 331    | Digital System Design                                                      | 3       |
| EEEN 332    | Digital System Design Lab                                                  | 1       |
| EEEN 333    | Linear Electronics I                                                       | 3       |
| EEEN 334    | Linear Electronics I Lab                                                   | 1       |
| CSCI 326    | Database Systems                                                           | 3       |
| CSCI 462    | Data Communications and Computer Networks                                  | 3       |
| CSCI 463    | Data Communications and Computer Networks Lab ( <b>Writing Intensive</b> ) | 1       |
| CENG 492    | Senior Design Project I                                                    | 2       |
| CENG 493    | Senior Design Project II                                                   | 4       |

#### Technical Electives

**6 Credit Hours**

| Course Code | Course Title                                       | Credits |
|-------------|----------------------------------------------------|---------|
| CSCI 415    | Introduction to Parallel Programming               | 3       |
| CSCI 450    | Information Security and Privacy                   | 3       |
| CENG 435    | Parallel Computer Architectures                    | 3       |
| CENG 437    | Introduction to Robotics                           | 3       |
| CENG 461    | Network Security                                   | 3       |
| CENG 401    | Network Servers & Architecture                     | 3       |
| CSCI 499    | Special Topics in Computing                        | 3       |
| EEEN 431    | Digital Circuit Design                             | 3       |
| EEEN 481    | Concepts of Multimedia Processing and Transmission | 3       |
| ENGR 399    | Undergraduate Research Project                     | 3       |

## BS in Computer Engineering Four-Year Sample Schedule

| First Year, First Semester   |                                                |         | First Year, Second Semester  |                                                                                                                      |         |
|------------------------------|------------------------------------------------|---------|------------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| Course Code                  | Course Title                                   | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| PHYS 110                     | University Physics I                           | 3       | ARAB 101<br>Or<br>ARAB 110   | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| PHYS 111                     | University Physics I Lab                       | 1       | CSCI 112                     | Introduction to Computer Programming                                                                                 | 3       |
| ENGL 101                     | Composition                                    | 3       | CSCI 113                     | Intro. to Computer Programming Lab                                                                                   | 1       |
| ENGR 107                     | Introduction to Engineering                    | 2       | MATH 114                     | Calculus II                                                                                                          | 3       |
| MATH 113                     | Calculus I                                     | 3       | PHYS 220                     | University Physics II                                                                                                | 3       |
| ENVS 102                     | Sustainability and Human-Environment Relations | 3       | PHYS 221                     | University Physics II Lab                                                                                            | 1       |
| UNIV 100                     | University First-Year Transition               | 1       | MEST 100                     | Introduction to Islam in World Culture                                                                               | 3       |
| Subtotal =                   |                                                | 16      | Subtotal =                   |                                                                                                                      | 17      |
| Second Year, First Semester  |                                                |         | Second Year, Second Semester |                                                                                                                      |         |
| Course Code                  | Course Title                                   | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| MATH 225                     | Discrete Mathematics                           | 3       | CSCI 215                     | Data Structures and Algorithms                                                                                       | 3       |
| CSCI 211                     | Object-Oriented Programming                    | 3       | EEEN 220                     | Signal & Systems I                                                                                                   | 3       |
| EEEN 280                     | Electric Circuit Analysis                      | 3       | EEEN 221                     | Signal & Systems I Lab                                                                                               | 1       |
| EEEN 281                     | Electric Circuit Analysis Lab                  | 1       | EEEN 331                     | Digital System Design                                                                                                | 3       |
| MATH 213                     | Calculus III                                   | 3       | EEEN 332                     | Digital System Design Lab                                                                                            | 1       |
| ENGR 200                     | Engineering Statistics                         | 3       | MATH 203                     | Linear Algebra                                                                                                       | 3       |
|                              |                                                |         | PHIL 100<br>Or<br>ENGL 200   | Critical Thinking and Reasoning or<br>Advanced Composition                                                           | 3       |
| Subtotal =                   |                                                | 16      | Subtotal =                   |                                                                                                                      | 17      |
| Second Year, Summer Semester |                                                |         |                              |                                                                                                                      |         |
| Course Code                  | Course Title                                   | Credits |                              |                                                                                                                      |         |
| ENGR 390                     | Internship I                                   | 3       |                              |                                                                                                                      |         |
| Third Year, First Semester   |                                                |         | Third Year, Second Semester  |                                                                                                                      |         |
| Course Code                  | Course Title                                   | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| CENG 335                     | Computer Architecture                          | 3       | CENG 315                     | Microprocessor                                                                                                       | 3       |
| CENG 336                     | Computer Architecture Lab                      | 1       | CENG 316                     | Microprocessor Lab                                                                                                   | 1       |
| CSCI 326                     | Database Systems                               | 3       | MATH 214                     | Elementary Differential Equations                                                                                    | 3       |
| EEEN 333                     | Linear Electronics I                           | 3       | CSCI 462                     | Data Comm. and Computer Networks                                                                                     | 3       |
| EEEN 334                     | Linear Electronics Lab I                       | 1       | CSCI 463                     | Data Comm. and Computer Networks Lab                                                                                 | 1       |
| CSCI 312                     | Operating System Fundamentals                  | 3       | UAES 200                     | Survey of United Arab Emirates Studies                                                                               | 3       |
|                              |                                                |         | UNIV 200                     | Innovation, Entrepreneurship, and Sustainability                                                                     | 3       |
| Subtotal =                   |                                                | 14      | Subtotal =                   |                                                                                                                      | 17      |
| Third Year, Summer Semester  |                                                |         |                              |                                                                                                                      |         |
| Course Code                  | Course Title                                   | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| ENGR 391                     | Internship II                                  | 3       |                              |                                                                                                                      |         |

| Fourth Year, First Semester |                                         |         | Fourth Year, Second Semester |                             |         |
|-----------------------------|-----------------------------------------|---------|------------------------------|-----------------------------|---------|
| Course Code                 | Course Title                            | Credits | Course Code                  | Course Title                | Credits |
| ENGR 450                    | Engineering Seminar                     | 1       | CENG 493                     | Senior Design Project II    | 4       |
| CSAI 484                    | Internet of Things System               | 3       | CENG 431                     | Embedded Systems Design     | 3       |
| Gen Ed                      | Social and Behavioral Sciences          | 3       | CENG 432                     | Embedded Systems Design Lab | 1       |
| CENG 411                    | Software Engineering                    | 3       |                              | Technical Elective I        | 3       |
| CENG 492                    | Senior Design Project I                 | 2       |                              | Technical Elective II       | 3       |
| CSAI 350                    | Introduction to Artificial Intelligence | 3       |                              |                             |         |
| Subtotal =                  |                                         | 15      | Subtotal =                   |                             | 14      |
| Total 132 Credits           |                                         |         |                              |                             |         |

## **Bachelor of Science in Computer Science**

### **Program Description**

The Bachelor of Science in Computer Science (CS) degree program provides a solid foundation in computing and technology related sciences, along with exploration of the various disciplines of liberal arts and humanities. By examining the latest industry tools and techniques and developing valuable experience and professional skills in areas such as object-oriented programming, database systems, and network security, students are equipped with the in-depth understanding of their chosen professions.

The CS program aims to develop and prepare individuals for rewarding, cutting- edge careers in software engineering, system administration, and management. Furthermore, the CS graduates will be well trained to handle the problems faced in industry, academia, and daily life.

***The Bachelor of Science in Computer Science is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The Bachelor of Science in Computer Science aims to provide students with high-quality education covering a broad and well-integrated knowledge in the concepts and methodologies underlying the analysis, design, and utilization of computer software, algorithms, and systems. The graduates of the program will be well versed in computer technology and its impact on the global society.

### **Program Educational Objectives**

A few years after graduation, our alumni will be:

1. Innovators in applied areas of computer science, effectively accomplishing software-related projects to meet market needs, and actively engaging in research.
2. Professionals making informed judgments and creating a positive technological impact with awareness of the ethical and legal issues of computer science.
3. Leaders in the computing field capable of working in teams, communicating effectively, and engaging in life-long learning and community service.

### **Program Learning Outcomes**

Graduates of the program will have the ability to:

1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions;
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline;
3. Communicate effectively in a variety of professional contexts;
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles;

5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline; and
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

### **Degree Requirements**

The BS in Computer Science (CS) requires the completion of 127 credits in the following areas:

| Area                                      | Credit |
|-------------------------------------------|--------|
| University General Education Requirements | 32     |
| School of Engineering Requirements        | 26     |
| CS Program Requirements                   | 66     |
| Free Electives                            | 3      |

### **University General Education Requirements**

### **32 Credit Hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Computer Engineering is CSCI 463 Data Communication and Computer Networks Laboratory.**

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition ( <b>Writing Intensive</b> )                                                                             | 3       |
| CSCI 112                | Introduction to Computer Programming                                                                                 | 3       |
| CSCI 113                | Introduction to Computer Programming Lab                                                                             | 1       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                        | 3       |
| PHIL 100 or<br>ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR<br>Advanced Composition ( <b>Writing Intensive</b> ) | 3       |
| MEST 100                | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                                  | 3       |
| UAES 200                | Survey of United Arab Emirates Studies                                                                               | 3       |
| MATH 113                | Calculus I                                                                                                           | 3       |
| Gen Ed                  | Social and Behavioral Sciences Course                                                                                | 3       |
| ENVS 102                | Sustainability and Human-Environment Relations                                                                       | 3       |



**School of Engineering Requirements****26 Credit Hours**

| Course Code | Course Title              | Credits |
|-------------|---------------------------|---------|
| PHYS 110    | University Physics I      | 3       |
| PHYS 111    | University Physics I Lab  | 1       |
| MATH 114    | Calculus II               | 3       |
| MATH 213    | Calculus III              | 3       |
| PHYS 220    | University Physics II     | 3       |
| PHYS 221    | University Physics II Lab | 1       |
| MATH 203    | Linear Algebra            | 3       |
| ENGR 200    | Engineering Statistics    | 3       |
| CSCI 390    | Internship I              | 3       |
| CSCI 391    | Internship II             | 3       |

**CS Program Requirements****66 Credit Hours****Core Courses****60 Credit Hours**

| Course Code | Course Title                                                             | Credits |
|-------------|--------------------------------------------------------------------------|---------|
| CSCI 104    | Introduction to Computing                                                | 3       |
| CSCI 211    | Object Oriented Programming                                              | 3       |
| MATH 225    | Discrete Mathematics                                                     | 3       |
| CSCI 215    | Data Structures and Algorithms                                           | 3       |
| CSCI 232    | Computer Organization                                                    | 3       |
| CSAI 350    | Introduction to Artificial Intelligence                                  | 3       |
| CSAI 351    | Data Science                                                             | 3       |
| CSCI 315    | Design and Analysis of Algorithms                                        | 3       |
| CSCI 326    | Database Systems                                                         | 3       |
| CSCI 312    | Operating System Fundamentals                                            | 3       |
| CSCI 372    | Compiler Design                                                          | 3       |
| CSCI 388    | Programming Languages                                                    | 3       |
| CENG 335    | Computer Architecture                                                    | 3       |
| CENG 336    | Computer Architecture Lab                                                | 1       |
| EEEN 331    | Digital Systems Design                                                   | 3       |
| EEEN 332    | Digital Systems Design Lab                                               | 1       |
| CENG 411    | Software Engineering                                                     | 3       |
| CSCI 462    | Data Communications and Computer                                         | 3       |
| CSCI 463    | Data Communications and Computer Networks <b>Lab (Writing Intensive)</b> | 1       |

| Course Code | Course Title             | Credits |
|-------------|--------------------------|---------|
| CENG 461    | Network Security         | 3       |
| CSCI 492    | Senior Design Project I  | 2       |
| CSCI 493    | Senior Design Project II | 4       |

#### **Technical Electives**

**6 Credit Hours**

| Course Code | Course Title                                       | Credits |
|-------------|----------------------------------------------------|---------|
| CENG 437    | Introduction to Robotics                           | 3       |
| CSCI 411    | Computer Graphics                                  | 3       |
| CSCI 412    | Computer Graphics Lab                              | 1       |
| CSCI 415    | Introduction to Parallel                           | 3       |
| CSCI 416    | Human Computer Interaction                         | 3       |
| CENG 435    | Parallel Computer Architectures                    | 3       |
| EEEN 481    | Concepts of Multimedia Processing and Transmission | 3       |
| CENG 401    | Network Servers & Architecture                     | 3       |
| CSCI 450    | Information Security and Privacy                   | 3       |
| CSCI 499    | Special Topics in Computing                        | 3       |
| ENGR 399    | Undergraduate Research Project                     | 3       |

#### **Free Electives**

**3 Credit Hours**

Students must complete one course (3 credits) of free electives.

## BS in Computer Science Four-Year Sample Schedule

| First Year, First Semester  |                                                  |         | First Year, Second Semester  |                                                                                                                   |         |
|-----------------------------|--------------------------------------------------|---------|------------------------------|-------------------------------------------------------------------------------------------------------------------|---------|
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                      | Credits |
| CSCI 104                    | Introduction to Computing                        | 3       | ARAB 101<br>Or<br>ARAB 110   | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                    | Composition                                      | 3       | CSCI 112                     | Introduction to Computer Programming                                                                              | 3       |
| MATH 113                    | Calculus I                                       | 3       | CSCI 113                     | Intro. to Computer Programming Lab                                                                                | 1       |
| PHYS 110                    | University Physics I                             | 3       | MATH 114                     | Calculus II                                                                                                       | 3       |
| PHYS 111                    | University Physics I Lab                         | 1       | PHYS 220                     | University Physics II                                                                                             | 3       |
| UNIV 100                    | University First-Year Transition                 | 1       | PHYS 221                     | University Physics II Lab                                                                                         | 1       |
|                             |                                                  |         | MEST 100                     | Introduction to Islam in World Culture                                                                            | 3       |
| Subtotal =                  |                                                  | 14      | Subtotal =                   |                                                                                                                   | 17      |
| Second Year, First Semester |                                                  |         | Second Year, Second Semester |                                                                                                                   |         |
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                      | Credits |
| CSCI 211                    | Object Oriented Programming                      | 3       | CSCI 215                     | Data Structures and Algorithms                                                                                    | 3       |
| MATH 213                    | Calculus III                                     | 3       | EEEN 331                     | Digital Systems Design                                                                                            | 3       |
| MATH 225                    | Discrete Mathematics                             | 3       | EEEN 332                     | Digital Systems Design Lab                                                                                        | 1       |
| CSCI 232                    | Computer Organization                            | 3       | MATH 203                     | Linear Algebra                                                                                                    | 3       |
| UNIV 200                    | Innovation, Entrepreneurship, and Sustainability | 3       | ENGR 200                     | Engineering Statistics                                                                                            | 3       |
|                             |                                                  |         | PHIL 100<br>Or<br>ENGL 200   | Critical Thinking and Reasoning or Advanced Composition                                                           | 3       |
| Subtotal =                  |                                                  | 15      | Subtotal =                   |                                                                                                                   | 16      |
| Second Year, Summer Session |                                                  |         |                              |                                                                                                                   |         |
| Course Code                 | Course Title                                     | Credits |                              |                                                                                                                   |         |
| CSCI 390                    | Internship I                                     | 3       |                              |                                                                                                                   |         |
| Subtotal =                  |                                                  | 3       |                              |                                                                                                                   |         |
| Third Year, First Semester  |                                                  |         | Third Year, Second Semester  |                                                                                                                   |         |
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                      | Credits |
| CENG 335                    | Computer Architecture                            | 3       | Gen Ed                       | Social and Behavioral Sciences                                                                                    | 3       |
| CENG 336                    | Computer Architecture Lab                        | 1       | CSCI 372                     | Compiler Design                                                                                                   | 3       |
| ENVS 102                    | Sustainability and Human-Environment Relations   | 3       | CSCI 462                     | Data Comm. and Computer Networks                                                                                  | 3       |
| CSCI 326                    | Database Systems                                 | 3       | CSCI 463                     | Data Comm. and Computer Networks Lab                                                                              | 1       |
| UAES 200                    | Survey of United Arab Emirates Studies           | 3       | CSCI 388                     | Programming Languages                                                                                             | 3       |
| CSCI 312                    | Operating System Fundamentals                    | 3       | CSCI 315                     | Design and Analysis of Algorithms                                                                                 | 3       |
| Subtotal =                  |                                                  | 16      | Subtotal =                   |                                                                                                                   | 16      |
| Third Year, Summer Session  |                                                  |         |                              |                                                                                                                   |         |
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                      | Credits |
| CSCI 391                    | Internship II                                    | 3       |                              |                                                                                                                   |         |
| Subtotal =                  |                                                  | 3       |                              |                                                                                                                   |         |

| Fourth Year, First Semester |                                         |         | Fourth Year, Second Semester |                          |         |    |
|-----------------------------|-----------------------------------------|---------|------------------------------|--------------------------|---------|----|
| Course Code                 | Course Title                            | Credits | Course Code                  | Course Title             | Credits |    |
| CSAI 350                    | Introduction to Artificial Intelligence | 3       | CSCI 493                     | Senior Design Project II | 4       |    |
| CENG 411                    | Software Engineering                    | 3       | CENG 461                     | Network Security         | 3       |    |
| CSCI 492                    | Senior Design Project I                 | 2       | CSAI 351                     | Data Science             | 3       |    |
|                             | Free Elective                           | 3       |                              | Technical Elective II    | 3       |    |
|                             | Technical Elective I                    | 3       |                              |                          |         |    |
| Subtotal =                  |                                         | 14      | Subtotal =                   |                          |         | 13 |
| Total 127 Credits           |                                         |         |                              |                          |         |    |

## **Bachelor of Science in Electrical and Electronics Engineering**

### **Program Description**

The Bachelor of Science in Electrical and Electronics Engineering provides graduates with a sound understanding of fundamental engineering principles and applications in the field.

Graduates are prepared for post-graduate study or employment and are able to demonstrate their ability to apply the knowledge and methodologies of electrical and electronics engineering essential for a successful career. The American University of Ras Al Khaimah's (AURAK) Bachelor of Electrical and Electronics Engineering aims to produce effective, well educated, and globally competitive electrical and electronics engineers for the region and the wider world. The program aims to be amongst the top in the GCC region in the fields of electrical and electronics engineering with highly qualified and dedicated faculty and staff. It also aims to have a strong link with the local and regional government, public-sector organizations, and private sector enterprises. Moreover, the program aims to provide service to the community by sharing the program's technical expertise and human resources.

***The Bachelor of Science in Electrical and Electronics Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The mission of the Electrical and Electronics Engineering program is to provide graduates with a sound understanding of fundamental engineering principles and applications in the field. Graduates will have a comprehensive understanding of subjects like signals and systems, digital logic, circuits and electronics, and the theory and applications of digital communications, particularly networks and wireless communications, in addition to power systems, control systems, and electric machines. Graduates will be prepared for both graduate study and employment. They will be able to demonstrate their ability to apply the knowledge and methodologies in power systems, electronics, and communications engineering essential for a successful career.

### **Program Educational Objectives**

The graduates of Electrical and Electronics Engineering Program, within a few years will be:

1. Become successful practitioners and innovators in the region and worldwide;
2. Continue their professional development throughout their careers; and
3. Work productively and ethically in teams, and function effectively in diverse cultural environments.

### **Program Learning Outcomes**

AURAK Electrical and Electronics Engineering graduates should have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;

2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. An ability to communicate effectively with a range of audiences;
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

### Degree Requirements

The BS in Electrical and Electronics Engineering (EEE) requires the completion of 135 credits in the following areas:

| Area                                      | Credit                                                |
|-------------------------------------------|-------------------------------------------------------|
| University General Education Requirements | 32                                                    |
| School of Engineering Requirements        | 34                                                    |
| EECE Department Requirements              | 69<br>(63 CH compulsory and 6 CH technical electives) |

### University General Education Requirements

### 32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Electrical and Electronics Engineering is EEEN 434 Linear Electronics II Laboratory.**

| Course Code          | Title                                                                                                             | Credits |
|----------------------|-------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101             | Composition ( <b>Writing Intensive</b> )                                                                          | 3       |
| CSCI 112             | Introduction to Computer Programming                                                                              | 3       |
| CSCI 113             | Introduction to Computer Programming Lab                                                                          | 1       |
| UNIV 100             | University First-Year Transition                                                                                  | 1       |
| UNIV 200             | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                     | 3       |
| PHIL 100 or ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR Advanced Composition ( <b>Writing Intensive</b> ) | 3       |
| MEST 100             | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                               | 3       |

|          |                                                                    |   |
|----------|--------------------------------------------------------------------|---|
| Gen Ed   | Social and Behavioral Science Course –<br>ECON 103 is recommended. | 3 |
| UAES 200 | Survey of United Arab Emirates Studies                             | 3 |
| MATH 113 | Calculus I                                                         | 3 |
| CHEM 211 | General Chemistry I                                                | 3 |

### School of Engineering Requirements

**34 Credit Hours**

| Course Code | Course Title                      | Credits |
|-------------|-----------------------------------|---------|
| ENGR 107    | Introduction to Engineering       | 2       |
| PHYS 110    | University Physics I              | 3       |
| PHYS 111    | University Physics I Lab          | 1       |
| MATH 114    | Calculus II                       | 3       |
| MATH 203    | Linear Algebra                    | 3       |
| MATH 213    | Calculus III                      | 3       |
| MATH 214    | Elementary Differential Equations | 3       |
| PHYS 220    | University Physics II             | 3       |
| PHYS 221    | University Physics II Lab         | 1       |
| ENGR 200    | Engineering Statistics            | 3       |
| ENGR 210    | Sustainability in Engineering     | 2       |
| ENGR 390    | Internship I                      | 3       |
| ENGR 391    | Internship II                     | 3       |
| ENGR 450    | Engineering Seminar               | 1       |

### EECE Department Requirements

**69 Credit Hours**

#### Core Courses

**63 Credit Hours**

| Course Code | Course Title                     | Credits |
|-------------|----------------------------------|---------|
| EEEN 220    | Signal & Systems I               | 3       |
| EEEN 221    | Signal & Systems I Lab           | 1       |
| EEEN 280    | Electric Circuit Analysis I      | 3       |
| EEEN 281    | Electric Circuit Analysis I Lab  | 1       |
| EEEN 282    | Electric Circuit Analysis II     | 3       |
| EEEN 283    | Electric Circuit Analysis II Lab | 1       |
| EEEN 305    | Electromagnetic Theory           | 3       |
| EEEN 331    | Digital System Design            | 3       |
| EEEN 332    | Digital System Design Lab        | 1       |
| EEEN 333    | Linear Electronics I             | 3       |
| EEEN 334    | Linear Electronics I Lab         | 1       |
| EEEN 350    | Electric Machines                | 3       |
| EEEN 351    | Electric Machines Lab            | 1       |
| EEEN 360    | Random Signal & Noise            | 3       |
| EEEN 412    | Power Systems                    | 3       |
| EEEN 431    | Digital Circuit Design           | 3       |
| EEEN 433    | Linear Electronics II            | 3       |
| EEEN 434    | Linear Electronics II Lab        | 1       |
| EEEN 437    | Power Electronics                | 3       |
| EEEN 451    | Control Theory                   | 3       |

| Course Code | Course Title                       | Credits |
|-------------|------------------------------------|---------|
| EEEN 452    | Control Lab                        | 1       |
| EEEN 460    | Communication Systems              | 3       |
| EEEN 461    | Communication Engineering Lab      | 1       |
| EEEN 464    | Digital Communication Systems      | 3       |
| EEEN 467    | Mobile and Wireless Communications | 3       |
| EEEN 492    | Senior Design Project I            | 2       |
| EEEN 493    | Senior Design Project II           | 4       |

#### Technical Electives

6 Credit Hours

| Course Code | Course Title                                       | Credits |
|-------------|----------------------------------------------------|---------|
| CENG 315    | Microprocessor                                     | 3       |
| EEEN 432    | Nanotechnology Fundamentals & Applications         | 3       |
| EEEN 435    | Introduction to Optical Electronics                | 3       |
| EEEN 462    | Data and Computer Communication                    | 3       |
| EEEN 466    | Digital Signal Processing                          | 3       |
| EEEN 472    | Antenna Theory and Design                          | 3       |
| EEEN 473    | Radio Frequency and Microwave Engineering          | 3       |
| EEEN 474    | Advanced Information Theory and Coding             | 3       |
| EEEN 481    | Concepts of Multimedia Processing & Transmission   | 3       |
| EEEN 499    | Special Topics in EEEN                             | 3       |
| EEEN 421    | Power System Protection                            | 3       |
| EEEN 422    | High Voltage Engineering                           | 3       |
| EEEN 423    | Electrical Energy Systems and Fault Analysis       | 3       |
| EEEN 425    | Smart Power Grid Systems Theory and Implementation | 3       |
| EEEN 426    | Renewable Energy Systems                           | 3       |
| ENGR 399    | Undergraduate Research Project                     | 3       |



## Electrical and Electronics Engineering Program EEE 4-Year Study Plan

| First Year, First Semester  |                                                         |         | First Year, Second Semester  |                                                          |         |
|-----------------------------|---------------------------------------------------------|---------|------------------------------|----------------------------------------------------------|---------|
| Course Code                 | Course Title                                            | Credits | Course Code                  | Course Title                                             | Credits |
| PHYS 110                    | Physics I                                               | 3       | ARAB 101<br>or<br>ARAB 110   | Arabic Language and Culture for Non-Native Learners I OR | 3       |
| PHYS 111                    | Physics I Lab                                           | 1       |                              | Arabic Language and Culture for Native Arabic Speakers I |         |
| ENGL 101                    | Composition                                             | 3       | CSCI 112                     | Intro. to Computer Programming                           | 3       |
| ENGR 107                    | Introduction to Engineering                             | 2       | CSCI 113                     | Intro. to Computer Programming Lab                       | 1       |
| MATH 113                    | Calculus I                                              | 3       | MATH 114                     | Calculus II                                              | 3       |
| UNIV 100                    | University First-Year Transition                        | 1       | PHYS 220                     | Physics II                                               | 3       |
| CHEM 211                    | General Chemistry I                                     | 3       | PHYS 221                     | Physics II Lab                                           | 1       |
|                             |                                                         |         | MEST 100                     | Introduction to Islam in World Culture                   | 3       |
| Subtotal =                  |                                                         | 16      | Subtotal =                   |                                                          | 17      |
| Second Year, First Semester |                                                         |         | Second Year, Second Semester |                                                          |         |
| Course Code                 | Course Title                                            | Credits | Course Code                  | Course Title                                             | Credits |
| EEEN 280                    | Electric Circuit Analysis I                             | 3       | EEEN 220                     | Signals & Systems I                                      | 3       |
| EEEN 281                    | Electric Circuit Analysis I Lab                         | 1       | EEEN 221                     | Signals & Systems I Lab                                  | 1       |
| UAES 200                    | Survey of United Arab Emirates Studies                  | 3       | EEEN 282                     | Electric Circuit Analysis II                             | 3       |
| MATH 203                    | Linear Algebra                                          | 3       | EEEN 283                     | Electric Circuit Analysis II Lab                         | 1       |
| MATH 213                    | Calculus III                                            | 3       | ENGR 200                     | Engineering Statistics                                   | 3       |
| EEEN 331                    | Digital System Design                                   | 3       | MATH 214                     | Elementary Differential Equations                        | 3       |
| EEEN 332                    | Digital System Design Lab                               | 1       | ENGR 210                     | Sustainability in Engineering                            | 2       |
| Subtotal =                  |                                                         | 17      | Subtotal =                   |                                                          | 16      |
| Second Year, Summer Session |                                                         |         |                              |                                                          |         |
| Course                      | Course Title                                            | Credits |                              |                                                          |         |
| ENGR 390                    | Internship I                                            | 3       |                              |                                                          |         |
| Third Year, First Semester  |                                                         |         | Third Year, Second Semester  |                                                          |         |
| Course Code                 | Course Title                                            | Credits | Course Code                  | Course Title                                             | Credit  |
| EEEN 305                    | Electromagnetic Theory                                  | 3       | EEEN 412                     | Power Systems                                            | 3       |
| UNIV 200                    | Innovation, Entrepreneurship, and Sustainability        | 3       | EEEN 433                     | Linear Electronics II                                    | 3       |
| EEEN 350                    | Electric Machines                                       | 3       | EEEN 460                     | Communication Systems                                    | 3       |
| EEEN 351                    | Electric Machines Lab                                   | 1       | EEEN 431                     | Digital Circuit Design                                   | 3       |
| EEEN 333                    | Linear Electronics I                                    | 3       | EEEN 451                     | Control Theory                                           | 3       |
| EEEN 334                    | Linear Electronics I Lab                                | 1       | EEEN 452                     | Control Lab                                              | 1       |
| EEEN 360                    | Random Signal and Noise                                 | 3       |                              |                                                          |         |
| Subtotal =                  |                                                         | 17      | Subtotal =                   |                                                          | 16      |
| Third Year, Summer Semester |                                                         |         |                              |                                                          |         |
| Course Code                 | Course Title                                            | Credits | Course Code                  | Course Title                                             | Credits |
| ENGR 391                    | Internship II                                           | 3       |                              |                                                          |         |
| Fourth Year, First Semester |                                                         |         | Fourth Year, Second Semester |                                                          |         |
| Course Code                 | Course Title                                            | Credits | Course Code                  | Course Title                                             | Credits |
| PHIL 100 or ENGL 200        | Critical Thinking and Reasoning OR Advanced Composition | 3       | EEEN 464                     | Digital Communication Systems                            | 3       |
| EEEN 434                    | Linear Electronics II Lab                               | 1       | ENGR 450                     | Engineering Seminar                                      | 1       |
| EEEN 437                    | Power Electronics                                       | 3       | EEEN 493                     | Senior Design Project II                                 | 4       |
| EEEN 492                    | Senior Design Project I                                 | 2       | EEEN 461                     | Communication Engineering Lab                            | 1       |
| EEEN 467                    | Mobile and Wireless Communications                      | 3       |                              | Technical Elective II                                    | 3       |
|                             | Technical Elective I                                    | 3       | ECON 103                     | Principles of Microeconomics                             | 3       |
| Subtotal =                  |                                                         | 15      | Subtotal =                   |                                                          | 15      |
| Total 135 Credits           |                                                         |         |                              |                                                          |         |

## **Bachelor of Science in Mechanical Engineering**

### **Program Description**

The mechanical engineering program is offered in the Mechanical and Industrial Engineering Department. The Bachelor of Science degree in mechanical engineering requires a minimum of 134 credit hours of coursework. Mechanical engineering is one of the largest, broadest, and oldest engineering disciplines. Mechanical engineers use the principles of energy, materials, and mechanics to design and manufacture machines and devices of all types. They create the processes and systems that drive technology and industry.

The key characteristics of the profession are its breadth, flexibility, and individuality. The career paths of mechanical engineers are largely determined by individual choices, a decided advantage in a changing world.

Mechanics, energy and heat, mathematics, engineering sciences, design and manufacturing form the foundation of mechanical engineering. Mechanics includes fluids, ranging from still water to hypersonic gases flowing around a space vehicle; it involves the motion of anything from a particle to a machine or complex structure.

***The Bachelor of Science in Mechanical Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The Bachelor of Science in the mechanical engineering program at the American University of Ras Al Khaimah (AURAK) educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of mechanical engineering. The graduates of the program will be well versed in technology and in social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of mechanical engineering and technologies, and a broad introduction into machine design, engineering materials, thermal sciences, energy and environment protection and mechanical systems. The program provides a strong background for graduate study in the diverse areas branching out of the mechanical engineering field. The technical focus is complemented with topics in general education leading to a well-rounded member of the global society.

### **Program Educational Objectives**

A few years after earning their degree, our graduates will:

1. Take leadership positions in energy and industrial sectors in response to regional and global market needs.
2. Pursue higher education, research and development, and/or other creative and innovative efforts in science, engineering and technology.
3. Create start-up companies that provide creative products and services for industry and society.

## Program Learning Outcomes

Graduates of the program will have the ability to:

- 1 Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
- 2 Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- 3 Communicate effectively with a range of audiences
- 4 Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- 5 Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
- 6 Develop and conduct appropriate experimentation, analyze and interpretation of data, and use engineering judgment to draw conclusions
- 7 Acquire and apply new knowledge as needed, using appropriate learning strategies

## Degree Requirements

The BS in Mechanical Engineering (MENG) requires the completion of 134 credits in the following areas:

| Area                                           | Credit |
|------------------------------------------------|--------|
| University General Education Requirements      | 32     |
| School of Engineering Requirements             | 32     |
| Mechanical Engineering Department Requirements | 64     |
| Technical Electives                            | 6      |

### University General Education Requirements

**32 credit hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Mechanical Engineering is MENG 313 Manufacturing Processes Laboratory.**

| Course Code             | Title                                                                                                                | Credits |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or<br>ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101                | Composition ( <b>Writing Intensive</b> )                                                                             | 3       |
| CSCI 112                | Introduction to Computer Programming                                                                                 | 3       |
| CSCI 113                | Introduction to Computer Programming Lab                                                                             | 1       |
| UNIV 100                | University First-Year Transition                                                                                     | 1       |
| UNIV 200                | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                        | 3       |

|                      |                                                                                                                   |   |
|----------------------|-------------------------------------------------------------------------------------------------------------------|---|
| PHIL 100 or ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR Advanced Composition ( <b>Writing Intensive</b> ) | 3 |
| MEST 100             | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                               | 3 |
| ECON 103             | Principles of Microeconomics                                                                                      | 3 |
| UAES 200             | Survey of United Arab Emirates Studies                                                                            | 3 |
| MATH 113             | Calculus I                                                                                                        | 3 |
| CHEM 211             | General Chemistry I                                                                                               | 3 |

### School of Engineering Requirements

**32 Credit Hours**

| Course Code | Course Title                      | Credits |
|-------------|-----------------------------------|---------|
| ENGR 107    | Introduction to Engineering       | 2       |
| PHYS 110    | University Physics I              | 3       |
| PHYS 111    | University Physics I Lab          | 1       |
| MATH 114    | Calculus II                       | 3       |
| MATH 213    | Calculus III                      | 3       |
| MATH 214    | Elementary Differential Equations | 3       |
| MATH 203    | Linear Algebra                    | 3       |
| PHYS 220    | University Physics II             | 3       |
| PHYS 221    | University Physics II Lab         | 1       |
| ENGR 210    | Sustainability in Engineering     | 2       |
| ENGR 450    | Engineering Seminar               | 1       |
| ENGR 390    | Internship I                      | 3       |
| ENGR 391    | Internship II                     | 3       |
| CHEM 212    | General Chemistry I Lab           | 1       |

### MENG Program Requirements

**70 Credit Hours**

#### Core Courses

**64 Credit Hours**

| Course Code | Course Title                      | Credits |
|-------------|-----------------------------------|---------|
| CIEN 211    | Statics                           | 3       |
| CIEN 212    | Mechanics of Materials            | 3       |
| CIEN 251    | Fluid Mechanics                   | 3       |
| CIEN 301    | Numerical Analysis                | 3       |
| EEEN 280    | Electric Circuits Analysis I      | 3       |
| EEEN 281    | Electric Circuits Analysis I Lab. | 1       |
| MENG 241    | Engineering Materials             | 3       |
| MENG 242    | Engineering Materials Lab.        | 1       |
| ENGR 200    | Engineering Statistics            | 3       |
| MENG 311    | Manufacturing Processes           | 3       |
| MENG 313    | Manufacturing Processes Lab.      | 1       |
| MENG 201    | Mechanical Engineering Drawing    | 3       |
| MENG 211    | Thermodynamics I                  | 3       |
| MENG 212    | Thermodynamics II                 | 3       |
| MENG 221    | Dynamics                          | 3       |
| MENG 231    | Engineering Measurements          | 2       |
| MENG 321    | Vibration and Control             | 3       |
| MENG 342    | Fluid Mechanics Lab.              | 1       |
| MENG 361    | Heat Transfer                     | 3       |

| Course Code | Course Title             | Credits |
|-------------|--------------------------|---------|
| MENG 362    | Thermal Sciences Lab.    | 1       |
| MENG 421    | Theory of Machines       | 3       |
| MENG 451    | Mechanical Design I      | 3       |
| MENG 452    | Mechanical Design II     | 3       |
| MENG 491    | Senior Design Project I  | 2       |
| MENG 492    | Senior Design Project II | 4       |

#### **Technical Electives**

**6 Credit Hours**

| Course Code | Course Title                             | Credits |
|-------------|------------------------------------------|---------|
| MENG 311    | Internal Combustion Engines              | 3       |
| MENG 441    | Turbo Machinery                          | 3       |
| MENG 453    | Computer Aided Design                    | 3       |
| MENG 455    | Finite Elements in Machine Design        | 3       |
| MENG 461    | HVAC and Refrigeration Systems           | 3       |
| MENG 462    | Design of Thermal System                 | 3       |
| MENG 463    | Energy Conversion and Management         | 3       |
| MENG 493    | Special Topics in Mechanical Engineering | 3       |
| MENG 321    | Engineering Economy                      | 3       |
| ENGR 399    | Undergraduate Research Project           | 3       |
| EEEN 426    | Renewable Energy Systems                 | 3       |
| EEEN 451    | Control Theory                           | 3       |

## BS in Mechanical Engineering Four-Year Sample Schedule

| First Year, First Semester  |                                        |         | First Year, Second Semester  |                                                                                                                   |         |
|-----------------------------|----------------------------------------|---------|------------------------------|-------------------------------------------------------------------------------------------------------------------|---------|
| Course Code                 | Course Title                           | Credits | Course Code                  | Course Title                                                                                                      | Credits |
| PHYS 110                    | University Physics I                   | 3       | ARAB 101<br>Or<br>ARAB 110   | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3       |
| PHYS 111                    | University Physics I Lab               | 1       | CSCI 112                     | Introduction to Computer Programming                                                                              | 3       |
| ENGL 101                    | Composition                            | 3       | CSCI 113                     | Intro. to Computer Programming Lab                                                                                | 1       |
| ENGR 107                    | Introduction to Engineering            | 2       | MATH 114                     | Calculus II                                                                                                       | 3       |
| MATH 113                    | Calculus I                             | 3       | PHYS 220                     | University Physics II                                                                                             | 3       |
| CHEM 211                    | General Chemistry I                    | 3       | PHYS 221                     | University Physics II Lab                                                                                         | 1       |
| CHEM 212                    | General Chemistry I Lab                | 1       | MEST 100                     | Introduction to Islam in World Culture                                                                            | 3       |
| UNIV 100                    | University First-Year Transition       | 1       |                              |                                                                                                                   |         |
| Subtotal =                  |                                        | 17      | Subtotal =                   |                                                                                                                   | 17      |
| Second Year, First Semester |                                        |         | Second Year, Second Semester |                                                                                                                   |         |
| Course Code                 | Course Title                           | Credits | Course Code                  | Course Title                                                                                                      | Credits |
| MENG 201                    | Mechanical Engineering Drawing         | 3       | CIEN 212                     | Mechanics of Materials                                                                                            | 3       |
| CIEN 211                    | Statics                                | 3       | MENG 221                     | Dynamics                                                                                                          | 3       |
| ENGR 210                    | Sustainability in Engineering          | 2       | CIEN 251                     | Fluid Mechanics                                                                                                   | 3       |
| MATH 213                    | Calculus III                           | 3       | MATH 214                     | Elementary Differential Equations                                                                                 | 3       |
| MENG 211                    | Thermodynamics I                       | 3       | MENG 212                     | Thermodynamics II                                                                                                 | 3       |
| MENG 231                    | Engineering Measurements               | 2       | PHIL 100<br>Or<br>ENGL 200   | Critical Thinking and Reasoning<br>Advanced Composition                                                           | 3       |
|                             |                                        |         |                              |                                                                                                                   |         |
| Subtotal =                  |                                        | 16      | Subtotal =                   |                                                                                                                   | 18      |
| Second Year, Summer Session |                                        |         |                              |                                                                                                                   |         |
| Course Code                 | Course Title                           | Credit  |                              |                                                                                                                   |         |
| ENGR 390                    | Internship I                           | 3       |                              |                                                                                                                   |         |
| Third Year, First Semester  |                                        |         | Third Year, Second Semester  |                                                                                                                   |         |
| Course Code                 | Course Title                           | Credits | Course Code                  | Course Title                                                                                                      | Credits |
| MENG 241                    | Engineering Materials                  | 3       | MENG 361                     | Heat Transfer                                                                                                     | 3       |
| MENG 242                    | Engineering Materials Lab.             | 1       | MENG 362                     | Thermal Sciences Lab                                                                                              | 1       |
| MATH 203                    | Linear Algebra                         | 3       | MENG 321                     | Vibration and Control                                                                                             | 3       |
| ECON 103                    | Principles of Microeconomics           | 3       | ENGR 200                     | Engineering Statistics                                                                                            | 3       |
| MENG 342                    | Fluid Mechanics Lab                    | 1       | MENG 311                     | Manufacturing Processes                                                                                           | 3       |
| UAES 200                    | Survey of United Arab Emirates Studies | 3       | MENG 313                     | Manufacturing Processes Lab.                                                                                      | 1       |
|                             |                                        |         | EEEN 280                     | Electric Circuit Analysis                                                                                         | 3       |
|                             |                                        |         | EEEN 281                     | Electric Circuit Analysis Lab                                                                                     | 1       |
| Subtotal =                  |                                        | 14      | Subtotal =                   |                                                                                                                   | 18      |
| Third Year, Summer Session  |                                        |         |                              |                                                                                                                   |         |
| Course Code                 | Course Title                           | Credits |                              |                                                                                                                   |         |
| ENGR 391                    | Internship II                          | 3       |                              |                                                                                                                   |         |

| Fouthr Year, First Semester |                         |         | Fourth Year, Second Semester |                                               |         |
|-----------------------------|-------------------------|---------|------------------------------|-----------------------------------------------|---------|
| Course Code                 | Course Title            | Credits | Course Code                  | Course Title                                  | Credits |
| MENG 451                    | Mechanical Design I     | 3       | MENG 452                     | Mechanical Design II                          | 3       |
| MENG 421                    | Theory of Machines      | 3       |                              | ME Technical Elective 2                       | 3       |
|                             | ME Technical Elective I | 3       | ENGR 450                     | Engineering Seminar                           | 1       |
| CIEN 301                    | Numerical Analysis      | 3       | UNIV 200                     | Innovation, Entrepreneurship & Sustainability | 3       |
| MENG 491                    | Senior Design Project I | 2       | MENG 492                     | Senior Design Project II                      | 4       |
| Subtotal =                  |                         | 14      | Subtotal =                   |                                               | 14      |
| Total 134 Credits           |                         |         |                              |                                               |         |

## **Bachelor of Science in Petroleum Engineering**

### **Program Description**

The Bachelor of Science in Petroleum Engineering program (PENG) was established in 2014 and started accepting students in Fall 2014. Over thirty students are currently enrolled in this program.

The PENG program is a unique four-year undergraduate program. The program covers areas of global issues, multidisciplinary problem solving and system analysis components across its curriculum to solve society and industrial most challenging problems. In addition, the program closely matches local professional opportunities and the direction of professions in petroleum engineering.

The Petroleum Engineering program has 133 credit hours over the duration of eight semesters including six credit hours of internship program. The curriculum of the program covers reservoir production and drilling engineering. The program was prepared by taking into consideration criteria set by the Accreditation Board for Engineering and Technology (ABET) and the Commission for Academic Accreditation (CAA) at the Ministry of Higher Education and Scientific Research, United Arab of Emirates.

The Petroleum Engineering program has very well-equipped laboratories. Some of these laboratories enable the students to visualize the various petroleum processes and how they are interrelated. Likewise, the program has advanced computation facilities, and the computation laboratories are equipped with a number of simulation software that is used by the students to enhance their understanding of the various petroleum engineering processes.

The Petroleum Engineering faculty members have graduated from top universities in the world and their expertise covers a broad range of technical areas, including reservoir engineering, drilling engineering, geothermal reservoir engineering, and reservoir economic evaluation.

***The Bachelor of Science in Petroleum Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.***

### **Program Mission**

The Bachelor of Science in Petroleum Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Petroleum Engineering. The graduates of the program will be well versed in technology and social and environmental issues.

The Bachelor of Science in Petroleum Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Petroleum Engineering. The graduates of the program will be well versed in technology and social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of Petroleum Engineering and technologies and a broad introduction into geophysics, petroleum geology, fluid properties, formation evaluation, drilling, reservoir engineering and simulation, artificial lift systems, completions, oil and gas facilities engineering. The program provides a strong background for graduate study in the diverse areas branching out of the Petroleum



Engineering field. The technical focus is complemented with topics in general education leading to a well-rounded member of a global society.

### **Program Educational Objectives**

The Bachelor of Science in Petroleum Engineering program is developed to prepare students with a strong understanding of basic sciences, engineering, and a systematic study of the physical behavior of oil, water, and gas in porous media, flow lines and surface facilities in order to exploit petroleum resources with maximum economic benefits by natural and enhanced recovery methods.

The Program Educational Objectives (PEOs) of the Bachelor of Science in Petroleum Engineering program describe the career and professional activities that the program prepares graduates to:

1. Attain a broad knowledge in petroleum engineering applications in order to achieve critical thinking skills required to design and analyze petroleum engineering problems for maximum economic recovery of oil and gas resources, taking into account safety, environmental and social impacts;
2. Pursue life-long learning through training and possible post graduate education, engage in research, and contribute to scientific progressions through participation and leadership in professional organizations; and
3. Commit to professional ethics, environmental safety, and public welfare.

### **Program Learning Outcomes**

AURAK has adopted ABET's student outcomes for the Petroleum Engineering Program. The outcomes for the program are:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. An ability to communicate effectively with a range of audiences;
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

### **Degree Requirements**

The Bachelor of Science in Petroleum Engineering program requires the completion of 133 credits in the following areas:

| Area                                       | Credit                                                 |
|--------------------------------------------|--------------------------------------------------------|
| University General Education Requirements  | 33                                                     |
| School of Engineering Requirements         | 31                                                     |
| Petroleum Engineering Program Requirements | 69 (63 hrs. compulsory and 6 hrs. technical electives) |

### University General Education Requirements

**33 Credit Hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Petroleum Engineering program is PENG 322 Drilling Fluids Laboratory.**

| Course Code          | Title                                                                                                             | Credits |
|----------------------|-------------------------------------------------------------------------------------------------------------------|---------|
| ARAB 101 or ARAB 110 | Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I | 3       |
| ENGL 101             | Composition ( <b>Writing Intensive</b> )                                                                          | 3       |
| CSCI 112             | Introduction to Computer Programming                                                                              | 3       |
| CSCI 113             | Introduction to Computer Programming Lab                                                                          | 1       |
| UNIV 100             | University First-Year Transition                                                                                  | 1       |
| UNIV 200             | Innovation, Entrepreneurship, and Sustainability ( <b>Writing Intensive</b> )                                     | 3       |
| PHIL 100 or ENGL 200 | Critical Thinking and Reasoning ( <b>Writing Intensive</b> ) OR Advanced Composition ( <b>Writing Intensive</b> ) | 3       |
| MEST 100             | Introduction to Islam in World Culture ( <b>Writing Intensive</b> )                                               | 3       |
| ECON 103             | Principles of Microeconomics                                                                                      | 3       |
| UAES 200             | Survey of United Arab Emirates Studies                                                                            | 3       |
| MATH 113             | Calculus I                                                                                                        | 3       |
| CHEM 211             | General Chemistry I                                                                                               | 3       |
| CHEM 212             | General Chemistry I Lab                                                                                           | 1       |

### School of Engineering Requirements

**31 Credit Hours**

| Course Code | Course Title                      | Credits |
|-------------|-----------------------------------|---------|
| ENGR 107    | Introduction to Engineering       | 2       |
| PHYS 110    | University Physics I              | 3       |
| PHYS 111    | University Physics I Lab.         | 1       |
| MATH 114    | Calculus II                       | 3       |
| MATH 203    | Linear Algebra                    | 3       |
| MATH 214    | Elementary Differential Equations | 3       |
| PHYS 220    | University Physics II             | 3       |
| PHYS 221    | University Physics II Lab.        | 1       |
| ENGR 200    | Engineering Statistics            | 3       |
| ENGR 390    | Internship I                      | 3       |

|          |                     |   |
|----------|---------------------|---|
| ENGR 391 | Internship II       | 3 |
| MENG 323 | Engineering Economy | 3 |

### Petroleum Engineering Program Requirements

**69 Credit Hours**

Program Requirements are 69 credits hours as follows:

#### Compulsory:

**63 Credit Hours**

| Course Code | Course Title                                                    | Credits |
|-------------|-----------------------------------------------------------------|---------|
| PENG 101    | Physical Geology                                                | 3       |
| PENG 202    | Introduction to Petroleum Engineering                           | 1       |
| PENG 231    | Mechanics of Materials                                          | 3       |
| CHEM 215    | Organic Chemistry I                                             | 3       |
| MENG 211    | Thermodynamics I                                                | 3       |
| CIEN 251    | Fluid Mechanics                                                 | 3       |
| PENG 351    | Reservoir Rock and Fluid Properties                             | 3       |
| PENG 321    | Reservoir Rock and Fluid Properties Laboratory                  | 1       |
| PENG 322    | Drilling Fluids Laboratory ( <b>Writing Intensive</b> )         | 1       |
| PENG 361    | Drilling Engineering I                                          | 2       |
| PENG 371    | Petroleum Reservoir Engineering                                 | 3       |
| PENG 381    | Well Logging                                                    | 3       |
| PENG 382    | Petroleum Production Engineering (Well Performance)             | 4       |
| PENG 422    | Fluid Flow in Porous Media Lab                                  | 1       |
| PENG 403    | Petroleum Property Evaluation                                   | 3       |
| PENG 462    | Drilling Engineering II                                         | 2       |
| PENG 472    | Applied Reservoir Geology                                       | 3       |
| PENG 473    | Reservoir Simulation                                            | 3       |
| PENG 482    | Natural Gas Engineering                                         | 3       |
| PENG 483    | Well Testing                                                    | 3       |
| PENG 484    | Stimulation and Intervention in Petroleum Production Operations | 3       |
| PENG 485    | Water Flooding                                                  | 3       |
| PENG 491    | Senior Design Project I                                         | 3       |
| PENG 492    | Senior Design Project II                                        | 3       |

#### Technical Electives:

**6 Credit Hours**

| Course Code | Course Title                                 | Credits |
|-------------|----------------------------------------------|---------|
| PENG 486    | Enhance Oil Recovery                         | 3       |
| PENG 442    | Transportation and Storage of Petroleum      | 3       |
| PENG 443    | Separation and Treatment of Petroleum Fluids | 3       |
| PENG 474    | Applied Reservoir Simulation                 | 3       |
| CHEN 404    | Safety and Environmental Impact              | 3       |
| PENG 493    | Special Topics in Petroleum Engineering      | 3       |
| ENGR 399    | Undergraduate Research Project               | 3       |

## Petroleum Engineering Four-Year Sample Schedule

| First Year, First Semester  |                                                  |         | First Year, Second Semester  |                                                                                                                      |         |
|-----------------------------|--------------------------------------------------|---------|------------------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| PHYS 110                    | University Physics I                             | 3       | ARAB 101<br>OR<br>ARAB 110   | Arabic Language and Culture for Non-Native Learners I OR<br>Arabic Language and Culture for Native Arabic Speakers I | 3       |
| PHYS 111                    | University Physics I Lab                         | 1       | MATH 114                     | Calculus II                                                                                                          | 3       |
| ENGL 101                    | Composition                                      | 3       | MEST 100                     | Introduction to Islam in World Culture                                                                               | 3       |
| ENGR 107                    | Introduction to Engineering                      | 2       | PHYS 220                     | University Physics II                                                                                                | 3       |
| MATH 113                    | Calculus I                                       | 3       | PHYS 221                     | University Physics II Lab                                                                                            | 1       |
| UNIV 100                    | University First-Year Transition                 | 1       | CSCI 112                     | Introduction to Computer Programming                                                                                 | 3       |
| CHEM 211                    | General Chemistry I                              | 3       | CSCI 113                     | Introduction to Computer Programming Lab                                                                             | 1       |
| CHEM 212                    | General Chemistry I Lab                          | 1       |                              |                                                                                                                      |         |
| Subtotal =                  |                                                  | 17      | Subtotal =                   |                                                                                                                      | 17      |
| Second Year, First Semester |                                                  |         | Second Year, Second Semester |                                                                                                                      |         |
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| ECON 103                    | Principles of Microeconomics                     | 3       | PHIL 100<br>OR<br>G ENGL200  | Critical Thinking and Reasoning OR<br>Advanced Composition                                                           | 3       |
| MENG 211                    | Thermodynamics I                                 | 3       | PENG 231                     | Mechanics of Materials                                                                                               | 3       |
| MATH 203                    | Linear Algebra                                   | 3       | CHEM 215                     | Organic Chemistry I                                                                                                  | 1       |
| UNIV 200                    | Innovation, Entrepreneurship, and Sustainability | 3       | PENG 202                     | Introduction to Petroleum Engineering                                                                                | 3       |
| PENG 101                    | Physical Geology                                 | 3       | MATH 214                     | Elementary Differential Equations                                                                                    | 3       |
|                             |                                                  |         | PENG 351                     | Reservoir Rock and Fluid Properties                                                                                  | 3       |
|                             |                                                  |         | PENG 321                     | Reservoir Rock and Fluid Prop. Lab.                                                                                  | 1       |
| Subtotal =                  |                                                  | 15      | Subtotal =                   |                                                                                                                      | 17      |
| Second Year, Summer Session |                                                  |         |                              |                                                                                                                      |         |
| Course Code                 | Course Title                                     | Credits |                              |                                                                                                                      |         |
| ENGR 390                    | Internship I                                     | 3       |                              |                                                                                                                      |         |
| Third Year, First Semester  |                                                  |         | Third Year, Second Semester  |                                                                                                                      |         |
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| ENGR 200                    | Engineering Statistics                           | 3       | PENG 382                     | Petroleum Production Engineering (Well Performance)                                                                  | 4       |
| MENG 323                    | Engineering Economy                              | 3       | PENG 371                     | Petroleum Reservoir Engineering                                                                                      | 3       |
| CIEN 251                    | Fluid Mechanics                                  | 3       | PENG 381                     | Well Logging                                                                                                         | 3       |
| PENG 361                    | Drilling Engineering I                           | 2       | PENG 462                     | Drilling Engineering II                                                                                              | 2       |
| PENG 322                    | Drilling Fluids Laboratory                       | 1       | PENG 472                     | Applied Reservoir Geology                                                                                            | 3       |
| UAES 200                    | Survey of United Arab Emirates Studies           | 3       |                              |                                                                                                                      |         |
| Subtotal =                  |                                                  | 15      | Subtotal =                   |                                                                                                                      | 15      |
| Third Year, Summer Session  |                                                  |         |                              |                                                                                                                      |         |
| Course Code                 | Course Title                                     | Credits | Course Code                  | Course Title                                                                                                         | Credits |
| ENGR 391                    | Internship II                                    | 3       |                              |                                                                                                                      |         |

| Fourth Year, First Semester     |                                 |         | Fourth Year, Second Semester |                                                                 |         |
|---------------------------------|---------------------------------|---------|------------------------------|-----------------------------------------------------------------|---------|
| Course Code                     | Course Title                    | Credits | Course Code                  | Course Title                                                    | Credits |
| PENG 482                        | Natural Gas Engineering         | 3       | PENG 483                     | Well Testing                                                    | 3       |
| PENG 403                        | Petroleum Property Evaluation   | 3       | PENG 484                     | Stimulation and Intervention in Petroleum Production Operations | 3       |
| PENG 485                        | Water Flooding                  | 3       | PENG 473                     | Reservoir Simulation                                            | 3       |
|                                 | Technical Elective I            | 3       |                              | Technical Elective II                                           | 3       |
| PENG 491                        | Senior Design Project I         | 3       | PENG 492                     | Senior Design Project II                                        | 3       |
| PENG 422                        | Fluid Flow in Porous Media Lab. | 1       |                              |                                                                 |         |
| Subtotal =                      |                                 | 16      | Subtotal =                   |                                                                 | 15      |
| Total 133 Semester Hour Credits |                                 |         |                              |                                                                 |         |







## GRADUATE PROGRAMS



## ADMISSION TO GRADUATE STUDIES

AURAK offers five Master Programs in response to the needs of Ras Al Khaimah, the region, and the international community. AURAK's programs meet the highest international standards, and our students will be well prepared for the professional challenges they will face in their careers. The programs engage relevant contemporary issues as well as crucial intellectual contexts. Students can expect to engage with one another and their professors in stimulating discussions and challenging assignments. They will study with experienced faculty from many of the best university in North America, Europe, Asia, and the Middle East.

### Degree Offerings

AURAK currently offers five (5) programs of graduate study leading to a master's degree. These are:

- Master of Education: Educational Leadership
- Executive Master of Business Administration
- Master of Business Administration
- Master of Science in Engineering Project Management
- Master of Science in Sustainable and Renewable Energy

### Application Process

#### *Matriculating Students*

Applications for all AURAK graduate programs are processed through the Department of Admissions under the guidance of the appropriate school and program of study.

Prospective students can apply online or in person. The online application is available on <https://apply.aurak.ac.ae>.

The following documents would be required to submit:

- Attested transcripts for bachelor's degree(s) and certificate(s);
- English proficiency scores (EmSAT/TOEFL/IELTS or equivalent);
- Two (2) recommendation letters;
- Updated resume;
- Valid passport copy, Emirates ID, and residence visa (If Applicable);
- Health insurance card;
- Health history form; and
- Two (2) passport photos.

**Incomplete applications are not processed.**



Upon receiving a complete application, the Department of Admissions coordinates with the appropriate school and program to determine if the applicant meets the requirements for admission to the program. The relevant Graduate Program Coordinator makes all final admissions decisions. The Department of Admissions will notify the applicant of the university's final decision and provide all relevant information.

### **Application Deadlines**

AURAK Admissions Application Deadline for Fall 2023 Semester

**11<sup>th</sup> of September 2023**

AURAK Admissions Final Application Deadline for Spring 2023 Semester

**21<sup>st</sup> of January 2024**

Note: International students must apply two weeks in advance of the stated deadline to meet visa timeline requirements.

International applicants (i.e., graduates of universities located outside the UAE) are required to submit completed application forms and all supporting documents to the Department of Admissions.

Graduates of universities outside the UAE are required to present an equivalency of their degree from the UAE Ministry of Education located in Abu Dhabi, UAE. The procedure for obtaining an equivalency can be obtained from the Ministry by calling +971 2 695 1300 or +971 2 642 8000.

Admitted international students who need visas for the UAE should submit the visa application form, which is available from the Department of Admissions, at least three weeks prior to the first day of class.

Admission is valid only for the semester immediately following the admission decision. If an admitted student would like to have the admission date delayed, he or she must submit a formal request in writing to the Department of Admissions. After consultation with the Graduate Program Coordinator and the Dean of the school in which the admitted student's degree program is located, the student will be informed of the university's decision concerning delay of admission.

### **University Requirements for Admission to Graduate Programs**

To be considered for admission, all applicants must meet the general university requirements for graduate admission. There are also program-specific requirements, information about which applicants will find in the section of the catalog in which the specific program is described.

**For admission to a graduate degree program at AURAK, an applicant must:**

- Hold a four-year bachelor's degree from an independently accredited university recognized by the UAE Ministry of Education (applicants with a bachelor's degree obtained outside the UAE must submit an equivalency of their degree from the UAE Ministry of Education); and
- Have attained the minimum score of one of the specified exams as indicated in the Graduate English Language Proficiency requirement table below. Note that the EmSAT is

the preferred assessment of English proficiency.

| Type of English Proficiency Exam | Minimum Score |
|----------------------------------|---------------|
| EmSAT                            | 1400          |
| IELTS                            | 6.0           |
| TOEFL                            | 550           |
| TOEFL (Internet-based)           | 79            |

Table: Graduate English Proficiency Requirement

### **Program-Specific Requirements**

- Master of Education in Educational Leadership:
  - A minimum of two (2) years of experience of teaching or equivalent work experience is required.
  - The minimum entry requirement for the Master of Education is a CGPA of 3.0 on a 4.0 scale.
- Executive Master of Business Administration:
  - A baccalaureate degree in Business Administration, with a minimum cumulative GPA of 3.0 on a 4.0 scale, or its established equivalent in the applicant's undergraduate degree program;
  - A baccalaureate degree, or its equivalent, with a major or minor in Business Studies, Marketing, Finance, Accounting, Economics, or Management/Business Information Systems with a minimum CGPA of 3.0 on a 4.0 scale or its recognized equivalent in the applicant's undergraduate degree program, or
  - A baccalaureate degree, or its equivalent, in a non-business field with a minimum of CGPA of 3.0 on a 4.0 scale, followed by a postgraduate certificate or diploma in a business-related field.
  - The students must also submit an experience certificate, which proves at least five years of work experience in a managerial position to be eligible.
  - Successful Personal Interview with School of Business Faculty.
- Master of Business Administration:
  - A baccalaureate degree in Business Administration, with a minimum cumulative GPA of 3.0 on a 4.0 scale, or its established equivalent in the applicant's undergraduate degree program;
  - A baccalaureate degree, or its equivalent, with a major or minor in Business Studies, Marketing, Finance, Accounting, Economics, or Management/Business Information Systems with a minimum CGPA of 3.00 on a 4.0 scale or its recognized equivalent in the applicant's undergraduate degree program, or

- A baccalaureate degree, or its equivalent, in a non-business field with a minimum of CGPA of 3.00 on a 4.0 scale, followed by a postgraduate certificate or diploma in a business-related field.
- This MBA program is geared toward applicants with or without work experience. Leveling courses may be required by the School of Business.
- Successful Personal Interview with School of Business Faculty .
- Master of Science in Engineering Project Management:
  - A baccalaureate degree in Engineering, Computer Science, or Information Technology, with a minimum CGPA of 3.0 on a 4.0 scale.
  - Qualified applicants requiring prerequisite courses may be required to take such courses in addition to their regular graduate program courses.
  - Students with no prior background in probability and statistics will be required to take ENGR 200 Engineering Statistics.
  - Students with no prior background in engineering economy will be required to take MENG 323 (Engineering Economy).
- Master of Science in Sustainable and Renewable Energy
  - A baccalaureate degree in engineering or a science discipline.
  - Completion of specific courses in the areas of thermodynamics, fluid mechanics and heat transfer.

### **Conditional Admission**

**English Language:** AURAK may conditionally admit students with a recognized Bachelor's degree and an EmSAT score of 1250 or its equivalent on another standardized test approved by the CAA. Such a student must meet the following requirements during the period of conditional admission or be subject to dismissal:

- must achieve an EmSAT score of 1400 or equivalent, by the end of the student's first semester of study.
- may take a maximum of six credit hours in the first semester of study, not including intensive English courses.
- must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in the first six credit hours of credit-bearing courses studied for the graduate program.

**CGPA:** AURAK may conditionally admit students with a recognized Bachelor degree and a minimum cumulative grade point average (CGPA) of 2.5 on a 4.0 scale or its established equivalent. Such a student must take a maximum of nine credit hours of courses studied for the graduate program during the period of conditional admission and must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of courses studied for the graduate program or be subject to dismissal.

For those prospective students who have a CGPA of 2.0 – 2.49, the students will have to complete remedial courses designed to prepare them for graduate-level coursework. The remedial courses

consist of three compulsory courses detailed in the table below.

|                    |                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compulsory Courses | MBFN 500 Business Graduate Foundation (3 CH)<br>EDUC 500 Foundations of Education (3 CH)<br>ENGR 500 Engineering Foundation for Graduate Students (3 CH) |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|

The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within two semesters will students be allowed to register for the graduate program for which they meet all other requirements. An unsuccessful student who has not completed the courses within two semesters will not be able to continue at the institution.

There may also be program-specific prerequisite courses for conditional admission students. Credits from these courses do not satisfy credit requirements for the degree and are not used to calculate the graduate CGPA.

#### **Non-Degree Seeking Graduate Student Admission**

A non-degree seeking graduate student is one who would like to take AURAK courses for academic credit but who does not seek a master's degree. Students are admitted to AURAK with non-degree status if they meet the normal requirements for graduate admission or receive permission from the Dean of the appropriate school of study. Complete applications should be submitted to the Department of Admissions.

A non-degree seeking graduate student may take a maximum of six credit hours at the graduate level.

#### **Visiting Students Admission**

A visiting student is one who is not formally admitted to AURAK, and who is not participating in a formal exchange agreement between AURAK and a partner university but is typically enrolled in a degree program at another university. Such a student may take courses at AURAK for transfer to the student's home institution. It is wise for visiting students to check with their home institutions. Checking for the transferability of credit hours from a course from AURAK to another institution is the responsibility of the student.

To be admitted as a visiting graduate student, a student must be enrolled in a graduate program at an accredited institution and be in good academic standing in his/her current institution. Students must also have attained the same minimum score on the English Proficiency Exam as students gaining admission to the Graduate Degree Programs at AURAK. Applicants seeking visiting graduate student status must submit to AURAK's Department of Admissions a complete application form and evidence of current enrollment in another university's graduate program, typically an official transcript. The application will be considered for admission. Visiting students

will be admitted as space allows. Registration in courses is subject to approval by the respective graduate program chair/coordinator.

Applicants seeking visiting student status must submit to AURAK's Department of Admissions the following documents:

- Transcripts for bachelor's degree and certificate;
- Evidence of enrollment in another university;
- An official letter from the university allowing the student to enroll for the course(s) at AURAK;
- English proficiency scores (TOEFL/IELTS or equivalent) – expired will also be accepted;
- Two (2) recommendation letters;
- Updated resume;
- Valid passport copy, Emirates ID, and residence visa (If Applicable); and
- Two (2) passport photos.

A complete application form is required with evidence of current enrollment in another university's graduate program, typically an official transcript. The application will be considered and visiting students will be admitted as space allows. Registration in courses is subject to approval by the relevant graduate program director.

### **Change of Status**

A student may apply at any time for a change of status from non-degree to degree status, or from visiting to degree status. To change status, a student should submit a complete application through the Department of Admissions. All admissions and program requirements in place at the time of the change of status request must be met to be admitted and to graduate. Grades earned in courses taken as a visiting student in AURAK will be counted towards the CGPA (CGPA).

Courses already taken at previous institution may be accepted with the approval of the **relevant school and program**. University rules and regulations governing transfer courses and credits will apply.

### **Transfer Credit Policy for Graduate Studies**

Applicants may be allowed to transfer up to 25% of the total program credit hours subject to review and recommendation of the program faculty. Partial transfer credit will not be awarded. Transfer credit may not be awarded for graduation projects or theses courses.

Grades earned in transferred courses do not count in the student's CGPA, though transferred credits count toward the cumulative earned hours and may apply towards meeting graduation requirements. Applicants must request that credit transfers be reviewed at the time of application.

## **GRADUATE ACADEMIC POLICIES**

Each student is responsible for knowing AURAK's rules, regulations, requirements, and academic policies. Both the graduate information in the catalog and the institutional website are repositories of policy statements. Corrections, changes, or interpretations may be communicated by other means including electronically. Any student in doubt about an academic matter should consult the Associate Provost for Academic Affairs, the Associate Provost for Research and Community Service, Department of the Registrar, Dean of the appropriate school, or their assigned faculty advisor. Students are subject to the university's stated policies regarding patents and copyrights.

### **Academic Advising**

Academic Advisement is a critical dimension of the AURAK student experience, one that involves both students and faculty. For students it helps them to make acceptable progress toward their degree, and for faculty it is an important aspect of their teaching responsibilities.

Each student is responsible for his or her progress towards his or her degree completion and for meeting all requirements for graduation. The advisement process is available to assist the student in completing his or her studies, but responsibility for doing so lies ultimately with each student.

Academic advising is the responsibility of full-time faculty members. A full-time faculty member will be assigned to serve as a student's academic advisor from each department where a degree or minor program is housed. Typically, advisees are distributed equally among the full-time faculty in the relevant academic departments. The advisor's responsibilities include:

- Meeting with each advisee in the two-week period before early registration each semester to guide the advisee in course selection;
- Following up on the student's course requests during early and regular registration to facilitate the student's successful registration;
- Meeting with each advisee early in the semester to ensure that the student has registered for the appropriate courses;
- Meeting with each advisee roughly at mid-semester to gauge the student's progress and advise accordingly; and
- Serving as a mentor to advisees to facilitate their academic success and to provide guidance when appropriate on their future studies and careers.

Academic advisors are expected:

- To be familiar with all academic requirements for graduation; and
- To collaborate with university colleagues to provide the strongest academic advisement possible.

### **Attendance Policy**

Please see the Student Attendance Policy section of this catalog.

## **Courses and Course Schedules**

Each course offered by the university has a designated course prefix (or code) and number. The course prefix represents the discipline or field of study; the number indicates the level of the course content.

Courses are offered at the discretion of the VPAASS/Provost, the Associate Provost of Academic Affairs, the Registrar, the respective deans, and department chairs. Not every course is offered every semester. The Registrar has primary responsibility for approving the course schedule.

## **Course Credit**

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the sixteen-week semester. Normally one credit represents fifty minutes of class instruction per week per semester.

## **Course Prerequisites and Corequisites**

Course prerequisites or corequisites state requirements for student entry into courses and reflect necessary preparation for attempting a course. It is the student's responsibility to be aware of these requirements as stated in the catalog, and to have taken prerequisites recently enough to be of value. The instructor of the course may drop students who have enrolled in a course for which they have not met the prerequisites.

## **Course Selection and Registration**

By mid-semester, the class schedule for the following semester is available through the Department of Registration and on the website. Students may select courses in consultation with their faculty/academic advisor, and they can then register online or submit a completed "Course Request Form" to the Department of the Registrar by the published deadlines.

## **Registration Criteria**

Only registered students may participate in AURAK classes. Full-time status requires that a graduate student enroll in 9 semester credit hours in Fall and Spring semesters, EXCEPT for Master of Education students in their first semester of study. In their first semester of study, the full-time course load for Master of Education students is 6 semester credit hours. Apart from the first-semester exception for Education students, graduate students enrolling in fewer than 9 semester credit hours in Fall and Spring semesters are considered to be enrolled on a part-time basis. However, please note the following:

- A student with a CGPA of less than 3.0 is limited to registering for six credit hours the following semester subject to the approval of the academic advisor.
- No more than six (6) credit hours are allowed during summer terms.
- A student with a CGPA of 3.50 or higher can enroll in 12 semester credit in the subsequent semester unless the student's program restricts registration to no more than nine credit hours in a semester.

## **Independent Study**

Independent study courses are not allowed for graduate programs.

## **Summer Courses Outside AURAK**

### **Requirements**

An enrolled student is eligible to apply to take courses at another college/university during the summer with the aim of transferring credits to AURAK provided the following conditions are met:

- The student must be in good academic standing at AURAK.
- The summer courses at the host university must not be taken as attempts to repeat AURAK courses in which F grades were previously earned.
- The host university must be recognized by the UAE Ministry of Education.
- The host university must provide learning experiences similar to those offered by AURAK.
- Contact/credit hours for the course at the host university must be equivalent to, or greater than, the contact/credit hours required for its equivalent course at AURAK.
- The language of instruction of the course(s) taken at the host university must be English (except for language courses conducted in other languages) unless otherwise approved by AURAK.
- The student must obtain approval from AURAK prior to registering the summer course(s).
- Some programs may reserve the right not to allow any courses to be taken at another college/university.

### **Final Semester Grades**

Final grades are made available on-line through the Student Portal. Official transcripts and official semester grade reports are obtained through the Department of the Registrar.

Students may access and print an unofficial record of their semester grades and other academic information from the Student Portal on the university website by logging onto their account. Official transcripts are available on request.

### **Change of Grade**

Once a final grade has been recorded, it can be changed only in cases of computational error, recording error, or pursuant of a successful appeal of grade. Additional work of any type submitted to improve a grade after the final grade has been assigned will not be accepted. All changes of final grades must be initiated, approved, and recorded by the second week of classes of the next regular semester (spring for fall grades and fall for spring grades).

### **Grade Appeal**

Although faculty members are generally the best judges of student performance, sometimes a student believes a grade is unfair. This procedure is therefore designed to specify the process through which students can appeal final course grades and the process through which faculty members can initiate a change in a final course grade following the official posting of students' grades in a course.

Where a student believes their final course grades to be in error the student must request a meeting with the member of faculty to ask that they reconsider the grade and ensure that the appeal is



submitted within the required timeline. Completed Grade Appeal Forms must be submitted within the following timeframe:

- For Fall Semester courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Spring Semester.
- For Spring Semester courses, completed Grade Appeal Forms must be submitted within three weeks of the “Grades Available” date listed in the Academic Calendar.

For Summer Session courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Fall Semester.

***The grade appeal will only be considered where:***

- There was an error or unfairness in the assessment of the exam grade.
- There was an error or unfairness in the evaluation process for determining the course grade.

The faculty member will meet formally with the student and a record of the meeting will be recorded on the Grade Appeal Review Form. In the meeting with the student, the faculty member must explain how the grade in question was arrived at. If the faculty member decides, based on the discussion with the student, that a change of grade is warranted, a Change of Grade Form will be completed by the faculty member, explaining the basis for the grade change and this will be forwarded to the Department Chair.

If the faculty member decides a change of grade is not warranted, the faculty member will complete the Grade Appeal Review Form, explaining why the original grade should not be changed and this will be forwarded to the Department Chair. The Change of Grade Form/Grade Appeal Review Form (as applicable) will then be reviewed in succession by the Department Chair, the Dean, and the Associate Provost for Student Affairs and a decision will be made.

The Associate Provost for Academic Affairs will render a decision only after the faculty member has responded to any requests for further consideration made by the Department Chair and the Dean. The decision by the Associate Provost for Academic Affairs will be final and not subject to further appeal. The Registrar will not change a student's final grade in a course for any other reason or through any other procedure.

Where an appeal is successful, the Associate Provost for Academic Affairs will forward the Change of Grade Form to the Registrar's Department so that the student's grade is changed on the Student Information System (SIS). A copy of the Grade Change Form will be filed in the student's record.

### **Grading System**

University course work is measured in terms of quantity and quality. One semester credit hour normally represents fifty minutes per week of lecture throughout a semester. The number of credits is a measure of quantity. The grade is a measure of quality. The university system for graduate grading is as follows:

| Grade | GPA Points | Percentage Scores |
|-------|------------|-------------------|
| A     | 4.0        | 95-100            |
| A-    | 3.7        | 90-94             |
| B+    | 3.3        | 86-89             |
| B     | 3.0        | 83-85             |
| B-    | 2.7        | 80-82             |
| C+    | 2.3        | 76-79             |
| C     | 2.0        | 70-75             |
| F     | 0          | 0-69              |

### **Grade Point Average (GPA)**

Grade or quality point values are assigned to letter grades as indicated in the grading system table. A quality or grade point score is computed by multiplying the value of a letter grade by the number of credits for the course. For example, a student receiving an A in a three-credit course earns twelve quality points. The GPA is computed by dividing the quality points earned by the number of credits graded A through F (GPA hours).

### **Additional Grade Notations**

#### *Satisfactory/Unsatisfactory (S/U):*

An “S” grade reflects satisfactory work and is considered passing in a course; a “U” grade reflects unsatisfactory work and is considered a failure. Courses evaluated with the “S” and “U” assessment have no effect on GPA.

#### *Incomplete (I):*

This grade is given to students who are passing a course but are unable to complete all the course work or the final exam for a verified reason beyond their control. The student must then complete all the requirements by the end of the next semester, not including any summer term, and the instructor must turn in the final grade by the end of the that semester’s grading deadline. Unless an explicit written extension is filed, the grade of “I” is changed to an “F” if the course requirements have not been fulfilled. Students who have filed their intention to graduate have only six weeks from the date of degree conferral to resolve any incomplete grades. An “I” grade is not calculated in a student’s GPA.

#### *In Progress (IP):*

The grade of “IP” indicates that the student is making progress in a course, writing a thesis or dissertation, or participating in an internship that extends beyond one semester or summer term. The grade of “IP” has no effect on the student’s GPA.

#### *Withdrawal (W):*

The grade of “W” on a course indicates that the student has dropped the course. While it has no effect on the GPA, dropped courses are part of attempted course credits that serve as the basis for the

student's credit level. A "W" for all courses in a given semester and the transcript notation "withdrawn" indicates that the student withdrew from AURAK.

### **Repeating a Course**

Normally, graduate courses cannot be repeated. With the recommendation of the dean of the student's school and with the approval of the Associate Provost for Academic Affairs, a graduate student may be allowed to repeat any course in which a grade of "C+," "C," or "C-" is received. The original grade and the new grade will appear on the transcript, but only the new grade will be calculated into the GPA.

No course may be taken more than twice.

### **Academic Progress / GPA Retention Levels**

#### **Satisfactory Progress**

To make satisfactory progress toward degree completion, the student must attain a CGPA of 3.0 ("B") or better.

#### **Academic Probation and Dismissal**

Graduate students receiving a grade of F in any given term will be subject to academic dismissal from the university. If the CGPA of a graduate student falls below 3.0 in any semester and no grade of F is earned, the student will be placed on Probation. Graduate students on probation will be allowed a maximum of 6 semester credit hours in the subsequent semester. Graduate students failing to earn a CGPA of 3.0 or higher in the subsequent semester will be subject to academic dismissal from the university. Probation will be removed at the end of any semester in which the student attains a CGPA of 3.00.

#### **Academic Dismissal**

A graduate student on probation who does not achieve good academic standing by the end of the regular semester following the term in which the CGPA fell below 3.00 will be dismissed from the university.

### **University Withdrawal**

Official withdrawal removes students from any academic program and cancels student status at AURAK. In order to return to the university, students need to apply for readmission through the Department of Admissions. Students who wish to withdraw have to complete a "Withdrawal Form" available at the Department of the Registrar and get clearance from all university departments. If the withdrawal process is completed satisfactorily and all financial obligations to the university are cleared, the effective date of withdrawal is noted on the student's permanent academic record. The effective date is the date used for calculating billing or refunds. No grades other than a "W" for the current semester are recorded.

### **Leave of Absence**

Occasionally, students are forced to leave the university for a semester or two because of circumstances beyond their control. Others find they simply need a break from studying. In such

circumstances, taking a leave of absence might be wise. Students who have an approved leave of absence for a semester or a year may register for the semester in which they plan to return without applying for readmission. Unless there are extenuating circumstances such as illness, a leave of absence is not normally given to a student who leaves the university during a term.

### **Fields of Study**

#### **Degree Offerings**

The American University of Ras Al Khaimah has three schools that offer both undergraduate and graduate degree programs. Graduate programs are listed below:

- Master of Education: Educational Leadership
- Executive Master of Business Administration
- Master of Business Administration
- Master of Science in Engineering Project Management
- Master of Science in Sustainable and Renewable Energy

### **Transferring from Non-Degree or Visiting to Degree Status**

Students may request a change of status from non-degree to degree status or from visiting to degree status by submitting a complete application through the Department of Admissions. All admissions requirements in place at the time of the change-of-status request must be met. Schools may choose to take into consideration the student's academic performance in the 500-level courses taken with AURAK.

Courses taken while under the non-degree or the visiting status may apply towards the degree program given the approval of the dean of the student's school. Grades earned in courses that are accepted will count in the CGPA. The university rules and regulations governing transfer courses and credits will apply.

The graduation requirements will be determined by the catalog that is effective when the student joins a degree program. Additional leveling courses may apply if appropriate.

### **Change of Program**

Students seeking to change their graduate degree program must complete the Change of Major Form available from the Department of the Registrar. Requests for a change of program should be submitted to the dean of the school of the program of the student's choice by the last day of the 12<sup>th</sup> week of classes of the fall or spring semester. The dean will forward the approved forms to the Department of Registration. Forms received by the Department of the Registrar by the end of the Add and Drop period will be effective as of the following semester.

To be eligible for a change of program, the graduate student must meet the requirements for admission to the new program. Please refer to the relevant program's catalog section for information on admission requirements.

## **Student Petitions and Appeals**

### **Students' Responsibility**

All official university communications are distributed through the AURAK-issued email address. These are considered official notifications. Students are responsible for checking their AURAK email accounts and for responding to or acting upon messages accordingly.

Students should keep their own records of all transactions with the university (e.g., registration schedules and forms, grade reports, payment records, etc.). It is also advisable to keep copies of all tests, digital files, papers and so forth submitted in fulfillment of course work. Students should keep copies of all course syllabi.

### **Appeal of Academic-Related Issues**

If a student wishes to discuss an issue pertaining to a course, instructor, or other academic-related issues, the student may direct his or her concern to the involved faculty member. If a satisfactory resolution of the situation is not reached, the student may appeal to the Department Chair, then to the Dean of the School, and finally to the VPAASS/Provost (or his/her designee), who has the final authority to decide the merit of the appeal.

If the instructor is no longer associated with the university, the VPAASS/Provost will appoint a faculty surrogate, who will assume the authority of the instructor. If the matter is not resolved at this point, the VPAASS/Provost will hear the case and make a decision that is not subject to further appeal. Academic appeal requests must be submitted no later than the end of the first day of orientation week of the following semester.

### **Degree Completion**

At AURAK, Master's degree programs will typically require at least one year of full-time study, or a minimum of 30 semester credits of course work (or equivalent) beyond the Bachelor's degree. These minimum credits shall not be inclusive of any non-credit bridge courses which may be required. Master's degrees offered by AURAK must include no less than 15 credit hours of courses (excluding thesis/dissertation credits) that are distinctive to that particular Master's program. AURAK also requires that all graduate students hold a CGPA of 3.0 in order to graduate and that Masters students complete a thesis.

AURAK reserves the right to change program or course requirements, grading procedures, or the requirements for acceptance into particular programs, academic standing, or graduation. These changes generally apply to all students enrolled at the time of implementation of the change and thereafter. When changes are made to degree requirements, however, students may elect to complete either the requirements in effect at the time of their admission or the new set of requirements, as detailed in the program requirements section of the catalog. Degree or program requirements do not include modality of program delivery (e.g., online or in-person delivery). With a change in modality of program delivery (e.g., shifting from in-person classes to online delivery of classes), the new modality of program delivery will typically apply only to students admitted following the change in modality.

For all degrees, requirements must be met as stated in the University Catalog. Candidates for graduate degrees who have been continuously enrolled may choose to graduate under the terms of any catalog in effect at or after their admission.

### **Master's Thesis**

In AURAK, a Master's degree will require a substantial thesis of at least six and no more than nine credit hours. A master's thesis documents research conducted by AURAK graduate students undertaking a master's degree program. Preparation of the thesis includes training in research methods, the development of the thesis is supervised by appropriately qualified and experienced, full-time faculty. The thesis is the culmination of the students' programs of study and is expected to reflect appropriate scholarly depth and rigor.

Students who do not complete their thesis in the final semester of the program of study will be required to maintain continuous enrollment in the thesis in succeeding semesters, with an In-progress (IP) grade assigned in the course designed to be the final thesis course until subsequent thesis coursework leads to the successful completion of the thesis. At that time, all IP grades originally assigned will be converted by the thesis supervisor to standard letter grades.

Where a student wishes to pursue an additional Master's degree from AURAK, the student must meet the admission requirements for the second degree and complete requirements for the additional degree. The requirements of the second degree must include at least 15 credit hours of courses (excluding thesis/dissertation credits) that are distinctive and not taken to meet requirements of the first degree.

It is AURAK policy to maintain master's theses in AURAK's Saqr Library so that they are available to other students and scholars. Students completing their thesis will be required to submit a bound copy of their theses to the Library.

### **Maximum Graduate Degree Completion Periods**

In order to graduate successfully from a graduate degree program after full admission to the program, the following periods for completion will be applied:

- Master's Degree program
  - One year degree program must be completed within four regular semesters.
  - Two year degree program must be completed within six regular semesters.

### **Academic Standing Requirement**

A student must be in good academic standing to be eligible for graduation.

### **Application for Graduation**

Candidates for graduate degrees must file an "Application for Graduation Form" with the Department of the Registrar during the registration period of the last expected term of study. Only after an "Application for Graduation Form" has been filed can the Department of the Registrar begin processing the necessary information for final certification for graduation.

Students who fail to complete all degree requirements by the end of the term for which they apply to graduate need not reapply for graduation. Their previous application will be automatically moved to the following semester.

### **Conferral of Degrees**

Only students who have successfully completed degree requirements and all thesis requirements, including corrections and final submission of the completed thesis to the library, by the end of the term for which they have applied to graduate are certified for conferral of a degree. Degrees are conferred at the end of the semester in which requirements have been met. Conferral of the degree is noted on the academic transcript of the graduate with the date of graduation.

### **Names on Degrees**

The names of AURAK students will be spelled in English exactly as they appear on their passports or identity cards when printed on degrees. If a name on a passport or an identity card does not appear in English, then the spelling of the name will be printed according to the personal preference of the student.

### **Attestation of Degrees and Transcripts**

The Department of the Registrar provides information relevant to the attestation of degrees and transcripts with the UAE Ministry of Education.

## ACADEMIC PROGRAMS

### Executive Master of Business Administration

#### Program Description

The Executive Master of Business Administration (EMBA) program is designed for mid-career and senior professionals who want to accelerate their career progression and develop higher level skills for senior leadership roles. An EMBA provides students with a comprehensive understanding of business operations and strategy and to be able to analyze and solve complex business problems.

In addition to traditional classroom learning, the program offers a range of experiential learning opportunities, such as case studies, collaborative online international learning, consulting projects and simulations to prepare students to navigate the complexities of the ever-changing business landscape.

The program employs a **blended learning mode of teaching**. This means that some courses are taught on-campus in a traditional in-person mode of instruction. Other courses are taught exclusively online, and a small number of courses are offered in a blended format, meaning that some class sessions are conducted on-campus, while others are conducted online. Overall, 50% of the program is taught on-campus, and 50% is taught online.

**The Executive Master of Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 9 requirements.**

#### School Mission Statement

The mission of the School of Business is to work with the AURAK 'family' to prepare tomorrow's business leaders to be thoughtful, creative, collaborative, and responsible with an ability to think critically, communicate their ideas clearly and take initiative in applying what they have learned. Backed by a diverse faculty committed to enrich their teaching and enhance their community impact by engaging in applied, cross-disciplinary, and pedagogical scholarship, the School offers programs based on the North American model and the cultural characteristics of the Gulf region in a student-centered teaching environment.

#### Program Aims

The overall aims of the program are:

1. To impart contemporary knowledge, and provide an integrated background in advanced concepts, frameworks and methodologies underpinning the high-level management of organizations.
2. To afford candidates the opportunity to concentrate on specific areas and issues within the business field. It also seeks to encourage life-long learning beyond the duration of the program.
3. To enhance professional communications skills and peer-based interaction, through its curriculum and associated group-based coursework.



## Program Learning Outcomes

The program learning outcomes for the Executive MBA are as follows:

- PLO1** Evaluate core business knowledge for the effective functioning of local and international organizations across business sectors.
- PLO2** Enhance strategic leadership role through application of program knowledge for analyses of corporate strategies.
- PLO3** Assess effectiveness of corporate governance, ethical functioning, and socially responsible organizational behavior.
- PLO4** Develop skills of collaborative decision making through teamwork and leadership to communicate complex business requirements to organizational personnel and stakeholders.
- PLO5** Produce critical and entrepreneurial thinking to support decision-making for business innovation and sustainability.
- PLO6** Integrate research principles and techniques with prior work experience to execute a major business project.
- PLO7** Combine personal reflection with program learning for professional development to executive level.

## EMBA Remedial Courses

If an EMBA applicant holds a baccalaureate degree, or its equivalent, in a non-business field with a minimum CGPA of 3.00, followed by a postgraduate certificate or diploma in a business-related field, he or she can qualify for full admission to the program, assuming all other admission requirements (such as English language proficiency) are met.

However, for those prospective students who have a CGPA of 2.0-2.4 and/or hold a degree from a non-business field, the students will have to complete remedial courses designed to prepare them for graduate-level coursework. The remedial courses consist of three compulsory courses as listed below.

|                    |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Compulsory Courses | EMFN 500 EMBA Foundation (3 CH)<br>EDUC 500 Foundations of Education (3 CH)<br>ENGR 500 Engineering Foundation for Graduate Students (3 CH) |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|

The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within two semesters will students be allowed to register for the EMBA Program. An unsuccessful student who has not completed the courses within two semesters will not be able to continue to the EMBA program and will have to leave the institution.

Where feasible, the foundational business course and the other remedial courses, which do not count toward the required credits for the degree, may be run in a condensed format (e.g., as condensed multi-weekend course modules) provided the learning outcomes remain unchanged and achievable.

AURAK will also accept candidates who have taken and passed the EMBA preparatory courses from other UAE Ministry or Higher Education-accredited universities, provided the courses taken have compatible syllabi and learning outcomes.

### Degree Requirements

The Executive Master of Business Administration requires the completion of 36 credits in the following areas:

| Area                             | Credits   |
|----------------------------------|-----------|
| I. Core Course Requirements      | 30        |
| II. Elective Course Requirements | 6         |
| <b>Total</b>                     | <b>36</b> |

### CORE COURSES

**30 Credits**

| EMBA Core Courses |                                                                           |                   |               |         |
|-------------------|---------------------------------------------------------------------------|-------------------|---------------|---------|
| Course Code       | Course Title                                                              | Prerequisite      | Teaching Mode | Credits |
| EMMG 510          | Leadership and Managing People                                            | Graduate Standing | Online        | 3       |
| EMAC 512          | Managerial and Financial Accounting                                       | Graduate Standing | On-Campus     | 3       |
| EMFN 514          | Managerial Finance                                                        | Graduate Standing | On-Campus     | 3       |
| EMMG 515          | Strategic Management                                                      | Graduate Standing | Online        | 3       |
| EMMK 518          | Marketing Management                                                      | Graduate Standing | Online        | 3       |
| EMMI 520          | Managing Information Systems Performance                                  | Graduate Standing | On-Campus     | 3       |
| EMIB 524          | International Business                                                    | Graduate Standing | Online        | 3       |
| EMGN 525          | Research Methods for Business                                             | Graduate Standing | On-Campus     | 3       |
| EMGN 555          | EMBA Investigative Thesis or Project I ( <b>Thesis Research Course</b> )  | EMGN 525          | Blended       | 3       |
| EMGN 556          | EMBA investigative Thesis or Project II ( <b>Thesis Research Course</b> ) | EMGN 555          | Blended       | 3       |

### ELECTIVE COURSES

**6 Credits**

| EMBA Elective Course Requirements |                                         |                           |          |
|-----------------------------------|-----------------------------------------|---------------------------|----------|
| Course Code                       | Course Title                            | Prerequisite/ Corequisite | Credits  |
| Elective 1                        | Any course from the EMBA Electives List | -                         | <b>3</b> |
| Elective 2                        | Any course from the EMBA Electives List | -                         | <b>3</b> |

| EMBA Elective Courses – Any Two Courses |                                                  |              |               |         |
|-----------------------------------------|--------------------------------------------------|--------------|---------------|---------|
| Course Code                             | Course Title                                     | Prerequisite | Teaching Mode | Credits |
| EMFN 560                                | International Corporate Finance                  | EMFN 514     | On-Campus     | 3       |
| EMFN 562                                | Investment Planning & Management                 | EMFN 514     | On-Campus     | 3       |
| EMFN 564                                | Financial Markets                                | EMFN 514     | On-Campus     | 3       |
| EMHR 590                                | Applied and Strategic Human Resources Management | EMMG 510     | Online        | 3       |
| EMHR 592                                | Managing Change and Innovation                   | EMMG 510     | Online        | 3       |
| EMHR 594                                | Employee Relations and Compensation Management   | EMMG 510     | Online        | 3       |

For Further Detailed information, Please Visit: <https://aurak.ac.ae/graduate-degrees/business/executive-master-of-business-administration>

### Executive Master of Business Administration - Two Year Plan

| Year 1, First Semester         |                                        |         | Year 1, Second Semester |                                          |         |
|--------------------------------|----------------------------------------|---------|-------------------------|------------------------------------------|---------|
| Course Code                    | Course Title                           | Credits | Course Code             | Course Title                             | Credits |
| EMMG 510                       | Leadership and Managing People         | 3       | EMGN 525                | Research Methods for Business            | 3       |
| EMFN 514                       | Managerial Finance                     | 3       | EMAC 512                | Managerial and Financial Accounting      | 3       |
| EMMG 515                       | Strategic Management                   | 3       | EMMK 518                | Marketing Management                     | 3       |
| Subtotal =                     |                                        | 9       | Subtotal =              |                                          | 9       |
| Year 2, First Semester         |                                        |         | Year 2, Second Semester |                                          |         |
| Course Code                    | Course Title                           | Credits | Course Code             | Course Title                             | Credits |
| EMGN 555                       | EMBA Investigative Thesis or Project I | 3       | EMGN 556                | EMBA Investigative Thesis or Project II  | 3       |
| EMIB 524                       | International Business                 | 3       | EMMI 520                | Managing Information Systems Performance | 3       |
| Elective 1                     | Any course from the EMBA Elective List | 3       | Elective 2              | Any course from the EMBA Elective List   | 3       |
| Subtotal =                     |                                        | 9       | Subtotal =              |                                          | 9       |
| Total 36 Semester Credit Hours |                                        |         |                         |                                          |         |

## **Master of Business Administration**

### **Program Description**

The Master of Business Administration (MBA) program prepares potential students to become effective business managers and leaders of the future. The program provides basic and advanced knowledge and skills in various aspects of business management.

AURAK's Master of Business Administration provides students with a well-rounded business education with, for example, courses to manage accounts, finance, information systems, international business, marketing, strategy, to undertake research, and to lead people. In addition to classroom learning, this program offers a range of experiential learning opportunities, such as case studies, collaborative online international learning, simulations, and consulting projects, preparing students for the challenges of a fast-paced business environment.

**The Master of Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 9 requirements.**

### **School Mission Statement**

The mission of the School of Business is to work with the AURAK 'family' to prepare tomorrow's business leaders to be thoughtful, creative, collaborative and responsible with an ability to think critically, communicate their ideas clearly and take initiative in applying what they have learned. Backed by a diverse faculty committed to enrich their teaching and enhance their community impact by engaging in applied, cross-disciplinary, and pedagogical scholarship, the School offers programs based on the North American model and the cultural characteristics of the Gulf region in a student-centered teaching environment.

### **Program Aims**

The overall aims of the program are:

1. To impart contemporary knowledge, and provide an integrated background in advanced concepts, frameworks and methodologies underpinning the high-level management of organizations.
2. To afford candidates the opportunity to concentrate on specific areas and issues within the business field. It also seeks to encourage life-long learning beyond the duration of the program.
3. To enhance professional communications skills and peer-based interaction, through its curriculum and associated group-based coursework.

### **Program Learning Outcomes**

On successful completion of this program the graduate will be able to:

- PLO 1** Evaluate core business knowledge for the effective functioning of local and international organizations across business sectors.
- PLO 2** Enhance strategic management roles by applying program knowledge to solve complex business problems, constructing viable and effective solutions in the process.

- PLO 3** Assess effectiveness of corporate governance, ethical functioning, and socially responsible organizational behavior.
- PLO 4** Cultivate teamwork, interpersonal, communication and leadership skills for communicating business requirements to organizational personnel and stakeholders.
- PLO 5** Produce critical and entrepreneurial thinking to support business innovation and sustainability decision-making.
- PLO 6** Integrate research principles and techniques to address business issues.
- PLO 7** Combine personal reflection with further learning for professional development at early mid career stage.

### **MBA Remedial Courses**

If an MBA applicant holds a baccalaureate degree, or its equivalent, in a non-business field with a minimum CGPA of 3.00, followed by a postgraduate certificate or diploma in a business-related field, he or she can qualify for full admission to the program, assuming all other admission requirements (such as English language proficiency) are met.

However, for those prospective students who have a CGPA of 2.0 – 2.49 and/or hold a degree from a non-business field, the students will have to complete remedial courses designed to prepare them for graduate-level coursework. The remedial courses consist of three compulsory courses detailed in the table below.

|                    |                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compulsory Courses | MBFN 500 Business Graduate Foundation (3 CH)<br>EDUC 500 Foundations of Education (3 CH)<br>ENGR 500 Engineering Foundation for Graduate Students (3 CH) |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|

The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within two semesters will students be allowed to register for the MBA. An unsuccessful student who has not completed the courses within two semesters will not be able to continue to the MBA program and will have to leave the institution.

Where feasible, the foundational business course and the other remedial courses, which do not count toward the required credits for the degree, may be run in a condensed format (e.g. as condensed multi-weekend course modules) provided the learning outcomes remain unchanged and achievable.

AURAK will also accept candidates who have taken and passed the MBA preparatory courses from other UAE Ministry of Higher Education-accredited universities, provided the courses taken have compatible syllabi and learning outcomes.

## Program Requirements

The Master of Business Administration requires the completion of 36 credits in the following areas:

| MBA DEGREE REQUIREMENTS |           |
|-------------------------|-----------|
| Area                    | Credit    |
| Core Courses            | 30        |
| Elective Courses        | 6         |
| <b>Total</b>            | <b>36</b> |

## Degree Requirements

| Core Course Requirements (30 Credits)        |                                                                        |                              |         |
|----------------------------------------------|------------------------------------------------------------------------|------------------------------|---------|
| Course Code                                  | Course Title                                                           | Prerequisite/<br>Corequisite | Credits |
| MBMG 510                                     | Leadership and Managing People                                         | Graduate Standing            | 3       |
| MBAC 512                                     | Managerial and Financial Accounting                                    | Graduate Standing            | 3       |
| MBFN 514                                     | Managerial Finance                                                     | Graduate Standing            | 3       |
| MBMG 515                                     | Strategic Management                                                   | Graduate Standing            | 3       |
| MBMK 518                                     | Marketing Management                                                   | Graduate Standing            | 3       |
| MBMI 520                                     | Managing Information Systems Performance                               | Graduate Standing            | 3       |
| MBIB 524                                     | International Business                                                 | Graduate Standing            | 3       |
| MBGN 525                                     | Research Methods for Business                                          | Graduate Standing            | 3       |
| MBGN 555                                     | MBA Investigative Thesis or Project I <b>(Thesis Research Course)</b>  | MBGN 525                     | 3       |
| MBGN 556                                     | MBA Investigative Thesis or Project II <b>(Thesis Research Course)</b> | MBGN 555                     | 3       |
| MBA Elective Course Requirements (6 Credits) |                                                                        |                              |         |
| Elective 1                                   | Any MBA Elective Course                                                | -                            | 3       |
| Elective 2                                   | Any MBA Elective Course                                                | -                            | 3       |

Further Detailed information, please visit: <https://aurak.ac.ae/graduate-degrees/business/master-of-business-administration>

### MBA Elective Course List – Any 2 Courses

| Course Code | Course Title                                     | Prerequisite | Credits |
|-------------|--------------------------------------------------|--------------|---------|
| MBFN 560    | International Corporate Finance                  | MBFN 514     | 3       |
| MNFN 562    | Investment Planning & Management                 | MBFN 514     | 3       |
| MBFN 564    | Financial Markets                                | MBFN 514     | 3       |
| MBHR 590    | Applied and Strategic Human Resources Management | MBMG 510     | 3       |
| MBHR 592    | Managing Change and Innovation                   | MBMG 510     | 3       |
| MBHR 594    | Employee Relations and Compensation Management   | MBMG 510     | 3       |

Further Detailed Information, please visit: <https://aurak.ac.ae/graduate-degrees/business/master-of-business-administration>

### Master of Business Administration - Two Year Plan

| Year 1, First Semester         |                                       |         | Year 1, Second Semester |                                          |         |
|--------------------------------|---------------------------------------|---------|-------------------------|------------------------------------------|---------|
| Course Code                    | Course Title                          | Credits | Course Code             | Course Title                             | Credits |
| MBMG 510                       | Leadership and Managing People        | 3       | MBGM 525                | Research Methods for Business            | 3       |
| MBFN 514                       | Managerial Finance                    | 3       | MBAC 512                | Managerial and Financial Accounting      | 3       |
| MBMG 515                       | Strategic Management                  | 3       | MBMK 518                | Marketing Management                     | 3       |
| Subtotal =                     |                                       | 9       | Subtotal =              |                                          | 9       |
| Year 2, First Semester         |                                       |         | Year 2, Second Semester |                                          |         |
| Course Code                    | Course Title                          | Credits | Course Code             | Course Title                             | Credits |
| MBGN 555                       | MBA Investigative Thesis or Project I | 3       | MBGN 556                | MBA Investigative Thesis or Project II   | 3       |
| MBIB 524                       | International Business                | 3       | MBMI 520                | Managing Information Systems Performance | 3       |
| Elective 1                     | Any MBA Elective Course               | 3       | Elective 2              | Any MBA Elective Course                  | 3       |
| Subtotal =                     |                                       | 9       | Subtotal =              |                                          | 9       |
| Total 36 Semester Credit Hours |                                       |         |                         |                                          |         |

## **Master of Education: Educational Leadership**

### **Program Description**

The Master of Education Degree is a thorough scholarly experience that provides educators with excellent learning opportunities in leadership theory and practice. Students will acquire practical knowledge in research, data-driven decision-making, and real-world leadership. Upon completing the M.Ed., graduates will have the necessary skills to become positive change agents capable of leading the continuous improvement of schools and student learning.

The program is designed to provide educational professionals with the opportunity to develop the necessary skills to become effective educational leaders, school administrators, cluster managers, educational researchers, employees of governmental agencies, and/or higher education professionals.

The M.Ed program codifies, in an intentional way, the value of learning communities. This is accomplished through the implementation of a Graduate Cohort. In a cohort model, all students take all the same courses at the same times, altogether. Cohort models are common in Graduate education programs across the United States, as this approach yields immensely valuable benefits. Notable, a cohort model allows the program to achieve the following goals: (1) to implement a meaningful progression of knowledge and skills while promoting rigor, and (2) to build an authentic learning community.

The Master of Education consists of a total of thirty-three (33) credit hours over four semesters. All courses are scheduled according to a pre-set timetable, and the language of instruction is English. Courses are offered in a face-to-face format with designated out-of-classroom experiences/activities conducted collaboratively or independently to complement course content and interactions. The program is collaborative with teaching and learning occurring through applied interactive lectures, discussions, case studies, and projects. A student-centered, active learning, and inquiry-based approach is consistently utilized, and the entire program incorporates reflective practice.

**The Master of Education: Educational Leadership is designed to satisfy QF Emirates Level 9 requirements. The award is overseen by the School of Arts and Sciences.**

### **Program Mission**

AURAK's Master of Education offers a transformational experience to graduate candidates as they actively apply theory, critical reflection and analysis, and best practice. M.Ed. students actively engage in practical diverse discourse on topics related to leadership, change agency, and governance within the educational systems of the UAE and surrounding region. Candidates obtain the requisite knowledge, skills, and tools to meaningfully contribute to positive change in addressing the dynamic challenges that occur when leading educational institutions, communities, faculty, and staff, in both their daily activities and in overall systemic change.

### **Program Educational Objectives**

Graduates of the Master of Education are prepared to actively contribute to K-12 schools, higher education institutions, and governmental entities in leadership roles such as instructional, administration and community building. The program is designed to prepare educational leaders with



the requisite knowledge, skills, and tools to meaningfully contribute to positive change in addressing the dynamic challenges that occur when leading educational institutions, communities, faculty, and staff, in both their daily activities and in overall systemic change.

The specific program goals of the Master of Education are to:

1. Prepare scholar-practitioners who engage in lifelong learning to meaningfully develop their careers in education;
2. Produce educators who apply in reflective practice to promote inclusion and diverse discourse grounded in the culture of the region;
3. Prepare scholar-practitioners who engage in data-driven decision making, to effect meaningful, research-based positive change in academic settings; and
4. Develop problem-solvers with advanced knowledge of education and specialized skills in education who contribute to overcoming challenges in education, academic settings, and at the intersection of education and industry.

### **Master of Education (MEd) Program Learning Outcomes**

Graduates of the Master of Education degree are expected to:

1. Assess highly specialized knowledge of leadership theories and/or methodologies as the basis for original thinking, encompassing reflective practice and ethical processes of a detailed body of knowledge of recent developments in the field of educational leadership;
2. Evaluate the specialized and advanced problem-solving skills required in educational leadership, research and analysis, while utilizing specialist communication and technology skills to present, explain and/or critique highly complex matters;
3. Construct a comprehensive, internalized, personal worldview, while accepting responsibility to society, and the sociocultural norms and relationships;
4. Take responsibility for leading strategic performance and development of professional teams, and self, in learning contexts that are complex and may require new strategic approaches within the ethical boundaries of educational leadership;
5. Critically reflect, and actively choose to enhance, leadership skills in the field of Education to contribute to, and implement, ethical leadership standards that advances professional knowledge and practice; and
6. Lead and participate in developing academic settings, and where education meets industry, demonstrating best practice in leadership.

### **Degree Requirements**

The following program requirements must be met for degree completion.

1. Successful completion of 33 credit hours, across 11 courses, identified below
2. Maintenance of a Grade Point Average of 3.00 or higher throughout the program
3. Successful submission of an original research thesis
4. Successful defense of an original research thesis

Assuming students do not interrupt their study, the M.Ed. degree is designed to be completed in 24 months (4 semesters). Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in AURAK's catalog. Students must meet specific standards to progress and do so within the maximum time allowed to complete the program; these standards and time allotment are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the degree may be re-evaluated for its current relevance.

The M.Ed. program requires thirty-three credit hours detailed as presented below.

| Program Course Requirements for the Master of Education |                                                                    |                     |
|---------------------------------------------------------|--------------------------------------------------------------------|---------------------|
| Course No.                                              | Title                                                              | No. of Credit Hours |
| EDUC 511                                                | Introduction to Applied Research <b>(Thesis Research Course I)</b> | 3                   |
| EDUC 515                                                | Educational Psychology for Inclusive Learning                      | 3                   |
| EDUC 525                                                | Theories and Applications of Leadership                            | 3                   |
| EDUC 545                                                | Budgeting and Resource Management                                  | 3                   |
| EDUC 620                                                | Evaluation of Instruction for Continuous Improvement               | 3                   |
| EDUC 640                                                | Inclusive Leading of Diverse Groups                                | 3                   |
| EDUC 660                                                | Organizational Change and Continuous Improvement                   | 3                   |
| EDUC 675                                                | Educational Policy and Community Engagement                        | 3                   |
| EDUC 680                                                | Applied Research I <b>(Thesis Research Course II)</b>              | 3                   |
| EDUC 685                                                | Analytics for Data-Driven Leadership                               | 3                   |
| EDUC 695                                                | Applied Research II                                                | 3                   |
| <b>TOTAL =</b>                                          |                                                                    | <b>33</b>           |

## Master of Education: Educational Leadership Two-Year Plan

| Year One, First Semester        |                                                               |              |
|---------------------------------|---------------------------------------------------------------|--------------|
| Course No                       | Course Title                                                  | Credit Hours |
| EDUC 511                        | Introduction to Applied Research (Thesis Course I) - 16 weeks | 3            |
| EDUC 525                        | Theories of Leadership and Administration – 16 weeks          | 3            |
| Semester Total =                |                                                               | 6            |
| Year One, Second Semester       |                                                               |              |
| EDUC 525                        | Theories and Applications of Leadership                       | 3            |
| EDUC 545                        | Budgeting and Resource Management                             | 3            |
| EDUC 680                        | Analytics for Data-Driven Leadership                          | 3            |
| Semester Total =                |                                                               | 9            |
| Year Two, First Semester        |                                                               |              |
| EDUC 620                        | Evaluation of Instruction for Continuous Improvement          | 3            |
| EDUC 675                        | Educational Policy and Community Engagement                   | 3            |
| EDUC 685                        | Applied Research I                                            | 3            |
| Semester Total =                |                                                               | 9            |
| Year Two, Second Semester       |                                                               |              |
| EDUC 640                        | Inclusive Leading of Diverse Groups                           | 3            |
| EDUC 660                        | Organizational Change and Continuous Improvement              | 3            |
| EDUC 695                        | Applied Research II                                           | 3            |
| Semester Total =                |                                                               | 9            |
| Total Program Credit Hours = 33 |                                                               |              |

# **Master of Science in Engineering Project Management**

## **Program Description**

The curriculum consists of thirty-six credit hours that provide the students with balanced coverage of project management: seven core courses, three elective courses and a master thesis. The master courses is designed to to showcase the graduate student's proficiency in applying the program learning outcomes to a real-world project. This comprehensive course integrates all fundamental aspects of project management, including planning, organizing, securing, managing, leading, and controlling resources, with the ultimate aim of accomplishing specific and attainable goals.

**The Master of Science in Engineering Project Management is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 9 requirements.**

## **Program Mission**

The Master of Science in Engineering Project Management (MSEPM) program at AURAK mission is to meet the needs of graduates in disciplines requiring an advanced understanding of the theoretical and practical principles of the project management function.

The MSEPM program requires students to apply principles of engineering and management science to contribute to the growing body of knowledge in project management. The program will prepare project managers from diverse engineering disciplines to efficiently manage complex projects in the UAE and the Gulf Region countries. The program provides students with excellent technical and managerial skills: Change Control, Cost Management, Documentation Management, Information / Communication Management, Procurement, Project Context / Environment, Quality Management, Reporting, Resource Management, Risk Management, Time Management / Scheduling / Planning, Stakeholder / Relationship Management, Team Building / Development / Teamwork, Work Content, and Scope Management to better satisfy customer needs. After completing the requirements for the degree, students will be able to apply engineering and management knowledge to successfully lead and execute engineering projects. Students will be trained to design, model, manage and control complex projects, and prepare to play leadership roles in industry and government.

## **Program Educational Objectives**

The Program Educational Objectives describe the expected accomplishments of graduates during their first few years after graduation. The program objectives have been derived from and support the mission statement of American University of Ras Al Khaimah. On successful completion of the Master of Science in Engineering Project Management program, a graduate should have:

1. Acquired skills in the initiation of projects, methods and techniques to control time cost and quality, resource management and long-term stewardship of assets;
2. Acquired a comprehensive understanding of the whole process of project management;
3. Gained project team leadership skills; and
4. Understood the establishment of project management methodologies.

## Program Learning Outcomes

The program learning outcomes of the MSEPM program are aligned with those of the Project Management Institute (PMI), Global Accreditation Center (GAC), and with level nine descriptors of the QF Emirates. Students who successfully complete this program will be able to:

- PLO 1:** Apply the principles of project management knowledge throughout the project life cycle within a broad range of scope and complexity;
- PLO 2:** Utilize the full range of project management tools and techniques to satisfy industry and stakeholders' requirements best;
- PLO 3:** Demonstrate professional responsibilities concerning people, resources, and the environment adherence to the project management code of ethics and professional conduct.
- PLO 4:** Integrate scope, schedule, budget, resource, quality, procurement, and risk processes in projects to meet stakeholders' requirements.
- PLO 5:** Direct projects with a clear understanding of project manager responsibilities of professional teams within each domain area of project management to motivate and influence team members and stakeholders.
- PLO 6:** Create (Design) effective project management communications and problem-solving techniques, project status reporting, conflict management, negotiation, and stakeholder engagement.
- PLO 7:** Conduct research using highly developed cognitive and creative skills and intellectual independence that significantly contributes to expanding project management knowledge.
- PLO 8:** Develop a project management framework by tailoring the processes based on project complexity and the competing constraints of scope, schedule, cost, resources, quality, and risk.

## Degree Requirements

The requirements of the MSEPM program are thirty-six (36) credit hours as follows: twenty-seven (27) credit hours are compulsory and nine (9) are electives.

### **Compulsory**

**27 Credit Hours**

| Course Code | Course Title                                        | Credits |
|-------------|-----------------------------------------------------|---------|
| MEPM 511    | Project Management Fundamentals                     | 3       |
| MEPM 512    | Engineering Contracts and Procurement               | 3       |
| MEPM 515    | Project Scheduling                                  | 3       |
| MEPM 521    | Project Cost Accounting and Finance                 | 3       |
| MEPM 532    | Engineering Management                              | 3       |
| MEPM 541    | Organizational Behavior and Project Team Management | 3       |
| MEPM 543    | Risk Management for Project Managers                | 3       |
| MEPM 590    | Thesis I ( <b>Thesis Research Course</b> )          | 3       |
| MEPM 591    | Thesis II ( <b>Thesis Research Course</b> )         | 3       |

**Electives Selected from the Following List****9 Credit Hours**

| Course Code | Course Title                                 | Credits |
|-------------|----------------------------------------------|---------|
| MEPM 513    | Project Implementation and Performance       | 3       |
| MEPM 514    | Global Projects Management                   | 3       |
| MEPM 516    | Project Quality Management                   | 3       |
| MEPM 522    | Financial Analysis and Decision Making       | 3       |
| MEPM 531    | Operation Research                           | 3       |
| MEPM 533    | Information Systems for Project Management   | 3       |
| MEPM 542    | Supply Chain Management for Project Managers | 3       |

**Master of Science in Engineering Project Management**  
**Two-Year Study Plan**

| Year 1, First Semester         |                                                     |         | Year 1, Second Semester |                                       |         |
|--------------------------------|-----------------------------------------------------|---------|-------------------------|---------------------------------------|---------|
| Course Code                    | Course Title                                        | Credits | Course Code             | Course Title                          | Credits |
| MEPM 511                       | Project Management Fundamentals                     | 3       | MEPM 512                | Engineering Contracts and Procurement | 3       |
| MEPM 532                       | Engineering Management                              | 3       | MEPM 515                | Project Scheduling                    | 3       |
| MEPM 521                       | Project Cost Accounting and Finance                 | 3       | MEPM 543                | Risk Management for Project Managers  | 3       |
| Subtotal =                     |                                                     | 9       | Subtotal =              |                                       | 9       |
| Year 2, First Semester         |                                                     |         | Year 2, Second Semester |                                       |         |
| Course Code                    | Course Title                                        | Credits | Course Code             | Course Title                          | Credits |
| MEPM 590                       | Thesis I                                            | 3       |                         | MEPM Elective (2)                     | 3       |
| MEPM 541                       | Organizational Behavior and Project Team Management | 3       |                         | MEPM Elective (3)                     | 3       |
|                                | MEPM Elective (1)                                   | 3       | MEPM 591                | Thesis II                             | 3       |
| Subtotal =                     |                                                     | 9       | Subtotal =              |                                       | 9       |
| Total 36 Semester Credit Hours |                                                     |         |                         |                                       |         |

## **Master of Science in Sustainable and Renewable Energy**

### **Program Description**

The Master of Science in Sustainable and Renewable Energy program is designed to provide the students with balanced coverage of advanced renewable energy technologies. Students will be exposed to the status of energy resources, their interaction with the environment, different renewable energy sources technologies, different techniques and technologies for energy management and energy conservation along with the economic aspects of renewable energy-based power generation. The curriculum developed for this Master's degree brings together the necessary tools to learn how to identify energy needs and propose the application of techniques for the saving and efficient use of energy as well as integrate the knowledge acquired in the field of sustainability and energy efficiency. The curriculum consists of 36 credit hours composed of six core courses, three elective courses and nine credits for the master thesis. The program can be completed in as little as two years. The program also targets full-time working students.

**The Master of Science in Sustainable and Renewable Energy is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 9 requirements.**

### **Program Mission**

The mission of the Master of Science in Sustainable and Renewable Energy program at the American University of Ras Al Khaimah (AURAK) is to provide a high-quality education for United Arab Emirates students in the key aspects of Renewable and Sustainable Energy. Thus, it will enable the students to take responsible, creative, challenging and stimulating posts in research and industry in this exciting field.

The Master of Science in Sustainable and Renewable Energy program is designed to close the strategic gap between the technical aspects of renewable energy and the policy drivers for sustainable development. The program aims also to address the social, economic, and environmental issues involved with renewable energy systems. In this way, students will not only learn about technical and operational details, but also about the policy context, market mechanisms, and business models for renewable energy technologies and installations.

After completing the requirements for the degree, students will be in an excellent position to start or continue a career in renewable and sustainable energy technologies. Students will be trained to design, model, manage and control complex projects and prepared to play leadership roles in industry and government.

### **Program Learning Objectives (PLOs)**

The Program goals describe the expected accomplishments of graduates during their first few years after graduation. The program goals have been derived from and support the mission statement of the American University of Ras Al Khaimah. On successful completion of the Master of Science in Sustainable and Renewable Energy, a graduate should have:

1. Acquired skills and competency in using renewable energy solutions in place of fossil fuel energy sources.
2. Acquired a comprehensive understanding of renewable and sustainable energy systems.
3. Teamwork skills in multidisciplinary projects and the identification of problems and proposal of solutions, including rigorous analysis and design of a process, component, or system.
4. Appreciation of the ethical and societal responsibilities entailed in the environmental engineering profession and the need for continuous education in the field and commitment to life-long learning.

### **Program Learning Outcomes**

The outcomes of the Master of Science in Sustainable and Renewable Energy program have developed the Student Outcomes (SOs) that resulted from an extensive outreach effort and analysis. The outcomes are based on program educational objectives and are in line with level 9 descriptors of the QF Emirates. Students who successfully complete this program will develop:

1. An ability to acquire and apply knowledge in the field of renewable and sustainable energy systems, their components and interactions between the components;
2. An ability to identify and critically evaluate current developments and emerging trends within the renewable energy sector;
3. An ability to contextualize renewable technology projects within a local, national and international framework, and consider social, ethical and environmental impacts and consequences;
4. An ability to analyze, formulate, and handle technical problems from a system perspective, with an overview on their life-cycle, from idea/need to specification, development, maintenance and termination, and the ability to set conditions, decide necessary resource consumption and manage procedures to solve and realize problems;
5. An ability to communicate effectively with peers, other engineering experts and the community using a variety of media and instruments of communication;
6. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts; AND
7. An ability to work effectively and proactively within cross-cultural, multi-disciplinary teams, to accomplish high-level engineering results.



## Degree Requirements

The requirements of the Master of Science in Sustainable and Renewable Energy are thirty-six (36) credit hours as follows: eighteen (18) credit hours are compulsory, nine (9) are electives, and nine (9) are for the thesis.

| Degree Requirements | Credits   |
|---------------------|-----------|
| Core Courses        | 18        |
| Elective Courses    | 9         |
| Thesis              | 9         |
| <b>Total</b>        | <b>36</b> |

### Core Requirements

18 Credit Hours

| Code     | Course Title                             | Credit |
|----------|------------------------------------------|--------|
| MEPM 511 | Project Management Fundamentals          | 3      |
| MSRE 511 | Energy Conversion                        | 3      |
| MSRE 512 | Energy Systems Modeling and Optimization | 3      |
| MSRE 513 | Energy Economics                         | 3      |
| MSRE 514 | Direct Research                          | 3      |
| MSRE 515 | Energy Efficiency and Management         | 3      |

### Electives from the Following List

9 Credit Hours

| Code     | Course Title             | Credit |
|----------|--------------------------|--------|
| MSRE 521 | Photovoltaics            | 3      |
| MSRE 522 | Wind Energy Systems      | 3      |
| MSRE 523 | Biofuels                 | 3      |
| MSRE 524 | Sustainable Desalination | 3      |
| MSRE 525 | Concentrated Solar Power | 3      |
| MSRE 526 | Green Buildings          | 3      |

### Master Thesis

9 Credit Hours

| Code                                 | Course Title                              | Credit |
|--------------------------------------|-------------------------------------------|--------|
| MSRE 590                             | Thesis I <b>(Thesis Research Course)</b>  | 3      |
| MSRE 591                             | Thesis II <b>(Thesis Research Course)</b> | 6      |
| <b>Program Total 36 Credit Hours</b> |                                           |        |

## Master of Science in Sustainable and Renewable Energy Two-Year Study Plan

| Year 1, First Semester         |                                          |         | Year 1, Second Semester |                                  |         |
|--------------------------------|------------------------------------------|---------|-------------------------|----------------------------------|---------|
| Course Code                    | Course Title                             | Credits | Course Code             | Course Title                     | Credits |
| MEPM 511                       | Project Management Fundamentals          | 3       | MSRE 513                | Energy Economics                 | 3       |
| MSRE 511                       | Energy Conversion                        | 3       | MSRE 515                | Energy Efficiency and Management | 3       |
| MSRE 512                       | Energy Systems Modeling and Optimization | 3       | MSRE 514                | Direct Research                  | 3       |
| Subtotal =                     |                                          | 9       | Subtotal =              |                                  | 9       |
| Year 2, First Semester         |                                          |         | Year 2, Second Semester |                                  |         |
| Course Code                    | Course Title                             | Credits | Course Code             | Course Title                     | Credits |
|                                | MSRE Elective (1)                        | 3       |                         | MSRE Elective (3)                | 3       |
|                                | MSRE Elective (2)                        | 3       | MSRE 591                | Thesis II                        | 6       |
| MSRE 590                       | Thesis I                                 | 3       |                         |                                  |         |
| Subtotal =                     |                                          | 9       | Subtotal =              |                                  | 9       |
| Total 36 Semester Credit Hours |                                          |         |                         |                                  |         |



# FACULTY LIST



## FACULTY LIST

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# COURSE DESCRIPTIONS



## COURSE DESCRIPTIONS

| Program Title                                | Program Code                                                                                         |
|----------------------------------------------|------------------------------------------------------------------------------------------------------|
| Accounting                                   | ACCT                                                                                                 |
| Architecture                                 | ARCH                                                                                                 |
| Arabic                                       | ARAB                                                                                                 |
| Artificial Intelligence                      | CSAI                                                                                                 |
| Biology, Bioinformatics, Biotechnology       | BIOL                                                                                                 |
| Business, Business Analytics (Undergraduate) | BUAN/BUSN/MGHR/MGMT/<br>MIST/MKTG/OPMT                                                               |
| Business (MBA/EMBA Graduate)                 | EMAC/EMFN/EMGN/EMHR/EMIB/<br>EMMG/EMMI/EMMK/<br>MBAC/MBFN/MBGN/MBHR/MBIB/<br>MBMG/MBMI/MBMK/<br>MBOM |
| Chemical Engineering                         | CHEN                                                                                                 |
| Chemistry                                    | CHEM                                                                                                 |
| Civil & Infrastructure Engineering           | CIEN                                                                                                 |
| Communication                                | COMM                                                                                                 |
| Computer Engineering                         | CENG                                                                                                 |
| Computer Sciences                            | CSCI                                                                                                 |
| Economics                                    | ECON                                                                                                 |
| Education                                    | EDUC                                                                                                 |
| Electrical and Electronics Engineering       | EEEN                                                                                                 |
| Electrical Engineering                       | ELEN                                                                                                 |
| Electronics and Communication Engineering    | ECEN                                                                                                 |
| Engineering                                  | ENGR                                                                                                 |
| Engineering Project Management               | MEPM                                                                                                 |
| English                                      | ENGL                                                                                                 |
| Environmental Science                        | ENVS                                                                                                 |
| Finance                                      | FNAN                                                                                                 |
| Geography                                    | GEOG                                                                                                 |
| History                                      | HIST                                                                                                 |
| Hospitality and Tourism                      | HTMT                                                                                                 |
| Industrial Engineering                       | IENG                                                                                                 |
| Information Technology                       | ITEC                                                                                                 |
| Interior Design                              | INDS                                                                                                 |
| Mathematics                                  | MATH                                                                                                 |
| Mechanical Engineering                       | MENG                                                                                                 |
| Middle Eastern Studies                       | MEST                                                                                                 |
| Petroleum Engineering                        | PENG                                                                                                 |
| Philosophy                                   | PHIL                                                                                                 |
| Physics                                      | PHYS                                                                                                 |
| Political Sciences                           | POLS                                                                                                 |
| Psychology                                   | PSYC                                                                                                 |
| Sociology                                    | SOCI                                                                                                 |

|                                             |      |
|---------------------------------------------|------|
| Statistics                                  | STAT |
| Sustainable and Renewable Energy (Graduate) | MSRE |
| University Studies                          | UNIV |

### **Course Descriptions**

Each course provides a course number, name, semester credit hours, prerequisites and corequisites if any, and a detailed course description. Course titles are followed by values in parentheses (0:0:0) separated by colons. The first number refers to the number of credit hours for the course; the second number refers to the number of lecture hours per week; the third number refers to the number of laboratory hours/studio hours/research or practicum hours per week.

### **Course Level Designations**

AURAK offers degree programs that provide a coherent course of study. The coursework in all undergraduate programs follows a logical pattern from the 100 through the 200, 300, and 400 levels. The distinctions among these levels are generally understood as follows:

#### **000-level course designation**

Prerequisite courses serving to prepare students for successful performance in college level courses.

#### **100-level course designation**

Courses with no prerequisites, or survey courses, or courses defining basic concepts or presenting the terminology of a discipline.

#### **200-level course designation**

Courses of intermediate university-level difficulty; courses with 100-level course(s) as prerequisite(s); or survey courses devoted to specific areas within a discipline.

#### **300-level course designation**

Courses of advanced university-level difficulty taken by majors and upper division students who have a clear interest and who are qualified in the subject.

#### **400-level course designation**

Advanced courses and/or seminars, tutorials and special courses for majors and upper-division students.

***For Masters Programs courses are offered at the 500 and 600 levels:***

**500-level course designation**

Courses that assume bachelor's level familiarity with the subject, typically serve as the core for graduate programs, and are the foundation for more advanced and focused study.

**600-level course designation**

Courses at a more advanced graduate level through which students attend to focused and specialized topics in the discipline.

The number designations of the course descriptions in this chapter have the following significance:

- The single number (BIOL 100) indicates the course is complete within a single semester, and the semester course may be taken separately with credit toward a degree.
- The double number separated by a comma indicates that the subject matter or content of the course extends through two semesters, but that either semester may be taken by itself.
- Unless otherwise specified, the first semester is not a prerequisite to the second semester.

## **COURSE DESCRIPTIONS**

### **ACCT 204            Principles of Financial Accounting**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ECON 103**

This is an introductory course in financial accounting which incorporates International Financial Reporting Standards (IFRS)/GAAP and emphasizes the use of accounting information in an ethical manner. Topics covered include: Basic accounting concepts and processes with particular emphasis on double entries; the accounting cycle; transaction analysis; accounting for assets, liabilities, owner's equity, expenses and revenues recognition, preparation of financial statements and ethical decision-making.

### **ACCT 205            Principles of Managerial Accounting**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ECON 103**

This is the first course in cost/managerial accounting. It discusses a range of introductory managerial accounting topics such as cost measurement and terminology, costs systems; analyzing cost-volume-profit relationships, job order and process costing, operating budgets, standard costing, introduction to capital budgeting and ethical managerial accounting-related decision-making.

### **ACCT 311            Managerial and Cost Accounting**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ACCT 205**

This course extends topics studied in Principles of Managerial Accounting (ACCT 205), as well as covers topics; pricing decisions, and joint product costing. The course examines the application of Managerial Management Accounting concepts and skills in decision making, of contemporary management philosophies and techniques in relation to JIT, quality, transfer pricing, performance measurement and questions the relevance of various aspects of traditional management accounting for internal management decision.

### **ACCT 331            Intermediate Accounting I**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ACCT 204**

The aim of this course is to develop the learner's understanding of financial accounting that comprises complex topics related to measuring and reporting of cash, inventories, investments, property, receivables, revenue recognition principles, plant, and intangible assets, and equipment. Emphasis is placed on both accounting theory and practice and on the development of professional opinion and significant thoughts.

### **ACCT 332            Intermediate Accounting II**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ACCT 331**

In this course, students examine the liability and equity sections of the balance sheet. Topics include current and other liabilities; applications to long-term liabilities; dilute securities and earnings per share (EPS); income taxes; pensions and other employee future benefits; leases; accounting changes and error analysis; statement of cash flows; and full disclosure in financial reporting.

### **ACCT 352            Taxation and Managerial Decisions**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ACCT 331**

The objective of this course is to develop a sound conceptual and technical foundation for the study of federal tax system and thus provide students with the tools necessary to stay up to date with ever-changing tax laws, where these apply. The course will imbue students with sufficient understanding of the tax environment to identify important issues when evaluating business transactions.

**(3:3:0)**

This course discovers the study and development of accounting information systems. The focus is on business processes covering many industries with an emphasis on accounting data modeling and internal control. Special topics including Extensible Business Reporting Language (XBRL) and commercial systems are incorporated throughout the course.

**Prerequisite(s): Senior Standing**

The Internship is designed for candidates on the undergraduate accounting major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation takes place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

**(3:3:0)**

This course will cover: [1] Study and evaluation of internal control; [2] audit planning and assessment of risk; [3] audit evidence and documentation; [4] professional ethics; [5] Application of auditing and [6] audit report.

**Prerequisite(s): ACCT 331**

The accounting project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the accounting specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

**(3:3:0)**

This course discovers a broad overview of the global financial and accounting environment and addresses three areas: financial reporting, financial analysis, and planning and control of multinational enterprises. In addition, the course analyzes how different businesses operating in global environments influence accounting practices.

**(3:3:0)**

This is an advanced subject and seminar course which covers major applications of Classical/current issues in Accounting. The topics to be covered will vary and will be announced by the end of the first week of commencing the course.

**(3:3:0)**

The course provides an introduction to the basics of the Arabic language. It aims to develop students' communicative competence in spoken as well as written Arabic. Intended to resonate with the experiences of non-native learners of Arabic who are visiting the UAE, the course's linguistic content is based within the contemporary culture of the region. The course is intended to stand-alone and not to serve as a prerequisite for any other university courses.

- ARAB 101      Arabic Language and Culture for Non- Native Learners I      (3:3:0)**  
**3 Semester Credit Hours**  
 Arabic for speakers of other languages helps develop students' communicative competence in the skills of speaking, listening, reading, and writing. The course provides for the acquisition of vocabulary, the study of grammar, and structures of spoken and written Arabic. Non-native speakers of Arabic gain insights into target language countries.
- ARAB 102      Arabic Language and Culture for Non- Native Learners II      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 101**  
 Beginner Level Arabic II is a direct continuation of Beginner Level Arabic I. This course is designed to enhance the reading, speaking, and listening skills. With this course, students can increase their vocabulary and improve their grammar in Arabic.
- ARAB 110      Arabic Language and Culture for Native Arabic Speakers I      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ENGL 101**  
 Arabic literature has developed many traditions though originating from a common source. The course is an introduction to representative texts from contemporary Arab writers, and their connections with the traditions of the past.
- ARAB 201      Intermediate Arabic I      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 102**  
 The course provides an introduction to academic sources of the language. This program is specially designed to teach Arabic to university level Arabic students.
- ARAB 202      Intermediate Arabic II      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 201**  
 Students engage in simple Arabic conversation on a range of everyday subjects, so students properly introduce themselves, and engage in simple conversation on a range of everyday topics. Building upon the basic foundation provided in Level I, topics include the definite article. Proper pronunciation and listening skills continue to be emphasized. In addition to readings and exposure to Arabic culture, students demonstrate further competence with structure and the pattern of words and sentences.
- ARAB 210      Arabic for Media: Native Speakers      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 110**  
 This course introduces Arabic language for Media: the language of print, broadcast, and digital media to Mass Communication native-speaking students. It aims to familiarize students with media vocabulary, text structure, and language usage. The course prepares Mass Communication students to apply requisite Arabic language skills and adhere to ethical norms in various mediums used in Mass Communication.
- ARAB 211      Arabic for Media: Non-Native Speakers      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 101**  
 This course introduces students to language used in Arabic media. The course covers the diverse media platforms and topics in the Arab world including weather news, deaths, special events, sports, advertisements, and economics. In each of those topics, the students will be equipped with requisite vocabulary and jargon used for discussing these in media.
- ARAB 311      Upper Intermediate Level Arabic Language and Culture for Native Arabic Speakers I      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Arabic Placement Test**  
 Upper Intermediate Level Arabic Language and Culture I is the first course in a four-course sequence for the native Arabic speaker. All courses in the sequence are taught in Modern Standard Arabic (MSA). The course is designed to provide native Arabic-speakers with higher level linguistic skills in writing, reading, speaking, and listening. Emphasis is placed on grammar review, vocabulary acquisition, and composition. This course and the subsequent sequence of courses are tailored particularly to students interested in Arabic-English translation in the English Language and Mass Communication Programs.

- ARAB 312      Upper Intermediate Level Arabic Language and Culture for Native Arabic Speakers II      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 311**  
 Upper Intermediate Level Arabic Language and Culture II is a direct continuation of ARAB 311. It is designed to provide Arabic-speaking Translation and Mass Communications majors with the linguistic skills (writing, reading, speaking, and listening) that serve as a solid foundation to the journalistic expression in Arabic. Emphasis is placed on grammar review, vocabulary acquisition, and composition.
- ARAB 411      Advanced Level Arabic Language and Culture for Native Arabic Speakers I      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s) (s): ARAB 312**  
 This is a direct continuation of Arabic 312. The course substantially expands students' existing vocabulary and capability of expression, both orally and in writing. Literary texts of increasing sophistication are used in the course.
- ARAB 412      Advanced Level Arabic Language and Culture for Native Arabic Speakers II      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 411**  
 Advance -Level Arabic Language and Culture for Native Arabic Speakers II is a direct continuation of ARAB 411 and represents the fourth course in the Arabic Language and Culture for native speakers' sequence. This course provides a survey of themes and genres of Arabic literature from the mid- 19th century to the present. While focus will be on content, students will continue their acquisition of MSA through written and oral assignments designed for their advanced level of competency.
- ARAB 420      Special Topics in Arab Culture and Civilization for Native Arabic Speakers      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARAB 412**  
 Supplementing the four-course sequence in Arabic Language and Culture for native speakers, ARAB 420 will be taught in MSA. The course critically examines issues, values, and institutions of the contemporary Arab world primarily through analysis and discussion of current events. While focus will be on content, students will continue their acquisition of Modern Standard Arabic through written and oral assignments designed for their level of competency.
- ARCH 121      Basic Design I      (4:0:12)**  
**4 Semester Credit Hours**  
 An introduction to the basic design components: subject, form, content, with a focus on their dynamic interrelation. Through observation, analysis, discussion of traditional and contemporary approaches in fine arts, architecture and interior design, students learn to apply the elements and principles of visual organization to create projects using various methods.
- ARCH 122      Basic Design II      (4:0:12)**  
**4 Semester Credit Hours**  
**Prerequisite(s): ARCH 121**  
 The course introduces basics of design process and theory as it relates to the 3-dimensional world. Students are involved in hand-on exploration of simple structures and spaces using diverse materials and textures. Space function, volume, mass are discovered through designing a small-scale residential space(s), where design elements/principles are applied.
- ARCH 221      Architectural Design Studio 1      (4:0:12)**  
**4 Semester Credit Hours**  
**Prerequisite(s): ARCH 122**  
 Introduction to the application of design principles in architecture and to the technical conventions and expressions. Focus on elementary architectural design as it relates to function, structure, site, land plot and the environment.
- ARCH 222      Architectural Design Studio 2      (4:0:12)**  
**4 Semester Credit Hours**  
**Prerequisite(s): ARCH 221**  
 Design of medium scale buildings with an integrated program and moderate site conditions. Emphasis on human centered design process, impact of human and climatic factors on design solutions. Exploration of sustainability and particularly natural ventilation, daylighting, building materials in regards to the design intent and environment.



**(3:2:3)**

**Prerequisite(s): INDS 122**

ARCH 231 Building Construction 1

**(3:2:3)**

**Prerequisite(s): INDS 122**

ARCH 232 Building Construction 2

**(3:2:3)**

**Prerequisite(s): ARCH 231**

ARCH 241 History and Theory of Architecture 1

**(3:3:0)**

ARCH 242 History and Theory of Architecture 2

**(3:3:0)**

**Prerequisite(s): ARCH 241**

ARCH 321 Architectural Design Studio 3

**(4:0:12)**

**Prerequisite(s):** ARCH 222

ARCH 322 Architectural Design Studio 4

**(4:0:12)**

**Prerequisite(s): ARCH 321**

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**(3:2:3)**

### 3 Semester Credit Hours

**Prerequisite(s): ARCH 232**

This course explores advanced building technologies and complex building structural configurations for long span buildings such as structural frames, space structure, suspended roof structure and air-supported structures. Offsite constructions, prefabricated and modular housing, envelopes to above listed types of building structures in relation to impact on environment are discussed.

**(3:2:3)**

**3 Semester Credit Hours**

**Prerequisite(s): ARCH 223, ARCH 331**

This course will lead the students to explore the principles of BIM used to visualize, document and communicate professional architectural working drawings respecting national, regional and international building codes and regulations.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ARCH 242**

The study of the development of Islamic art and architecture and its characteristics since the dawn of Islam and the impact of social, cultural, environmental and functional factors on architecture in the Moslem world. Analysis of selected examples of historical monuments from Islamic countries and interpretations of contemporary Islamic Architecture.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): ARCH 341**

The formation of modern theories and trends in contemporary art and architecture since the industrial revolution until today. Analysis of works by the pioneers of modern movements in architecture. Regional and vernacular architecture with special regards to environmental, social and technological issues. Selected examples of characteristic contemporary architecture in the UAE.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** ENVS 102

The course addresses the relationship of the built environment to natural environment through a whole system approach, with a focus on sustainable design of buildings. The emphasis is on local green design approaches, energy efficiency, renewable energy, and the appropriate use and conservation of resources, including materials, water, and land.

## ARCH 390      Internship I

**3 Semester Credit Hours**

**Prerequisite(s): 90 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

## ARCH 391      Internship II

**3 Semester Credit Hours**

**Prerequisite(s):** Completion of 130 credits and a CGPA of 2.0 or higher

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

**(4:0:12)**

**4 Semester Credit Hours**

**Prerequisite(s): ARCH 322**

Introduce students to the integrated design process with multiple stakeholders and multidisciplinary approaches (structural, mechanical, environmental) through buildings with shared functions. Via independent study and creative solutions students

explore approaches and processes of integrated design process and produce environmentally comprehensive and human-centered design solutions.

## ARCH 422 Architectural Design Studio 6

**(4:0:12)**

**4 Semester Credit Hours**

**Prerequisite(s): ARCH 421**

Design of a complex site and buildings through an integrated process taking in consideration the boundaries and exploring their mutual reciprocities. Emphasis on professional design thinking approaches that reflect student's personal values and ethical attitude that address issues of sustainability and technology in response to the project's functional requirements and regional context.

**ARCH 449      Technology of the Built Environment**

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s):** ARCH 223

This is an advanced level course that will allow students to discover the fundamentals of Parametric modeling for design applications in Architecture. Students will investigate rule-based and parametric design concepts and techniques in the context of a generative modeling environment: Dynamo for Revit.

## ARCH 454      Urban Planning

**(3:1:6)**

### 3 Semester Credit Hours

**Prerequisite(s):** ARCH 322

Introduction to the environmental, social and economic aspects necessary in the formation of an urban planning process; the characteristics of the urban fabric and the development of cities and their urban elements into integrated, harmonious and functional unity; social services and infrastructure and their necessary provision according to norms and standards.

## ARCH 456 Sustainable Housing

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): ENVS 102**

The course begins with a section on society and housing, exploring how sustainable housing practices build community. The second part of this course focuses on the relationship between society and the environment. We explore questions about how community growth has influenced the environment and how natural events affects our communities.

## ARCH 457      Urban Design

**(3:1:6)**

**3 Semester Credit Hours**

**Prerequisite(s):** ARCH 463

This course focuses on the definition, development and designing urban space concepts that both challenges and demonstrates students' knowledge and skills in urban design. Through advanced investigation of the history, theory and practice of urban design approaches, students learn the principles of place making, connections between people and places within the spatial layers of a city.

**ARCH 459      Conservation of Historic Environment**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** ARCH 341

This course aims to provide students with knowledge about development of historic conservation theory and practice. Students will learn about the principal schools of thoughts, defining practices and new approaches to definition and assessment of cultural heritage. Students will explore a range of subjects and issues that affect contemporary heritage conservation practice, including its historical and philosophical underpinnings and its relation to sustainable environment.

## ARCH 463 Landscape Architecture

**(3:1:6)**

**3 Semester Credit Hours**

**Prerequisite(s):** ARCH 322

The natural environmental equilibrium affecting different site scales. History of gardens, evolution through civilizations of the Mediterranean region & plant species in UAE. Landscape as gardens, open spaces & gathering areas. Introduction to environment-friendly concepts, energy conservation implementation on building sites. The influence of local climates to landscape designs.

- ARCH 466      Building Utilities II: Illumination, Acoustics, and Electrical Building Services      (3:2:3)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MENG 468**  
 This course focuses on sustainable-green architecture design supported by the computer simulation of building energy use, natural ventilation, and daylighting. The simulation process utilizes research and simulation software to produce design evidence of the sustainability criteria. The course also may prepare undergraduate students for their further graduate studies in sustainability.
- ARCH 473      Environmental Control      (3:3:0)**  
**3 Semester Credit Hours**  
 The course focuses on sustainable-green architecture design supported by the computer simulation of building energy use, natural ventilation, and daylighting. The simulation process utilizes research and simulation software to produce design evidence of the sustainability criteria. This course also may prepare undergraduate students for their further graduate studies in sustainability.
- ARCH 485      Professional Practice      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARCH 422**  
 An overview of the architectural practice complexities in historic and contemporary context; the roles and responsibilities of the architect with emphasis on the characteristics of best practices; the architects in the office from starting a practice, marketing and strategic planning of practice, methods of project delivery to design contracts and agreements and AIA documents as well as ethics and existence in the society.
- ARCH 487      Project Management      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ARCH 336**  
 The course covers key components of project management, including project integration, project scope management, project time and cost management, construction project, contract documents and types of specification depending on project nature, quantity surveying and quality management, human resource considerations, communications, risk management, and procurement management.
- ARCH 521      Architectural Design Studio 7      (4:0:12)**  
**4 Semester Credit Hours**  
**Prerequisite(s): ARCH 422**  
 Contextual design of a group of buildings through complex design approaches and integrated urban design framework that might urban infill, upgrading existing built environment, or heritage buildings conservation via reuse, etc. Focus of demographics of the site, socio-economic, environmental conditions for users' wellness and life quality improvement.
- ARCH 591      Graduation Project Thesis Research      (2:1:3)**  
**2 Semester Credit Hours**  
**Prerequisite(s): ARCH 422**  
 The course is oriented towards systematic analysis of the graduation project on selected topic, its functional and spatial relationships, location and setting, physical and human environments, use of case studies, and provision of synthesis and alternative concepts as design solutions. The outcome is presented in both forms, written as a report and visual as posters, to a jury of professionals. **Writing Intensive Course**
- ARCH 592      Graduation Project Design      (5:0:15)**  
**5 Semester Credit Hours**  
**Prerequisite(s): ARCH 521 and ARCH 591**  
 Under the supervision of a faculty advisor(s), students independently develop design solutions on investigated architectural topic of personal interest selected for ARCH 591. Through final project jury presentation and portfolio, students demonstrate their ability to professionally develop design solutions and solve problems related to spatial, functional, aesthetic, social, environmental, technical and other aspects of the built environment.

|                                                                                                                                                                                                                                                                                                                                          |                                        |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------|
| <b>BIOL 095</b>                                                                                                                                                                                                                                                                                                                          | <b>Introductory Biology</b>            | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                           |                                        |                |
| This course is offered for new science students without high school science backgrounds. The course provides an introduction to basic concepts in biology, cell energetics, genetics, diseases and biotechnology. This course is not for degree credit.                                                                                  |                                        |                |
| <b>BIOL 100</b>                                                                                                                                                                                                                                                                                                                          | <b>Humankind in a Biological World</b> | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                           |                                        |                |
| In this course, students will learn how human beings interact with, affect, and are affected by other living organisms. Ways in which human activities have had an impact on other life on earth, humankind and disease, and the development of scientific thought how we could solve this issue are examined.                           |                                        |                |
| <b>BIOL 112</b>                                                                                                                                                                                                                                                                                                                          | <b>University Biology I</b>            | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                           |                                        |                |
| <b>Corequisite(s): BIOL 113 University Biology I Laboratory</b>                                                                                                                                                                                                                                                                          |                                        |                |
| Introduction to cell chemistry, metabolism, and genetics.                                                                                                                                                                                                                                                                                |                                        |                |
| <b>BIOL 113</b>                                                                                                                                                                                                                                                                                                                          | <b>University Biology I Lab.</b>       | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                            |                                        |                |
| <b>Corequisite(s): BIOL 112</b>                                                                                                                                                                                                                                                                                                          |                                        |                |
| Introduction to Cell Structure & Function and Basics on plant biology.                                                                                                                                                                                                                                                                   |                                        |                |
| <b>BIOL 114</b>                                                                                                                                                                                                                                                                                                                          | <b>University Biology II</b>           | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                           |                                        |                |
| <b>Corequisite(s): BIOL 115</b>                                                                                                                                                                                                                                                                                                          |                                        |                |
| This course is intended for science majors and pre-professionals in life sciences and provides an introduction to Genetics, Microbiology and animal form and function.                                                                                                                                                                   |                                        |                |
| <b>BIOL 115</b>                                                                                                                                                                                                                                                                                                                          | <b>University Biology II Lab.</b>      | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                            |                                        |                |
| <b>Corequisite(s): BIOL 114</b>                                                                                                                                                                                                                                                                                                          |                                        |                |
| This course encompasses the basic techniques in Biology for science majors and pre-professionals in life sciences that accompany BIOL 114.                                                                                                                                                                                               |                                        |                |
| <b>BIOL 230</b>                                                                                                                                                                                                                                                                                                                          | <b>General Microbiology</b>            | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                           |                                        |                |
| <b>Prerequisite(s): BIOL 114</b>                                                                                                                                                                                                                                                                                                         |                                        |                |
| <b>Corequisite(s): BIOL 231</b>                                                                                                                                                                                                                                                                                                          |                                        |                |
| This course covers the classification of microorganisms, microbial metabolism, microbial growth, microbial genetics, structure of eubacteria, archaea, bacteria, fungi, and viruses, specific defenses, and diseases.                                                                                                                    |                                        |                |
| <b>BIOL 231</b>                                                                                                                                                                                                                                                                                                                          | <b>General Microbiology Lab</b>        | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                            |                                        |                |
| <b>Corequisite(s): BIOL 230</b>                                                                                                                                                                                                                                                                                                          |                                        |                |
| This course focuses on laboratory techniques involved in culturing, staining, and identifying microorganisms.                                                                                                                                                                                                                            |                                        |                |
| <b>BIOL 250</b>                                                                                                                                                                                                                                                                                                                          | <b>Biochemistry I</b>                  | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                           |                                        |                |
| <b>Prerequisite(s): CHEM 111 or CHEM 120</b>                                                                                                                                                                                                                                                                                             |                                        |                |
| <b>Corequisite(s): BIOL 251</b>                                                                                                                                                                                                                                                                                                          |                                        |                |
| Structure and chemical behavior of biochemical compounds, levels of protein structure, steady state enzyme kinetics and activities, protein purification, protein functions. Structure and functions of Carbohydrates, Nucleotides and Nucleic Acids, DNA-Based Information Technologies. Lipids and Biological Membranes and Transport. |                                        |                |
| <b>BIOL 251</b>                                                                                                                                                                                                                                                                                                                          | <b>Biochemistry I Lab</b>              | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                            |                                        |                |
| <b>Corequisite(s): BIOL 250</b>                                                                                                                                                                                                                                                                                                          |                                        |                |
| This experimental course will introduce students with hand practice of some major biochemistry techniques. This course has two parts, part A is wet lab where students learn and practice separation, identification, and quantization techniques that                                                                                   |                                        |                |

exploit properties of biological molecules. To analyze results, data and write reports to develop analytical reasoning and problem-solving skills. The second part B is dry/computer lab where students will use Geneious to study the 3D structures of biological macromolecules from PDB files (Protein Data Bank).

## BIOL 270      General Genetics

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): BIOL 112 or BIOL 114**

**Corequisite(s): BIOL 271**

This course focuses on the basic concepts of heredity and modern developments in Genetics.

## BIOL 271      General Genetics Lab.

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): BIOL 270**

This course covers basic laboratory techniques in Genetics that accompany the course BIOL 270.

## BIOL 322      Microbial Genetics

**(3:3:0)**

3 Semester Credit Hours

**Prerequisite(s): BIOL 270**

This course covers the basic concepts of microbial genetics including gene structure, gene expression and its control, and evolution of bacterial genetic diversity. The emphasis will be on topics such as mechanisms of genetic exchange, recombination, plasmids, analysis of bacteria and phage genomes, mutagenesis, and DNA repair. The course will also discuss the common methods using bacteria in gene cloning and identification and isolation of mutants

## BIOL 330      Applied and Industrial Microbiology

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): BIOL 230**

**Corequisite(s): BIOL 331**

This course covers the biology of microorganisms of ecological and industrial significance and includes topics such as food production, spoilage and preservation, fermentation technology, waste disposal, water purification, biodeterioration, and decomposition.

## BIOL 331      Techniques in Applied and Industrial Microbiology Lab.

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): BIOL 330**

This course includes laboratory exercises that illustrate applied methodologies in microbiology, including isolation of commercially useful strains, and discusses the production and purification of industrial products.

## BIOL 350 Biochemistry II

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** CHEM 250

This course covers the bioenergetics, Glycolysis, Gluconeogenesis, the Citric Acid Cycle and the oxidative phosphorylation and photophosphorylation. Lipid, amino acids and nucleotides Biosynthesis. DNA and RNA metabolism and gene expression regulation.

## BIOL 356      Virology

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): BIOL 230**

This course provides an overview of the characteristics of major families of viruses; the intrinsic properties of viruses that cause disease, their development, and life cycles; interaction with host cells; genetics; and tumor-inducing properties and epidemiology.

## BIOL 360      Developmental Biology

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): BIOL 270**

This course exposes students to the principles of embryonic development and differentiation in animal species at the cellular, molecular, tissue, and whole organism levels.

- BIOL 380                      Biotechnology and Genetic Engineering                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): BIOL 270**  
**Corequisite(s): BIOL 381**  
This course focuses on the theory and applications of Biotechnology to real-world problems, emphasizing significance and societal applications.
- BIOL 381                      Biotechnology Lab Methods and Techniques                      (1:0:3)**  
**1 Semester Credit Hour**  
**Corequisite(s): BIOL 380**  
This course focuses on the fundamental practical techniques used in Biotechnology research and industry.
- BIOL 390                      Internship                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites(s): Prior completion of 90 credits and CGPA of 2.0 or higher**  
This is a supervised training experience for biotechnology-related duties for a duration of 240 hours at an approved biotechnology or biotechnology-related organization under the guidance of a designated site supervisor in coordination with a faculty supervisor.
- BIOL 411                      Drug Discovery, Design and Development                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): BIOL 350**  
This course will explore the process of drug development, from target identification to final drug registration. It will present drug development as a process involving target selection, lead discovery using computer-based methods and combinatorial chemistry/high-throughput screening. Safety evaluation, bioavailability, clinical trials, and the essentials of patent law will also be discussed. Along the way, you will learn about molecular recognition, computer aided drug design, and toxicology as applied to the development of new medicines.
- BIOL 420                      Molecular Biology                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): BIOL 270**  
**Corequisite(s): BIOL 421**  
This course focuses on the basic concepts of structure and function of genetic material at the molecular level. Topics such as DNA replication, chromosome structure, gene expression and mutation, DNA repair, and transposable elements are covered in molecular detail.
- BIOL 421                      Molecular Biology and Recombinant DNA Lab.                      (1:0:3)**  
**1 Semester Credit Hour**  
**Corequisite(s): BIOL 420**  
This course focuses on the techniques in modern molecular biology techniques such as restriction mapping, isolation of DNA and RNA from eukaryotic cells, Southern hybridization, reverse transcription, and polymerase chain reaction (PCR and RT-PCR).
- BIOL 423                      Advanced Molecular Biology and Forensic Sciences                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): BIOL 270**  
This is for forensic DNA practitioners, those interested in a career in forensic DNA and all students interested in broadening their understanding of the genetic principles that underlie the forensic application of DNA typing methods.
- BIOL 424                      DNA Fingerprinting and Serology Lab                      (1:0:3)**  
**1 Semester Credit Hour**  
**Prerequisite(s): BIOL 270**  
**Corequisite(s): BIOL 423**  
Laboratory course to accompany BIOL 423 (Advanced Molecular Biology and Forensic Sciences). Focusing on the basic techniques used in forensic DNA laboratories, this course will introduce the students to the most recent DNA fingerprinting techniques and serological analysis.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                    |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------|
| <b>BIOL 425</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Advanced Genetics</b>                           | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                |
| <b>Prerequisite(s): BIOL 270</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |                |
| This course discusses the principles of genetics and the role of genetic variations on the expressed phenotype in unicellular and multicellular organisms. The topics include genome structure and function, gene mutations, recombination, transposable elements, experimental genetics analysis, population genetics and diseases.                                                                                                                                                                                                                                                                                                                                                              |                                                    |                |
| <b>BIOL 430</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Cell Culture Theory and Technology</b>          | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                |
| <b>Prerequisite(s): BIOL 270</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |                |
| <b>Corequisite(s): BIOL 431</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                    |                |
| This course aims at providing a strong foundation in the structure, growth, and function of plant and animal cells as well as the technology involved in cell and tissue culture, cell preservation, protoplast culture and fusion, cell cloning and fusion, monoclonal antibody production, breeding and genetic engineering. The applications of stem cells to therapy will also be emphasized in this course.                                                                                                                                                                                                                                                                                  |                                                    |                |
| <b>BIOL 431</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Cell Culture Techniques Lab.</b>                | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                    |                |
| <b>Corequisite(s): BIOL 430</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                    |                |
| This course deals with the principles and practical applications of animal tissue culture. Molecular techniques or manipulation and maintenance of cells in culture using animal systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                    |                |
| <b>BIOL 434</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Cell Communication and Signal Transduction</b>  | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                |
| <b>Prerequisite(s): BIOL 270</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |                |
| This course focuses on how cells communicate with each other and their microenvironment, the signaling events that result from these interactions, and their translation into cellular responses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                    |                |
| <b>BIOL 435</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Stem Cell Biology and Regenerative Medicine</b> | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                |
| <b>Prerequisite(s): BIOL 270</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |                |
| This course provides a broad overview of the biological principles influencing stem cell populations and emphasizes the functional role of stem cells in regulating normal development and contributing to disease, as well as examines the therapeutic potential of stem cells in regenerative medicine.                                                                                                                                                                                                                                                                                                                                                                                         |                                                    |                |
| <b>BIOL 436</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Immunology</b>                                  | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                |
| <b>Prerequisite(s): BIOL 270</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |                |
| This course focuses on the structure and function of immunoglobulins, the role of cell-mediated immunity, the protective role of the immune system, and disease and injury related to malfunctions of the immune system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    |                |
| <b>BIOL 440</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Medical Microbiology</b>                        | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                    |                |
| <b>Prerequisite(s): BIOL 230</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |                |
| <b>Corequisite(s): BIOL 441</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                    |                |
| This course focuses on mechanisms of microbial pathogenesis, the host response, and the scientific approaches that are used to investigate these processes. It provides students with: (1) in depth-understanding of the pathogenic microorganisms and particularly address the fundamental mechanisms of their pathogenicity. (2) the necessary foundation to identify and characterize microorganisms as well as a foundation in the methods to prevent, diagnose and treat infectious diseases. (3) opportunities for a student to develop diagnostic skills in microbiology, including the identification of and interpretation of laboratory tests for the diagnosis of infectious diseases. |                                                    |                |
| <b>BIOL 441</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Medical Microbiology Lab.</b>                   | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                    |                |
| <b>Corequisite(s): BIOL 440</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                    |                |
| The course covers a collection of techniques used for culture, isolation, and identification of common microorganisms. The topics aim to provide students with practical training on standardized methods for microbial identification including staining                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                    |                |



and microscopy, antibiotic susceptibility, and biochemical tests. In addition, the course provides in depth emphasis on some methods of disinfection, sterilization, and antimicrobial therapy.

## BIOL 442      Molecular Diagnostics

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): BIOL 380**

**Corequisite(s): BIOL 443**

This course covers a collection of techniques used to analyze biological markers in the genome and proteome to diagnose human disease. The course will examine and understand modern tools employed to identify variations in the genetic material or proteins associated with specific health conditions or disease. Topics include the most commonly used molecular tests to diagnose disease such as DNA and RNA isolation, amplification, hybridization, PCR, DNA, and RNA sequencing, ELISA, and flow cytometry. In addition, the course provides in depth emphasis on specimen handling, advantages, and disadvantages of molecular diagnostics, and quality assurance in the molecular lab.

## BIOL 443      Molecular Diagnostics Lab

**(1:0:3)**

**1 Semester Credit Hour**

**Prerequisite(s): BIOL 380 and BIOL 381**

**Corequisite(s): BIOL 442**

This course covers the basic concepts and techniques that are used in the diagnosis of disease. The students will be trained to have hands-on experience on several assays such as nucleic acid isolation, PCR, gel electrophoresis, restriction fragment length polymorphisms analysis, and ELISA. In addition, the course will prepare students to evaluate the controls for validating results, standardize tests, and discuss quality control for diagnostic laboratory.

## BIOL 450 Bioethics

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): 90 credits**

This course examines selected ethical issues arising from the application of modern biotechnology to microorganisms, plants, animals, and humans. It also discusses the ethics and misconduct in research, authorship, and peer-review. **(Writing Intensive Course)**

## BIOL 462      Crime Scene Investigation

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): BIOL 270**

This course will introduce students to crime scene investigation procedures and techniques. With a focus on crime scene processing, this course discusses the most recent topics on database technologies and DNA analyses.

## BIOL 464 Criminal Justice

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Completion of 90 credit hours**

This course will introduce students to the real world of criminal justice with a focus on topics related to understanding crimes, roles of the police, and the court system.

## BIOL 473      Bioinformatics and Computational Biology

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): BIOL 270**

This course is designed to provide instruction in the scientific concepts and computer skills currently used in searching biological databases, comparing sequences, looking at protein and DNA characteristics and answering biological and biomedical questions using databases.

## BIOL 480 Food Biotechnology

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): BIOL 270**

This course introduces students to the impact of Biotechnology on food production, the genetic tools applied in Food Biotechnology, the improvement of microbes used in food production by modern biotechnological approaches, and also discusses related ethical, legal, and regulatory issues.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                             |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------|
| <b>BIOL 481</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Bioprocessing Technology in Pharmaceuticals Industry</b> | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): BIOL 380</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                             |                 |
| This course provides an overview of the main elements, theory and application of biotechnology procedures, and bioprocess design settings and operations related to the development of biopharmaceutical products including good manufacturing practices and bioprocessing equipment standards. The course lays emphasis on fermentation, harvest, primary and final purification, media and buffer preparation, equipment cleaning and sterilization, and critical process utilities.                                                                                                                                                                                                                                                                  |                                                             |                 |
| <b>BIOL 491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Senior Seminar</b>                                       | <b>(2:2:0)</b>  |
| <b>2 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): 90 credits</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                             |                 |
| This is a capstone course and involves student participation in seminars on various Biotechnology topics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                             |                 |
| <b>BIOL 492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Special Topics in Biotechnology</b>                      | <b>(2:2:0)</b>  |
| <b>2 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): Minimum 90 credits and Minimum CGPA 2.0</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                             |                 |
| This course focuses on the major advances in genomics, proteomics, and metabolomics and their significance in biotechnological applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                             |                 |
| <b>BIOL 493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Research Methodology in the Biological Sciences</b>      | <b>(2:2:0)</b>  |
| <b>2 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): Senior Status/Standing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| This course provides in-depth knowledge of research design and methodology and to train the student in writing a research proposal and critically reviewing scientific literature. The emphasis will be on interpreting and evaluating recent studies from the scientific literature and communicating biological findings in a written and oral format to a range of audiences. The course focus is to train students to critically review literature, identify a research question, lay out a detailed experimental plan including the statistical analysis, and discuss anticipated results.                                                                                                                                                         |                                                             |                 |
| <b>BIOL 494</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Senior Project in Cell and Molecular Biotechnology</b>   | <b>(4:0:12)</b> |
| <b>4 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): Senior Status/Standing BIOL 493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                             |                 |
| Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Cell and Molecular Biotechnology and present their research findings in both oral and written formats.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                 |
| <b>BIOL 495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Senior Project in Medical Biotechnology</b>              | <b>(4:0:12)</b> |
| <b>4 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): Senior Status/Standing BIOL 493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                             |                 |
| Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Medical Biotechnology and present their research findings in both oral and written formats.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                             |                 |
| <b>BIOL 496</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Senior Project in Forensic Biology</b>                   | <b>(4:0:12)</b> |
| <b>4 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): Senior Status/Standing BIOL 493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                             |                 |
| Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Forensic Biology and present their research findings in both oral and written formats.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                 |
| <b>BUAN 301</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Analytics for Decision Making</b>                        | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                             |                 |
| <b>Prerequisite(s): OPMT 311</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                             |                 |
| This course is an introductory survey course in business analytics and is geared towards providing a broad coverage of descriptive, predictive, and prescriptive analytics techniques and applications within a business context. The course aims to introduce students to foundational concepts, methodologies, and tools required to understand the emerging role of business analytics in organizations, with a focus on decision making for efficiency, profitability, and sustainability. Through hands-on practice tutorials and assignments, students will also be introduced to computer-based modeling approaches to facilitate formulation, resolution, and interpretation of the results, and the process of transforming data into actions. |                                                             |                 |

**BUAN 311 Business Data Mining**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): BUAN 301**

Data Mining (DM) is the non-trivial process of extracting implicit, novel, and useful information from large volume of data. It has emerged as a unique combination of several fields of science and technology including statistics, database systems, computer programming, machine learning, and artificial intelligence. This course is designed to provide undergraduate students with an understanding of data mining techniques, including data pre-processing, classification, clustering and association rules, as well as their applications in business intelligence, customer relationship management, marketing, credit scoring, churn and survival analysis. The students will also be introduced to Big Data and Big Data analytics. Students gain hands-on experience with the Rapid Minder software application. They will perform pre-processing and apply data mining techniques to analyze the selected data to discover useful and actionable patterns and information.

**BUAN 312 Visual Analytics**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): BUAN 301**

This course provides an in-depth view of working with data to extract and present valuable information. Students will learn to collect, clean, manipulate, analyze, and visualize data from various sources correctly and efficiently. Through hands-on application, students will gain an understanding of what problems can occur when dealing with a variety of data and what solutions exist. This course introduces data visualization as an analytical tool, a medium of communication, and the basis for interactive information dashboards. Students will learn best practices in data visualization, sharpen analytical skills, and learn how to design dashboards for use by stakeholders. Computing is a major component of this course, and students will learn a number of in-demand technical skills.

**BUAN 390 Internship in Business Analytics**  
**3 Semester Credit Hours**

**Prerequisite(s): Senior Standing**

The Internship is designed for candidates on the undergraduate business analytics major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

**BUAN 401 Business Data Engineering**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): MIST 301, BUAN 311**

This course introduces students to the foundations of database management systems (DBMS), data warehousing fundamentals, practices, and technologies, and the challenges faced by businesses in managing large amounts of data and making meaningful use of this data for informed decision making. The course focuses on the basics of data pipelines, data pipeline flows and associated business use cases, and how organizations derive value from data and data engineering. Throughout the course, students will be introduced to various computing platforms for developing, deploying, configuring a wide range of data science applications for different business use-cases.

**BUAN 411 Business Intelligence & Big Data Analytics**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): MIST 301, BUAN 311**

As an intermediate-level course in business analytics, this course covers foundational theories and practices concerned with business intelligence and big data analytics. The course focuses on business applications of big data and provides an introduction to supporting technologies for big data analytics including MapReduce, Hadoop, Spark, and their ecosystems. The course also provides an introduction to the data analytics lifecycle to address business challenges that leverage big data. Examples of various types of BI use-cases for business decision-making will be discussed to allow students to identify and apply appropriate tools and methodologies to real-world business challenges and to facilitate data-driven decision-making.

**BUAN 444 Business Analytics Project Based Internship**  
**3 Semester Credit Hours**

**Prerequisite(s): BUAN 301**

The business analytics project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the business analytics specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

**BUSN 301 Business Communications (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s): ENGL 101**

This course introduces fundamentals of business models and writing as a learning tool. The interrelationships between accounting, finance, information systems, marketing, and operations are the subject of several "learning by writing" deliverables. The course teaches business students effective methods for communicating business-related issues to clients, management, and fellow employees. **(Writing Intensive Course)**

**BUSN 304 Business Law (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s): Junior Standing**

This course provides exposure to general legal environment of business, emphasizing government regulatory process, business ethics, and social responsibility. Regulatory topics include torts and crimes, product liability, intellectual property and cyber law, contracts, and issues related to employment and competition.

**BUSN 306 International Business (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s): ECON 104**

This course offers a multi-disciplinary approach to global economic issues from the viewpoint of managing international business. It introduces salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It provides a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business.

**BUSN 307 Research Methodology (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s): Junior Standing**

This course provides essential underpinning support for research elements of higher-level business courses. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of research work in the business field, and the generation of research proposals.

**BUSN 312 Public Policy and Business Ethics (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s): ECON 103**

This course introduces the policy and ethical dimensions of business as they relate to the various stakeholders inside and outside the organization. It includes topics such as actors, institutions and ideas in the policy process, business ethical theory, ethical decision making, ethical leadership, ethical and policy dilemmas, corporate social responsibility, and environmental sustainability.

**BUSN 390 Internship**  
**3 Semester Credit Hours**

**Prerequisite(s): Senior Standing**

The Internship is designed for candidates on the undergraduate finance, marketing, human resource management and accounting programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the

beginning of the internship and will serve to introduce the student to real-world business practices, culture, and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

**BUSN 391 International Internship**

**3 Semester Credit Hours**

**Prerequisite(s): Senior Standing**

This course is designed for candidates on the Bachelor of Science in Business Administration to provide a field experience in which students are able to apply knowledge and skills gained through course work, and to demonstrate their capacity to function successfully in a professional setting. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues in international settings. Through this internship, the students will learn, and apply real-world management skills on an international level, and throughout the process, the student will complete the internship report to fulfil the course requirements. The vast majority of the work and value of this course, however, will come in the form of the international, intercultural, and interpersonal experiences gained throughout the period in the international workplace.

**BUSN 498 Strategic Management**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Senior Standing**

This course is designed to help students apply functional knowledge gained from earlier courses and integrate these with more practical information and real-life experiences. The course takes the general management point of view, emphasizing the creation, implementation, and evaluation of strategy in organizations. Students adopt the viewpoint of upper management, strategize, and make essential "Big Picture" decisions, using standard strategic analysis frameworks. Students develop expertise in the analysis of complex business situations and in clearly presenting strategic findings both orally and in writing.

**CENG 315 Microprocessors**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 112 and EEEN 331**

Microprocessors as components in a computer system; programmer's view of a microprocessor's architecture; microprocessor instruction set; assembly language programming; interrupts; input and output; interfacing a microprocessor to memory and I/O devices from the programmer's view. At the end of the course, the students should be able to program a modern microprocessor in assembly or C language and perform hardware I/O interfacing.

**CENG 316 Microprocessors Lab**

**(1:0:3)**

**1 Semester Credit Hour**

**Prerequisite(s): CSCI 112**

**Corequisite(s): CENG 315**

This course introduces microprocessors in a laboratory environment. Topics include: assembly language programming; hardware interface and design with common microprocessor peripherals, uVision Simulation, Software Setup, GPIO: LEDs and Joystick, LCD Display, Keypad, LED in Assembly, Stepper Motor in C and Assembly, System timer, PWM, ADC/Infrared Sensor and DAC / Digital Music.

**CENG 335 Computer Architecture**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): EEEN 331**

**Corequisite(s): CENG 336**

This course covers details of microprocessor design including the instruction set architecture, memory design, and data path and control design. The course also emphasizes memory performance related concepts such as associativity and multi-level caching. Additional topics include virtual memory and performance speed-up techniques using pipelining, multithreading, and multiprocessing.

**CENG 336 Computer Architecture Lab**

**(1:0:3)**

**1 Semester Credit Hour**

**Prerequisite(s): EEEN 331**

**Corequisite(s): CENG 335**

This course covers modern computer system architecture and computer design principles. A Hardware Description Language is used to design basic components of a microprocessor datapath and control. Additional topics covered include Adders, MUX, Counters, ALU, registers/shift registers, RAM, pipelining, and cache memory.

**(3:3:0)**

This course discusses socket programming, client-server applications, peer-to-peer networks and applications, Web servers, datacenters and load balancing, CDNs, IPV6, NAT, anycasting, DHCP, multimedia networking, network operation and management and basic Internet measurements.

**(3:3:0)**

This course examines in detail the software development process. Topics include concepts such as software processes, software lifecycle models, requirement engineering, system modeling, architectural design, software design implementation, software testing, software evolution, and software reuse.

**(3:3:0)**

This course introduces the design and analysis of embedded computer systems that interact with the physical world. The course provides an integrated approach to the development of low-power systems with hardware, software, sensors, actuators, processors and networks. Topics include hardware and software architectures, assembly and C programming, real-time design, interrupts, multitasking, embedded software tools, and embedded systems performance. Students will work on a comprehensive project to design, implement, and evaluate a prototype embedded system.

**(1:0:3)**

In this lab course, students will design microcontroller-based embedded systems for various applications. Topics covered include interfacing to sensors and actuators from both hardware and software perspectives, assembly and C programming for embedded systems I/O, real-time design, and interrupts. Additionally, students will implement and evaluate a complete embedded system.

**(3:3:0)**

This course provides students with a deep understanding of parallel computing architectures including multicore CPUs and GPUs. The course emphasizes trade-offs involved in design of modern parallel computing systems, and techniques necessary to effectively utilize these machines. Topics covered include data-level parallelism, thread-level parallelism, GPU architectures, cache coherence, memory consistency, interconnection networks, reconfigurable computing and domain-specific parallel programming systems.

**(3:3:0)**

This course examines information security services and mechanisms in a network context. Topics include Cryptographic tools, User Authentication, Authorization and Access Control, Database Security, Malicious Software, Denial of Service Attacks, Firewall and Intrusion Prevention, Software Security, Operating System, Internet Security Protocols and Standards. Stack and Buffer Overflow, Security Auditing, Forensics, Ethical and Legal aspects, and Wireless Network Security.

**(2:0:6)**

**Corequisite(s): ENGR 450**

This is the first course of a capstone project that requires students to develop, design, and implement a solution to an engineering problem under the supervision of a faculty advisor. Students are required to consider the ethical, social, and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical and professional conduct.

**CENG 493      Senior Design Project II      (4:0:12)**

**4 Semester Credit Hours**

**Prerequisite(s): CENG 492**

Implementation of the project for which preliminary work was done in CENG 492. Project includes designing and constructing software and/or hardware, conducting experiments or studies, and testing and validating a complete system. At the end of the term, each team presents to a committee information related to its project in both written and oral formats.

**CHEM 095**      **Introductory Chemistry**      **(3:3:0)**

### 3 Semester Credit Hours

This course is an introduction to the basic chemistry notions such as atomic structure, molecules and mole concept, periodic classification of elements, naming of compounds, chemical formulas and equations and its balancing, the basic understanding of simple compounds of carbon and its applications, the influence of chemistry in the environment, and its scope for application in other fields of study. This course is not for degree credit. Placement in this course is based on the student's score on the Chemistry Placement Test.

|                 |                                   |                |
|-----------------|-----------------------------------|----------------|
| <b>CHEM 100</b> | <b>Chemistry in Everyday Life</b> | <b>(3:3:0)</b> |
|-----------------|-----------------------------------|----------------|

### 3 Semester Credit Hours

This course focuses on chemistry and its impact in everyday life. Basic chemical principles, the impact, and practical everyday applications are examined. Environmental concerns including energy, resource consumption and pollution as well as the crucial role involved in making informed opinions and better consumer choices in our society is addressed.

**CHEM 101**      **Chemistry in Everyday Life Lab**      **(1:0:3)**

### 1 Semester Credit Hour

A chemistry appreciation course for non-science majors. Covers basic chemistry concepts with technical/fundamental aspects. Emphasizes the practical chemical world of human beings and the chemical nature of everyday products. Hands-on laboratory sessions provide the opportunity to put the scientific method into action allowing further exploration and reinforcement of chemical principles.

**CHEM 111**      **Principles of General Chemistry**      **(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): CHEM 095 or Appropriate Score on Chemistry Placement Test**

**Corequisite(s): CHEM 112**

This course introduces students to the fundamental principles and laws of chemistry. This course covers coordination compounds, acid-base equilibrium, chemical equilibrium, solubility and complex-ion equilibria, thermochemistry, gas laws, and solutions. Introduction to organic chemistry, electrochemistry and nuclear chemistry.

**CHEM 112**      **Principles of General Chemistry Lab**      **(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): CHEM 111**

This course introduces chemistry lab techniques and provides hands-on experience in conducting chemical reactions that accompany the course CHEM111. It provides experience in basic techniques for the synthesis of coordination complexes and physical chemistry-based experiments.

**CHEM 120      Analytical Chemistry      (3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s):** CHEM 095 or Appropriate Score on Chemistry Placement Test

**Corequisite(s): CHEM 121**

This course emphasizes quantitative chemical analysis and underlying principles. Specific analytical techniques or concepts covered are: basic understanding of analytical chemistry including experimental error, statistical analysis of data, quality assurance, calibration methods, introduction of titrations, and polyprotic acid base equilibria. This course will introduce spectroscopic, separation, gravimetric, and combustion methods for quantitative chemical analysis.

- CHEM 121          Analytical Chemistry Lab** **(1:0:3)**  
**1 Semester Credit Hours**  
**Corequisite(s): CHEM 120**  
 Laboratory techniques and experiments arranged to accompany course Analytical Chemistry (CHEM 120). The aim of this course is to teach students the basic principles of analytical chemistry. The course covers the preparation of reagent, statistical evaluation of replicated data measurements and studies the validity of results. In addition, the analysis of unknowns using different analytical qualitative and quantitative methods is also enclosed.
- CHEM 211          General Chemistry I** **(3:3:0)**  
**Prerequisite: CHEM 095 or Appropriate Score on Chemistry Placement Test**  
**3 Semester Credit Hours**  
 This course forms the foundation of chemical concepts: basic facts and principles of chemistry, including atoms, molecules, ions, chemical reactions, gas theory, thermochemistry, electrochemistry, chemical kinetics and equilibrium, molecular geometry, and states of matter
- CHEM 212          General Chemistry I Lab** **(1:0:3)**  
**1 Semester Credit Hour**  
**Corequisite(s): CHEM 211**  
 This is a Laboratory class that is a companion class to CHEM 211 with laboratory techniques and reactions arranged to reinforce concepts. This laboratory course provides experience in physical-chemical measurement, synthesis-based experiments, qualitative, and quantitative analysis.
- CHEM 213          General Chemistry II** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CHEM 211**  
**Corequisite(s): CHEM 214**  
 This course covers the foundations of chemical concepts including: solutions, chemical reactions, acid base equilibrium, acids and bases, quantum theory of atoms, solubility and complex-ion equilibrium, chemistry of main group elements, polymer chemistry, and introduction of organic and nuclear chemistry, transition elements and coordination complexes.
- CHEM 214          General Chemistry II Lab** **(1:0:3)**  
**1 Semester Credit Hour**  
**Corequisite(s): CHEM 213**  
 This course introduces Laboratory techniques and reactions arranged to accompany CHEM-213. This laboratory course provides experience in physical /chemical measurement, synthesis-based experiments, qualitative, and quantitative analysis.
- CHEM 215          Organic Chemistry I** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CHEM 111 or CHEM 120 or CHEM 211**  
**Corequisite(s): CHEM 216**  
 This course provides a foundation in Organic Chemistry and includes the theoretical and synthetic aspects of the chemistry of carbon compounds.
- CHEM 216          General Organic Chemistry Lab I** **(1:0:3)**  
**1 Semester Credit Hour**  
**Corequisite(s): CHEM 215**  
 This course provides hands-on laboratory experience in techniques and chemical reactions in organic chemistry that accompany CHEM 215. The course includes basic techniques for the preparation, isolation, purification, inter-conversion, and spectroscopic study of common classes of organic compounds.
- CHEM 217          Organic Chemistry II** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CHEM 215**  
**Corequisite(s): CHEM 218**  
 This course focuses on aromatic organic chemistry, stereochemistry, organic chemistry reaction mechanisms, bio-molecules and the methods used for the synthesis of organic compounds.



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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|----------------|
| <b>CHEM 218</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>General Organic Chemistry Lab II</b>              | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                      |                |
| <b>Corequisite(s): CHEM 217</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                      |                |
| This course provides practical experience in common techniques for synthesis, separation, purification, and identification of organic molecules such as NMR, Mass Spectrometry, and Fourier Transform Infrared spectroscopy, as well as the qualitative analysis of functional groups.                                                                                                                                                                                               |                                                      |                |
| <b>CHEM 315</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Physical Chemistry</b>                            | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |                |
| <b>Prerequisite(s): CHEM 215</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      |                |
| <b>Corequisite(s): CHEM 316</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                      |                |
| This course will introduce the concepts of physical chemistry, properties of gases, first and second law of thermodynamics, physical transformation, phase diagram, chemical equilibrium, atomic structure and spectra, molecular structure, spectroscopy, chemical reactions, statistical thermodynamics, kinetics and dynamics of chemical reactions.                                                                                                                              |                                                      |                |
| <b>CHEM 316</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Physical Chemistry Lab.</b>                       | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                      |                |
| <b>Corequisite(s): CHEM 315</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                      |                |
| Introductions to the techniques and reasoning of experimental chemistry. Lab techniques and reactions arranged to accompany Physical Chemistry.                                                                                                                                                                                                                                                                                                                                      |                                                      |                |
| <b>CHEM 411</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Medicinal Chemistry</b>                           | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |                |
| <b>Prerequisite(s): CHEM 217</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      |                |
| This course examines principles of structural, physical, and physical-organic chemistry, including mechanistic considerations involved in synthetic organic chemistry, bioorganic chemistry, and the design of chemotherapeutic agents. This course includes mechanisms of action and factors that influence drug action within specific drug classes of pharmaco-dynamic and chemotherapeutic nature, drug structures, chiral drugs, and structure-activity relationships of drugs. |                                                      |                |
| <b>CHEM 420</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Forensic Chemistry</b>                            | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |                |
| <b>Prerequisite(s): CHEM 120</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      |                |
| <b>Corequisite(s): CHEM 421</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                      |                |
| Foundation of forensic chemistry: applications of the principles, methods, and instrumentation of chemistry to forensic problems. Covers topics most commonly applicable to the functions of a crime laboratory professional.                                                                                                                                                                                                                                                        |                                                      |                |
| <b>CHEM 421</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Forensic Chemistry Lab</b>                        | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                      |                |
| <b>Prerequisite(s): CHEM 121</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      |                |
| <b>Corequisite(s): CHEM 420</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                      |                |
| This course introduces laboratory techniques to the forensic analysis of trace evidences according to established forensic procedures. The focus of the course will be on identifying and understanding the nature of the samples, sample preparation methods, chemical and instrumental methods for analysis.                                                                                                                                                                       |                                                      |                |
| <b>CHEN 201</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Principles of Chemical Engineering</b>            | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |                |
| <b>Prerequisite(s): CHEM 211</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      |                |
| This course introduces the students to chemical engineering profession and basic calculations in mass and energy balance; phase equilibrium; and process flow sheeting. It includes applications on reactive and non-reactive chemical processes. Computer programs are used to implement these topics.                                                                                                                                                                              |                                                      |                |
| <b>CHEN 302</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Computer Applications in Chemical Engineering</b> | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                      |                |
| <b>Prerequisite(s): CHEN 201</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      |                |
| This course will focus on computer applications in chemical engineering including available software packages. Students will be introduced to the applications of software packages such as E-Z solve, Polymath, MATLAB, COMSOL, MATHCAD, Hysys, Aspen Plus for solving systems of algebraic and differential equations related to chemical engineering processes.                                                                                                                   |                                                      |                |

- CHEN 312            Chemical Engineering Thermodynamics            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CHEM 315**  
 This course aims to introduce the principles of Chemical Engineering Thermodynamics and illustrate their application. The content comprises the fundamentals of thermodynamics, such as thermodynamic properties (energy, entropy, enthalpy, heat capacity, etc.), the first and second law of thermodynamics (energy and entropy balance), heat of reactions, thermodynamics of ideal and real gases and liquids, vapor-liquid equilibrium and thermodynamics of chemical reaction processes. In addition, to the use of software tools to solve different thermodynamic problems.
- CHEN 351            Chemical Reaction Engineering            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CHEN 201 and CHEM 315**  
 This course provides a detailed analysis to the principles of chemical kinetics, and reactor analysis and design. The course covers kinetics of homogeneous and heterogeneous reactions, design of isothermal reactors such as Batch, Continuous Stirred Tank Reactor (CSTR) and Plug Flow Reactor (PFR). Other topics include data collection and handling, non- isothermal reactor design and multiple reactions. The last part of the course considers homogeneous and heterogeneous catalytic reactions. Further, the course introduces the link between the theoretical part and the laboratory scale chemical reactors through physical tour of the Chemical Engineering Laboratory facility.
- CHEN 352            Chemical Reaction Lab            (1:0:3)**  
**1 Semester Credit Hour**  
**Corequisite(s): CHEN 351**  
 This course covers experiments that deal with chemical reactions over different set of reactors including batch, tubular and continuous stirred tank reactors. Performing theoretical calculations and experimentation on real and virtual reactor systems for homogeneous phase as well as catalytic reactions.
- CHEN 371            Heat and Mass Transport            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CIEN 251**  
 This course introduces the physical origins and the governing laws for heat and mass transfer. The principal topics covered include identification of the driving forces for heat and mass diffusion, development of transport models from first principles, steady state and transient solutions, fundamentals and engineering treatment of convection heat and mass transfer, heat transfer with phase change, radiation heat transfer in addition to interfacial mass transfer and continuous stage-wise contact operations. Further, the course introduces the link between the theoretical and experimental design of heat and mass transfer equipment.
- CHEN 372            Heat and Mass Transport Lab            (1:0:3)**  
**1 Semester Credit Hour**  
**Corequisite(s): CHEN 371**  
 This course is composed of a set of selected experiments which demonstrate and apply the concepts of fluid, heat and mass transfer. The course aims to develop student's basic knowledge in: volumetric properties of pure fluids, principles and concepts related to heat transfer, principles of steam and power systems, principles of fluid flow (laminar flow and turbulent flow) and principles of heat exchangers. The lab also includes an open-ended design of experiment.
- CHEN 403            Fundamentals of Biochemical Engineering            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): BIOL 112 and CHEN 351**  
**Corequisite(s): CHEN 371**  
 This course aims to apply the principles of Chemical Engineering in bioprocess, in which biocatalysts, i.e. cells and enzymes, play key roles in producing new bioproducts and chemicals. The purpose of this course is to introduce the fundamental principles of biochemical engineering and present a wide spectrum of potential technological applications. This course includes introduction to biochemical and microbiological applications to commercial and engineering processes, including industrial fermentation, ultrafiltration and resulting waste treatment. In addition to introduction to enzyme kinetics, cell growth, energetics and mass transfer.

**(3:3:0)**

Sources of hazard in chemical and petroleum industry, management of safety and loss prevention, safety programs and safety rules, typical and sources of pollutant in chemical and petroleum operations, environmental protection, rules and regulation.

**(3:3:0)**

**Prerequisite(s): CHEN 351 and CHEN 371**

This course aims at introducing different techniques for petroleum refining. Topics include refinery feed stocks and products, field processes, crude distillation, coking and thermal processes, catalytic reforming and cracking, hydro- processing, and solvent treating processes. Solve petroleum-refining problems including material balances and process economics. Conduct lab experiments to characterize crude oils and final petroleum products. Students will do a case study of a typical refinery.

**(3:3:0)**

**Prerequisite(s): CHEN 371**

This course introduces the natural gas properties including real gas mixtures behavior in addition to flow of gas in reservoir, wellbore and surface pipelines. The course also deals with natural gas water systems, natural gas condensate systems, hydrate formation and inhibition, separation processes, field treatment of natural gas, absorption and adsorption processes, natural gas dehydration, sweetening and sulfur recovery, designing and sizing of the main equipment and performing nodal analysis for gas production system.

**(3:3:0)**

**Prerequisite(s): CHEN 351**

This course will provide a comprehensive overview of homogenous and heterogeneous catalytic science and technology for different chemical engineering industry. Course modules will cover the preparation and characterization of catalyst systems, kinetics and mechanism of catalyzed reactions including adsorption, desorption and surface reactions. Emphasis will be placed on understanding the design, synthesis, characterization and industrial application of various catalytic systems.

**(1:0:3)**

**Corequisite(s): CHEN 472**

This course aims to deepen the students' knowledge of the unit operations with a focus on distillation, absorption, liquid-liquid extraction, membrane and drying processes. This course requires students to integrate and apply knowledge from previous chemical engineering courses to design experiments, collect data, analyse results and make recommendations. Students will also work on developing oral and written technical communication skills. **(Writing Intensive Course)**

**(3:3:0)**

**Prerequisite(s): CHEN 371**

This course aims to introduce the fundamental principles of Separation Processes through examining both equilibrium-controlled separation processes as well as separation processes that involve both mass transport and equilibrium considerations. In order to probe the key concepts in depth, the course will focus primarily on binary and multi-component distillation, leaching, absorption, membranes and liquid-liquid extraction. Furthermore, Computer instruction will be employed throughout the course to illustrate important characteristics of these separation systems.

**(3:3:0)**

**Prerequisite(s): CHEN 371**

This course covers water resources, chemistry of saline water, modeling and analysis of single effect desalination combined with mechanical vapor compression and thermal vapor compression, modeling of multiple effect desalination (MED), modeling of single and multistage desalination processes, reverse osmosis. Also, cogeneration, scale formation, pre and post treatment operations will be discussed.

**(3:3:0)**

**Prerequisite(s): Senior Standing**

**(3:3:0)**

**Pre-requisite(s): CHEN 351 and CHEN 371**

**(3:3:0)**

**Prerequisite(s): MENG 241 and MENG 323**

**Corequisite(s): CHEN 472**

**(3:3:0)**

**Prerequisite(s): CHEN 302 and CHEN 351**

**(1:0:3)**

**Prerequisite(s): CHEN 312**

**Corequisite(s): CHEN 472**

**(3:3:0)**

**Prerequisite(s): CHEN 481**

**(3:3:0)**

**Prerequisite(s): ENGR 200**

**Corequisite(s): Senior Standing**

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**(3:0:9)**

The Senior Design Project entails team work involving development of a two-semester long workflow that includes preparing and presenting a proposal for a major design work. The design work may involve physic-chemical process design and testing, or simulation workflow design related to one or more chemical engineering activities, such as, reaction, separation, mass or heat transfer and dynamic and control– with due consideration of economic, environmental and societal impacts. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, communication, with consideration of ethical and professional conduct.

**(3:0:9)**

The Senior Design Project entails team work involving development of a two-semester long workflow that includes designing, implementing and evaluate the propose solution to a certain engineering problem. The design work may involve process design and testing, or simulation workflow design related to one or more chemical engineering activities, such as, reaction, separation, mass or heat transfer and dynamic and control – with due consideration of economic, environmental and societal impacts. The SDP-II entails finalizing the tasks planed in SDP-I. Students must present their work in a detailed technical report and defend their project in oral presentation.

**(3:3:0)**

A specific topic in Chemical Engineering that is not covered in other program courses is presented in a course format.

**(3:1:6)**

Introduction to computer graphics. Includes the following topics: geometric construction; line convention; orthographic projections, isometric projections; oblique projections; Descriptive projections; dimensioning, and sectional views. Computer-aided design and problem-solving techniques. Uses computer software AutoCAD in data analysis, data display and visualization techniques. Prepare drawings in civil and infrastructure engineering.

**(3:3:0)**

Vectors, force systems (2D and 3D), equilibrium of particles and rigid bodies (2D and 3D), structures (trusses, cables, frames and machines), distributed forces (centroids and centers of mass), internal forces (shearing force and bending moment diagrams), friction, and moment of inertia.

**(3:3:0)**

Types of loads, axial stress and strain of determinate and indeterminate system, normal and bending moment diagrams, torsion of determinate and indeterminate system, bending of beams, combined stresses, shearing stress and strain, Mohr's circle of stress and strain, thin walled pressure vessels.

**(3:3:0)**

Equilibrium of a rigid body, force systems, geometric properties of various shapes, internal forces in beams, trusses, and frames, analysis of three-hinged and tied arches, construction of shear force and bending moment diagrams for beams and frames, mechanical properties of materials, axial, shear, and bending stresses in members, stability, determinacy and indeterminacy of beams and trusses.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                            |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------|
| <b>CIEN 241</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Infrastructure Management</b>           | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |                |
| An introduction to infrastructure management systems, including management process, data collection technologies, interdependence, benchmarking and best practices. Other related issues, such as, resilience security of infrastructure systems are addressed.                                                                                                                                                                                                                                                  |                                            |                |
| <b>CIEN 251</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Fluid Mechanics</b>                     | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |                |
| <b>Prerequisite(s): MATH 213 &amp; CIEN 211 (Civil Engineering) and MATH 114 and PHYS 110 (Chemical, Mechanical, and Petroleum Engineering)</b>                                                                                                                                                                                                                                                                                                                                                                  |                                            |                |
| Properties of fluids, flow regimes, pressure and force calculations under hydrostatic conditions, manometers, buoyancy and stability of floating and submerged bodies, elementary fluid dynamics, conservation equations: mass, energy and momentum, continuity and Bernoulli equations. Boundary layer theory, fluid flow in conduits and determination of head loss in pipes.                                                                                                                                  |                                            |                |
| <b>CIEN 261</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Surveying</b>                           | <b>(1:0:3)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                            |                |
| <b>Prerequisite(s): CIEN 201 (for Civil and Infrastructure Engineering); INDS 122 (for Architecture Students)</b>                                                                                                                                                                                                                                                                                                                                                                                                |                                            |                |
| Principles of surveying by conducting tests on distance measurements, levels and theodolites, directions and angular measurements, topographic surveys, areas and volumes; traverse surveys; setting out horizontal and vertical curves, training on Total Station.                                                                                                                                                                                                                                              |                                            |                |
| <b>CIEN 265</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Geographic Information System (GIS)</b> | <b>(2:1:3)</b> |
| <b>2 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |                |
| <b>Prerequisite(s): CSCI 112</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                            |                |
| This course exposes students to the principles of GIS (hardware, software, people, data, and methods) and its environmental and infrastructure management applications. Subjects also include the acquisition and compilation of data from maps, field surveys, air photographs and satellite images. The course includes hands-on GIS state-of-the-art software.                                                                                                                                                |                                            |                |
| <b>CIEN 301</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Numerical Analysis</b>                  | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |                |
| <b>Prerequisite(s): MATH 214 and CSCI 112</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                            |                |
| Matrix Algebra: Matrix operations, inverse, determinant, Eigen vectors, Eigen Values and solutions of systems of linear equations. Application of computers to solve engineering problems using various numerical methods, mathematical modeling and error analysis, solution of algebraic and differential equations, numerical differentiation and integration and curve fittings. Presents application examples from civil and infrastructure engineering, industrial engineering and mechanical engineering. |                                            |                |
| <b>CIEN 311</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Structural Analysis</b>                 | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |                |
| <b>Prerequisite(s): CIEN 212</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                            |                |
| Structural forms, reactions, determinate structures, degree of determinacy, shear and moment diagrams for beams and frames, influence lines for beams, deflections (principle of moment area theorems, virtual work and conjugate-beam methods), Analysis of indeterminate structures by moment force method, slope-deflection method and distribution method.                                                                                                                                                   |                                            |                |
| <b>CIEN 316</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Structures for Architecture 2</b>       | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |                |
| <b>Prerequisite(s): CIEN 216</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                            |                |
| Introduction to concrete, structural steel, and masonry as construction materials. Design of rectangular reinforced concrete beams reinforced concrete one-way slabs, and simple design of reinforced concrete short columns according to ACI 318 design code. Design of simple masonry walls. Design of structural steel beams and structural steel columns according to LRFD design method.                                                                                                                    |                                            |                |
| <b>CIEN 321</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Reinforced Concrete Design</b>          | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                            |                |
| <b>Prerequisite(s): CIEN 311 and CIEN 331</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                            |                |

Properties of concrete and steel, cracked and uncracked section analysis, strength design, stress block, design for bending and shear, singly, doubly reinforced sections, rectangular sections, and T-sections, load cases and moment envelopes, bond requirements, development length and bar cutoffs, one-way solid and one-way ribbed slabs, design of short columns.

**CIEN 331                      Construction Materials**

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): CIEN 212 and CHEM 211**

Cement (types, manufacture, properties and hydration), aggregates, fresh concrete, hardened concrete (strength, strength development, shrinkage, creep), concrete in severe environment (hot and cold), durability, mix design by ACI and DoE methods, use of masonry, fiber reinforcement and metal form decking and structural steel joists.

**CIEN 332                      Construction Materials Lab.**

**(1:0:3)**

**1 Semester Credit Hour**

**Prerequisite(s): CIEN 212 and CHEM 211**

**Corequisite(s): CIEN 331**

Introduction to testing & specifications, concrete and mortar tests, aggregate testing, fresh and hardened concrete testing, non-destructive tests, design & testing of concrete mixes, brick testing, tests on steel reinforcing bars.

**CIEN 333      Geotechnical Engineering**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CIEN 212 and CHEM 211**

Index and classification of soils, water flow in soils (one- and two-dimensional water flow), soil stresses, soil compaction, distribution of stresses in soil due to external loads, consolidation and consolidation settlement, shear strength of soils, slope stability.

**CIEN 334**      **Civil Engineering Testing and Materials**

**(1:0:3)**

**1 Semester Credit Hour**

**Prerequisite(s): CIEN 331, CIEN 333**

**Corequisite(s): CIEN 361**

Tests on soils: specific gravity, grain size distribution, consistency limits, coefficient of permeability (constant and falling head), consolidation test, direct shear and tri-axial and CBR. Tests on fluids: Center of pressure, orifice and jet flow, Pressure variation in flowing fluid, momentum principle, flow through pipelines and closed conduits. Tests on asphalt properties, and marshall mix design. **(Writing Intensive Course)**

**CIEN 351      Environmental Engineering**

**(2:2:0)**

**2 Semester Credit Hours**

**Prerequisite(s): CIEN 251, CHEM 211**

Introduces principles of environmental engineering management and design pertaining to water supply and treatment, wastewater treatment, solid waste management, air pollution control, noise pollution measurement and control, and environmental impact assessment. Includes case studies from UAE.

## CIEN 361 Highway Engineering and Design

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CIEN 331**

Pavement types; Materials used in flexible pavement layers (soil, aggregate, and asphalt); Calculation of Equivalent Standard Axle Loads (ESAL); flexible pavement thickness design, Hot Mix Asphalt (HMA) design; Highway construction operations; highway performance.

## CIEN 362 Transportation Engineering

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CIEN 261**

Role of transportation engineering; Transportation system issues and challenges; Modes of transportation, Main components of highway, mode of transportation (driver, pedestrian, traffic, road); Geometric design of highways and highway facilities; Highway functional classification and special facilities; Intersection design and control. Introduction to rail, air, and water transportation.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                  |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------|
| <b>CIEN 421</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Structural Steel Design</b>                   | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 311</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Introduction to steel structures and practical design methods, steel sections, load factors and load combinations, design of various steel elements using LRFD-method, design of tension and compression members, beam design: Compact section criterion, lateral-torsional buckling, lateral supports, and various design aspects of beams. Beam-column elements, design of column base plates, design of simple bolted (or welded) steel connections. |                                                  |                |
| <b>CIEN 422</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Advanced Reinforced Concrete Design</b>       | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 321</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Structural layout, estimation of dead and live loads, serviceability, deflections and crack control, design for torsion, design of frames, moment redistribution, slender columns, approximate methods for two-way slabs, detailing of reinforcement.                                                                                                                                                                                                   |                                                  |                |
| <b>CIEN 424</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Bridge Design</b>                             | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 321</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Covers design of highway bridges; history, classification and aesthetics of bridge structures; design philosophy; loading, girder distribution factors; and load combinations; design of concrete deck slab; design of reinforced concrete T-beam and box girders bridges; and design of piers, bearings and abutments. Introduces pre-stressed concrete bridges.                                                                                       |                                                  |                |
| <b>CIEN 431</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Foundation Engineering</b>                    | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 333</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Site investigation, bearing capacity of shallow foundation, distribution of stresses in soils, settlement of shallow foundation, factors to be considered in foundation design, introduction to deep foundation, lateral earth pressure and retaining walls.                                                                                                                                                                                            |                                                  |                |
| <b>CIEN 440</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Infrastructure Financing</b>                  | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 241</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Fundamentals of infrastructure project financing, Public Private Partnerships (PPP), Build Own Transfer (BOT), variation models, financing structures, framework, bonds, life cycle analysis, sponsors and risk management.                                                                                                                                                                                                                             |                                                  |                |
| <b>CIEN 441</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Construction Management</b>                   | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 331</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Basic concepts of construction project management, Construction planning, project time Management, project cost management, project quality management, value engineering and project life cycle, construction process optimization, construction contracts, contracting methods, project specifications, bidding, procurement methods and contractor applications for payment procedures.                                                              |                                                  |                |
| <b>CIEN 442</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Construction Planning and Scheduling</b>      | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 441</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Principles of planning, monitoring, and controlling construction projects. Developing schedules using bar charts, precedence diagrams, program evaluation and review techniques (PERT), and linear schedules. Resource histograms and s-curves. Resource allocation and resource leveling. Schedule constraints. Earned value concept. Includes MS project and Primavera Project Planner software.                                                      |                                                  |                |
| <b>CIEN 444</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Construction Cost Analysis and Estimating</b> | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                  |                |
| <b>Prerequisite(s): CIEN 441</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |                |
| Perceptions of construction cost, engineering economic analysis, risk and uncertainty, range estimating, cost fundamentals, types of cost estimating, estimating construction materials cost, estimating construction labor cost, direct versus indirect costs, estimating construction equipment cost, cost of concrete structures, estimating project cost, time/cost trade-off analysis and contractor general requirements.                         |                                                  |                |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                         |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------|
| <b>CEN 451</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Infrastructure Systems</b>           | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| <b>Prerequisite(s): CEN 351 and CEN 362</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                         |                |
| Role of planning, system demands and networking in infrastructure systems, energy systems, water and waste water infrastructure, transportation systems, waste disposal and resources conservation, smart growth and effects of infrastructure on the environment, models of creating sustainable future development, planning, design and architecture in sustainable communities, and integrated infrastructure system models.                                                                                      |                                         |                |
| <b>CEN 454</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Water and Wastewater Supply</b>      | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| <b>Prerequisite(s): CEN 351</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                         |                |
| Sources of water, requirements for water supply projects, population's studies, rates of water consumption, variation in water demand, collection and distribution of water, distribution networks. Sources of wastewater, quantities and quality, sewage collection works, sewage purification works and disposal, primary treatment, secondary treatment, activated sludge system, design of sewer systems.                                                                                                         |                                         |                |
| <b>CEN 455</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Solid Waste Management</b>           | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| <b>Prerequisite(s): CEN 351</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                         |                |
| Evolution of solid waste management, integrated approach to solid waste management, sources, composition, and properties of solid waste, physical, chemical and biological properties of MSW and household hazardous wastes. Waste handling, separation, storage and collection. Building a sustainable future, application of life-cycle analysis to waste management systems, reuse technologies, energy recovery from liquid and solid wastes and product recovery from oily wastes.                               |                                         |                |
| <b>CEN 456</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Sustainable Urban Building Sites</b> | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| <b>Prerequisite(s): CEN 351</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                         |                |
| Concept of sustainability, how sustainability applies to infrastructure projects and programs, measuring sustainability, identification and design of sustainable technologies associated with water and energy management for infrastructure projects, green buildings and sustainable housing, sustainable transportation, energy, use of materials and waste management and water use. Includes case studies in sustainability.                                                                                    |                                         |                |
| <b>CEN 462</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Advanced Pavement Design</b>         | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| <b>Prerequisite(s): CEN 361</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                         |                |
| Pavement types, Pavement materials; subgrade stabilization methods; Principles of mix design using SUPERPAVE; Analysis of stresses in flexible and rigid pavement, Design methods of highway flexible and rigid pavements; Overlay design, Computer applications.                                                                                                                                                                                                                                                     |                                         |                |
| <b>CEN 464</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Airport Planning and Design</b>      | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| <b>Corequisite(s): CEN 362</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| Examines airport master planning, forecasting air travel demand and sustainable design of airports, including lighting, terminal facilities, noise-level control, aircraft control, airspace utilization and automobile parking.                                                                                                                                                                                                                                                                                      |                                         |                |
| <b>CEN 491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Senior Design Project (1)</b>        | <b>(2:0:6)</b> |
| <b>2 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |                |
| <b>Prerequisite(s): Senior Standing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                         |                |
| This is the first course of a capstone project that requires students to develop, design and implement a solution to an engineering problem under the supervision of a faculty advisor. Students are required to consider ethical, social and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resources, procurement, communication with consideration of ethical and professional conduct. |                                         |                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                          |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------|
| <b>CEN 492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Senior Design Project (2)</b>                         | <b>(4:0:12)</b> |
| <b>4 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| <b>Prerequisite(s): CEN 491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                          |                 |
| Continuation of the senior design project I. The project is a multidisciplinary interaction for infrastructure design and management that includes system analysis and inculcates sustainable engineering principles. Includes use of engineering software's especially project management software such as MS project, Primavera Project Planner and CYCLONE, writing a technical report and developing project drawings, specifications and details.                                       |                                                          |                 |
| <b>CEN 493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Special Topics in Civil Engineering</b>               | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| <b>Prerequisite(s): Department Consent</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                          |                 |
| Special up-to-date topic in the civil and infrastructure engineering.                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                          |                 |
| <b>COMM 101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Interpersonal Communication and Group Interaction</b> | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| The course presents the principles to develop appropriate and effective communication strategies in one-to-one and small group communication settings. It emphasizes analyzing and assessing communication skills to create and sustain effective communication in personal and professional relationships.                                                                                                                                                                                  |                                                          |                 |
| <b>COMM 102</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Reading Image and Film</b>                            | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| <b>Prerequisite(s): ENGL 101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                          |                 |
| This course allows learners to study aesthetic effects and strategies of persuasion in contemporary visual images. Learning engagement includes images in electronic media, film, photographs, cartoons, advertisements, and public performances and events.                                                                                                                                                                                                                                 |                                                          |                 |
| <b>COMM 111</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Introduction to Mass Communication</b>                | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| This course surveys the history and characteristics of mass communication as a field and set of intertwining industries and professions. Learners explore the role of mass media in modern society by considering the impact of technology, culture, government, and economics. Trends are considered in historical context.                                                                                                                                                                 |                                                          |                 |
| <b>COMM 112</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Introduction to Public Relations</b>                  | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| This course provides learners with basic knowledge about the theories, concepts, and best practices in communication within the public relations field.                                                                                                                                                                                                                                                                                                                                      |                                                          |                 |
| <b>COMM 113</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Introduction to Digital Media</b>                     | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| Introduction to Digital Media, allows students to learn and understand the basics of Digital Media. The course content discusses various technologies that will aid students to learn the language of visual imagery. The applications that will be focused on industry standard applications for many graphic design positions. In this course, students will build a blog as a means of communicating and presenting their work as producing print and digital layouts to a wide audience. |                                                          |                 |
| <b>COMM 212</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Media Writing</b>                                     | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| <b>Prerequisite(s): COMM 111 or COMM 112 or COMM 113</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                          |                 |
| This course introduces learners to the basic norms, values, standards and practices for writing for the mass media.                                                                                                                                                                                                                                                                                                                                                                          |                                                          |                 |
| <b>COMM 213</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Public Relations Writing</b>                          | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          |                 |
| <b>Prerequisite(s): COMM 111 or COMM 112 or COMM 113</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                          |                 |
| Learners improve their media-related writing skills with a focus on standard, professional presentation of information and messages in the public relations setting.                                                                                                                                                                                                                                                                                                                         |                                                          |                 |

- COMM 214      New Media and Digital Culture** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 This course examines the cultural impact of new digital technologies, such as the Internet and new telephone and audiovisual media. Students survey the origins of digital communication and the Internet; and they are introduced to contemporary scholarship on digital technologies, the Internet, and the institutions that control these technologies.
- COMM 215      Feature Writing** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 This course is designed to give students experience in recognizing and producing high-quality feature articles. Course materials and lectures will cover the basics of writing the newspaper and magazine feature story. Students will be exposed to, and write in, a diverse variety of approaches and techniques.
- COMM 221      Communication Analysis and Criticism** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 This course investigates media through a variety of theories and methods. The focus will be on some of the dominant critical perspectives that have contributed to our understanding of media and its roles in society. The course provides the basic vocabulary and concepts used to analyze different types of media.
- COMM 222      Intercultural Mass Communication** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 This course explores the communication strategies and techniques within a specific cultural milieu and how those strategies and techniques differ among various cultural milieu, and learners practice in a variety of communication modes how best to accommodate their rhetorical strategies in communicating to both intracultural and intercultural audiences.
- COMM 223      Globalization and Media Cultures** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 Learners in this course apply communication strategies to and investigate real-world case study challenges with a global worldview of various issues of global media cultures. Students learn how to develop a *Weltanschauung* from which to best communicate in modern technologies with world audience.
- COMM 224      Visual Storytelling** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 This course combines storytelling arts with social, mobile, and digital media technologies. In this course, students apply digital storytelling theory and techniques to write, produce and publish digital stories. They integrate images, text, video and audio to create digital stories, and acquire competency in the use of digital media applications.
- COMM 225      Visual Communication and Digital Photography** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ENGL 101 or COMM 111**  
 This course explores photography, photographic composition and aesthetics as a form of visual communication. Through a blend of theoretical and practical approaches, students will develop an aesthetic and creative understanding of photography as well as gain knowledge of technical photographic terms and digital photography skills.
- COMM 226      Graphic Communication Design and Print Production** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ENGL 101 or COMM 111**  
 This course is designed to expose students to study and practice graphic communication design and print production fundamentals. In this course, students apply layout theory and varying techniques to write, produce, and publish news stories in the production process. They integrate images, and text, to create a page makeup, with the use of different digital media applications.

- COMM 227      Film and Video Editing** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): ENGL 101 or COMM 111**  
 This course is designed to provide students with the essential skills and knowledge required for video and film editing. Students will learn the principles and techniques of video editing, including the use of industry-standard software tools and technology.
- COMM 311      Media Law and Ethics** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 Learners engage the philosophical underpinnings of ethics and the core principles of journalism and mass communication to develop an understanding and appreciation of the field's normative ethical values. Students will learn how to apply an ethical decision-making framework to a variety of challenges.
- COMM 321      Theories of Media Communication** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 212 or COMM 222 or COMM 223**  
 Learners identify the major concepts, issues, and theories of media communication, and learners identify and use communication theories in a variety of best practices to demonstrate effective use of the theories learned in the course.
- COMM 322      Digital Resources and Content** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 111 or COMM 112 or COMM 113**  
 This course is designed to present fundamentals of open /data collecting, gathering, cleaning, analyzing, and create data visualizations. Students will learn how database journalism turns to data journalism; and how data became a strong role in transferring more efficient growing online resources, tools, and techniques.
- COMM 323      News Reporting** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 212 or COMM 215**  
 This course will engage learners in the values, norms and professional practices newsgathering and writing. Emphasis is on traditional reporting methods, including interviewing and observation, and on the ethical, reliable presentation of news in print/digital formats.
- COMM 334      Digital Journalism** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 212 or COMM 213 or COMM 222**  
 This course introduces students to the multimedia production skills and techniques associated with broadcast media. Learners will integrate text, audio, photos, and video to produce broadcast journalism content that is attractive, accessible, easy to navigate, and appropriate for the platform and the audience.
- COMM 337      Public Relations Cases** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 212 or COMM 213 or COMM 222**  
 This course gives learners the context, knowledge, and skills to examine and critically analyze real-world public relations problems and cases. Students will study and apply communication and public relations theories to case studies and problems. Learners examine the four steps included in the design of public relations programs that include research, planning, implementation, and evaluation.
- COMM 344      Public Relations Theory** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 212 or COMM 213 or COMM 222**  
 This course introduces students to the public relations process, focusing on the strategic planning of PR programs, public relations tactics, media relations management, crisis management, and event management. Students will learn how to develop goals, design strategies, and select tactics that can better serve an organization's goals.

**COMM 390      Internship in Mass Communication**  
**3 Semester Credit Hours**

**Prerequisite(s):** Prior completion of 90 credits

The Internship for Mass Communication is eight weeks (240 hours), and provides exposure to practices, challenges, and opportunities in real-world Media settings. Students apply real-world work strategies throughout the internship, and engage in self-reflection, and incorporate employer feedback to enhance their professional skills in creating an initial record of employment.

**COMM 391      Communication Research Methodology** **(2:2:0)**  
**2 Semester Credit Hours**

**Prerequisite(s):** COMM 212 or COMM 222 or COMM 223

This course provides an introduction to research methods and the philosophical underpinnings of research inquiry in the field of communication. It includes the topic adherence, overviews of quantitative, qualitative, and mixed-method research methodologies, a range of alternative research methods, including observation, archival research, questionnaire surveys, case studies, and experimentation, research design, data collection, and data analysis, the ethical implications of research with human and non-human subjects, and appropriate connections between research questions and methodologies.

**COMM 421      Social Media: Audiences and Messages** **(3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s):** COMM 311 or COMM 321 or COMM 334

Learners explore social media as a unique platform for interaction with multiple audiences and learn how to plan strategies to engage with key stakeholders and disseminate persuasive, effective messages.

**COMM 422      Streaming** **(3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s):** COMM 334

This advanced course covers the technical aspects of live video streaming, podcasting and radio production including live production techniques, audio recording, mixing, and editing techniques.

**COMM 423      Interactive Multimedia** **(3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s):** COMM 334

The course will examine the history, aesthetics, and cultural implications of interactivity through media. Students will gain experience and critical analysis skills by engaging with interactive multimedia works. Students will learn design strategies and development techniques for interactive media and will have the opportunity to explore the latest innovations in this area.

**COMM 424      Capstone: Multimedia Storytelling** **(3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s):** COMM 423

**Prerequisite(s):** COMM 321 or COMM 323 or COMM 334 or COMM 391

this course, learners apply skills in newsgathering, writing, and multimedia to produce an individual and a group project of publication quality that demonstrates their proficiency in multimedia storytelling.

**COMM 425      Media Management and Entrepreneurship** **(3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s):** COMM 212 or COMM 223

This course introduces students to key concepts, models, and thinkers in media management and examines the practice of and emerging models in entrepreneurial journalism. The course aims to equip learners with analytical tools to understand the current state of media and develop new models for the future.

**COMM 427      Capstone: Media Production in Public Relations** **(3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s):** COMM 321 or COMM 323 or COMM 334 or COMM 391

The course provides learners with practical instruction and experience in a variety of media modes of communication. Students focus on producing clear, persuasive messages in attractive formats.

- COMM 450      Selected Topics in Communication      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 212 or COMM 222 or COMM 223**  
 This course examines current and emerging issues in communication theory, research, and practice, focusing on specific media and public relations topics, trends, and challenges. Students will explore various topics, ideas, and issues that are not typically covered in standard courses. The course will provide an opportunity for students to engage in critical analysis and discussion, as well as research and writing, on a range of contemporary communication issues (**Writing Intensive Course**).
- COMM 492      Mass Media and Public Opinion      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): COMM 311 or COMM 321 or COMM 323 or COMM 391**  
 This course is designed to prepare students to be informed and critical consumers of polls and media coverage of them, to introduce students to basic theories and findings regarding the influence of mass media on public opinion, to provide students with firsthand experience in conducting and writing about public opinion research.
- CSAI 350      Introduction to Artificial Intelligence      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CSCI 215 and ENGR 200**  
 This course provides an introduction to the different sub-areas of Artificial Intelligence (AI). In addition, students learn basic concepts, methods, and algorithms of AI and how they can be used to solve practical AI problems. The topics include classical and adversarial search & heuristic, knowledge representation, probabilistic reasoning, convex optimization methods, Bayesian methods, reinforcement learning, and supervised and unsupervised learning techniques. Particular focus will be placed on real-world applications of the material.
- CSAI 351      Data Science      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CSAI 350 and MATH 203**  
 This course provides an introduction to data science and highlights its importance in real world context. Topics include data science concepts, project lifecycle, tools & programming environment, fundamentals of Python programming, numerical processing, data visualization, exploratory data analysis, data preprocessing, parameter optimization, model performance evaluation, and applications of machine learning algorithms in Python (i.e., Naïve Bayes, k-Nearest Neighbors, Linear/Multiple/Logistic Regressions, Decision Trees, and Clustering Applications), natural language processing, and real-world data science case studies.
- CSAI 390      Internship I      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher**  
 This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.
- CSCI 391      Internship II      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Completion of 90 credits and a CGPA of 2.0 or higher**  
 This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.
- CSAI 450      Machine Learning      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CSAI 350 and CSAI 351**  
**Corequisite(s): CSAI 451**  
 This course introduces fundamental concepts of machine learning, and provides students with knowledge and understanding of the methods, mathematics, and algorithms used in machine learning. Topics include statistical learning concepts, linear &

quadratic discriminant analysis, resampling methods, model selection and regularization, regression & smoothing splines, generalized additive models, regression trees, bagging and boosting, support vector machines, principal components analysis, k-means clustering, hierarchical clustering, and neural networks.

## CSAI 451      Machine Learning Lab

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): CSAI 450**

This course, which is conducted within a laboratory environment, aims to familiarize students with several techniques used in machine learning. The topics covered include Linear Regression, Classification, Resampling, Linear Model Selection, Tree-Based Methods, Support Vector Machines, and Neural Networks. **(Writing Intensive Course)**

## CSAI 452      Natural Language Processing

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): CSAI 351**

This course introduces the fundamental concepts and techniques of natural language processing (NLP). Topics include text corpora and conditional frequency distributions, lexical resources and WordNet, raw text processing and regular expressions, text normalization and lemmatization, structured natural language processing (NLP) programs, part-of-speech tagging, automatic tagging, n-gram, & transformation-based tagging, document and sequence classification, maximum entropy classifiers and modeling linguistic patterns, information extraction, linguistic structure, named entity recognition, & relation extraction, grammatical structure & context free grammar, context free grammar parsers & dependency grammar, and feature based grammars.

## CSAI 453 Data Visualization

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSAI 350**

Data visualization is an essential skill required in today's data-driven world. This course presents principles and techniques to design and create data visualization based on gathered data and the goals of the task at hand. Topics include the value of visualization, data, tasks, validation, marks and channels, design guidelines, tables, networks and trees, spatial, temporal and textual data, interaction and navigation, and data reduction.

## CSAI 480 Big Data

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** CSCI 326

This course provides an in-depth coverage of various topics in big data from data generation, storage, management, transfer, to analytics. Topics include Hadoop ecosystem, Hadoop architecture, YARN & HDFS, Sqoop, Apache Hive & Impala, Apache Flume, Apache Spark, resilient distributed datasets (RDDs) in Spark, map reduce in Spark, Spark applications, Spark parallel processing, SparkSQL, workflow management with Apache Oozie, machine learning in Spark, and big data applications.

## CSAI 481 Computer Vision

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** CSAI 450

This course provides introduction to fundamental topics in computer vision and the application of statistical estimation techniques in this field. It is intended to give students a good basis for work in this important field. Topics include image representation, image processing, image analysis, image segmentation, object tracking, 3D shape reconstruction, feature detection and tracking, object detection, and convolutional neural networks.

## CSAI 482 Data Mining

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSAI 350**

Data Mining studies algorithms and computational paradigms that allow computers to find patterns and regularities in datasets, then perform prediction/forecasting and generally improve their performance through interaction with data. The course introduces the fundamental concepts of data mining techniques. Topics include data preparation, data classification, performance measures, association rule mining, clustering, outlier detection, collaborative filtering and data mining case studies.

**(3:3:0)**

The course covers basic and advanced techniques necessary to define, design, and implement Information Retrieval (IR) systems, including the following topics: basic IR models, basic tokenizing, indexing, and implementation of vector-space retrieval, experimental evaluation of IR, query operations and languages, text representation, web search, text categorization, language-model based retrieval, recommender systems, information extraction, and integration.

**(3:3:0)**

The Internet of Things is an emerging technology in which interconnected devices are embedded within everyday objects enabling them to send and receive data. This course introduces core concepts and networking protocols for IoT applications. Topics covered include: IoT architectures, smart objects, sensing and actuation, displaying information, WPANs, WBANs, IoT clouds, security and privacy for IoT systems, IoT robustness and reliability, and IoT applications.

**(3:3:0)**

The course provides an introduction to neural networks and deep learning. The topics to be covered include basic conceptual understanding of neural networks, deep learning programming environments, shallow neural networks, radial basis function networks, recurrent neural networks, convolutional neural networks, and deep reinforcement learning.

**(3:3:0)**

This course gives instructors the opportunity to cover the latest developments and contemporary issues in Artificial Intelligence. Instructors will provide a detailed course outline at the beginning of the semester.

**(3:3:0)**

This course introduces fundamentals of robot modelling and control. Topics include forward and inverse kinematics, Jacobians, trajectory design, motion planning, in addition to relevant applications and technologies such as mobile robotics, industrial robotics and computer vision.

**(2:2:0)**

The course develops student understanding about various non-technical issues related to the field of computing. The course covers the field from economic, legal, ethical, and professional perspectives. Covered topics include code of conduct, ethics, intellectual property, cyber security, privacy, risk management, and teamwork.

**(3:3:0)**

This course serves as an introduction to the field of computer science and the computer's various layers. The course provides exposure to the following layers: information, hardware, programming, operating systems, applications, and communications. Additional topics include ethics, security, privacy, the impact of computing, and widely used software applications.

**(3:3:0)**

This course introduces students to computers and programming languages and more specifically the C++ language. Additionally, students learn to use computer programming as a problem-solving tool. The topics covered include basic



operations, data types, input/output, selection statements, control structures, arrays, functions, and strings, knowledge representation, neural networks and natural language, data summarization and visualization.

**CSCI 113                    Introduction to Computer Programming Lab                    (1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): CSCI 112**

This course introduces the use of computer programming as a problem-solving tool in laboratory environment. Topics in procedural programming include, simple data types, input/output selection statements, control loops, testing, debugging, and programming environments.

**CSCI 211                    Object Oriented Programming                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 112**

This course is an introduction to object-oriented programming principles and techniques using Java. Topics include Java elementary programming, and Java object-oriented features such as methods, objects, classes, access modifiers, constructors, immutable objects & classes, abstraction, encapsulation, inheritance, polymorphism, dynamic binding, object castings, abstract and interface classes, and exception handling.

**CSCI 215                    Data Structures and Algorithms                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 211 and MATH 225**

This course introduces data structures and various fundamental computer science algorithms. The course covers abstract data-type concepts, stacks, queues, lists, and trees. Several sorting and searching algorithms are covered. Additional topics include an introduction to graphs and their implementation and running time and time complexity measurement.

**CSCI 232                    Computer Organization                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 112**

This course provides a programmer's view of the execution of programs in computer systems. Topics covered include instruction sets, machine-level code, assembly language, performance evaluation and optimization, memory organization and management, address translation, and virtual memory.

**CSCI 312                    Operating System Fundamentals                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215**

This course covers the principles, components, and design of modern operating systems, focusing on the UNIX platform. Topics include system structure, process concept, multithreaded programming, process scheduling, synchronization, atomic transaction, deadlocks, memory management, and file system.

**CSCI 315                    Design and Analysis of Algorithms                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215**

This course introduces the design and analysis principles for various algorithms. The topics covered include searching algorithms, dynamic programming, greedy algorithms, Huffman coding, graph traversing algorithms, shortest path algorithms, linear programming, and NP-completeness.

**CSCI 326                    Database Systems                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 211**

This course is an introductory course on database management systems. The goal of the course is to present a comprehensive introduction to the use of data management systems. Some of the topics covered are the following: The Entity-Relationship Model, the Relational Data Model, the SQL language, the database design, and the database integrity and security.

**CSCI 372                    Compiler Design                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 232**

The course introduces the theory and practice of compilation. Topics include compiler architecture, components, phases, software tools, lexical roles and specifications, regular expressions, syntax roles and specifications, context-free grammars,

top-down parsing, bottom-up parsing, LR parsers & parse trees, syntax directed translation, syntax tree, abstract syntax tree, and finite automata.

**CSCI 388                      Programming Languages                      (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215**

This course is an introduction to basic concepts in the design of programming languages. The course focuses on programming languages within the functional and logic programming paradigms such as Scheme and Prolog. Topics include history of programming languages, language design criteria, functional programming, syntax, logic programming, semantics, and object-oriented principles.

**CSCI 390                      Internship I**

**3 Semester Credit Hours**

**Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

**CSCI 391                      Internship II**

**3 Semester Credit Hours**

**Prerequisite(s): Completion of 90 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

**CSCI 411                      Computer Graphics                      (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215**

**Corequisite(s): CSCI 412**

This course introduces interactive computer graphics and drawing algorithms. The topics covered include graphics output primitives and their implementations, two-dimensional and three-dimensional geometric transformations and viewing, hierarchical modeling, computer animation, spline representations, visible-surface detection methods, illumination models, and surface-rendering methods, and texturing and surface-detail methods.

**CSCI 412                      Computer Graphics Lab                      (1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): CSCI 411**

This course introduces computer graphics and drawing algorithms in a laboratory environment. The topics covered include graphics output primitives and their implementations, two-dimensional and three-dimensional geometric transformations and viewing, hierarchical modeling, computer animation, spline representations, visible-surface detection methods, illumination models, and surface-rendering methods, and texturing and surface-detail methods.

**CSCI 415                      Introduction to Parallel Programming                      (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215**

This course is an introduction to parallel programming principles and techniques. Topics include parallel computing memory architecture, memory organization, parallel programming models, parallel program design, performance evaluation, thread-based parallelism, process-based parallelism, message passing, asynchronous programming, and heterogeneous programming.

**CSCI 416                      Human Computer Interaction                      (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215**

This course provides an introduction to and overview of the field of human-computer interaction (HCI). The topics include usability principles, predictive evaluation, design management processes, graphic design, understanding users' requirements

gathering, task analysis, handling errors & help, prototyping & UI software, interaction styles, user models, evaluation, and universal design.

**CSCI 450                    Information Security and Privacy                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215 or Instructor permission**

This course is a survey of information security considerations as they apply to information systems analysis, design, and operations. Topics include information security vulnerabilities, threats, and risk management. Furthermore, the course introduces several cryptographic algorithms in addition to the privacy and secrecy of statistical databases and e-government applications.

**CSCI 462                    Data Communications and Computer Networks                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 112**

This course introduces computer networks. Topics include layering approach, functions of different layers, Internet applications (HTTP, DNS), reliable and unreliable transport (TCP and UDP), routing and IP addressing, data link layer services and protocols, and Ethernet.

**CSCI 463                    Data Communications and Computer Networks Lab                    (1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): CSCI 462**

This course provides students with hands on training on design, troubleshooting, modeling and evaluating of computer networks. Topics include network addressing, Address Resolution Protocol (ARP), basic troubleshooting tools, IP routing, and route discovery. Additionally, student will perform network modeling, simulation, and analysis using Packet tracer and WireShark analyzer. **(Writing Intensive Course)**

**CSCI 492                    Senior Design Project I                    (2:0:6)**

**2 Semester Credit Hours**

**Prerequisite(s): Senior standing**

This is the first course of a capstone project that requires students to develop, design, and implement a solution to a computing problem under the supervision of a faculty advisor. Students are required to consider the ethical, social, and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical and professional conduct.

**CSCI 493                    Senior Design Project II                    (4:0:12)**

**4 Semester Credit Hours**

**Prerequisite(s): CSCI 492**

Implementation of the project for which preliminary work was done in CSCI 492. Project includes designing and constructing software and/or hardware, conducting experiments or studies, and testing and validating a complete system. At the end of the term, each team presents to a committee information related to its project in both written and oral formats.

**CSCI 499                    Special Topics in Computing                    (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CSCI 215**

This course gives instructors the opportunity to cover the latest developments and contemporary issues in computing. Instructors will provide a detailed course outline at the beginning of the semester.

**ECON 103                    Principles of Microeconomics                    (3:3:0)**

**3 Semester Credit Hours**

This course introduces microeconomics in the context of current problems. This course explores how market mechanism allocates scarce resources among competing uses; uses supply, demand, production, and distribution theory to analyze problems.

**ECON 104                    Macroeconomic Policy                    (3:3:0)**

**3 Semester Credit Hours****Prerequisite(s): ECON 103**

This course introduces students to the applications of macroeconomics in the context of current problems to business decision making and economic policy. Students will be introduced to the economic way of thinking and how key concepts; theories and methods of modern economic analysis can be applied to everyday economic issues and problems. Key areas covered include the national economy; national income accounting; business cycles; inflation; economic growth and stability; unemployment; and monetary and fiscal policy. Particular emphasis is placed on reviewing contemporary economic issues and how economics permeates almost every aspect of business, highlighting economics as the fundamental discipline underpinning the study of most business specializations.

**EDUC 500 Foundations of Education****(3:3:0)****3 Semester Credit Hours**

This course examines historical, philosophical, social, and psychological foundations of education. Students will explore the nature of teaching and learning as well as UAE education policy.

**EDUC 511 Introduction to Applied Research****(3:3:0)****3 Semester Credit Hours**

This course provides students with introduction to quantitative, qualitative, and mixed-methods research design in the educational sciences. This is a practical course designed to guide students through the development of their applied research proposals into a gap in practice. **Thesis Research Course I**

**EDUC 515 Educational Psychology for Inclusive Learning****(3:3:0)****3 Semester Credit Hours**

This course provides an overview of child and adolescent development, teaching and learning theories, strategies, and approaches for inclusive education. Students will utilize covered concepts as the framing for the development of a personal philosophy of teaching and learning.

**EDUC 525 Theories and Applications of Leadership****(3:3:0)****3 Semester Credit Hours**

This course provides students with conceptual and practical knowledge in leadership theories, with an emphasis on application in real world education settings. Students will acquire skills grounded in various leadership theories and use this conceptual knowledge to create an individual, metacognitive Leadership Profile.

**EDUC 545 Budgeting and Resource Management****(3:3:0)****3 Semester Credit Hours**

This course provides students with a broad overview of school finance, budgeting, and the ways in which resources are allocated. Students develop an understanding of the economic, social, and legal principles that inform how schools fund day-to-day operations.

**EDUC 620 Evaluation of Instruction for Continuous Improvement****(3:3:0)****3 Semester Credit Hours**

This course will provide students with the fundamental concepts of instructional evaluation, observation, peer coaching and mentoring. The observation findings will be used to by students to design, create, present, and evaluate staff development.

**EDUC 640 Inclusive Leading of Diverse Groups****(3:3:0)****3 Semester Credit Hours**

This leadership course is designed to prepare students to lead diverse groups and/or teams of individuals in schools and where education meets industry. Throughout the duration of this course, students will engage in meaningful dialogue centered on these authentic, real-world scenarios.

**EDUC 660 Organizational Change and Continuous Improvement****(3:3:0)****3 Semester Credit Hours**

This course provides students with the knowledge and skills to carry out continuous school improvement initiatives which includes determining needs, planning for improvement, understanding challenges and roadblocks to initiatives, and implementing and evaluating results toward established goals.

**EDUC 675 Educational Policy and Community Engagement****(3:3:0)**

In this course students explore the governance of schools and the building of constructive relationships between schools and communities from socio-economical and sociological perspectives grounded in educational research.

**(3:1:6)**

**3 Semester Credit Hours**

**Prerequisite(s): EDUC 511**

This field-based course provides students with the tools needed to undertake data collection and analysis in real-world settings. Throughout the duration of this course, students will undertake data collection and analysis for their IRB-approved research project. **Thesis Research Course II**

**(3:3:0)**

### 3 Semester Credit Hours

This course focuses on the role of research in the work of school leaders; ways leaders use research to identify opportunities for improvement and change; relationship between theory, research, and practice; role of research and benchmarking in school improvement; and strategies for change in education.

**(3:1:6)**

**3 Semester Credit Hours**

**Prerequisite(s):** EDUC 680

This research capstone prepares students in the Master of Education program to finalize their original research for submission to an academic journal. This course is designed to enable students with requisite skills in disseminating research.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): MATH 113**

**Corequisite(s): EEEN 221**

Introduction to methods of representing continuous-time signals and systems, and interaction between signals and systems. Analysis of signals and systems through differential equations and transform methods; Laplace and Fourier transforms; frequency response of systems; stability in time and frequency domains. Presents application examples from communications, circuits, control, and signal processing.

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): EEEN 220**

Laboratory course to accompany EEEN 220. In this course, the student will acquire hands-on experience with programming in MATLAB. Topics include representation of different signals, system linearity and time invariance, analysis of a first - order system, implementing matched filter for Barker codes, response of second - order systems and damping ratio, synthesis periodic signals

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): PHYS 220**

Basic circuit concepts and DC analysis, circuit analysis techniques, circuit theories, fundamental operation of operational amplifiers and their applications, transient and steady state analysis of RL, RC, and RLC circuits and basic AC analysis.

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): EEEN 280 or MENG 231**

Laboratory course to accompany EEEN 280. In this course, students will experimentally verify circuit analysis concepts under DC excitation and transient response. They will use different measurement instruments and build DC electric circuits.

**(3:3:0)**

**3 Semester Credit Hours****Prerequisite(s): EEEN 280**

Review of AC sinusoidal circuit analysis with active and reactive power. Covers magnetically coupled inductors and ideal transformers, three phase circuits, Laplace transform, application of Laplace transform in circuit analysis, passive and active filter analysis and design, two port networks.

**EEEN 283            Electric Circuit Analysis II Lab****(1:0:3)****1 Semester Credit Hour****Prerequisite(s): EEEN 281****Corequisite(s): EEEN 282**

Laboratory course to accompany EEEN 282. In this course, students will experimentally verify circuit analysis concepts under steady state AC excitation. Students will use different measurement instruments and build AC electric circuits.

**EEEN 305            Electromagnetic Theory****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): PHYS 220 and MATH 214**

Course uses vector algebra and vector calculus. Covers topics related to electrostatic and magnetostatic fields, electric and magnetic properties of media, electric boundary value problems, Maxwell's equations, electromagnetic waves and plane wave propagation, Poynting theorem and transmission line theory.

**EEEN 320            Signal and Systems II****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 220 and MATH 203****Corequisite(s): EEEN 321**

Methods of representing and analyzing discrete-time signals and systems. Studies effects of converting from continuous-time to discrete time and presents Z-transform as convenient analysis tool. Emphasizes powerful concept of frequency response of systems developed in first semester. Studies random signals in continuous and discrete time. Presents application examples from communications, circuits, control, and signal processing.

**EEEN 321            Signal and Systems II Lab****(1:0:3)****1 Semester Credit Hour****Corequisite(s): EEEN 320**

Laboratory course to accompany EEEN 320. In this course, the student will acquire hands-on experience with programming in MATLAB. The experiments cover sound signals processing using MATLAB, smoothing data, difference equations, and discrete time Fourier transformation.

**EEEN 331            Digital System Design****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): PHYS 220****Corequisite(s): EEEN 332**

This course covers principles of digital logic and digital system design. Topics include number systems; Boolean algebra; analysis, design, and minimization of combinational logic circuits; analysis and design of synchronous and asynchronous finite state machines; and an introduction to VHDL and behavioral modeling of combinational and sequential circuits.

**EEEN 332            Digital Systems Design Lab****(1:0:3)****1 Semester Credit Hour****Prerequisite(s):****Corequisite(s): EEEN 331**

Laboratory course to accompany EEEN 331. In this course, the student will acquire hands-on experience with basic logic components, combinational and sequential logic circuits, and the use of VHDL.

**EEEN 333            Linear Electronics I****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 280**

Principles of operation and application of electron devices and linear circuits. Topics include semiconductor properties, diodes, bipolar and field effect transistors, biasing, amplifiers, frequency response, operational amplifiers, and analog design.

**EEEN 334            Linear Electronics I Lab****(1:0:3)**

**1 Semester Credit Hour****Prerequisite(s): EEEN 281****Corequisite(s): EEEN 333**

Laboratory course to accompany EEEN 333. In this course, the student will acquire hands-on experience with basic electronic components and circuits. Topics covered include: Semiconductor diodes, rectification, Zener diodes, BJT and FET transistors and Amplifiers.

**EEEN 350            Electric Machines****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 282****Corequisite(s): EEEN 351**

The general theory of electro-mechanical motion devices relating to electric variables and electromagnetic forces. Basic concepts and operational behavior of DC motors, induction and brushless DC Motors, and stepper motors used in control applications.

**EEEN 351            Electric Machines Lab****(1:0:3)****1 Semester Credit Hour****Corequisite(s): EEEN 350**

Laboratory course to accompany EEEN 350. In this course, students will acquire hands-on experience with the characteristics of dc motors and dc generators (separate, series, shunt and compound). They will learn to find the parameters of transformers and evaluate their performance characteristics. The starting, speed control and performance of 3-phase induction motors are also studied. **(Writing Intensive Course)**

**EEEN 360            Random Signal and Noise****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ENGR 200 and EEEN 220**

Introduction to set theory, probability, random variables, and random processes. Modelling various types of noise encountered in communication systems as random processes. Analyzing the system response in the presence of random noise processes and calculating the power spectral density.

**EEEN 412            Power Systems****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 350**

Electric Power Systems, Elements of a Power Systems; The analysis of power systems starting with the calculation of line resistance, line inductance, and line capacitance of power transmission lines; Analysis of power systems in terms of current, voltage, and active/reactive power; Per-Unit Quantities; Load Flow Study; Economic Dispatch; Symmetrical Components; Fault Study; System Protection; System transient and Stability issues.

**EEEN 421            Power System Protection****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 412**

Power systems protection schemes for transmission and distribution networks, connection and standards of current and voltage instrument transformers for protection and metering applications, Protective relays, Protection of generators; Differential protection - Problems with differential protection - Biased differential protection - Biased differential protection of generator - Over current and earth fault protection, P; Buchholz relay - Biased differential protection of transformers - Harmonic restraint – Harmonic blocking - Other transformer protections, Protection of transformers, Protection of transmission lines.

**EEEN 422            High Voltage Engineering****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 282**

The components of power system and their characteristics. Fundamental electric field calculations (Laplacian fields) in insulation systems of simple geometries, introduction to gas discharge physics, Townsends theory of electric breakdown in air and Paschens law and its implications on gas insulation strength. Experimental techniques applied in high voltage engineering.

**EEEN 423            Electrical Energy Systems & Fault Analysis****(3:3:0)**

**3 Semester Credit Hours****Prerequisite(s): EEEN 412**

Energy and power; forms of energy; energy conversion from energy sources including wind, solar, tidal, bio-fuel, wave, hydro, nuclear and fossil fuel. Structure of a modern power system: operating charts, voltage control, and matrix representation of transmission lines. Two port network representation of transmission lines, per unit system, fault analysis: symmetrical components, transformers: construction, operation, connections, and relevant calculations. Load flow analysis: network matrix representation, Gauss-Seidel and Newton-Raphson solution techniques. AC/DC conversion: converter types, dc transmission, advantages compared to AC transmission. Over-voltages: switching and fault over-voltages, Bewley Lattice diagrams, switchgear principles, current chopping, insulation coordination. Modal component theory: wave propagation.

**EEEN 425 Smart Power Grid Systems Theory & Implementation****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 412**

This course explores a set of emerging concepts, technologies, applications and business models, and the related trade-off decisions involved in transforming the traditional centralized power grid into a climate and renewable energy-friendly "Smart Grid." A cross-disciplinary approach intended to deepen individual areas of expertise in the context of multidisciplinary teamwork. Basic Smart Grid literacy, applications of this knowledge base to specific "real world" case studies.

**EEEN 426 Renewable Energy Systems****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): Senior standing**

The course aims to introduce a general engineering/science audience to the basic concepts of renewable energy. In the interest of time some mathematical criteria will be covered, e.g. Betz limit for wind, limit of efficiency of WEC point absorber. Each lecture contains several examples from real world applications and in-progress industrial developments.

**EEEN 431 Digital Circuit Design****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 331 and EEEN 333**

Analysis and design of discrete and integrated switching circuits. Topics include transient characteristics of diodes, bipolar, and field-effect transistors; MOS and bipolar inverters; non-regenerative and regenerative circuits; TTL, ECL, IIL, NMOS, and CMOS technologies; semiconductor memories; VLSI design principles; and SPICE circuit analysis.

**EEEN 432 Nanotechnology Fundamentals & Applications****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): Senior Standing**

Students will be provided an introduction to the basic principles of the subject of nanotechnology with minimum mathematics involved. The applications of nanotechnology to various fields such as environment, health, alternative energy and electronic/optics will be addressed and explained. Students will be required to do research projects and presentations in class.

**EEEN 433 Linear Electronics II****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EEEN 333.**

Differential amplifiers, feedback circuits, power amplifiers, feedback amplifier frequency response, analog integrated circuits, operational amplifier systems, oscillators, wide band and microwave amplifiers, and computer-aided design.

**EEEN 434 Linear Electronics II Lab****(1:0:3)****1 Semester Credit Hour****Prerequisite(s): EEEN 334.****Corequisite(s): EEEN 433**

Laboratory course to accompany EEEN 433. In this course, the student will acquire hands-on experience with Electronic Amplifiers, active filters and oscillators. Topics covered include: Cascade amplifiers, differential amplifier, active filters, oscillators, and feedback amplifier concepts. **(Writing Intensive Course)**



**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): EEEN 305**

**Corequisite(s): EEEN 333**

Introduces optoelectronic devices for generation, detection, and modulation of light. Topics include electro-optic modulators, gas, solid state and semiconductor lasers, photodetectors, and detector arrays.

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): EEEN 333**

Course examines the application of electronics to energy conversion and control. The subject covers modern power semiconductor devices e.g., diodes, thyristors, MOSFETS, and other insulated gate devices; Static and switching characteristics, gate drive and protection techniques; Various DC-DC, AC-DC and DC-AC converter circuit topologies, their characteristics and control techniques; Analysis of input and output waveforms of these circuits; and their applications. Utility interference and Harmonic issues for power electronics circuits.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): EEEN 220 or MATH 214**

Introduction to feedback control systems; Block diagram and signal flow Graph representation; Mathematical modeling of physical systems; Stability of linear control systems; Time-domain and frequency-domain analysis tools and performance assessment; Lead and lag compensator design; Multi input multi output systems; Routh, Nyquist; Bode and root locus diagrams; Introduction to state variable techniques; state transmission matrix and state variable feedback.

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): EEEN 451**

Laboratory course to accompany EEEN 451. In this course, the student will acquire hands-on experience with feedback control circuits/systems. Topics covered include; Introduction in control systems, DC motor speed controller, open loop/closed, Proportional Integral Derivative PID controller and position control.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): EEEN 220 and ENGR 200**

Introduction to analog and digital communications. Topics include review of important concepts from signals and systems theory and probability theory; Gaussian processes and power spectral density; digital transmission through additive white Gaussian channels; sampling and pulse code modulation; analog signal transmission and reception using amplitude, frequency and phase modulation; and effects of noise on analog communication systems.

**(1:0:3)**

**1 Semester Credit Hour**

**Prerequisite(s): EEEN 460**

**Corequisite(s):** EEEN 464.

Laboratory course to follow EEEN 460 and accompany EEEN 464. In this course, the student will acquire hands-on experience with fundamental blocks of Analog and Digital communication systems. Topics covered include: Amplitude and Angle Modulation and demodulation, sampling and reconstruction, PCM Encoding & PCM Decoding and digital modulation and demodulation.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** EEEN 220 and ENGR 200

Introduction to modern data communications and computer networks. Topics include point-to-point communication links and transmission of digital information, modems, and codecs; packet switching, multiplexing, and concentrator design; multi-access and broadcasting; local area and wide area networks; ISDN; architectures and protocols for computer networks; OSI reference model and seven layers; physical interfaces and protocols; and data link control layer and network layer.

- EEEN 464            Digital Communication Systems            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EEEN 460**  
**Corequisite(s): EEEN 461**  
 Introduces digital transmission systems. Topics include quantization, digital coding of analog waveforms, PCM, DPCM, DM, base band transmission, digital modulation schemes, ASK, FSK, PSK, MSK, QAM, pulse shaping, inter symbol interference, partial response, voice band and wideband modems, digital cable systems, regenerative repeaters, clock recovery and jitter, multi path fading, digital radio design, optimal receiver design, MAP receiver, and probability of error.
- EEEN 466            Digital Signal Processing            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EEEN 320 and STAT 346**  
 This course provides a thorough treatment of digital signal processing including the fundamental theorems and properties of discrete-time linear systems, filtering, sampling, and discrete-time Fourier Analysis.
- EEEN 467            Mobile and Wireless Communications            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EEEN 460**  
 Cellular systems design fundamentals, fading and multipath channels, Modulation techniques for mobile radio systems, Diversity and combining techniques for mobile radio systems, multiple access techniques for mobile systems, Mobile systems and standards.
- EEEN 472            Antenna Theory and Design            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EEEN 305**  
 Course provides the fundamental knowledge in the theory and design of antennas. The theory of electromagnetic radiation is introduced and the fundamental antenna properties and parameters are explained. Standard antenna characterization parameters such as impedance, far-field radiation pattern, gain, directivity, bandwidth, beam width, polarization, efficiency, antenna temperatures are studied. The electromagnetic theory behind antenna operation and an overview of different antenna systems such as monopoles, dipoles, wire antennas and loop antennas etc... are discussed. The principles of analysis and design of antenna arrays are discussed.
- EEEN 473            Radio Frequency and Microwave Engineering            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EEEN 305**  
 This course covers a broad range of topics in the field of radio frequency (RF) and microwave engineering. This includes transmission lines, waveguides, impedance matching, microwave resonators, RF filters, RF amplifiers, and passive RF and microwave devices (mixers, diplexers, etc.). Furthermore, RF/microwave communications link design will be provided.
- EEEN 474            Advanced Information theory and coding            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EEEN 460**  
 Advanced topics in information theory and coding. The course is divided into two main parts, namely, Source coding and data compression, and channel coding and error detection/correction codes. The first part covers, entropy, amount of information source coding techniques, Shannon Fano, Huffman, and Lempel-Ziv codes. The second part covers binary symmetric channels, Z-channels, and E-channels, channel capacity, mutual information, linear block codes and convolutional codes, Viterbi decoders and cyclic redundancy check codes.
- EEEN 481            Concepts of Multimedia Processing and Transmission            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EEEN 220 or CSCI 462**  
 The course introduces the fundamentals of signal processing and communications for multimedia applications. It covers various topics relating to audio, image and video processing, storage and transmission. It discusses the human visual and hearing systems and relates them to image and sound digitization processes. The course also covers various lossless and lossy methods for audio, image and video compression. In addition, it gives the student hands on experience on applying the presented processing techniques using suitable software packages.

**EEEN 492          Senior Design Project I**  
**2 Semester Credit Hours**

**(2:0:6)**

**Prerequisite(s): Senior Standing**

This is the first course of a capstone project that requires students to develop, design, and implement a solution to an engineering problem under the supervision of a faculty advisor. Students are required to consider the ethical, social and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical and professional conduct.

**EEEN 493                      Senior Design Project II**  
**4 Semester Credit Hours**

**(4:0:12)**

**Prerequisite(s): EEEN 492**

Implementation of project from EEEN 492. Project includes designing and constructing hardware, writing required software, conducting experiments or studies, and testing complete system. Requires oral and written reports during project and at completion.

**EEEN 499                  Special Topics in EEEN**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): Senior Standing**

Advanced and emerging topics in electronics and communication engineering. Topics are announced through the Schedule of Classes.

**EMAC 512      Managerial and Financial Accounting**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): Graduate Standing**

The course deals with managerial and financial accounting techniques and practices. The course examines the impact of cost and cost allocation on business performance analysis, using a variety of costing schemes. It presents accounting as an information system construct, and looks at methods for recording, presenting and analyzing accounting information. The course also looks at budgeting, ratio analysis and other fiscal management measures. The course discusses the impact, on Management and Financial Accounting, of contemporary management philosophies and techniques including JIT practices, quality assurance and performance measurement.

**EMFN 514      Managerial Finance**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): Graduate Standing**

The course provides detailed critical treatment of the theory and practice of financial management within organizations. Topics covered include financial modeling, breakeven analysis, investment decision-making (NPV, IRR, etc.), capital budgeting and structure, working capital management, analysis of financial statements, and risk analysis. There is additional treatment of managerial economics.

**EMFN 560      International Corporate Finance**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisites:** EMFN 514

This course focuses on corporate finance at a global environment context. It explores advanced concepts and practices applicable to multinational company's financing and investment decision- makings. Topics covered include financial analysis for investment decisions in an international setting, international money operations and capital markets, management of foreign exchange risk, direct foreign investment, funding of international projects, political risk analysis, currency derivatives and swap markets..

**EMFN 562 Investment Planning and Management**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): EMFN 514**

The course deals with the advanced analysis of equity securities and investment portfolios, predicated on relevant market hypothesis and capital market theory. The course looks at investment risk-return trade-off, asset pricing models, and stock price behavior. The course adds emphasis on stocks, bonds, and financial futures and options.

- EMFN 564      Financial Markets and Institutions      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EMFN 514**  
The course deals with financial markets, primarily global equity markets. It looks at markets for handling government debt instruments, and exchange-traded and over-the-counter financial derivative instruments such as futures, options, swaps, and asset-backed securities. There is detailed treatment of derivatives theories, derivatives exchanges, and valuation of derivatives, including standard and other non-standard options on a variety of underlying assets, in relation to relevant financial markets. The course also discusses emerging financial markets, and the effective management of risks emanating from these markets.
- EMGN 525      Research Methods for Business      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Graduate Standing**  
This course provides the necessary underpinning support for the EMBA dissertation and general business domain research. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of business research documents and EMBA dissertations.
- EMGN 555      EMBA investigative Thesis or Project I      (3:0:9)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EMGN 525**  
EMGN 555 is the first part of the EMBA investigative Thesis or Project course, serving to integrate the various topics studied in the School of Business EMBA program. Students undertake a work-based study that focuses on a chosen business /management topic. The outcome for EMGN 555 is to submit a research introduction, literature review, and research methodology that will be used for the completion of this course. Satisfactory performance of the students in this course will move the student into EMBA 556.
- EMGN 556      EMBA Investigative Thesis or Project II      (3:0:9)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EMGN 555**  
EMGN 556 is the second part of the six-credit hour research course, serving to integrate the various topics studied in the School of Business EMBA program. Students undertake a work-based study that focuses on a chosen business/management topic. The outcome is a thesis, an extended piece of work necessitating investigation of secondary data and empirical fieldwork, through collection of primary data, pertaining to an organization and/ or a business domain problem. Analysis of collected data is expected to generate critical commentary and recommendations.
- EMHR 590      Applied and Strategic Human Resource Management      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EMMG 510**  
The course deals with key areas of applied and strategic human resource management, essential for providing supervisors and human resource specialists with a thorough understanding of the strategic role of human resource management and the techniques available to management for making effective use of the human resources of an organization. The material is delivered from the perspective of strategic and applied HRM practice in both the UAE/GCC and internationally.
- EMHR 592      Managing Change and Innovation      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EMMG 510**  
This course discusses issues related to the management of change in organizations. The different types of change typically encountered in organizations are characterized using well- defined analysis frameworks. The effective management of corporate innovation is also considered, looking in detail at the ideas generation, development and diffusion phases.
- EMHR 594      Employee Relations and Compensation Management      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): EMMG 510**  
This course deals with legal and regulatory issues pertaining to employee relations from several perspectives. The compensation management aspect focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the

UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of payment, and incentive systems will be covered.

## EMIB 524 International Business

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): Graduate Standing**

The course looks in-depth at salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It offers a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business practice. The course additionally places stress on international business case analysis, within the UAE/GCC context.

**EMMG 510      Leadership and Managing People**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Graduate Standing**

This course provides an in-depth study of concepts relating to leadership and people management within organizations. The work integrates theory, research and applications, with an emphasis on context-driven learning. Students apply principles of leadership and people management to their own occupational situations and will investigate topical case studies. The material is delivered from the perspective of people-management practice in both the UAE/GCC and internationally.

**EMMG 515 Strategic Management**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Graduate Standing**

The course deals with the formulation, implementation and evaluation of strategies designed to give organizations a competitive edge. Corporate, business and operational – level strategy types are treated, and the use of various strategy analysis frameworks and models are discussed. Additional emphasis is placed on strategy formulation in the context of business practice in the UAE/GCC.

**EMMI 520      Managing Information Systems Performance**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Graduate Standing**

The course studies techniques and practices for analyzing business information systems performance with emphasis on support for business processes. The course gives a strategic and integrated view of the exploitation of information and communication technology, with particular emphasis on e-business for improving business performance and business efficiency. The course discusses IT-driven business strategy, and reviews IT-related factors that are necessary for enhancing the performance of organizations in the current competitive global business environment.

**EMMK 518      Marketing Management**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Graduate Standing**

This course emphasizes managerial aspects of marketing, including developing marketing strategies and plans, and integrating specific elements of the marketing process. The course additionally places stress on case analysis, as a means for learning the material and for students to develop problem-solving, group work, judgement and decision-making skills.

**ENGL 099**      **Foundations of College English**

**(3:3:0)**

**3 Semester Credit Hours –**

**Prerequisite(s):** Requisite score on the English Placement Test, IETLS, or TOEFL

This course prepares learners to achieve proficiency in standard written and spoken English by conforming to correct grammar, spelling, diction, and punctuation. This course emphasizes critical and creative thinking, reading, writing, speaking, and listening skills in preparation for the TOEFL or IELTS exam. This course is not for degree credit.

**ENGL 101      Composition**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** ENGL 099, Requisite score on the English Placement Test, IETLS, or TOEFL

The course provides intensive practice in drafting, revising and editing expository essays for a variety of audiences. It studies the logical, rhetorical, and linguistic structures of expository prose, and develops the ability to think critically and apply critical thinking skills of analysis, synthesis, and evaluation of texts from different genres. (**Writing Intensive Course**)

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): ENGL 101**

This course is designed to provide an overview of principles to develop effective presentations for public and professional settings while integrating appropriate technologies. Course content and assignments emphasize audience analysis, effective and coherent composition development, research strategies and skills, and presentation delivery methods to strengthen confidence and credibility.

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s):** Completion of a minimum of 45 credit hours and ENGL 101

This course is designed to build on the general writings skills and techniques students have acquired in ENGL 101 and other university courses, and to prepare students for completing advanced level writing, analysis, and research. (***Writing Intensive Course***)

**(2:1:3)**

**2 Semester Credit Hours**

**Prerequisite(s):** Math Placement Test score qualifying student for MATH 113

This course introduces functions of the engineer, professionalism, engineering ethics, problem-solving techniques, presentation of technical information, estimation, analysis, and design. The course will include a workshop where students develop basic skills in hand filing, turning, welding, carpentry, sheet metal fabrication, metal forming, and basic electric circuit connections.

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** Math 114

The course introduces principles of statistics and probability for undergraduate students in Engineering. The course covers the basic concepts of probability, discrete and continuous random variables, probability distributions, expected values, joint probability distributions, and independence. The course also covers statistical methods and topics including data summary and description techniques, sampling distributions, hypothesis testing, and regression analysis.

**(2:2:0)**

**2 Semester Credit Hours**

This course introduces engineering students to the contemporary issue of sustainability and how it should guide engineering practice and design. Topics covered include sustainability concepts, SDGs, climate change, environmental issues, life cycle assessment, sustainable engineering systems (Infrastructure, buildings, clean energy and smart cities, and sustainable business management).

**ENGR 390**      **Internship I**

**3 Semester Credit Hours**

**Prerequisite(s):** Completion of 60 credits and a CGPA of 2.0 or higher

This is one of two supervised field experiences of professional-level duties where each is for 240 to 320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

## ENGR 391      Internship II

**3 Semester Credit Hours**

**Prerequisite(s):** Completion of 90 credits and a CGPA of 2.0 or higher

This is one of two supervised field experience of professional-level duties where each is for 240 to 320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designate site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

- ENGR 399 Undergraduate Research Project (3:0:9)**  
**3 Semester Credit Hours**  
**Prerequisite(s): A minimum of 90 C.H., 3.0 CGPA or Department Chair approval**  
 Student in this course will be engaged in a creative research project under the guidance of the faculty member from school of engineering. The student will prepare a paper suitable for publications in peer review journal, conference or scientific gathering. Fixed credit hours; 3 credits are assigned.
- ENGR 450 Engineering Seminar (1:1:0)**  
**1 Semester Credit Hour**  
**Prerequisite(s): Senior Standing**  
 Engineering ethics, professionalism, the role of engineers in society, current topics, and employment opportunities.
- ENGR 500 Engineering Foundation for Graduate Students (3:3:0)**  
**3 Semester Credit Hours**  
 This course is designed to provide a foundation in engineering principles, including strategic and systems thinking, statistical analysis of scientific data and sustainability. It is required for students who do not meet the CGPA for enrollment in graduate programs. The course aims to provide students with the skills and education required to pursue a Master's program.
- ENVS 101 Energy and Environmental Science (3:3:0)**  
**4 Semester Credit Hours**  
 The course is an inter-disciplinary study of environmental disruption and management, natural environmental systems, and the human impact on them. Other topics include energy procurement and use, waste management, water resources and water pollution, acid rain, global warming and ozone depletion.
- ENVS 102 Sustainability and Human-Environment Relations (3:3:0)**  
**3 Semester Credit Hours**  
 This course is designed to present the student with an overview of environmental issues. Humans are closely related to their environment and depend upon it to supply all the necessary resources to maintain life and to provide inputs for industrial activities.
- FNAN 300 Principle of Business Finance (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite ACCT 204**  
 This course teaches students the fundamental principles of finance. It covers the financial decisions that financial managers should make to maximize shareholders' wealth or a firm's value. These financial decisions include investment, financing, dividend, and working capital decisions. Students are taught how to apply financial theories to real life financial situations, financial ratio analysis, time value of money, stock valuation, bond valuation, capital budgeting and risk and return. It also discusses the difference between profit maximization and shareholders' wealth maximization. The course entails calculations and arriving at conclusions that help financial managers make financial decisions and measure financial performance.
- FNAN 303 Financial Analysis, Forecasting, and Valuation (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites: ACCT 204**  
 This course teaches students a range of techniques for examining, analyzing, and interpreting financial statements, and applying the information for both forecasting and making operational or a strategic financial decision that maximize a firm's value. The course aims at providing comprehensive understanding of the role of financial statements play and their impact on financial decision-making. Topics covered in this course include financial statement analysis, analysis of risk and return, development of financial models, financial forecasting, and firm valuation.
- FNAN 304 Principles of Risk Management and Insurance (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites: FNAN 303**  
 This course introduces students to the techniques of risk management and insurance. The course also discusses the theory and application of risk and insurance to real world financial decisions. Different disciplines define risk differently. However, in finance, we define risk as the possibility or chances of not achieving the firm's set objective of maximizing shareholder wealth or firm value. Students will study how financial managers identify, measure, control, and monitor risk in order to eliminate, reduce, transfer or avoid it in their operational and strategic financial decision-making.

**(3:3:0)**

**Prerequisites: FNAN 300**

**FNAN 390      Internship in Finance**

**Prerequisites: Senior Standing**

**FNAN 401      Advanced Financial Management**

**(3:3:0)**

**Prerequisites: FNAN 303**

**FNAN 411      Investment Analysis and Portfolio Management**

**(3:3:0)**

**Prerequisites: STAT 100, FNAN 303**

**FNAN 412      Futures and Options Markets**

**(3:3:0)**

**Prerequisite** OPMT 311, FNAN 303, FNAN 411

**FNAN 421      Money and Capital Markets**

**(3:3:0)**

**Prerequisite(s): FNAN 303**

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encourage critical thinking and problem-solving skills of students.

**FNAN 431 International Financial Management (3:3:0)**

**3 Semester Credit Hours**

**Prerequisite FNAN 303**

The course focuses on the management of contemporary financial operations of international corporations. The topics include foreign exchange risk, political risk, returns and risks of international projects, international money and capital markets, financial accounting, capital structure, and cost of capital.

**FNAN 444 Finance Project-Based Internship**

**3 Semester Credit Hours**

**Prerequisite(s): FNAN 303**

The finance project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the finance specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

**GEOG 100 World Regional Geography (3:3:0)**

**3 Semester Credit Hours**

This course explores the major regions of the world through a critical examination of key geographic concepts. Physical and environmental factors of each region, grounded in each region's history, culture, and society, are discussed. This frames a larger exploration of globalization, which also takes into account political and economic systems.

**GEOG 101 Introduction to Urban Social Geography (3:3:0)**

**3 Semester Credit Hours**

This course provides an overview of urban social geography. The student will learn about the history of cities, how they effect the lives of urban dwellers, and the importance of cities in a globalized world. Environmental concerns will also be addressed.

**HIST 100 Contemporary Middle Eastern History (3:3:0)**

**3 Semester Credit Hours**

This course examines the major issues affecting the Middle East in the 21<sup>st</sup> century while reviewing the origins and development of the modern Middle East. Students will develop an understanding of the social, economic, and political foundations that set the stage for the region this century.

**HIST 101 Ancient History of the Arabian Peninsula (3:3:0)**

**3 Semester Credit Hours**

This course concentrates on the geographical background of the Arabian-peninsula including location, descriptions of its provinces, routes, flora and fauna. It also focuses on the commercial importance and its political situation during the period from third Millennium B.C. to the rise of Islam or to the seventh century A.D.

**HIST 111 Introduction to World History (3:3:0)**

**3 Semester Credit Hours**

Analytical approach to an overview of world history that surveys the major features of principal existing civilizations of the world, as originally formed and as altered by key global processes, including forces of modernity.

**HIST 281 Survey of Middle Eastern Civilization I (3:3:0)**

**3 Semester Credit Hours**

Survey of Middle Eastern history from rise of Islam to present, emphasizing processes that led to emergence of economic, cultural, social, and political institutions that characterize region today. HIST 210 surveys the period from rise of Islam in 570 to medieval period (ca. 1258).

- HIST 282                      Survey of Middle Eastern Civilization II                      (3:3:0)**  
**3 Semester Credit Hours**  
 Second half of the survey of Middle Eastern history. HIST 282 surveys the period from about 1258 to the present.
- HTMT 311                      Survey of the Hospitality & Tourism Industry                      (3:3:0)**  
**3 Semester Credit Hours**  
 A survey of the history, trends, organizational structure, and economic impact of the hospitality and tourism industry on UAE economy. Some study of the problems originating in the operation and management of various segments of the hospitality industry in UAE will be introduced.
- HTMT 312                      Sustainable Tourism                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): HTMT 311**  
 The course will introduce the issues associated with the sustainable management of destinations and tourist attractions. Students will: learn the variety of ways that sustainability can manifest in the hospitality and tourism industry in UAE; understand the roles of international stakeholders involved in sustainable tourism; analyze case studies in key areas related to social, economic, and environmental impacts and consequences; and learn broad perspectives on how tourism relates to ecosystems, poverty, health, education, and social equity.
- HTMT 321                      Hospitality Operations I - F&B Management                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): HTMT 311**  
 This course involves the concepts and applications of management in food and beverage operations in various types of hospitality firms across UAE. Topics such as menu development, beverage management, catering, service, sanitation, foodservice design, and cost controls will be presented. Management approaches will be developed to provide quality products and services.
- HTMT 322                      Hotel Operations                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): HTMT 311**  
 This course will offer a thorough overview of the management of various lodging properties, taking a generalist view of lodging management, moving from department to department in a typical hotel. While this course will be general in nature, we will take a much more strategic perspective, identifying and considering issues of concern to general managers of all types of lodging properties, with a particular focus on profit maximization (yield/revenue management) and distribution channel management. As such, we will consider the broader political, economic, social, and technological environments and trends and their impact on lodging operations. For example, the recent global economic downturn has had a devastating effect on the lodging industry. Thus, the recession will form a basic issue that we will deal with throughout the semester.
- HTMT 390                      Internship in Hospitality and Tourism Management                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Senior Standing**  
 The Internship is designed for candidates on the undergraduate hospitality and tourism management major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.
- HTMT 411                      Destination Management                      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): HTMT 311**  
 The course will introduce the various issues associated with the management of a convention visitor's bureau (CVB). The course will analyze the mission, structure, and business activities of organizations that develop and promote a complex tourism destination in UAE. This includes the management functions of membership services, visitor services, financial and marketing concerns, research activities, and the strategic planning and evaluation of these efforts. Topics covered will include developing the following consumer segments: meetings/conventions, pleasure/tours, and festivals/special events.

- HTMT 412 Meetings and Conventions (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Prior Completion of 84 Credit Hours**  
 A course dealing with the many issues impacting the management of large convention and exposition centers. The course is taught from an organizational marketing base. Topics include meeting site selection, program planning and budgeting, legal issues and insurance problems, housing, food and beverage arrangements, transportation, exposition management, and audio-visual services."
- HTMT 444 Hospitality & Tourism Management Project Based Internship (3:2:3)**  
**3 Semester Credit Hours**  
**Prerequisite(s): HTMT 311**  
 The hospitality and tourism management project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem- solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the hospitality and tourism management specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.
- INDS 111 Color Theory and Rendering (3:2:3)**  
**3 Semester Credit Hours**  
**Prerequisite(s): None**  
 The aim of this course is to introduce color theory as a component of the design process that influences human perception of the interior space, behavior, performance and wellbeing and therefore influences design solutions. It offers skills in color selection and rendering for diverse spaces function and design concept communication.
- INDS 112 Design Communication 1: Sketching and Drawing for Interiors (3:2:3)**  
**3 Semester Credit Hours**  
 This course introduces students to the principles and techniques of interior design's form and space representation and communication. The course makes the students familiar with manual drawing tools and modes of documenting to develop students' sketching, technical drawing and ideation skills.
- INDS 121 Interior Design Process, Human Factors and Ergonomics (3:3:0)**  
**3 Semester Credit Hours**  
 The aim of this course is to provide knowledge in design process steps, components, human factors, anthropometrics and ergonomics and how they influence design solutions and contribute to the human wellbeing. It offers skills in gathering, analyzing human-centered evidence, application of ergonomics and other human factors in inclusive, universal design.
- INDS 122 Design Communication 2: Digital Media for Interiors (3:2:3)**  
**3 Semester Credit Hours**  
**Prerequisite(s): INDS 112**  
 This course introduces students to the most common 2D and 3D software used in the field of interior design to create and develop graphical and technical presentations. Emphasis will be on giving the students tools to digitize a sketches, drawings and concepts to examine design idea in terms of accuracy and workability of schematic design.
- INDS 211 Interior Design Studio 1 (4:0:12)**  
**4 Semester Credit Hours**  
**Prerequisite(s): ARCH 122**  
**Corequisite(s): INDS 111 and ARCH 223**  
 This course introduces a small-scale residential environment interior design project. A scenario-based design brief focuses on relationships between the users' needs and space's features leads students through the early phases of the design process (pre-design, schematic design, design development) with an aim of meeting space functional requirements and user(s) needs.
- INDS 212 History of Art and Interior Design 1 (3:3:0)**  
**3 Semester Credit Hours**  
 The course aims to provide knowledge in history as it relates to decorative arts, architecture, interior design, furniture, material culture and understanding of how the design of the built environment changes along with historical periods that are influenced by the social, political, physical changes in design of the built environment.

**(3:2:3)**

**Corequisite(s): INDS 112**

**(4:0:12)**

**Corequisite(s): INDS 223**

**(3:3:0)**

**(3:2:3)**

**(3:3:0)**

**(4:0:12)**

**(3:2:3)**

- INDS 313            Environmental Systems 1: Indoor Comfort and Human Wellbeing            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): INDS 224**  
**Corequisite(s): INDS 312**  
The course aims to explore environmental control systems, types, size, their integration into interior design project, and awareness of these systems in relation to indoor thermal comfort, indoor air quality, environmental and human wellbeing. The course focuses on water, waste, HVAC, electrical, fire safety, conveyance, and communication and security systems.
- INDS 321            Interior Design Studio 4            (4:0:12)**  
**4 Semester Credit Hours**  
**Prerequisite(s): INDS 311**  
**Corequisite(s): INDS 322 and INDS 323**  
This design studio deals with contemporary complex interior design project within the hospitality industry and will enable the students to explore the combination of wellness, aesthetics, functionally and operational aspects in designing city restaurants, cafes, and hotels.
- INDS 322            Furniture Design            (3:2:3)**  
**3 Semester Credit Hours**  
**Prerequisite(s): INDS 121 and INDS 312**  
The aim of this course is to provide knowledge and skills in materials, construction methods and details for furniture design and making. The course analyzes materials for furniture design in relation to their impact on environment, healthy and safe living and efficient work within the interior space.
- INDS 323            Environmental Systems 2: Design with light and sound            (3:2:3)**  
**3 Semester Credit Hours**  
**Prerequisite(s): INDS 313**  
The course aim to provide knowledge in space's lighting and acoustical design for space quality enhancement in relation to safety, comfort, performance, and wellbeing. The course suggests skills in light and sound integration into design of residences, stages, exhibitions, theaters, halls, etc. with the application of high-tech materials and technologies.
- INDS 351            Parametric Design            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s) INDS 312 or ARCH 336**  
This course is an advanced level course that will allow students to discover the fundamentals of Parametric modeling for design applications in Interior Design and Architecture. Students will investigate rule-based and parametric design concepts and techniques in the context of a generative modeling environment: Dynamo for Revit.
- INDS 352            Sketching and Painting            (3:3:0)**  
**3 Semester Credit hours**  
This course aims to improve students' skills in sketching and painting and focuses on two-dimensional media. Students continue to explore and experiment various techniques in art making. Two-dimensional media art works such as still life, compositions, collage are practiced with the application of various two-dimensional media.
- INDS 353            Landscape and Garden Design            (3:3:0)**  
**3 Semester Credit Hours**  
This course aims to introduce principles of landscape design throughout all its phases. Students study types of landscape, materials and its relationship with the context, environmental and human wellbeing and apply various representation techniques in landscape design projects.
- INDS 390            Internship I            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher**  
This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

**INDS 391            Internship II****3 Semester Credit Hours****Prerequisite(s): Completion of 97 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

**INDS 411            Interior Design Studio 5****(4:0:12)****4 Semester Credit Hours****Prerequisite(s): INDS 321**

This course extends the scope of interior design within a medium scale project of a healthcare environment. Students explore design approaches related to this complex and sensitive environment with respect to safety, efficiency, codes and regulations and that will satisfy all end users (including patients, staff, and visitors)' needs and wellness.

**INDS 412            Graduation Final Thesis Research****(2:1:3)****2 Semester Credit Hours****Prerequisite(s): INDS 321****Corequisite(s): INDS 411**

The course aims to prepare students for Graduation Project Design that follows next semester. It offers students opportunity to consider their evolving interest, targeted career plan and select a project theme, gather, analyze, systemize relevant to the theme data, and develop a design brief, project statement, design concept and program. **(Writing Intensive Course)**

**INDS 414            Professional Practice 1: Codes and Regulations****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): INDS 323**

The aim of this course is to equip students with knowledge of regional and international laws, codes, standards and guidelines that influence the encounter of the inhabitants with interior spaces and skills in application of laws, codes, standards and guidelines concerning built environment, sustainability, wellness, inclusive and universal design.

**INDS 421            Graduation Project Design****(5:0:15)****5 Semester Credit Hours****Prerequisite(s): INDS 412****Corequisite(s): INDS 422**

The graduation project is a capstone self-directed course under the supervision of one the program's senior faculty member where the project identified during the Graduation Final Thesis Research course is implemented. Students apply gained knowledge and skills in developing a holistic design approach to a design project.

**INDS 422            Professional Practice II: Ethics and Project Management****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): INDS 414**

The aim of the course is to provide awareness of the context of interior design practice, the influence of local and global context on design practices, understanding of the interior design profession principles, processes, responsibilities and its value to the society, interior design management, business formation, practices, and ethics.

**INDS 451            Portfolio Design****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): None**

This course aims to assist and mentor students in assembling a comprehensive portfolio of their academic work, enhancing students' graphic techniques, and provides skills in organizing the layouts of their projects in both book and digital formats.

**INDS 452            Photography****(3:3:0)****3 Semester Credit Hours**

This course aims to provide knowledge in types of cameras and skills in camera operating, taking pictures in different contexts, enhancing the pictures, processing and organizing pictures and application of photographs in interior design presentations.

- INDS 453            Advanced BIM: Scheduling and Coordination            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): INDS 312 or ARCH 336**  
 The course aims to introduce technologically based collaboration of interior design with multiple disciplines and provides knowledge and training on collaboration tools and methods using Building Information Modelling applications. It focuses on developing, reading, exploring, and coordinating working drawings and executive documents to avoid conflicts and ensure construction documents consistency.
- INDS 454            Research Methods for Interior Design            (3:3:0)**  
**3 Semester Credit Hours**  
 This course aims to introduce research processes in design related tasks within the framework of Evidence-Based Design Approach. It offers a review of design research methods such as literature review, sampling procedures, questionnaires interviews, and observations.
- ITEC 103            Fundamentals of Information Technology            (3:3:0)**  
**3 Semester Credit Hours**  
 The course focuses on the nature and uses of computers with an introduction to word processing, spreadsheets, presentation software and effective presentation information, in addition to related lab projects. Additional topics include computer systems organizations, communications and networking, legal and ethical issues, computer security, the Internet, and trending technologies.
- MATH 091            Pre-algebra            (3:3:0)**  
**3 Semester Credit Hours**  
 This course prepares students to take MATH 101 (Numbers and Data interpretation). It enables them to gain a command in operations with integers, fractions, decimals and percent, geometric figures and their measures, and pre-algebra topics including properties of rational numbers, operations of rational numbers, simplification of polynomials and equation-solving techniques. This course is not for degree credit.
- MATH 093            Intermediate Algebra            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Math Placement Test**  
 This course covers basic algebraic skills including factoring, solving basic linear and quadratic equations, the rules of exponents and radicals, and basic concepts of functions with specific examples, such as polynomial and rational functions. The course serves as a prerequisite to MATH 108. This course is not for degree credit.
- MATH 095            Pre-calculus            (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Math Placement Test**  
 This course is a prerequisite to MATH 113. It reviews mathematical skills essential to studying calculus. Topics include inequalities, absolute values, graphs, functions, exponential and logarithmic functions, and trigonometry. This course is not for degree credit.
- MATH 101            Numbers and Data Interpretation            (3:3:0)**  
**Prerequisite(s): MATH 093 or Appropriate Score on the Placement Test**  
**3 Semester Credit Hours**  
 This course meets the quantitative reasoning requirement, one of the Foundation requirements of the University General Education program. Topics include Logic Mathematics, Sets and Counting, probability and descriptive statistics. This course is broad in scope, emphasizing applications.
- MATH 102            Mathematics in Civilization            (3:3:0)**  
**3 Semester Credit Hours**  
 Throughout the history of mankind, the mathematical spirit has been a driving force in the development of the civilized world. This course creates an understanding of the impact of mathematical ways of thinking on the arts and sciences, on the development of technologies, and on the study of history. The goal is to illuminate the role of mathematics in the development of Eastern and Western civilizations and, thereby, to shed on human civilization from a mathematical point of view.

**(3:3:0)**

**Prerequisite(s): MATH 093 or Appropriate Score on Math Placement Test**

**MATH 111**      **Calculus with Life Sciences Applications**

**(3:3:1)**

**Prerequisite(s): MATH 095 or Appropriate Score on Math Placement Test**

MATH 113      Calculus I

**(3:3:1)**

**Prerequisite(s):** MATH 095 or Appropriate Score on Math Placement Test

MATH 114      Calculus II

**(3:3:1)**

**Prerequisite(s): MATH 113**

**MATH 203**      **Linear Algebra**

**(3:3:0)**

**Prerequisite(s): MATH 113**

MATH 213      Calculus III

**(3:3:0)**

**Prerequisite(s): MATH 114**

MATH 214 Elementary Differential Equations

**(3:3:0)**

**Prerequisite(s): MATH 114**

MATH 225 Discrete Mathematics

**(3:3:0)**

**Prerequisite(s): MATH 113**

**MBAC 512      Managerial and Financial Accounting**

**(3:3:0)**

**Prerequisite(s): Graduate Standing**

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course also looks at budgeting, ratio analysis and other fiscal management measures. The course discusses the impact, on Management and Financial Accounting, of contemporary management philosophies and techniques including JIT practices, quality assurance and performance measurement.

## MBFN 500 Business Graduate Foundation

**(3:3:0)**

### 3 Semester Credit Hours

The course is designed to provide a foundation in business for Graduate students who graduated from a non-business discipline. This course offers a comprehensive introduction to aspects of business. This course aims to equip students with business education and skills to successfully complete the business graduate program. **This course does not count toward degree credit.**

## MBFN 514 Managerial Finance

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): Graduate Standing**

The course provides detailed critical treatment of the theory and practice of financial management within organizations. Topics covered include financial modeling, breakeven analysis, investment decision-making (NPV, IRR, etc.), capital budgeting and structure, working capital management, analysis of financial statements, and risk analysis. There is additional treatment of managerial economics.

**MBFN 560      International Corporate Finance**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** MBFN 514

This course focuses on corporate finance at a global environment context. It explores advanced concepts and practices applicable to multinational investing, financing and dividend decision making that maximize firm value. Topics covered include financial analysis for investment decisions in an international setting, international money operations and capital markets, management of foreign exchange risk, direct foreign investment, funding of international projects, political risk analysis, currency derivatives and swap markets.

**MBFN 562 Investment Planning and Management**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): MBFN 514**

The course deals with the advanced analysis of equity securities and investment portfolios, predicated on relevant market hypothesis and capital market theory. The course looks at investment risk-return trade-off, asset pricing models, and stock price behavior. The course adds emphasis on stocks, bonds, and financial futures and options.

## MBFN 564 Financial Markets

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite MBFN 514**

The course deals with financial markets, primarily global equity markets. It looks at markets for handling government debt instruments, and exchange-traded and over-the-counter financial derivative instruments such as futures, options, swaps, and asset-backed securities. There is detailed treatment of derivatives theories, derivatives exchanges, and valuation of derivatives, including standard and other non-standard options on a variety of underlying assets, in relation to relevant financial markets. The course also discusses emerging financial markets, and the effective management of risks emanating from these markets.

## MBGN 525      Research Methods for Business

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Graduate Standing**

This course provides the necessary underpinning support for the MBA dissertation and general business domain research. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of business research documents and MBA dissertations.

## MBGN 555 MBA Investigative Thesis or Project I

**(3:0:9)**

**3 Semester Credit Hours**

**Prerequisite(s): MBGN 525**

MBGN 555 is the first part of the MBA Investigative Thesis or Project course, serving to integrate the various topics studied

in the School of Business MBA program. Students undertake a work-based study that focuses on a chosen business/management topic. The outcome for MBGN 555 is to submit a research introduction, literature review and research methodology that will be used, for the completion of this course. Satisfactory performance of the students in this course will move the student into MBGN 556.

## MBGN 556      MBA Investigative Thesis or Project II

**(3:0:9)**

### 3 Semester Credit Hours

**Prerequisite MBGN 555**

MBGN 556 is the second part of the three-credit hour research course, serving to integrate the various topics studied in the School of Business MBA program. Students undertake a work-based study that focuses on a chosen business/ management topic. The outcome is a thesis, an extended piece of work necessitating investigation of secondary data and empirical fieldwork, through collection of primary data, pertaining to an organization and/or a business domain problem. Analysis of collected data is expected to generate critical commentary and recommendations.

**MBHR 590      Applied and Strategic Human Resource Management**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): MBMG 510**

The course deals with key areas of applied and strategic human resource management, essential for providing supervisors and human resource specialists with a thorough understanding of the strategic role of human resource management and the techniques available to management for making effective use of the human resources of an organization. The material is delivered from the perspective of strategic and applied HRM practice in both the UAE/GCC and internationally.

## MBHR 592 Managing Change and Innovation

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s):** MBMG 510

This course discusses issues related to the management of change in organizations. The different types of change typically encountered in organizations are characterized using well- defined analysis frameworks. The effective management of corporate innovation is also considered, looking in details at the ideas generation, development and diffusion phases.

**MBHR 594 Employee Relations and Compensation Management**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): MBMG 510**

This course deals with legal and regulatory issues pertaining to employee relations from several perspectives. The compensation management aspect focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of payment, and incentive systems will be covered.

**MBIB 524      International Business**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Graduate Standing**

The course looks in-depth at salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It offers a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business practice. The course additionally places stress on international business case analysis, within the UAE/GCC context.

**MBMG 510      Leadership and Managing People**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): Graduate Standing**

This course provides an in-depth study of concepts relating to leadership and people management within organizations. The work integrates theory, research, and applications, with an emphasis on context-driven learning. Students apply principles of leadership and people management to their own occupational situations and will investigate topical case studies. The material is delivered from the perspective of people-management practice in both the UAE/GCC and internationally.

- MBMG 515 Strategic Management (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Graduate Standing**  
 The course deals with the formulation, implementation and evaluation of strategies designed to give organizations a competitive edge. Corporate, business and operational – level strategy types are treated, and the use of various strategy analysis frameworks and models are discussed. Additional emphasis is placed on strategy formulation in the context of business practice in the UAE/GCC.
- MBMI 520 Managing Information Systems Performance (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Graduate Standing**  
 The course studies techniques and practices for analyzing business information systems performance with emphasis on support for business processes. The course gives a strategic and integrated view of the exploitation of information and communication technology, with particular emphasis on e-business for improving business performance and business efficiency. The course discusses IT-driven business strategy, and reviews IT-related factors that are necessary for enhancing the performance of organizations in the current competitive global business environment.
- MBMK 518 Marketing Management (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): Graduate Standing**  
 This course looks in-depth at techniques for crafting effective marketing strategies, for a variety of market environments. Customer-focused marketing strategy design implementation and management are discussed using contemporary frameworks within a market-driven setting. The course emphasizes managerial aspects of marketing, including integrating specific elements of the marketing process. The course additionally places stress on case analysis, within the UAE/GCC context.
- MENG 201 Mechanical Engineering Drawing (3:1:6)**  
**3 Semester Credit Hours**  
 The course introduces the fundamental rules of an international language that enables ideas to be expressed and communicate in an easy and clear way through visual illustration. Includes the following topics: geometric construction; line convention; orthographic projections, Position student feet on the first step of the engineering design based on the visual representation. The course covers the following topics: isometric projections; oblique projections; Perspective projections; dimensioning, and sectional views. Computer software design, prepare parts drawing and assembly it using the software CREO.
- MENG 211 Thermodynamics I (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CHEM 211**  
 This course will consider the fundamental science of classical thermodynamics and its practical applications. Problem solving will be emphasized, including problem formulation, analytic, and computational solutions. Topics include the first law of thermodynamics, work, heat, properties of substances and state equations, the second law of thermodynamics and applications to engineering systems.
- MENG 212 Thermodynamics II (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MENG 211**  
 This course is a continuation of MENG 211 - Thermodynamics I. It provides more depth in the study of cycles, with applications to gas power and refrigeration cycles, and vapor and combined power cycles; mixtures of gases and vapors, psychrometrics, chemical reactions, and energy analysis.
- MENG 221 Dynamics (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CIEN 211**  
 Kinematics of a Particle, Kinetics of a Particle: Force and Acceleration, Kinetics of a Particle: Work and Energy, Kinetics of a Particle: Impulse and Momentum, Planar Kinematics of a Rigid Body, Planar Kinetics of a Rigid Bodies.

**(2:2:0)**

**MENG 241      Engineering Materials**

**(3:3:0)**

**Pre-requisite(s): CHEM 211**

**MENG 242      Engineering Materials Lab**

**(1:0:3)**

**Corequisite(s): MENG 241**

**MENG 311      Internal Combustion Engines**

**(3:3:0)**

**Prerequisite(s): MENG 212**

**MENG 312      Manufacturing Processes**

**(3:3:0)**

**3 Semester Credit Hours**

**Pre-requisite(s): MENG 241**

**MENG 313      Manufacturing Processes Lab**

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): MENG 312**

This laboratory course provides an introduction to manufacturing processes experimentation. Experiments include Oxy, Arc, Spot welding as well as mechanical fastening by riveting, screwing and assembling, metal fabrication and sheet metal, machining processes of milling, turning and CNC. Advance manufacturing technologies of LASER and FDM are introduced. (Writing Intensive Course).

- MENG 321          Vibration and Control** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MATH 214, MENG 221**  
 Introduction to Vibration, Oscillatory Motion, Free Vibration, Forced Vibration, Rotating Unbalance, Multiple-Degree-of-Freedom Systems, Introduction to control, block diagrams, modeling of systems, state space representation, Laplace transform, solution of linear systems, stability, input/output description, PID controllers, transfer function methods, tracking.
- MENG 323          Engineering Economy** **(3:3:0)**  
**3 Semester Credit Hours**  
**Pre-requisite(s): MATH 113**  
 Principles of economic analysis and methods in engineering including: time value of money, discounted cash flow techniques equivalence, economic measures of worth, single and multiple alternatives evaluation and selection, replacement decisions, cost estimation, equipment depreciation, the use of Minimum Attractive Rate of Return MARR and Benefit/cost analysis.
- MENG 342          Fluid Mechanics Lab** **(1:0:3)**  
**1 Semester Credit Hour**  
**Prerequisite(s): MENG 231**  
**Corequisite(s): CIEN 251**  
 This course is composed of a set of selected experiments about general fluid mechanics. The experiments will be either performed in groups by the students or demonstrated by the instructor. Individual class work will be strongly encouraged as well as teamwork. The lab also includes an open-ended design of experiment.
- MENG 361          Heat Transfer** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MENG 212 and MATH 214**  
 Introduction to heat transfer mechanisms, heat conduction equation, steady heat conduction including the thermal resistance networks, transient heat conduction, lumped systems, fundamental of convection and thermal boundary layers, external and internal forced convection, natural convection, boiling and condensation, thermal radiation, and heat exchangers.
- MENG 362          Thermal Sciences Lab** **(1:0:3)**  
**1 Semester Credit Hour**  
**Prerequisite(s): MENG 211 and MENG 231**  
**Corequisite(s): MENG 361**  
 This course is composed of a set of selected experiments which demonstrate and apply the concepts of thermodynamics and heat transfer. The experiments will be either performed in groups by the students or demonstrated by the instructor. Individual class work will be strongly encouraged as well as teamwork. The lab also includes an open-ended design of experiment.
- MENG 421          Theory of Machines** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MENG 221**  
 This course focuses on the kinematic and kinetic analysis of mechanisms. It introduces the fundamental concepts, definitions and terminologies in mechanisms, basic mechanisms and applications, linkages and mobility, dynamic analysis of cams, gears and gear trains, velocity and acceleration analysis in mechanisms, and static and inertia force analysis of machinery.
- MENG 422          Building Utilities II: Illumination, Acoustics, and Electrical Building Services** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): PHYS 110, MENG 468**  
 This course explores building electrical system and controls, lighting and acoustic design and assist integrating them into architectural design of the building. Evaluates systems types, components, and installation and maintenance procedures as it relates to building regulations and systems impact human's indoor comfort and on environment.
- MENG 441          Turbo Machinery** **(3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): CIEN 251**

Turbomachinery classifications and terminology. Implementation of dimensional analysis for predicting performance of turbomachines and designing engineering systems. Understand the fundamentals of energy transfer between rotating rotors and fluid flow. Demonstrate the ability to construct velocity diagrams for various turbomachines (axial-flow compressors and turbines, radial-flow compressors and turbines, pumps, fans, blowers, hydraulic turbines) and their relation to design. Perform elementary analysis for determining input/output work of various turbo devices. Design and selection of turbomachines for various engineering applications.

**MENG 451      Mechanical Design I**

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): CIEN 212, MENG 241**

Introduction to Mechanical Engineering Design, Materials Properties, Load and Stress Analysis, Deflection and Stiffness, Failure Prevention, Fatigue Failure, Design of Mechanical Elements, Screws Fasteners and Nonpermanent Joints, Mechanical Springs.

**MENG 452      Mechanical Design II**

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): MENG 451**

This course is a continuation to the machine design I course. Students will be introduced to the analysis and design concepts of various types of machine elements that include: bearings (journal and anti-friction); spur, helical and bevel gears; flexible drives and flywheels; clutches and brakes.

**MENG 453      Computer Aided Design**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): MENG 451**

This is an upper-year mechanical engineering course. It introduces students to the analytical basis to CAD software and the three main ways to represent an entity, namely wireframe, surface and solid modeling. The course aims at introducing the concept and importance of CAD as part of the design process. Also it focuses on mathematical representation and manipulation of geometry. The course introduces students to Computer-Aided Mechanical Design (CAMD) tools and their applications to mechanical systems design.

## MENG 455 Finite Elements in Machine Design

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): MENG 451**

The objective of this course is to learn how to design and analyze structural components of machine system, especially using the finite element method. The course exposes students to analytical and numerical methods for computing stresses and strains in structures, use of finite element software for static structural analysis and the application of design and failure criteria to ensure that mechanical components can carry the design load without failure. Another important area of the course is to make the students recognize the importance of self-education and life learning.

## MENG 461 HVAC and Refrigeration Systems

**(3:3:0)**

### 3 Semester Credit Hours

**Prerequisite(s): MENG 361 and MENG 212**

Review of psychrometry. Air conditioning processes. Thermal comfort, inside and outside design conditions. Ventilation and infiltration. Heating and cooling load calculations. Solar radiation. Water heating systems layout and design. Air systems design. Under floor heating. Review of vapor compression and absorption cycles; compressors, condensers, evaporators, expansion devices; refrigerants (including new ones); cooling towers; components of an absorption cycles, controls.

**MENG 462      Design of Thermal Systems**

**(3:3:0)**

**Semester Credit Hours**

**Prerequisite(s): MENG 361 and CIEN 251**

Application of principles of fluid mechanics heat transfer and thermodynamics in the component design of thermal systems. Examples are drawn from power generations, environmental control, and industrial processes such as design and sizing of piping systems, piping networks, pumps sizing and selection, and heat exchangers selection and performance evaluation. Students work individually and on group to conduct assignments and projects for integration of these components in the design of thermal systems.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                    |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------|
| <b>MENG 463</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Energy Conversion and Management</b>                            | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |                 |
| <b>Prerequisite(s): MENG 361</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                    |                 |
| An introduction to the basic technical and economic criteria for the design of efficient energy conversion systems, including traditional as well as alternative power systems. To discuss strategies for increased energy efficiency and more environmentally sound operation. To assess design alternatives and selection criteria based on long-term economic viability and overall energy management strategies.                                                                                                                                                                                                                                                                                                     |                                                                    |                 |
| <b>MENG 468</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Building Utilities I: HVAC and Mechanical Building Services</b> | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |                 |
| <b>Prerequisite(s): PHYS 110, ARCH 336</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                    |                 |
| This course explores building systems such as heating, ventilating and air-conditioning, water supply, drainage, fire safety, vertical transportation, building control systems, and assist integrating them into architectural design of building. Evaluates systems types, components, and installation and maintenance procedures as it relates to building regulations, systems impact on indoor comfort and environment.                                                                                                                                                                                                                                                                                            |                                                                    |                 |
| <b>MENG 491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Senior Design Project I</b>                                     | <b>(2:0:6)</b>  |
| <b>2 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |                 |
| <b>Prerequisite(s): Senior Standing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                    |                 |
| This is the first course of a capstone project that requires students to develop, design, and implement a solution to an engineering problem under the supervision of a faculty advisor. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical, economical, social and professional conduct. In this course, students will learn several personal and professional skills such as teamwork, time management, leadership and oral and written presentations.                                                                                                          |                                                                    |                 |
| <b>MENG 492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Senior Design Project II</b>                                    | <b>(4:0:12)</b> |
| <b>4 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |                 |
| <b>Prerequisite(s): MENG 491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                    |                 |
| The Mechanical Engineering Senior Design Projects Program coordinates the completion of the second half of the capstone design sequence required of all Mechanical Engineering seniors. Students apply engineering design methodology, using both analysis and synthesis, to solve open-ended problems. The range of design problems considered spans other engineering fields as well as non-engineering disciplines. At the end of the term, each student design team is expected to present information related to their project in both written and oral formats.                                                                                                                                                    |                                                                    |                 |
| <b>MENG 493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Special Topics in Mechanical Engineering</b>                    | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |                 |
| <b>Prerequisite(s): Dept. Approval</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                    |                 |
| Special up-to-date topic in one of the mechanical engineering streams of applied mechanics or thermal sciences.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                    |                 |
| <b>MEPM 511</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Project Management Fundamentals</b>                             | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |                 |
| The fundamentals of project management including: overview and concepts of project management (principles, body of knowledge, strategies); planning successful projects (defining, specifying, delivery options, scheduling, budgeting); implementing (organizing the team, work assignments, team building, effective leadership); executing (performance measurement, maintaining the schedule, adjustments/mid-course corrections, record keeping, status reporting, communications, managing conflict, time management); and closeout (performance measurement, maintaining the schedule, adjustments/mid-course corrections, record keeping, status reporting, communications, managing conflict, time management). |                                                                    |                 |
| <b>MEPM 512</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Engineering Contracts and Procurement</b>                       | <b>(3:3:0)</b>  |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                    |                 |
| This course presents fundamental concepts and techniques for project procurement. Students are introduced to the PMBOK Guide four-step procurement process and expected to develop an in-depth understanding of project contracting, negotiation, and procurement execution.                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                 |

- MEPM 513      Project Implementation and Performance      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MEPM 511**  
This course examines various topics related to project initiation and performance, Key performance indicators and criteria for measuring project success. The course includes using project performance software. Topics include: team building and management, organization structure, performance and success measures and earned-value technique. Other topics include: work breakdown structure, stakeholder management and project communication management.
- MEPM 514      Global Projects Management      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MEPM 511**  
This course provides an overview of the global project management process with specific emphasis on cross-cultural considerations, environmental factors, challenges, benefits & risks of global projects and leadership in global projects. The course also includes best practices that apply to global project management.
- MEPM 515      Project Scheduling      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MEPM 511**  
The course will prepare students to master project scheduling. Project scheduling methods are covered including: network construction, forward pass and backward pass calculations, activity slakes or floats calculations, critical path method (CPM), critical chain scheduling, Program Evaluation Review Technique (PERT), resource loading and resource leveling, Project crashing, and scheduling risk analysis. Students will have the opportunity to learn and use one of the project management software's for project scheduling.
- MEPM 516      Project Quality Management      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): MEPM 511**  
This course covers the fundamentals of quality control and management, quality principles, quality control techniques, quality control tools (Check Sheets, Histograms, Pareto Diagram, Cause and Effect Diagram, Scatter Diagram, Flow Process Charts and Control Charts), Control Charts for Variables and for Attributes, Lot-by-Lot acceptance sampling, acceptance sampling system, quantitative techniques, quality decision-making techniques. Examples and case studies on project quality management are presented and discussed from a wide variety of engineering discipline.
- MEPM 521      Project Cost Accounting and Finance      (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisite(s): None**  
This course covers the fundamentals of project cost accounting and finance. It reviews the fundamentals of accounting; examines cost accounting principles, applications, and the impact on profitability. It also examines the principles of project costing and covers the elements and introduces a framework for using an effective project cost system. Moreover, it introduces a framework for how projects are financed.
- MEPM 522      Financial Analysis and Decision Making      (3:3:0)**  
**3 Semester Credit Hours**  
This course covers some of the theory and practice of decision-making, as well as basic procedures for the application and interpretation of financial statement analysis. Topics covered in this context will include the concept of time value of money, project analysis and evaluation, cost of capital, stock valuation and capital budgeting.
- MEPM 531      Operation Research      (3:3:0)**  
**3 Semester Credit Hours**  
This course introduces the modeling techniques, theory and computation of linear programming. Topics include: The structure of linear programming, basic feasible solutions, simplex method, sensitivity analysis and linear programming duality. Linear programming modeling techniques for different applications: blending problem, inventory problem, minimum cost network flow problem, transportation problem, capital budgeting and fixed charge problems. Basic integer programming, dynamic programming and queuing theory methodologies and techniques will be introduced.



**MEPM 532      Engineering Management**  
**3 Semester Credit Hours**

**(3:3:0)**

The fundamentals of engineering management including: overview and concepts of engineering management (Introduction to engineering management, Management cycle includes (Planning, organizing, Leading and controlling. Different aspects of engineering managements will be introduced; like managing research and development, managing engineering design and managing production activities and operations. Market management and service activities for engineers will be clarified as well. Engineers as managers and Engineers code of ethics concepts and elements will be clarified. Finally, the globalization concept and the new challenges for engineering managers in the future will be discussed.

**MEPM 533      Information Systems for Project Management**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): MEPM 511**

This course will assist the student in understanding the challenges, opportunities and risks involved in information technology management. It introduces the management information systems foundations; current trends; MIS technology fundamentals; its applications to business functions and management practice and in an ethical manner. The impact of information technology on project management will be stressed since it affects the planning process. This course will also cover various system applications for specific project and business functions and their importance to today's manager.

**MEPM 541      Organization Behavior and Project Team Management**  
**3 Semester Credit Hours**

**(3:3:0)**

This course deals with human behavior as individual and as groups in organizations. Individual-level characteristics such as personality, attitudes and values, perceptions and judgment, motivations, career development and ethics are emphasized. Topics include group formation, development, structure, leadership, diversity, and dynamics, as well as the processes of communication, decision making, power, and conflict. Class sessions and assignments are intended to help participants acquire skills and analytic concepts to improve organizational relationships and project team management effectiveness.

**MEPM 542      Supply Chain Management for Project Managers**  
**3 Semester Credit Hours**

**(3:3:0)**

This course focuses on management and improvement of supply chain processes and performance. It will be valuable for project managers who would like to pursue a career in consulting or take a position in operations, marketing or finance functions in a manufacturing or distribution firm. We explore important supply chain metrics, primary tradeoffs in making supply chain decisions, and basic tools for effective and efficient supply chain management, production planning and inventory control, order fulfillment and supply chain coordination.

**MEPM 543      Risk Management for Project Managers**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): MEPM 511**

This course addresses fundamental issues, principles, and theory of project risk management and planning. It covers the quantitative and qualitative approaches to identifying, analyzing, assessing, and managing risks inherent to engineering projects. Other topics include risk response strategies and planning.

**MEPM 590      Thesis I**  
**3 Semester Credit Hours**

**(3:0:9)**

**Prerequisite(s): Department Approval**

A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. The thesis work must make a significant contribution to knowledge in the field of infrastructure engineering. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

**MEPM 591      Thesis II**  
**3 Semester Credit Hours**

**(3:0:9)**

**Prerequisite(s): MEPM 590**

Continuation of the Master Thesis I. A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. The thesis work must make a significant contribution to knowledge in the field of engineering project management. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

- MEST 100 Introduction to Islam in World Culture (3:3:0)**  
**3 Semester Credit Hours**  
 This course is the introductory course on Islamic Studies that is required for all majors. The course provides an introduction to the basic sources and historical contexts for the origins of Islam. (*Writing Intensive Course*)
- MGHR 301 Human Resource Management (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites: MGMT 301**  
 This course is a broad survey of key areas of human resources management, designed to provide line students wishing to become future supervisors and/or staff human resource specialists with a thorough understanding of the role of human resource management and the techniques available to management in making effective use of the human resources of an organization. The material is delivered from the perspective of HRM practice in both the UAE/GCC and internationally.
- MGHR 302 Compensation Management (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites: MGHR 301**  
 The Compensation Management course focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of pay, and incentive systems will be covered.
- MGHR 304 Human Resource Recruitment and Selection (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites: MGHR 301**  
 The Human Resource recruitment and selection course focuses on employee staffing, training, and development processes in organizations. Topics covered include human resource planning, job analysis for selection, recruitment methods, selection methods, assessing training needs, developing and delivering training, evaluating training outcomes, career planning and career management. Multiple recruitment and selection are investigated, and the course looks in detail at how these HRM elements are typically implemented in the UAE/GCC region.
- MGHR 390 Internship in Human Resource Management (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites: Senior Standing**  
 The Internship is designed for candidates on the undergraduate human resource management major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.
- MGHR 403 Cross Cultural and International Human Resource Management (3:3:0)**  
**3 Semester Credit Hours**  
**Prerequisites: MGHR 301**  
 The course introduces students to the field of international human resource management (I-HRM). Topics covered include: HR planning for international operations, the impact of national cultures on I-HRM, including standardized cultural models; Divergent and Convergent International-HRM practices; Integration versus Differentiation; International people-resourcing; HR cultural adaptation and adoption issues; Organizational Culture and I-HRM practice; comparative I-HRM studies featuring country comparisons. A number of contemporary I-HRM models are also discussed and utilized, with particular reference to I-HRM within and out of the UAE/GCC.

**MGHR 431 Employee Relations****(3:3:0)****3 Semester Credit Hours****Prerequisites: MGHR 301**

The course is aimed at students who are interested in pursuing a career in human resource management. The course focuses on legal and regulatory issues that affect employee relations in the GCC and UAE. Students will compare and contrast policies and procedures related to employee benefits and possible labor violations. They will also examine the core issues and applications of UAE/GCC labor regulations in managing employee regulations. At the end of this course, students will have developed an understanding of individual and collective labor relations in the UAE/GCC region and the appropriate methods of dealing with employee disputes.

**MGHR 444 HRM Project Based Internship****3 Semester Credit Hours****Prerequisite(s): MGHR 301**

The HRM project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the HRM specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

**MGHR 461 Diversity in Organizations****(3:3:0)****3 Semester Credit Hours****Prerequisite(s): MGMT 301**

The course enables the student to understand the role of diversity in human affairs particularly in the workplace. The course discusses and analyzes the role of diversity in the workplace, along with the procedures and practices of diversity management. Students will be able to apply these concepts to discussions of specific dimensions of diversity, including gender, race-ethnicity, sexual orientation, religion and disability. It also offers students the opportunity to develop their critical thinking on topics such as identity, relationships across difference and bias, and equality of opportunity in organizations around the world and how they relate to organizational issues of equality of opportunity, inclusion, and effectiveness.

**MGHR 462 Strategic Human Resource Management****(3:3:0)****3 Semester Credit Hours****Prerequisite MGHR 301**

This course will explore the contribution strategic Human Resource Management (HRM) makes towards the development and support of sustainable organizational strategies. On successful completion of this unit students will have the confidence to contribute to strategic decision-making in an HR context. This will be based on strong theoretical and applied foundations which will add value to an organization's HR function and role.

**MGHR 463 Negotiations in Organizations****(3:3:0)****3 Semester Credit Hours****Prerequisite MGHR 301**

This course focuses on developing the conflict management and negotiating skills of the learner. By the end of this course, the learner will be able to evaluate negotiation situations, strategize and plan upcoming negotiations, and engage in more effective negotiations. The course content will contextualize conflict management and negotiation to the local UAE/GCC environment, looking at cross-cultural influences. Consequently, to assist the learner in developing effective negotiation and conflict management skills, this course emphasizes experiential learning through student participation in a variety of exercises and role-plays.

**MGHR 464 Training and Development****(3:3:0)****3 Semester Credit Hours****Prerequisite MGHR 301**

The purpose of this course is to provide students a firm understanding of human resource training and development systems in today's business environment. A constant theme setting the foundation for this course will be on the various kinds of changes facing organizations, with a focus of UAE/GCC organizations, and how these changes relate to human resource training and development. Topics will cover assessing training needs, developing and delivering training, evaluating training outcomes, career planning and career management. Aspiring HR professionals will gain essential knowledge to effectively manage employee training and development systems in a variety of companies. The course looks in detail as to how these

HRM elements are typically implemented in the UAE/GCC region.

**MGMT 301 Principles of Management (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisites: Sophomore Standing**

This course presents essential management theories and concepts. It examines the nature of managerial work under a range of business models and under rapidly changing business conditions. Managerial functions and activities such as planning, organizing, leading, and controlling are examined in depth, and in the context of current organizational practices and scenarios.

**MGMT 401 Organizational Behavior (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite Senior Standing**

This course applies essential organizational behavior and leadership theories and concepts to organizational practices. It examines the complexities of human behavior in a range of organizational work settings under rapidly changing business environment. Individual personality, group/team dynamics, motivation, performance, communication, and leadership concepts are examined in depth, and in the context of current organizational practices.

**MIST 301 Introduction to Business Information Systems (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s): Sophomore Standing**

This course introduces fundamentals of computer hardware, software, networking, Internet and its technology components. The discussions are centered on the role of technology in contemporary business, and include basic relational storage concepts, with hands-on experience in building business database applications and web sites.

**MKTG 301 Principles of Marketing (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisites: ECON 103**

This course examines marketing principles, concepts, strategies, tactics, and analytical tools used by profit and nonprofit organizations to market ideas, products, or services to selected target groups. The course emphasizes how to promote, distribute, and price firm's offerings in dynamic economic, social, political, and international environments.

**MKTG 311 Sales Management (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisite(s): MKTG 301**

The course familiarizes students with marketing –sales interfaces including sales force role and capabilities, personal selling strategies, organizational relationships, and responsibilities of sales managers including training, motivating, and evaluating sales force.

**MKTG 312 Consumer Behavior (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisites: MKTG 301**

The Consumer Behavior course studies the purchasing behavior of consumers and the exchange process involved in acquiring, consuming and disposing of goods, services, experiences and ideas. In this course students learn to deal with an unstructured situation by the usage of a buyer decision – making model and buyer behavior concepts enabling them to identify important considerations and their possible resolutions.

**MKTG 313 Integrated Marketing Communications (3:3:0)**  
**3 Semester Credit Hours**

**Prerequisites: MKTG 301**

This course involves an in-depth study on the application of advertising and other forms of marketing communication with an emphasis on their role in marketing planning. The course includes a study of the identification of relevant data to analyze the marketing situation, the development of a product position, marketing and marketing communications objectives, creative strategy, media planning, and evaluation. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

**MKTG 332      Retailing and E-Commerce****(3:3:0)****3 Semester Credit Hours****Prerequisite MKTG 301**

Examination of retailing as a specialized economic and social institution within the distribution process and as it relates to society, fashion, and overall marketing activities. The planning and implementing of store and non-store (catalog, Internet) retail marketing strategies are addressed. Critical decision alternatives, variables, forces, and processes are considered from a managerial perspective. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

**MKTG 351      Marketing Research Techniques & Applications****(3:3:0)****3 Semester Credit Hours****Prerequisites: MKTG 301 & OPMT 311**

The course introduces the student to tools used in marketing research which are both quantitative and qualitative. Quantitative tools used are questionnaires, experiments, and conjoint analysis and qualitative tools used are focus groups and interviews. In the process, both primary and secondary research may be used. The course emphasizes problem formulation skills and takes a managerial perspective on research methodology that focuses on the kinds of decisions that each method can support. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

**MKTG 390      Internship in Marketing****3 Semester Credit Hours****Prerequisites: Senior Standing**

The Internship is designed for candidates on the undergraduate marketing major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

**MKTG 407      International Marketing****(3:3:0)****3 Semester Credit Hours****Prerequisite MKTG 301**

The course is multidisciplinary approach to international marketing from viewpoint of business management, examines major marketing issues affecting companies operating in a global environment. Students understanding of economics, political and cultural differences among nations as they affect marketing opportunities and operations and develop skills to identify international marketing opportunities.

**MKTG 412      Marketing of Services****(3:3:0)****3 Semester Credit Hours****Prerequisites: MKTG 301**

This course introduces the student to various aspects of Services Marketing. Specifically, the course examines the unique characteristics of services marketing, management and evaluation of service quality, techniques for service recovery and improvement, and services marketing in global environments. Particular emphasis is placed on analyzing the service encounter and the perception of service quality by service recipients and the service providers. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

**MKTG 444      Marketing Project Based Internship****3 Semester Credit Hours****Prerequisites: MKTG 301**

The marketing project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the marketing specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

**MKTG 471          Marketing Strategy**  
**3 Semester Credit Hours**

**(3:3:0)**

**Prerequisite(s): MKTG 301**

The course looks at the managerial aspects of marketing, including analyzing and developing marketing strategies and presents an overview in the study of the role of marketing strategy in modern businesses. Concepts related to the planning, execution, and evaluation of marketing strategies are emphasized. General topics include marketing planning and strategy, competitive analysis and advantage, and evolutionary markets among others. Specifically, the course topics include strategy definitions and related components: (i) objectives, (ii) synergy, (iii) scope, (iv) resource strategy, (v) strategy, and (vi) competitive advantage. The topics also include business unit strategy: (i) Porter's typology, (ii) Miles & Snow's typology, as well as marketing strategies: (i) generic, (ii) growth, (iii) leadership, (iv) target marketing and positioning, (v) 4P strategy, and (vi) international strategy. Marketing strategy throughout the life cycle is the central theme. A variety of teaching methods are utilized. Case studies introduce students to real-world issues and analytics. Lectures and readings deliver theoretical materials. Current events provide a link of theoretical materials to the business environment. Finally, class projects deliver leadership, analytical skills, and group decision-making skills. Links to the UAE environment will be made whenever possible.

**MSRE 511          Energy Conversion**  
**3 Semester Credit Hours**

**(3:3:0)**

Forms of energy. Energy needs and available sources of energy. Conversion of thermal energy into mechanical energy, including power, and heat engine cycles, internal and external combustion systems and turbines. Basic technical and economic criteria for the design of efficient energy conversion systems, including traditional as well as alternative power systems.

**MSRE 512          Energy Systems Modeling and Optimization**  
**3 Semester Credit Hours**

**(3:3:0)**

Review of renewable energy resources; solar energy and photovoltaic, wind energy, wave energy, biomass energy conversion, fuel cells and batteries. Hydroelectric power and geothermal energy. Modeling methodology including system conceptualization. Model construction and validation (computational accuracy). Model evaluation and calibration. Simulation of energy and environmental systems. Optimization techniques; Classical direct search-for-optimum methods, Golden Mean, Conjugate Gradients, Modified Newton Method. Methods for constrained optimization such as Lagrange Multipliers, Search methods, Linear and Dynamic Programming. Use of software packages.

**MSRE 513          Energy Economics**  
**3 Semester Credit Hours**

**(3:3:0)**

Comprehensive understanding of energy markets. Technological, cost, and environmental fundamentals of energy sources and environmental systems. Economic principles underlying the supply and demand for energy in a modern economy, through considerations of topics such as energy demand at the individual and economy-wide level, the supply of renewable and non-renewable energy resources, the rationale for energy policy and energy policy instruments. Advanced tools to analyze how energy and environmental policies affect the demand and supply of different types of energy.

**MSRE 514          Direct Research**  
**3 Semester Credit Hours**

**(3:0:9)**

**Prerequisite(s):** Department Approval

Under the guidance of an engineering faculty member, the Directed Research Project provides the student with a meaningful research experience. It requires that the student conduct a research topic or issue of significance to the field of renewable and sustainable energy.

**MSRE 515          Energy Efficiency and Management**  
**3 Semester Credit Hours**

**(3:3:0)**

Techniques and approaches adapted to improve the efficiency of energy generation, utilization, conversion, transport, storage and management. Energy audits. Energy conservation opportunities for efficiency improvements in different sectors: industrial, commercial, transportation and domestic. Economic evaluation of energy conservation opportunities using engineering economic formulas, simple pay-back analysis, and life-cycle cost models. Short- and long-term planning. Restructuring and Privatization. Models of electricity industry, contract issues, markets and transmission pricing. Deregulation of the energy market around the world. Electrical demand control and power factor correction. Load forecast and generation side management.

**(3:3:0)**

Introduction to solar radiation and some important related concepts like blackbody, solar spectrum, irradiance, irradiation, air mass and peak sun hour. Semiconductor and P-N junctions. PV cells properties, design, interconnection and module fabrication. Modules and arrays, stand-alone PV schemes with battery energy storage and grid-connected PV schemes.

**(3:3:0)**

Basic characteristics of wind, site characterization, statistical methods of wind analysis, wind resources assessment, fundamental principles of wind energy utilization, aerodynamics, mechanical and electrical design aspects. Wind machine technologies and wind turbines performance analysis. Wind farm planning and design. Wind power integration into the power systems, and the environmental impact of wind power utilization.

**(3:3:0)**

Fundamentals and applications of biofuels and bioenergy produced from biomass including processes of production, availability and attribution of biofuel/bioenergy production. Types of biomass raw material and methods of selection. First, second and third generation of biofuels. Methods of biomass conversion to fuel including thermochemical conversion, biochemical conversion and catalytic conversion. Production of hydrogen, biodiesel and microbial fuel cells. Environmental impacts of biofuel production. Economics and life-cycle analysis of biofuel. Value-added processing of biofuel residues.

**(3:3:0)**

Basic science and technology of water desalination to ensure sustainable water supply. Water production via desalination within the water-energy-cost nexus, evaluation of renewable energy- powered desalination processes, power-desalination cogeneration analysis, evaluation and applications of novel desalination systems, such as thermal desalination, membrane distillation and forward osmosis. Recent technological improvements for enhanced desalination processes, and fouling issues in current technologies. Assessing economic feasibility and the environmental impact of new desalination processes.

**(3:3:0)**

Introduction to solar energy, solar radiation; review of the basics of thermodynamics and heat transfer, power plant technologies; types of CSP systems including CSP parabolic trough systems, CSP dish technology, CSP Fresnel technology and solar tower; heat storage systems; hybridization; secondary use of CSP systems; operation and maintenance of CSP systems; power quality control and grid integration; CSP plant project planning: economic, social and environmental considerations and site assessment.

**(3:3:0)**

Green building initiatives, their origin, characteristics of a green building, certification of green buildings rating systems, criteria for rating. Policies and drivers that are leading to the more widespread uptake of green building technologies; green building codes, policies and planning from the past, present and future from around the globe. Integrated design: urban micro-climate design, passive and active architectural interventions.

**(3:0:9)**

**Prerequisite(s):** Department Approval

A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. Topic to be agreed in consultation with a supervisor. The project will be of suitable complexity for results to be published for an expert audience. This is the first part of the master thesis course. **Thesis Research Course**

**(6:0:18)**

**Prerequisite(s): MSRE 590**

Continuation of the Master Thesis I. A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. Topic to be agreed in consultation with a supervisor. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

**(3:3:0)**

The course examines principal aspects of organizational operations in various settings. It emphasizes the application of planning and decision-making activities associated with managing operations, with particular focus on production and service operations. It uses analytical models to describe key operations planning and control activities.

**(3:3:0)**

This is a course in applied operations research for business management. Business situations are represented by analytical mathematical models solved by the effective application of the methods mathematical programming and probabilistic process analysis. This course introduces operation research and management sciences techniques for supporting business management decisions. Specific mathematical programming and probabilistic topics include linear programming, transportation models, network flow models, decision analysis, inventory models, queuing models, and Monte Carlo simulation.

**(3:3:0)**

A comprehensive study of the concepts, processes, and strategies used in the development and management of global supply chains. Supply-chain management (SCM) is a systems approach to managing the entire flow of information, materials services from raw material suppliers through factories and warehouses to the final end-customer. SCM represents a philosophy of doing business that stresses processes and integration. Specific topics include global supply chain management, procurement, electronic commerce, information technologies, and logistics activities. This course will be taught through the use of textbook materials, outside readings, and case analysis.

**(3:3:0)**

Origin of the Earth and its shells. Composition of the Earth's crust and oceans, and their Geological characteristics. Primary and secondary structures. Internal Geological processes. Plate Tectonics and the relation of Geological events to it. External Geological processes. Stratigraphic columns, details of the Geological Time Scale and case studies of Geological ages and their palaeogeographic distribution. Climate. Important biological aspects.

**(1:1:0)**

The course provides an overview and history of the petroleum industry and petroleum engineering, including nature of oil and gas reservoirs, petroleum exploration and drilling, formation evaluation, well completions and production, surface facilities, reservoir mechanics, and improved oil recovery. It introduces the importance of ethical, societal, and environmental considerations and current events on activities in the petroleum industry. It also introduces students to professional society and university resources that aid career development.

**(3:3:0)**

This course covers external and internal forces in structures and/or machines, including conditions of equilibrium, systems of force, moments of inertia and friction. It also covers an introduction to Petroleum Rock Mechanics: Elasticity and Rock strength, Rock properties from logs, Stresses around wellbore and borehole failure criteria, Hydraulic Fracturing.

**(1:0:3)**

This course deals with the measurement of fundamental properties of reservoir rocks and fluids. Rock properties include porosity, irreducible water saturation, residual oil saturation, absolute permeability, capillary pressure, and relative permeability. Fluid properties include oil distillation, oil composition of one of oil fractions, oil density at room conditions and at high pressure and temperature conditions, oil viscosity at high pressure and temperature, surface and interfacial tensions,



flash liberation process, estimation of bubble-point pressure at reservoir temperature, and oil-formation-volume factor and solution gas/oil ratio at pressures below the bubble-point pressure.

**PENG 322      Drilling Fluids Laboratory**

**(1:0:3)**

**1 Semester Credit Hour**

**Corequisite(s): PENG 361**

This course deals with lab measurements of cement and mud properties. Mud preparation, mud rheology, filtration, wall building and resistivity, mud weight control, drilling fluid contamination test, oil well cementing experiment. (Writing Intensive Course)

## PENG 351      Reservoir Rock and Fluid Properties

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): CHEM 215**

Fundamental properties of reservoir rocks, namely porosity, compressibility, permeability, electric conductivity, fluid saturation, surface forces and capillary pressure, effective and relative permeability, and their applications in volumetric and flow calculations in petroleum reservoirs. Fundamental properties of reservoir fluids, namely oil, natural gas, and formation water. The course covers hydrocarbon phase behavior, equations of state, formation volume factors, gas solubility, types of vaporization, K values, and gas separation.

## PENG 361 Drilling Engineering I

**(2:2:0)**

**2 Semester Credit Hours**

**Prerequisite(s): PENG 231**

**Corequisite(s): CIEN 251**

This course introduces basic drilling techniques and drilling fluid properties. Topics include rock characteristics, drilling fluids, mud weight calculations, components of rotary drilling rig, drilling hydraulics, drilling bits, factors affecting rate of penetration, cementing operations.

**PENG 371      Petroleum Reservoir Engineering**

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): PENG 351**

This course deals with material balance (MB) techniques to estimate reserves. Topics include generalized MB equations, fluid drive mechanisms, selection of PVT data, water influx, flow through porous media, displacement of oil, fractional flow, and oil recovery by internal drive.

## PENG 381 Well Logging

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): PENG 351**

Logging is an essential operation to determine the key formation parameters, such as porosity, permeability, reservoir thickness, water saturation. All types of well logging techniques such as Logging while Drilling, Cased Hole Logging, and Production Logging are included. Logs are also run to gain information about well casing and cementing.

**PENG 382      Petroleum Production Engineering (Well Performance)**

**(4:4:0)**

**4 Semester Credit Hours**

**Prerequisite(s): CIEN 251**

This course covers basic well performance calculations necessary for the design and analysis of naturally flowing and artificially lifted wells. Topics include Inflow Performance Relationship (IPR), Tubing Performance Relationship (TPR), Flow line Performance Relationship (FPR), Choke Performance Relationship (CPR), Gas-Lift, Electric Submersible Pumps (ESP), and production forecasting.

## PENG 403 Petroleum Property Evaluation

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): PENG 371 and MENG 323**

Applications of reservoir engineering techniques, reserve calculations, decline curve analysis, rate of return calculations to project design and evaluation.

**(1:0:3)**

**Corequisite(s): PENG 485**

This course deals with the design aspects of oil displacement by another fluid in rock samples. It builds on the experiences of students obtained in lab measurements of individual reservoir rock and fluid properties in PENG 321 to create an integrated lab measurement of all properties needed to analyze oil displacement by a displacing fluid. The displacing fluid can be chosen to study the relative permeability and displacement efficiency of water flooding, gas flooding, or any enhanced oil recovery fluids (acidic water, microbial water, polymer solution, or steam) using cores, fractured cores (sand packs and glass beads may be considered as alternatives) in one-dimensional geometry or packed layers in two-dimensional geometry.

**(3:3:0)**

**Prerequisite(s): CIEN 251 and PENG 382**

This is an elective and a senior level course on production engineering. The course deals with analysis and design of surface piping and storage facilities of crude oil and natural gas. Topics include fluid flow and pressure losses in pipes, pipeline design, selection and sizing liquid pumps and gas compressors, corrosion in pipes, other transportation methods, and storage of petroleum and its products.

**(3:3:0)**

**Prerequisite(s): PENG 382**

This is an elective and a senior level course on production engineering. The course covers nature of fluids produced from oil and gas reservoirs, phase behavior of water/hydrocarbon systems, quantitative prediction of water content in light hydrocarbon systems, oil water separations, oil-water emulsions, treatment of oil field waters, oil and gas separations, flash calculations, separator sizing and design, heater-treater design, oil skimmers selection and design.

**(2:2:0)**

**Prerequisite(s): PENG 361**

This course deals with additional topics in drilling engineering, namely casing design and landing, directional and horizontal drilling, drilling problems and fishing operations, well control problems and solutions.

**(3:3:0)**

**Prerequisite(s): PENG 101**

**Corequisite(s): PENG 381**

Oil distribution in the world and in the UAE: geology of reservoirs, which includes the formation of reservoir rocks, cap rocks, source rocks and the environments of depositions: petrophysical parameters of parameters of reservoir fluids: oil field waters, crude oil and natural gas; reservoir conditions: pressure, temperature and their effects on oil maturation, migration and accumulation: oil generation. Oil migration: types of oil traps: methods of exploration.

**(3:3:0)**

**Prerequisite(s): PENG 371 and MATH 203**

This course covers fundamental concepts of reservoir simulation to model single-phase flow in petroleum reservoirs. Topics include reservoir engineering concepts, mathematical concepts, derivation of reservoir flow equations, finite difference approximations, and their solutions, and applications to predict reservoir performance.

**(3:3:0)**

**Prerequisite(s): PENG 473**

This course covers advanced topics in reservoir simulation. These include reservoir fluid flow equations in multiphase, multidimensional flow, up-scaling of rock properties, pseudo functions, vertical equilibrium, analysis of data for consistency, history matching, and applications to field cases.

**(3:3:0)**

This course introduces the properties of natural gas, covers flow of gas in reservoir, wellbore and surface pipelines. Based on such knowledge, students are able to run nodal analysis for gas production system design. This course also covers gas compression and gas operation issues.

**(3:3:0)**

This course covers reservoir characterization by pressure test analysis. Topics include fluid flow equations in porous media under transient and pseudo-steady state flow conditions, pressure buildup and pressure drawdown tests, average reservoir pressure, type curve matching, well testing of heterogeneous reservoirs, pressure derivatives analysis technique, multiple well testing, and test design and instrumentation.

**(3:3:0)**

Topics include well completions, perforations, wellbore damage sources and detection, hydraulic fracturing, fracturing fluids, acid/rock interactions, and acid treatment of oil wells, design and evaluation of treatments, evaluation by nodal system analysis.

**(3:3:0)**

This is a senior level course on reservoir engineering series courses. The course covers the reservoir engineering aspects of water flooding. Topics include introduction to reservoir forces and concepts of surface and interfacial tension, wettability, capillary pressure, relative permeability; trapping and mobilization of residual oil; fluid distribution and frontal displacement theory, concept of mobility ratio, flood patterns and areal sweep efficiency considerations, pattern injection rates and pressures; characterization of reservoir heterogeneity; vertical and volumetric sweep efficiency, waterflood performance prediction models; waterflood pilot test; designing, monitoring and evaluating a waterflooding operation; water for water flooding, its sources and treatment.

**(3:3:0)**

This course covers chemical and thermal method of EOR. Specific topics include interfacial tension, entrapment and mobilization of oil in porous media, residual oil, miscibility, adsorption at solid/liquid interfaces, surfactants and micro-emulsions, miscible gas flooding, polymer flooding, thermal methods, and the effect of reservoir heterogeneity.

**(3:0:9)**

The Senior Design Project entails team work involving development of a two-semester long workflow that includes preparing and presenting a proposal for a major design work. Throughout the course students are required to develop, design and implement a solution to one of the petroleum engineering problem under the supervision of a faculty advisor. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical, economic, social and professional conduct.

**(3:0:9)**

Continuation of phase (1).

**(3:3:0)**

A specific topic in Petroleum Engineering that is not covered in other program courses is presented in a course format.

**(3:3:0)**

**(3:3:0)**

**(3:3:0)**

**(3:3:0)**

**(3:3:0)**

**(1:0:3)**

**(3:3:0)**

**(1:0:3)**

This course is intended to accompany Physics 220. It includes experiments on electricity, magnetism and RLC circuits.

Particular emphasis is placed on three aspects of experimentation: laboratory technique, data analysis (including the treatment of statistical and systematic errors) and written communication of experimental procedures and results.

## PHYS 222 University Physics III

**(3:3:0)**

**3 Semester Credit Hours**

**Prerequisite(s): PHYS 220**

**Corequisite(s):** PHYS 223

This course covers important theoretical bases and experiments that support the development of modern physics with applications in various technological and scientific fields. Topics include physical optics, thermodynamics, special relativity, quantum physics, atomic and molecular physics and nuclear physics. The course's focus is on applications that have led to key discoveries.

## PHYS 223 University Physics III Lab

**(1:0:3)**

**1 Semester Credit Hour**

**Prerequisite(s):** PHYS 221

**Corequisite(s):** PHYS 222.

This course is intended to be taken with Physics 222. It includes experiments on optics, thermodynamics and atomic physics. Particular emphasis is placed on three aspects of experimentation: laboratory technique (including both the execution and the documentation of an experiment); data analysis (including the treatment of statistical and systematic errors, as well as computer-aided analysis of experimental data); and written communication of experimental procedures and results.

**POLI 100**                      **Contemporary Global Issues**

**(3:3:0)**

**3 Semester Credit Hours**

This course gives students a broad, interdisciplinary, comprehensive, and balanced, introduction to key debates, theories, major trends, and topics, relating to Globalization and global developments.

**POLI 101**      **Politics of Scarcity**

**(3:3:0)**

**3 Semester Credit Hours**

The problems of scarcity and security are as much political as they are economic or technological. This course identifies the political aspects of global economic exchange and distribution, flows of labor and capital, and international cooperation, global security and conflict.

## POLI 102 State and Society in the UAE

**(3:3:0)**

**3 Semester Credit Hours**

The course traces the history of the UAE, the establishment of the federation and the development of the UAE as a nation with significant global impact. It covers contemporary life, the economy, society, population, political system, social customs and traditions, and current changes.

## PSYC 100 Introduction to Psychology

**(3:3:0)**

**3 Semester Credit Hours**

This course provides an overview of major areas in the field of psychology including: the history of psychology, psychology research methods, organization of human brain and biological basis of behavior, sensation, perception, learning, language, intelligence, emotion, motivation, developmental psychology, personality theories /assessment, stress, abnormal behavior and therapies, social psychology.

## PSYC 102 Self and Society

**(3:3:0)**

**3 Semester Credit Hours**

Study of how society shapes us as individuals and how we as individuals shape society. Explores how we experience life as individuals and as members of cultural systems that shape our personalities, behavior, and perceptions of the world.

**SOCI 100      Introduction to Sociology**

**(3:3:0)**

**3 Semester Credit Hours**

Introduction to basic sociological concepts. Examines aspects of human behavior in cultural framework including: individual and group interaction, social mobility and stratification, status and class, race and gender relations, urbanism, crime and criminology, and social change and reform.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                         |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------|
| <b>SOCI 101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Contemporary Social Issues</b>                       | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                         |                |
| The course is an introduction to basic sociological concepts and examines aspects of human behavior in a cultural framework including: individual and group interaction, social mobility and stratification, status and class, race and gender relations, urbanism, crime and criminology, and social change and reform.                                                                                                                                                              |                                                         |                |
| <b>STAT 100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Introductory Probability and Statistics</b>          | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                         |                |
| This course introduces students to the basics of probability theory and statistical inference with examples and applications in sciences. At the end of this course, students will acquire the necessary quantitative competency in the program.                                                                                                                                                                                                                                      |                                                         |                |
| <b>STAT 210</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Probability and Statistics for Life Sciences</b>     | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours - Prerequisite(s): STAT 100</b>                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                         |                |
| This course involves the use of probability and descriptive and inferential statistical techniques in interpreting biological data.                                                                                                                                                                                                                                                                                                                                                   |                                                         |                |
| <b>STAT 346</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Probability for Engineers</b>                        | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours - Prerequisite(s): MATH 114</b>                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                         |                |
| The course introduces principles of statistics and probability for undergraduate students in Engineering. The course covers the basic concepts of probability, discrete and continuous random variables, probability distributions, expected values, joint probability distributions, and independence. The course also covers statistical methods and topics including data summary and description techniques, sampling distributions, hypothesis testing, and regression analysis. |                                                         |                |
| <b>UAES 200</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Survey of United Arab Emirates Studies</b>           | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                         |                |
| This course presents the UAE from multiple perspectives in an attempt to expose the students to the distinct qualities of the UAE. The purpose of this course is to give students a broad, interdisciplinary, and comprehensive introduction to key features and issues in the UAE's historical, cultural, political and economic landscapes. <b>(Writing Intensive Course)</b>                                                                                                       |                                                         |                |
| <b>UNIV 100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>University First-Year Transition</b>                 | <b>(1:1:0)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                         |                |
| This course is designed to assist students with the transition from secondary school to university life. Students in this course transition to university life by focusing on academic adjustment, by developing decision-making skills, and by learning about services and opportunities for involvement.                                                                                                                                                                            |                                                         |                |
| <b>UNIV 200</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Innovation, Entrepreneurship, and Sustainability</b> | <b>(3:3:0)</b> |
| <b>3 Semester Credit Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                         |                |
| The course aims at equipping next generation of leaders with an innovative and entrepreneurial mindset and related core skills. The course explains key qualities of entrepreneurs and analyses how entrepreneurs make utilize innovation in creating new ventures. Design-thinking and business model innovation are key frameworks that underpin the course. <b>(Writing Intensive Course)</b>                                                                                      |                                                         |                |
| <b>UPAS 100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>University Preparation for Academic Success</b>      | <b>(1:1:0)</b> |
| <b>1 Semester Credit Hour</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                         |                |
| The focus of this course is on the development and application of college-level study skills, personal success strategies, and the use of campus resources that enhance individual student achievement. Designed for matriculated students who are at risk of being placed on Academic Probation 1 due to having a CGPA of less than 2.0. Graded as Satisfactory/Unsatisfactory.                                                                                                      |                                                         |                |



## GET IN TOUCH

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